Рахимов Р.Х.
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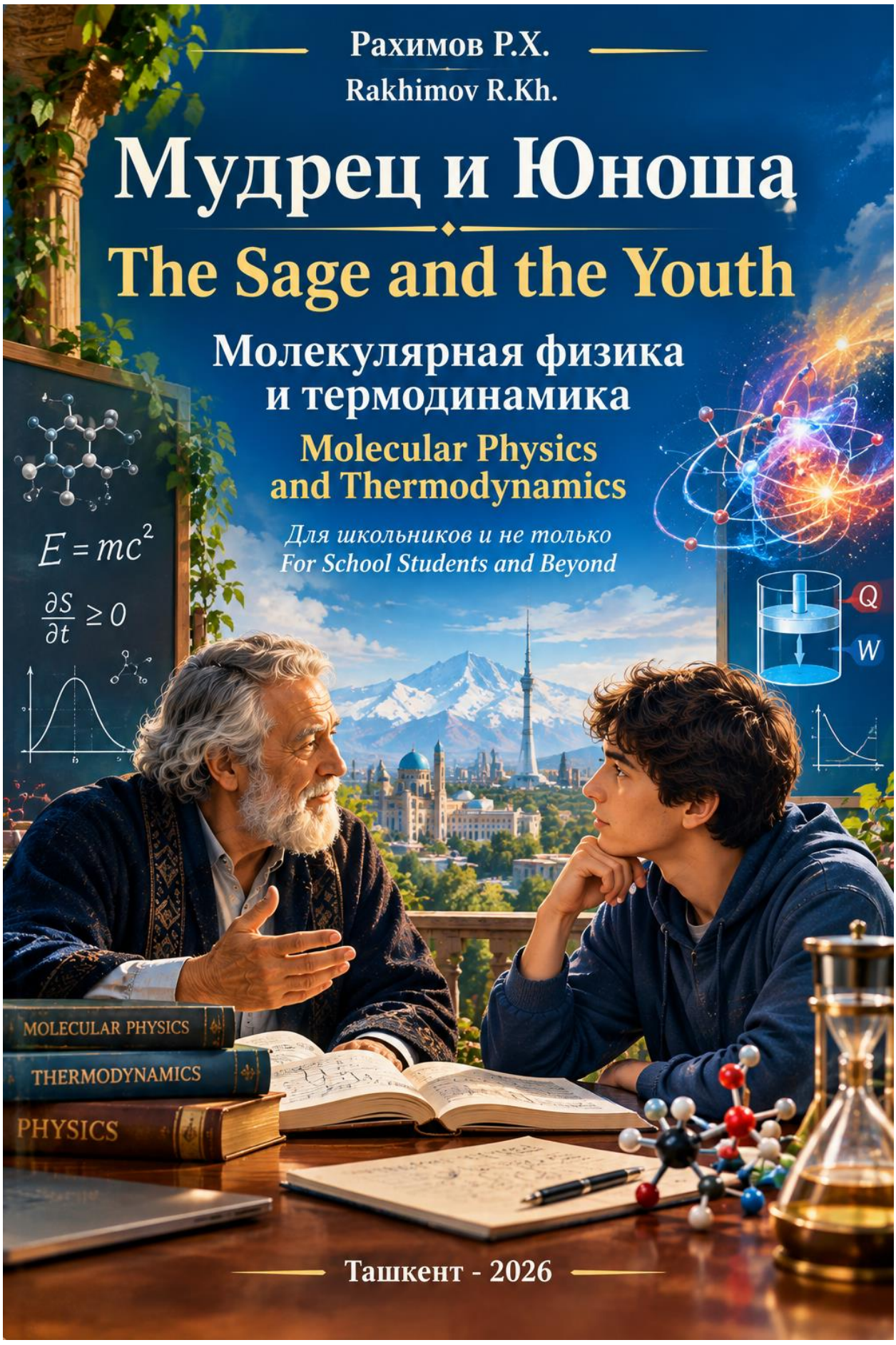
Мудрец и Юноша

The Sage and the Youth

Молекулярная физика и термодинамика

Molecular Physics and Thermodynamics

Для школьников и не только
For School Students and Beyond



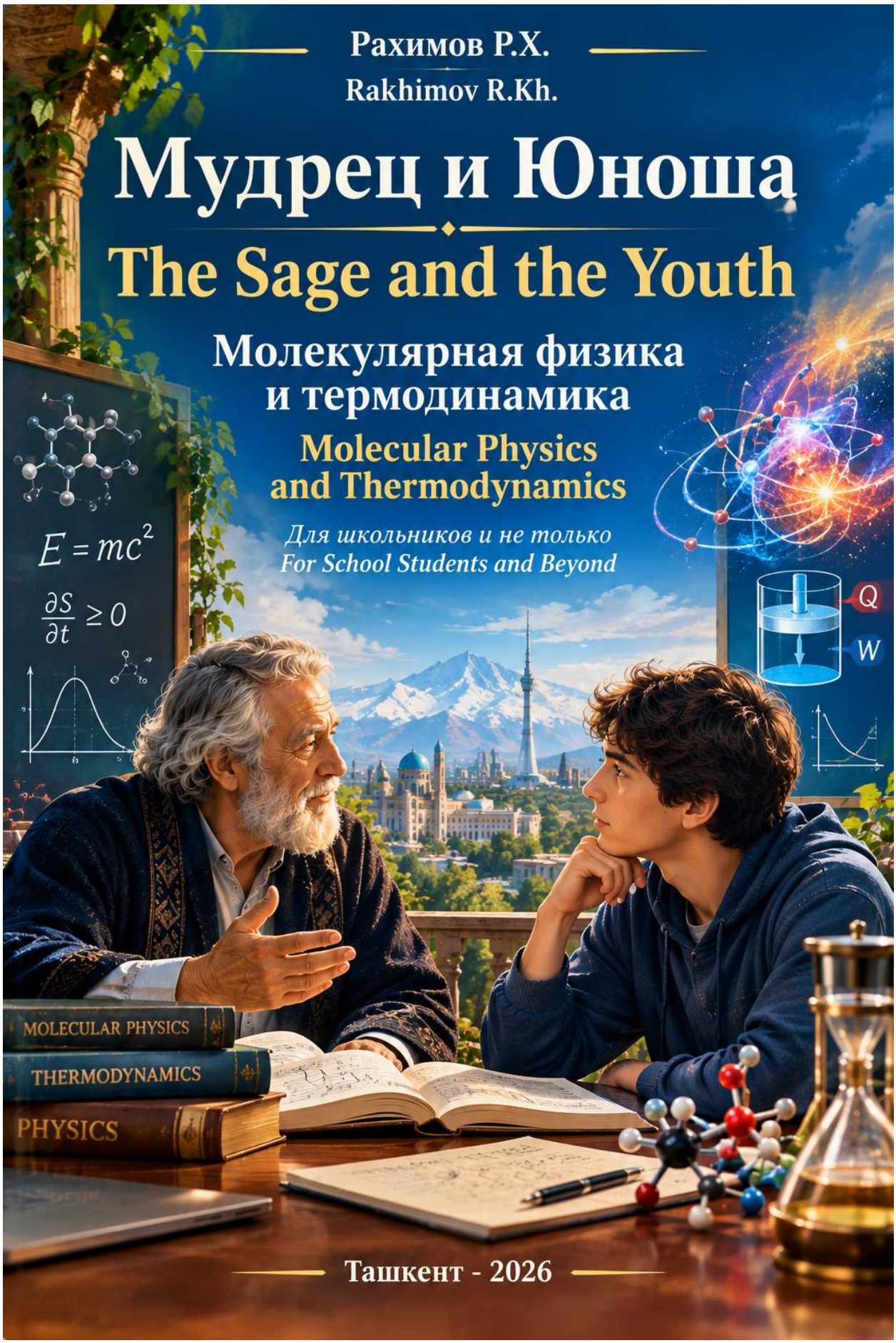
Ташкент - 2026

The Sage and the Young Man

Molecular physics and thermodynamics

For schoolchildren and not only

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Preface

Conversations about heat, energy, and time in simple terms

This book is about one of the most practical and philosophical areas of physics. Practical because steam engines, engines, refrigerators, dryers, power plants, and a significant part of modern technology are built on the laws of heat. Philosophical because concepts that changed the worldview emerged from reflections on heat: conservation of energy, entropy, direction of time, quantum constant.

The story is told in the form of a dialogue between the Sage and the Young Man, as in other books in the series. The Young Man is not shy about asking "why exactly this way?" He checks the formulas with numbers and finds weak points in the reasoning. The Sage does not provide ready-made answers but leads the interlocutor to them through questions. We wanted the school-reader to not just memorize the formula but to see how a person arrived at it — and where they made mistakes along the way.

Who is this book for?

For high school seniors and first-year students, for physics teachers, and for anyone who wants to understand how thermal phenomena work. Knowledge of school-level physics and mathematics at the level of exponents, logarithms, and simple formulas will be required. Everything else is explained along the way; at the end of each chapter, there are problems with answers, and we recommend solving them rather than just reading.

How do we distinguish knowledge from a hypothesis?

In science, it is important not only what is **claimed but also how thoroughly** it has been tested. Therefore, in the book, different statements are marked differently. Colored blocks with labels indicate the basis on which we say what we say:

Label	Value	How to read this
[UV]	Established fact	Repeatedly verified by experience, it is part of accepted science.
[G]	Hypothesis	A reasonable assumption, yet unconfirmed or disputed.
[GA]	Author's hypothesis	The idea developed by the author of the book and his colleagues requires independent verification, and the section specifies how to verify it.
[OD]	Justified addition	Calculation or conclusion made along the way based on established laws; not a citation from a source.
[ERROR]	Delusion / mistake	A common or historical misconception and how it was corrected.
[NI]	Not independently confirmed	A tradition, legend, or fact that we could not verify from the primary source.

We adhere to a simple rule: a **hypothesis should not look like a fact, and a fact should not be hidden behind the word "possibly."** The reader has the right to know where established physics ends and the search begins.

The peculiarity of this book: limits and speed

The main idea of the book is simple. Any process has two questions. **The first: what is possible** and what is the limit? This is addressed by thermodynamics: the three laws, entropy, **free energy. The second:** how fast and by what means? This is addressed by kinetics: barriers, diffusion, energy transfer, and radiation. Many technologies—drying, ceramic synthesis, water decomposition—are constrained by the second question: the limit is known, and the gain is achieved by how closely one can approach it. We paid special attention to this distinction in chapters 9–13.

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Introduction

Two roads to warmth

> *The young man brought two items to the bench in the yard: a thermometer and a school physics textbook, opened to the page "Molecular Physics." "It says two things here," he said. "That heat is the movement of molecules. And that heat transfers from hot to cold, but never the other way around. One doesn't seem to follow from the other. Why are they on the same page in the textbook?"*

1. Two ways to look at the same thing

Sage: You have posed a question that science has been answering for two centuries. There are two ways to describe thermal phenomena.

The sage: The **first is phenomenological, it** is called thermodynamics. **It does** not look inside the substance. Pressure, volume, temperature, heat, work are measured; general laws are found that are true for a steam engine, a star, and a refrigerator. Its strength lies in the fact that it does not depend on the structure of the substance and therefore is almost timeless.

The sage: The **second is molecular-kinetic.** It starts with molecules: it calculates their speeds, collisions, and probabilities, and derives from this pressure, temperature, heat capacity, and viscosity. Its strength lies in explaining "why."

Young man: Which one of them is correct?

Sage: Both. They describe the same thing from different sides, and the most beautiful parts of the book are the places where they meet. When Boltzmann linked entropy, conceived in thermodynamics, with the probability of molecular states, the two roads merged. And when Planck and Einstein tried to reconcile the ends in heat capacity and radiation, quantum physics was born.

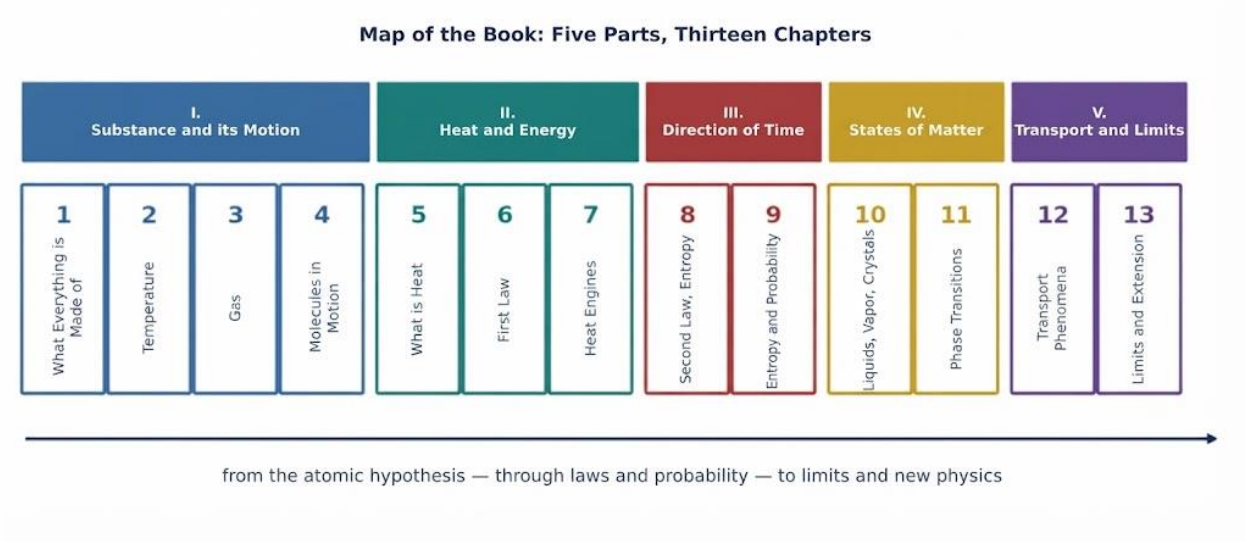


Fig. 1. Book map: five parts and thirteen chapters.

2. Two questions for any process

Sage: If you remember only one thought from the book, let it be this. To any process — cooling tea, drying laundry, melting ice, decomposition of water — two different questions can be asked.

Question	What do we respond with?	Examples
What is possible? What is the limit?	Thermodynamics: three laws, entropy, free energy	Minimum energy for the evaporation of one kilogram of water; maximum efficiency of the machine; minimum work for the decomposition of water.
How quickly? By what means?	Kinetics and transfer: barriers, diffusion, radiation	Drying rate; sintering time of ceramics; moisture diffusion rate in the material.

Young man: So, thermodynamics says "yes or no," but it doesn't answer "when"?

Sage: Exactly. The first principle will not tell us how quickly the laundry will dry, and the second will not tell us how long it will take for the tea to cool down. But once the limit is reached, no technology will improve the result. We must strive to approach it: faster, with fewer losses. This path is the work of an engineer and a scientist, and we will return to it in every chapter.

3. How each chapter is structured

- **Dialogue. The** Sage and the Young Man discuss the topic step by step. The Young Man counts the numbers himself, so they can be verified.
- **Blocks with labels** indicate the status of statements (see the preface).
- **The** drawings are calculated, constructed based on data and formulas; the captions indicate where the drawing is schematic or the data is rounded.
- **Problems with** answers at the end of the chapter.
- **A** bridge to the next chapter: a question that remains unanswered.

4. What is useful to know in advance

The school course of physics (pressure, work, energy, mechanics) and mathematics (exponents, roots, logarithms). Formulas are presented before your eyes, and the necessary numbers are collected in a table of constants at the end of the book. If something is unclear, that is a good sign: it means you have encountered an "extra question," from which discoveries begin.

> **[OD] Justified Addition — on "D Students" and "A Students"**
> The series "The Sage and the Young Man" is based on the idea that the value of a student is determined not only by quick answers but also by the ability to stop where the rule has ceased to work and ask "why." This does not negate the benefits of excellent study: both types of thinking are needed in science. In this book, the Young Man is precisely such a "poor student" in a good sense: he asks questions where others have already agreed.

Sage: Ready? Then let's start with the very first question: what is everything made of, and why did it take two thousand years to prove the existence of atoms?

CHAPTER 1

What is everything made of?

Atomic hypothesis: from Democritus's conjecture to Perrin's experiments

Morning. On the table, there is a piece of school chalk and a glass of water. The Sage extends the chalk to the Young Man.

The sage: Break it in half.

Young man: Ready.

Sage: Now half — again in half. And again. Do you think this process can continue infinitely?

Young man: With fingers — no, soon there will be dust left. But if you take a sharp knife, then maybe you can... Or not? I don't know.

Sage: This "I don't know" is where our book begins. You posed a question that people have pondered for twenty-five **centuries: can matter be divided endlessly, or is there a final, indivisible piece somewhere?**

1. Greek conjecture

The sage: In the 5th century BC, Greek thinkers Leucippus and his student Democritus decided that one cannot divide endlessly. Sooner or later, you will reach the smallest particles that can no longer be divided, and they called them **atoms** — in Greek "indivisible." Everything in the world, Democritus said, consists of atoms and void, and the differences between things are differences in the shape and arrangement of atoms.

Young man: So, he knew about atoms?

The sage: **He suspected** — and that is quite different. He had no measurements or experiments, only reasoning. His opponent, Aristotle, thought otherwise: matter is continuous, like water in a river, and consists of four elements — earth, water, air, and fire. Aristotle's authority was immense, and for almost two thousand years his view was considered correct.

The sage: But the idea did not die. In the 1st century BC, the Roman poet Lucretius wrote the poem "On the Nature of Things," where he presented the doctrine of atoms. And he provided a beautiful argument: look at the dust particles in a sunbeam — they are forever dancing and colliding. This, he said, is a visible echo of the invisible movement of atoms.

> **DELUSION**
>
> “Dust particles dance in the beam due to the impacts of atoms.” Beautiful, but incorrect. Dust particles that are visible to the naked eye are too large, and the impacts of individual molecules do not move them. They dance due to air currents: it never remains still because it is warmer in some places and colder in others. The tremor from molecular impacts exists, but it is thousands of times smaller, and it can only be seen under a microscope. Interestingly, Lucretius grasped the essence, **although** the example he provided was unsuccessful.

2. Chemists discover atoms "through the back door"

The sage: Further in history — a long pause. The idea of the Renaissance survived in the 17th century: the Englishman Robert Boyle explained the properties of matter as "corpuscles" — small particles — in his 1661 book "The Skeptical Chemist." But the real breakthrough came from chemistry. Chemists weighed substances before and after reactions, and an astonishing discovery was made.

The sage: Antoine Lavoisier showed in 1789 that in a chemical reaction, the total mass does not change. Joseph Proust established in the early 19th century that substances combine in strictly defined proportions: water is always 1 part hydrogen to 8 parts oxygen by mass, no matter where you take it from — from a river or from a kettle.

Young man: So what? Why does this speak about atoms?

Sage: Think. If a substance were solid, like dough, it could be combined in any proportions: a little more, a little less. But if it consists of identical "bricks," then only whole bricks can combine — one oxygen atom gets two hydrogen atoms, and "one and a half" does not exist. The Englishman John Dalton stated this directly between 1803 and 1808: each chemical element has its own atoms, identical to each other, while different elements have different atoms, with different masses.

> **HISTORICAL FACT**
>
> Dalton did not know how many atoms were in a water molecule and decided that the simplest was one atom of hydrogen and one atom of oxygen, that is, HO. He was mistaken: in fact, it is H_2 O. He was also wrong in another respect: he did not accept the law of the French chemist Gay-Lussac that gases react in simple volumetric ratios (two volumes of hydrogen and one volume of oxygen produce two volumes of water vapor). An Italian, Amedeo Avogadro, helped clarify this. In **1811, he proposed that equal volumes of any gases under the same conditions contain an equal number** of molecules, and that the molecules of simple gases—hydrogen, oxygen—consist of two atoms. The idea was correct, but it was not accepted for almost fifty years. Only in 1860, at a congress of chemists in Karlsruhe, did the Italian Stanislao Cannizzaro convince his colleagues.

Young man: Fifty years — to recognize the right idea?

Sage: Unfortunately, this happens. Not because scientists are foolish, but because Avogadro's idea was not confirmed by anything other than convenience. Direct, independent evidence was needed. It was to be found.

3. How small is an atom? Oil film

Sage: Let's try to estimate the size of a molecule ourselves. An old experiment: a drop of oil thrown onto water spreads into a thin film. Benjamin Franklin observed this in 1774 at a pond: a teaspoon of oil calmed the waves over an area of about half an acre. However, he did not calculate the thickness of the film, although he could have.

Sage: Let's do this for him. We'll take a drop with a volume of 1 cubic millimeter, that is, 10^{-9} m³. Let it spread into a film with an area of 1 m². What is the thickness?

Young man: The volume is equal to the area multiplied by the thickness. Therefore, the thickness h = V / S = 10^{-9} m³ / 1 m² = 10^{-9} meters. This is one billionth of a meter! One nanometer.

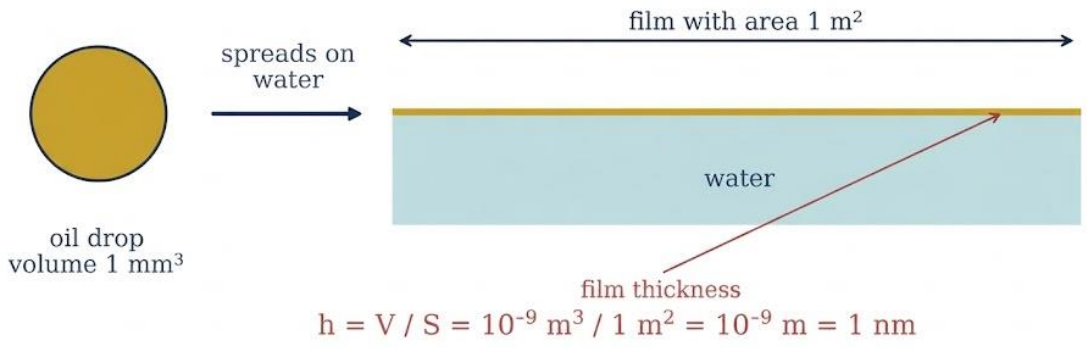


Fig. 1.1. A droplet spreads into a film: the thinner the layer, the smaller the "brick" from which it is composed.

Sage: Now the main reasoning. The film cannot be thinner than one molecule — otherwise, the molecules would have to be "cut." Thus, the oil molecule is no larger than a nanometer. In 1890, Lord Rayleigh conducted such an experiment carefully, with oleic acid dissolved in alcohol, and obtained a layer thickness of about 1–2 nm. Modern measurements show: the length of the oleic acid molecule is on the order of 2 nm.

> **LET'S REMEMBER**
> This is an example of an important technique **in physics: order of magnitude estimation. We** do not know the exact volume of the droplet and the area of the spot, but even a rough estimate gives the correct order of size — billionths of a meter.

4. How many molecules are in a drop of water?

Sage: Now we can move from sizes to counting. Chemists have long devised a convenient unit — the **mole.** Historically, it is the amount of substance that contains as many particles as there are atoms in 12 grams of carbon-12. The number of particles in one mole is **called Avogadro's number:**

$$N_A \approx 6.022 \cdot 10^{23}\ \text{mol}^{-1}$$

Sage: This number is so huge that it's hard to imagine. Let's estimate. The mass of one mole of water is 18 grams (16 for oxygen and 2 for two hydrogen atoms). A drop of water is about 0.05 grams. How many molecules are in it?

Young man: First, how many moles: $0.05 / 18 \approx 2.8 \cdot 10^{-3}$ moles. Then I multiply by Avogadro's number: $2.8 \cdot 10^{-3} \cdot 6.02 \cdot 10^{23} \approx 1.7 \cdot 10^{21}$ molecules.

Sage: That's right. It's almost two sextillion. A rough estimate of the number of grains of sand on all the beaches of Earth is about 10^{19} . So in a **single drop of water, there are more than two hundred times as many molecules as there are grains of sand on all the beaches** of the planet. Now let's find the size of a single water molecule from the density.

Young man: How?

Sage: The density of water is 1000 kg/m³. One mole is $18 \cdot 10^{-3}$ kg. Therefore, the volume of one mole: $V = 18 \cdot 10^{-3} / 1000 = 1.8 \cdot 10^{-5}$ m³, that is, 18 cm³. Let's divide by the number of molecules:

$V_1 = 1.8 \cdot 10^{-5} / 6.02 \cdot 10^{23} \approx 3.0 \cdot 10^{-29}\ m^3$

Young man: If we assume that each molecule occupies a small cube, then the side of the cube is the cubic root of $3 \cdot 10^{-29}$... This is approximately $3 \cdot 10^{-10}$ meters. Three tenths of a nanometer!

The sage: Exactly: the diameter of a water molecule is about 0.3 nm. Compare it with oil: fat molecules are long chains, which is why the film made from them is thicker. The famous comparison belongs to physicist Richard Feynman: if you enlarge an apple to the size of the Earth, then the atoms in it will be about the size of the original apple. A precise calculation gives more of a cherry, but the order is the same.

> **TRY IT YOURSELF**
>
> Check it on the calculator yourself. How many molecules are in a glass of water with a volume of 200 ml? (The mass is 200 g, which is 200 / 18 ≈ 11 moles.) Now imagine that you poured this glass into the ocean and the molecules mixed perfectly. How many molecules from your glass will end up in the next glass that someone scoops from the ocean? To do this, estimate the number of glasses in the ocean: the volume of the ocean is about $1.3 \cdot 10^9$ km³ = $1.3 \cdot 10^{18}$ m³, while a glass is $2 \cdot 10^{-4}$ m³. Hint: the answer is on the order of a thousand molecules.

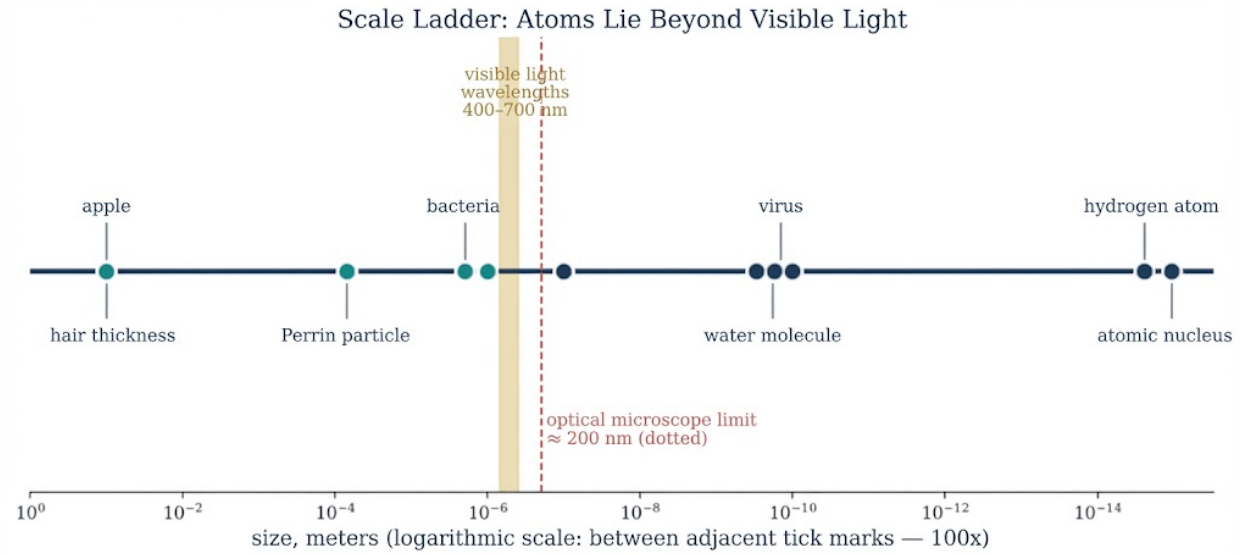


Fig. 1.2. A ladder of sizes: from an apple to an atomic nucleus. Atoms and molecules are smaller than the wavelength of light — therefore, an optical microscope will not show them.

Young man: I understand why no one has seen atoms! They are smaller than the wavelength of light, and light, as we learned in the optics book, cannot "show" details smaller than its wavelength.

Sage: Exactly. The limit of a conventional optical microscope is about 200 nanometers; it was established by Ernst Abbe in 1873. Atoms are a thousand times smaller. Therefore, atoms cannot be seen, and perhaps they do not exist? Many serious scientists thought so.

5. The dispute over the reality of atoms

> **DELUSION**
>
> "Atoms are just a convenient invention of chemists, and since they cannot be seen or counted, they do not exist." Thus reasoned the Austrian physicist Ernst Mach and the German chemist Wilhelm Ostwald, a future Nobel Prize laureate, at the end of the 19th century. This was not foolishness, but a position: science, they

> believed, should only speak about what can be measured. Ostwald built a theory where "energy" replaced atoms. Today it is clear that they were wrong in their conclusion, but right in their demand: prove **it through experimentation.**

The sage: And the experience was found. It came from a strange place: from a botanist.

6. The tremor in a drop of water

The sage: In 1827, Scottish botanist Robert Brown examined the pollen of the Clarkia plant under a microscope in a drop of water and noticed that the tiny grains inside the pollen were constantly shaking and wriggling. At first, he thought this was a sign of life. But then he checked: particles of coal, glass, and rocks also shake — that is, inherently "non-living" particles. Therefore, the cause is not life. But what is it?

Young man: Maybe the water is flowing? Or is the microscope lamp heating the drop?

The sage: These versions were tested and rejected: the trembling does not stop, no matter how long you wait, and is not related to the flow of the liquid. An explanation was found only almost eighty years later. In 1905, twenty-six-year-old Albert Einstein, and independently in 1906, Polish physicist Marian Smoluchowski showed: a particle in a liquid receives impacts from invisible molecules from all sides. If there were few molecules, the impacts would be rare and random, and the particle would jerk noticeably. If the substance were solid, there would be no impacts at all. But in reality, there are an enormous number of impacts, but by chance, there are slightly more from one side than from the other, and the particle wanders unevenly.

The sage: Einstein derived a formula that can predict how far a particle displaces over time t. For motion along one axis:

$$\langle x^2 \rangle = 2Dt, \quad D = kT / (6\pi\eta r)$$

Sage: Here D is the diffusion coefficient, k is the Boltzmann constant ($1.38 \cdot 10^{-23}$ J/K), T is the temperature in kelvins, η is the viscosity of the liquid, r is the radius of the particle. Note: the formula includes k, and through it, Avogadro's number (k = R/NA, where R is the gas constant). Thus, by measuring **the tremor, one can find out how many molecules are in** a mole. That is why this is such a remarkable prediction.

Young man: Let's calculate it ourselves! A particle with a radius of 0.5 micrometers in water at 20°C. The viscosity of water is 10^{-3} Pa·s, T = 293 K.

Sage: Count.

Young man: $D = 1.38 \cdot 10^{-23} \cdot 293 / (6\pi \cdot 10^{-3} \cdot 0.5 \cdot 10^{-6})$. Numerator: $4.04 \cdot 10^{-21}$. Denominator: $6\pi \approx 18.85$, multiplied by 10^{-3} — 0.01885, multiplied by $0.5 \cdot 10^{-6}$ — $9.4 \cdot 10^{-9}$. Thus, $D \approx 4.3 \cdot 10^{-13}$ m²/s. And the displacement per second $\sqrt{(2Dt)} = \sqrt{(8.6 \cdot 10^{-13})} \approx 9 \cdot 10^{-7}$ m — less than a micrometer!

Sage: Exactly. In one second, a particle shifts approximately by its own size, and in one minute — by $\sqrt{60} \approx 8$ times further, which is about 7 micrometers. This is already clearly visible under a microscope with several hundred times magnification. The jitter that Brown observed is a direct consequence of the fact that water is made up of molecules.

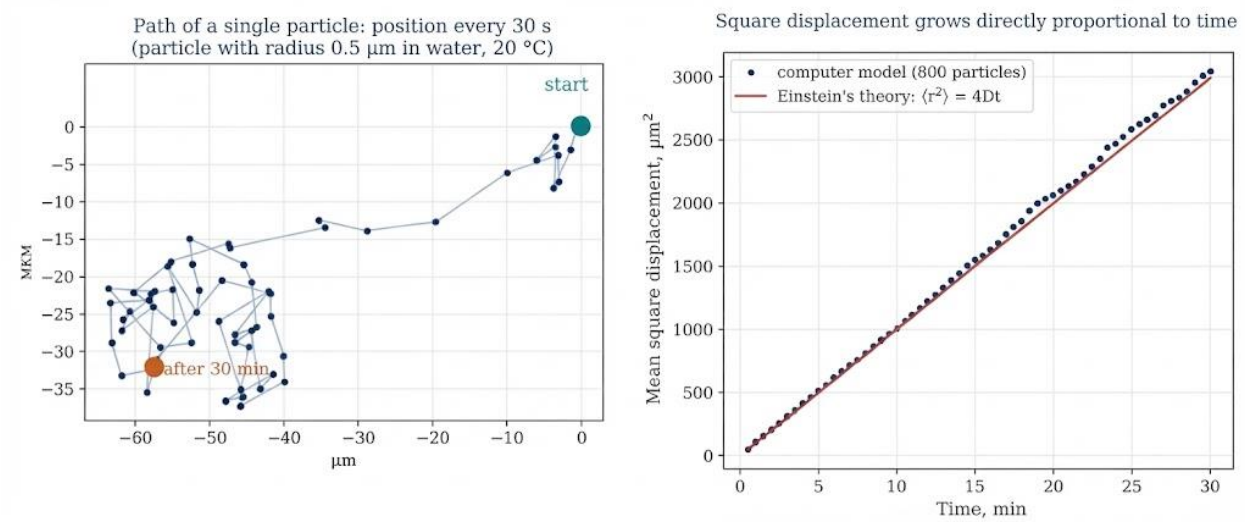


Fig. 1.3. On the left — a computer model of the particle's path (position every 30 s, as in the famous Perrin drawings). On the right — the displacement squared grows proportionally to time, as predicted by Einstein.

The sage: Notice an important thing: the path of a particle is not straight, but broken and "drunken." The average distance from the start grows not proportionally to time, but **as its square root:** for four times more time — twice as far. This is the imprint of random walks. We will encounter it many more times — in diffusion, in thermal conductivity, in the scent of perfume spreading through the room.

7. Perrin considers molecules

The sage: Einstein's prediction needed to be tested. This task was solved by the French physicist Jean Perrin. In 1908-1909, he took an emulsion of tiny resin balls — gum arabic and mastic — of the same size (he learned to select them by centrifugation) and patiently sketched the positions of the particles through a microscope at equal time intervals. He then measured the displacements, calculated D using Einstein's formula, and substituted everything else into it — the size of the particles, viscosity, temperature. From the formula, he obtained Avogadro's number:

$$N_A = RT / (6\pi\eta r \cdot D)$$

Young man: So he counted the molecules by observing the tremor on the visible particles?

The sage: Yes. And he obtained a value in the range of approximately 6.5 to 7.2 · 10^{23} — that is, the same order of magnitude and differing by only a few dozen percent from the modern value of 6.022 · **10^{23}.** The main point: his number coincided with the number obtained by completely different methods: from the viscosity of gases, from the radiation of heated bodies, from radioactive decay. The coincidence of independent methods is the strongest evidence in science.

> **HISTORICAL FACT**
>
> In 1909, Wilhelm Ostwald, who had recently doubted atoms, publicly acknowledged that the atomic hypothesis was now "experimentally confirmed." In 1926, Perrin received the Nobel Prize in Physics for this work. Ernst Mach remained a skeptic until the end of his life, and this is also part of the history of science: one must be able to acknowledge defeat, but not everyone succeeds in this.

> **TRY IT YOURSELF**

> Do you want to see Brownian motion for yourself? You will need a microscope with at least 400× magnification. Dilute a drop of milk with water one hundred times — to a barely noticeable whiteness — drop it onto a slide and cover it with a coverslip. Among the seemingly stationary particles, find the small, oscillating ones. These are fat droplets in the milk, pushed by water molecules. Try to sketch the position of one of them every 30 seconds.

8. Atoms are visible

Sage: Well, could one see **an atom with** their own eyes? Not just a few decades passed. In 1912, the German physicist Max von Laue and his colleagues passed X-rays through a crystal and obtained an ordered pattern of spots on a plate. This showed that the crystal is an ordered lattice of atoms, and that X-rays have a wavelength on the order of the size of atoms. In 1913, father and son Bragg deciphered the structure of table salt: it is a lattice of sodium and chlorine ions, like a chessboard in three dimensions.

The sage: The first images of individual atoms were obtained by Erwin Müller: his field ion microscope (1951) allowed for the observation of atoms at the tip of a tungsten needle — resolution sufficient for individual atoms was achieved by the mid-1950s. In 1981, Gerd Binnig and Heinrich Rohrer created the scanning tunneling microscope: a very fine needle "feels" the surface at a distance of fractions of a nanometer. It showed atoms as distinct bumps on the surface, and in 1986, it was awarded the Nobel Prize. In 1989, Don Eigler at IBM even arranged three letters — IBM — on a nickel plate using thirty-five xenon atoms.

Young man: I am glad. So, we no longer just believe in atoms — we see them.

Sage: We see them, count them, and move them. But do not forget that in the image of the scanning tunneling microscope, you are not seeing a "photograph" in the usual sense, but a map constructed from the electric current between the tip and the surface. This is another reminder: to "see" atoms, we need a theory and an instrument that we trust.

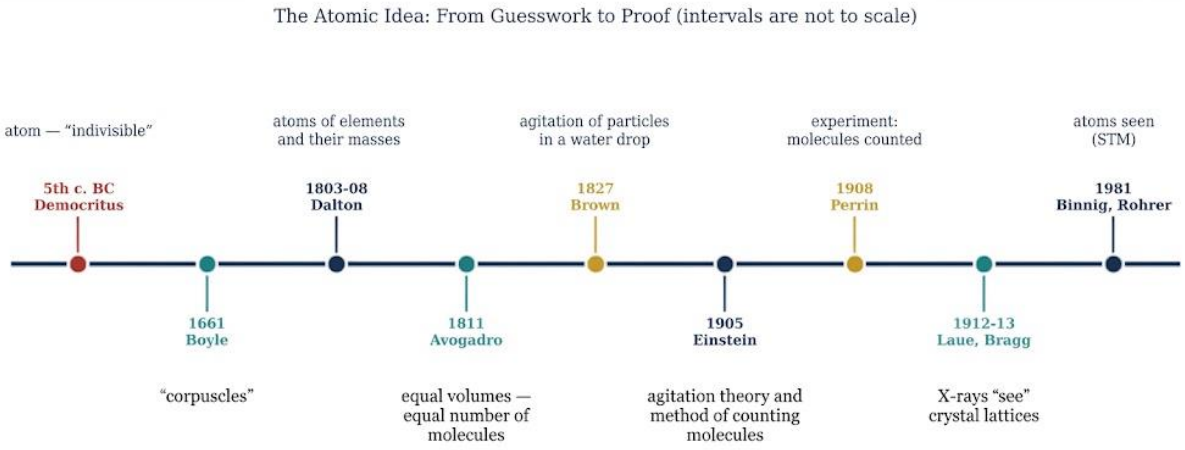


Fig. 1.4. Twenty-five centuries of progress: from conjecture to direct observation.

9. Solid body: lattice of atoms and grains

The sage: Now let's return to the solid bodies that surround us. In a crystal, atoms are arranged not randomly, but in a strictly ordered manner: in rows, layers, repeating cells. If such order is maintained

throughout the entire piece of substance, it is a **single crystal:** a piece of diamond, a large crystal of salt, a silicon wafer in a processor. But most of the solid bodies around us — metals, stones, ceramics, bricks — are structured differently.

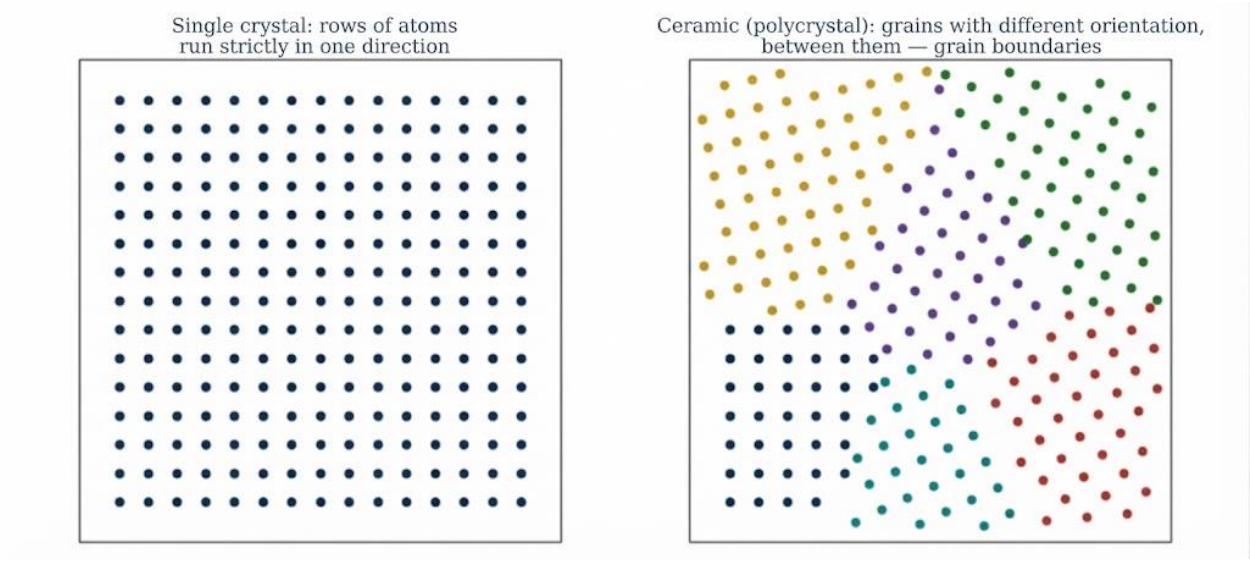


Fig. 1.5. On the left - a single crystal (order throughout the sample). On the right - a polycrystal, for example, ceramics: small crystalline grains, each with its own lattice orientation, separated by boundaries. The illustration is schematic: different grains are shown in different colors.

The sage: They consist of many tiny crystals — grains, each of which is a crystal on its own, but they are oriented differently, and between them lie **transitional layers —** grain boundaries. Such **bodies are called** polycrystals. Ceramics are exactly such a material: when small crystalline grains are heated to a high temperature, they sinter, resulting in a strong solid mass.

Young man: So, ceramics is a whole country divided by borders into small crystalline states?

Sage: Well said. And the boundaries between "states" often determine many properties of the material: strength, conductivity, how it conducts heat and sound.

> **HISTORICAL FACT**
>
> In Uzbekistan, in Parkent near Tashkent, there is a Large Solar Furnace: a system of mirrors collects sunlight into a single point, and the power of the light flux reaches about one megawatt. This is convenient for producing ceramics: the only "heater" is light, and at its focus, the substance can be heated and melted without fuel and electrodes. It is in such a furnace that the author of this book and his colleagues produce functional ceramics, which we will discuss further.

> **AUTHOR'S HYPOTHESIS**
>
> Further in the book, we will return more than once to the boundaries of grains. The author of the book suggests that energy barriers arise at the boundaries between ceramic grains, and the energy accumulated in the form of lattice vibrations overcomes them and "fires" — and that this is the key to the unusual properties of functional ceramics (the so-called impulse tunneling effect, ITE). This is **the author's hypothesis,** not an established fact; it is being tested, and we promise to honestly separate the verified from the assumed in this book each time.

The young man takes a piece of chalk and looks at it with new eyes.

Young man: So, in this flour there are thousands of billions of billions of atoms, they are arranged in a lattice, and I cannot see or feel them. It's still incomprehensible!

Sage: It is incomprehensible as long as you think in images. But we already know how to count: we estimated the sizes, counted the number of molecules in a drop, and observed their tremor. Science is the ability to make the incomprehensible measurable. Now we need to understand what these atoms "know" when we say "hot" and "cold." What is temperature? This is what the next chapter is about.

WHAT WE HAVE LEARNED

- The idea of atoms originated in Ancient Greece as a conjecture (Democritus) and was proven only in the early 20th century.
- Chemistry led to atoms "from the back door": the laws of constant composition and multiple proportions (Lavoisier, Proust, Dalton), Avogadro's hypothesis.
- The size of a molecule can be estimated by the oil film (≈ 1 nm) and by the density of the substance and Avogadro's number (water — about 0.3 nm).
- Avogadro's number $N_A \approx 6.022 \cdot 10^{23}$ mol^{-1}. There are about 10^{21} molecules in a drop of water.
- Brownian motion (Brown, 1827) was explained by Einstein and Smoluchowski: a particle is pushed by invisible molecules. The square of the displacement grows directly proportional to time: $\langle x^2 \rangle = 2Dt$.
- Perrin (1908–09) measured Avogadro's number through Brownian motion, which coincided with other methods. Skeptics (Ostwald) acknowledged the existence of atoms.
- Atoms are smaller than the wavelength of light; they are "seen" by X-ray radiation (Laue, Bragg) and scanning tunneling microscopy (Binnig, Rohrer).
- Most solid materials are polycrystals: grains with boundaries between them. We will return to this.

QUESTIONS FOR REFLECTION

- Estimate the thickness of the film if a drop with a volume of 2 mm³ spread over an area of 0.5 m². Does the order of magnitude match what we obtained above?
- How many times further will the Perrin particle move (in terms of average displacement) in four minutes compared to one minute? Why not four times?
- Why do you think Mach and Ostwald doubted atoms, even though chemists had been using them for a hundred years? Did they have reasonable grounds for their skepticism?

CHAPTER 2

What is temperature?

From the sensation of "warm-cold" to the average energy of molecules and absolute zero

The sage places three bowls on the table: one with cold water, another with warm water, and the third with very warm water, but not hot. The young man looks at them suspiciously.

Sage: Place your left hand in cold water and your right hand in very warm water. Hold them for a minute. Then, simultaneously lower both hands into the middle bowl.

The young man performs. After a few seconds, he raises his eyebrows in surprise.

Young man: Strange! The left hand is warm in the middle bowl, while the right one is cold. But the water is the same!

The sage: This experience was described by the English philosopher John Locke more than three hundred years ago. He showed the **main point: our sensations do not measure temperature, but rather how quickly heat leaves us or comes to us.** Warm water warms a hand accustomed to cold, while it cools a hand accustomed to warmth. An instrument that gives a new result every time is of no use. We need an instrument more honest than the palm.

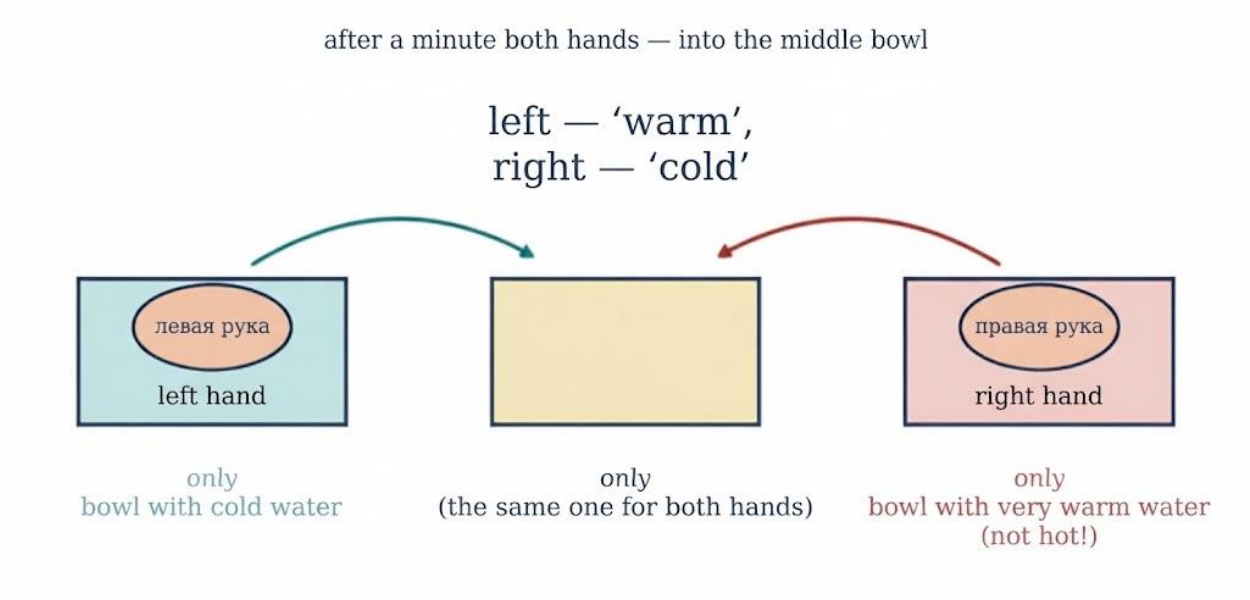


Fig. 2.1. Locke's experiment: the same average bowl feels warm at times and cold at others.

> **TRY IT YOURSELF**
> Repeat Locke's experiment yourself. For safety, the cold water should be from the refrigerator (not icy), and the "warm" water should not be hotter than what you can comfortably tolerate with your elbow (around 40 °C). Boiling water should not be used for this experiment.

Young man: But what if you touch a metal door handle and a wooden door in the same room? The metal feels colder, even though they are at the same temperature.

Sage: A magnificent example. Metal quickly conducts heat away from the skin, while wood does so slowly. The sensation of "cold" turns out to be a measure of the rate of heat transfer, rather than temperature. We will return to this topic—thermal conductivity—in the chapter on transport phenomena.

1. The first thermometers and scales

The sage: The first device that showed the change in "degree of heat" was built, as is commonly believed, by Galileo in the 1590s — a glass tube with a bulb, submerged open end in water. When the air in the bulb was heated, it expanded and pushed the water down. This was a "thermoscope": it indicated whether it was warmer or colder, but did not measure in numbers. Moreover, it was also affected by atmospheric pressure.

The sage: Real thermometers with a scale appeared in the 18th century. Two things were needed: a convenient substance whose expansion is noticeable (mercury, dyed alcohol), and reference **points that everyone** would agree on. The German master Daniel Fahrenheit made the first reliable mercury thermometers in 1714. He took the temperature of a mixture of ice, water, and ammonia as zero—the lowest

temperature he could achieve—and accepted the freezing point of pure water as 32. Later, the scale was refined so that the boiling point of water corresponded to 212.

The sage: The Swede Anders Celsius proposed a simpler scale in 1742: 100 divisions between the freezing and boiling points of water. There is one peculiarity here.

> **DELUSION**
> "The Celsius scale was invented as it is now: 0 — freezing, 100 — boiling." Not quite. Celsius himself set 0 at the boiling point of water and 100 at the freezing point: for him, the colder it was, the higher the number! The scale was flipped upside down after his death — in the 1740s, this was done by Martin Strömer and Carl Linnaeus (historians debate who exactly came up with the idea). And rightly so: this way "a higher number means warmer" aligns with our intuition.

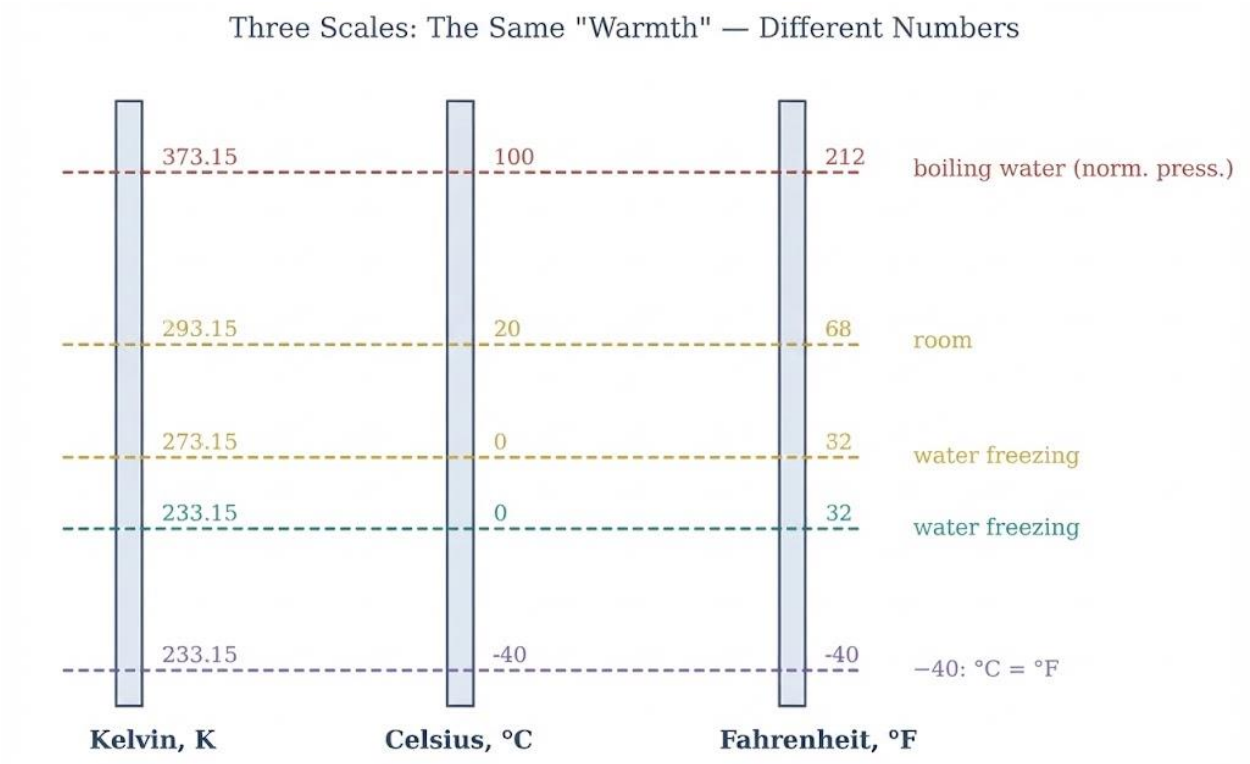


Fig. 2.2. Three temperature scales. Note: at −40 degrees, the Celsius and Fahrenheit scales show the same number.

Young man: Isn't it possible to do without "imaginary" reference points like the boiling point of water?

Sage: A good question. The imaginary points depend on the properties of water, on pressure, on purity. Scientists have been searching for a scale that does not depend on any substance. We will return to it when we reach absolute zero.

2. What does "equally warm" mean?

Sage: Let's think about what a thermometer does. You dip it in water and wait. The column of mercury first rises, then stops. Why does it stop?

Young man: Because the mercury heated up to the temperature of the water?

Sage: Exactly. Bodies in contact exchange heat until thermal equilibrium is reached: after that, **heat no longer** flows between them. Then we say they have the same temperature. And now an important thought,

which, strangely enough, physicists formulated later than the rest. Let body A be in thermal equilibrium with the thermometer, and body B is also. Then A and B are in equilibrium with each other, even if they have not been brought into contact.

> **LET'S REMEMBER**
>
> **The zeroth law of thermodynamics states** that if two bodies are each in thermal equilibrium with a third body, then they are in equilibrium with each other. This is precisely why a thermometer works at all: it is a "mediator" between bodies that cannot be brought into direct contact. This principle was termed "zeroth" by the English physicist Ralph Fowler in the 1930s: it was recognized after the first and second laws, but logically it should precede them.

3. Temperature and the dance of molecules

Young man: Alright, the thermometer shows a number. But what does this number actually measure? What is "hot" in the water?

Sage: And here we will need the previous chapter. We found out that water consists of molecules, and they are in continuous motion. Physicists of the 19th century — Clausius, Maxwell, Boltzmann **— understood: the higher the temperature, the faster the molecules move** chaotically. More precisely: temperature is proportional to the average kinetic energy of the translational motion of the molecules. For a rarefied gas:

$$\langle\ m v^2 / 2\ \rangle\ = (3/2) \cdot k \cdot T$$

Sage: Here T is the temperature in kelvins, $k = 1.38 \cdot 10^{-23}$ J/K is the Boltzmann constant, and the factor 3/2 is related to the fact that a molecule can move in three independent directions. Let's calculate. For room temperature 20°C, that is T = 293 K, what is the average energy of a molecule?

Young man: $(3/2) \cdot 1.38 \cdot 10^{-23} \cdot 293 \approx 6.1 \cdot 10^{-21}$ J. This is very little!

Sage: For one molecule — yes. In electronvolts (the energy gained by an electron when passing through a potential difference of one volt; 1 eV = $1.6 \cdot 10^{-19}$ J) this is about 0.038 eV — approximately 1/26 eV. Now, knowing the energy, let's find the speed. A nitrogen molecule — the main component of air — has a mass of 28 atomic mass units, that is

$m = 28 \cdot 1.66 \cdot 10^{-27} \approx 4.65 \cdot 10^{-26}$ kg.

From the formula for average energy, we obtain the so-called root mean square speed:

$$v = \sqrt{(3kT / m)} = \sqrt{(3 \cdot 1.38 \cdot 10^{-23} \cdot 293 / 4.65 \cdot 10^{-26})} \approx 510\ m/s$$

Young man: Five hundred meters per second! That's faster than the speed of sound in air.

Sage: The speed of sound in air is about 343 m/s, which is approximately two-thirds of the speed of molecules. And this is not a coincidence: sound is transmitted from molecule to molecule in collisions, and cannot travel faster than the molecules fly. (The exact multiplier is $\sqrt{(\gamma/3)}$, approximately 0.68 for air, we will prove this later.)

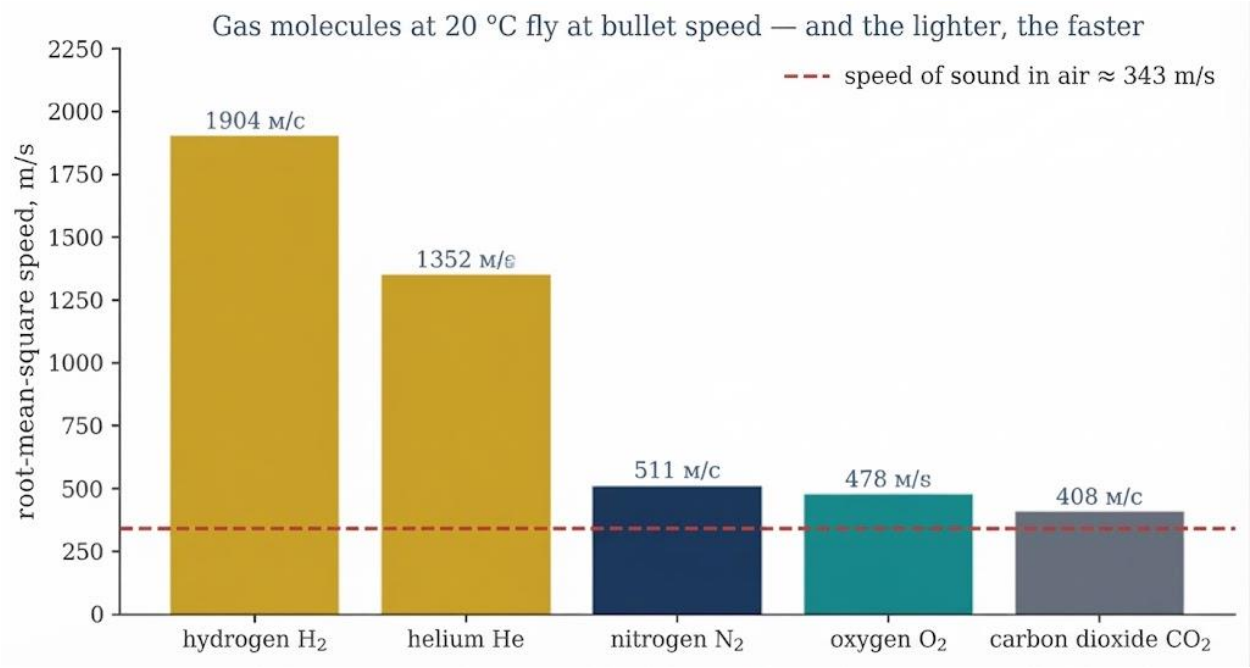


Fig. 2.3. Mean square speeds of molecules of different gases at 20°C. The lighter the molecule, the faster it moves at the same temperature.

Young man: Why do lighter molecules move faster?

Sage: Because at a given temperature, the average **energy of all molecules is the same, regardless** of mass. And energy is $m v^2 / 2$. Thus, the smaller the mass, the greater the speed. Hydrogen is 14 times lighter than nitrogen, so it is faster by $\sqrt{14} \approx 3.7$ times: 1900 m/s compared to 510 m/s. This is why light gases, hydrogen and helium, gradually leave the Earth's atmosphere: the fastest molecules in the upper layers of the atmosphere reach the second cosmic speed of 11.2 km/s, while the Earth retains heavy nitrogen and oxygen.

LET'S REMEMBER

Temperature characterizes chaotic motion. **If all** the air in the carriage is rushing along with the carriage at a speed of 100 km/h, the temperature of the air does not change: the ordered kinetic energy increases, while the temperature depends only on the disordered motion.

DELUSION

"Temperature is present in every molecule." No. Asking what the temperature of a single molecule is is about as meaningless as asking what the average income of one person is "on average across the country." Temperature is a property of a vast number of particles, a **statistical quantity.** One molecule can be fast or slow, but "hot" or "cold" — only the entire crowd can be described that way.

Young man: If temperature is a measure of molecular motion, then does it mean that at very low temperatures the molecules will almost stop?

Sage: Let's find out. Let's return to thermometers.

4. Absolute zero

Wise man: In 1702, the Frenchman Guillaume Amontons noticed that the air pressure in a closed vessel is almost strictly proportional to the temperature: when heated, the pressure increased, and when cooled, it decreased. If we plot this dependence on a graph, we will get a straight line. Let's construct such graphs for different amounts of gas.

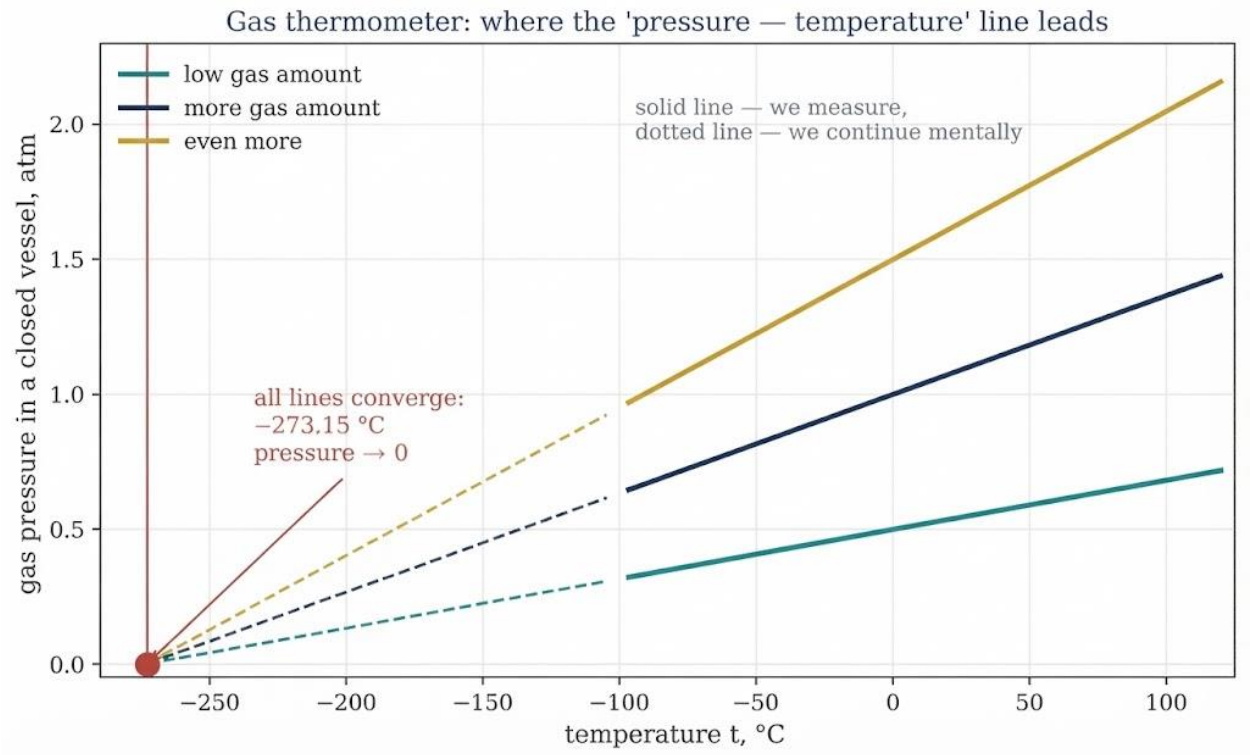


Fig. 2.4. Gas pressure at constant volume as a function of temperature. No matter what gas is taken and how much of it, the lines converge at one point — at −273.15°C.

Young man: All lines converge at a single point! And at that point, the pressure is equal to zero.

The sage: Yes, at a temperature close to −273°C. And now imagine this: pressure is created by the impacts of molecules against the walls. If the pressure were to become zero — the molecules would stop hitting the wall, that is, they would come to a halt. Of course, real gases would turn into liquids before that, and one has to continue the line of thought, but for very rarefied gases, it works perfectly. There is nowhere to go below this temperature. This **temperature is called** absolute zero, and the scale measured from it was proposed in 1848 by the British physicist William Thomson, later Lord Kelvin. The degree is the same as that of Celsius, and the zero point is absolute zero:

$$T\,(K) = t\,(°C) + 273.15$$

HISTORICAL FACT

Since 2019, the kelvin is defined differently: not by the triple point of water, as before (273.16 K), but through the Boltzmann constant, the value of **which has** been fixed: $k = 1.380649 \cdot 10^{-23}$ J/K exactly. That is, temperature is directly linked to the energy of molecules — to what we derived above. The constant that previously needed to be measured now defines the unit itself.

DELUSION

> "At absolute zero, all motion ceases, and molecules freeze." This is incorrect, and here's why. Quantum mechanics, which we will discuss at the end of the book, prohibits a particle from being simultaneously completely motionless and precisely localized. There remains what is called zero-point **motion, and no** amount of cold can eliminate it. The most striking consequence: liquid helium remains liquid even at absolute zero if the pressure is normal; its atoms lack the "rest" needed to arrange themselves into a crystal. To freeze helium, it must be compressed to about 25 atmospheres.

Sage: It is impossible to reach absolute zero in a finite number of steps — this is one of the strangest statements in physics, and we will discuss it in the last chapter (the third law of thermodynamics). However, it is possible to get close: laboratories have learned to cool clouds of atoms to billionths of a kelvin.

5. Temperature ladder

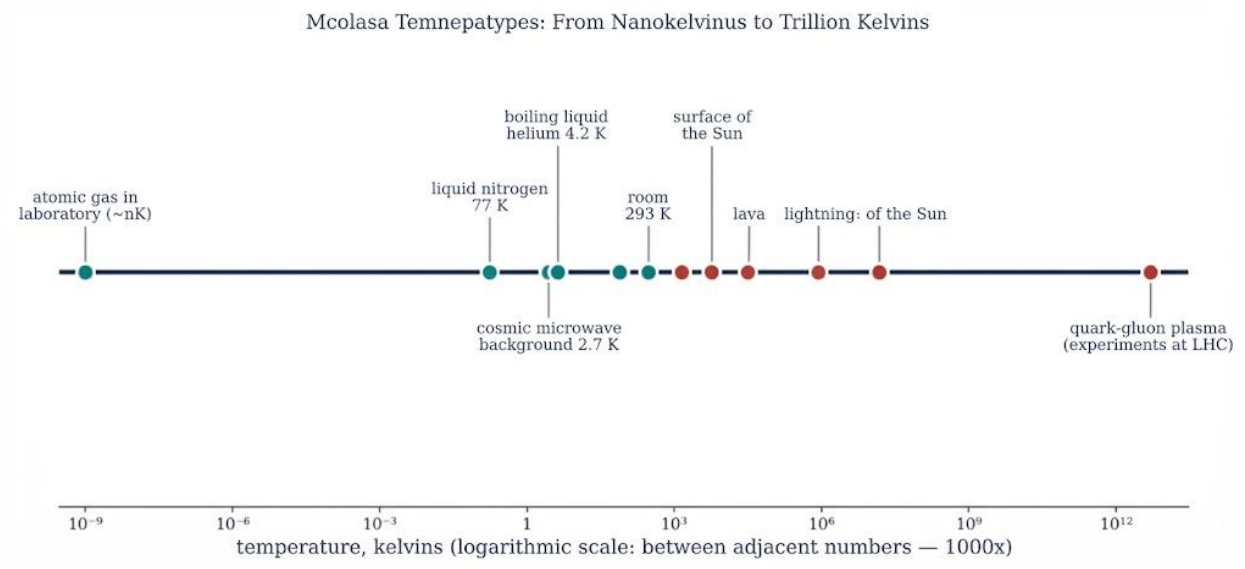


Fig. 2.5. Temperatures occurring in nature and in laboratories. The scale is logarithmic: between adjacent numbers — a thousand times.

Sage: Look at the ladder. There seems to be no upper limit to temperature, but the higher you go, the more unusual the substance becomes. At the center of the Sun — about 15 million kelvins, there atoms are broken down into nuclei and electrons. Even higher, in experiments at the Large Hadron Collider, plasma is produced in which even protons and neutrons do not survive: it consists of quarks and gluons, and the temperature is on the order of 5 trillion kelvins.

Young man: But how do you measure a temperature that is above the melting point of any thermometer? Not with mercury in the center of the Sun!

Sage: For this, temperature is measured **by luminescence.** Heated bodies emit light, and its color depends on the temperature. Let's figure out how this works.

6. The color of a heated body

The sage: Potters have known this for a very long time. The English potter Josiah Wedgwood noted in the late 18th century: no matter what substance is placed in the kiln — clay, metal, stone — at the same temperature, they all begin to glow the same red color. Today, this temperature is called the Draper point: about 525°C. Wedgwood also invented one of the first methods for measuring temperature in the kiln: he

took pieces of clay and observed how much they shrank during firing — the hotter it was, the greater the shrinkage. This was his famous "pyrometer" (1782).

Sage: As it heats up, the glow changes color: dull red, then orange, yellow, and finally dazzling white — this is how the blacksmith determines the temperature of the metal "by eye." The law that relates the temperature of a body to the wavelength at which its radiation is maximized was discovered in 1893 by the German physicist Wilhelm Wien:

$$\lambda_{max} \cdot T = b \approx 2.898 \cdot 10^{-3}\ m \cdot K$$

Young man: So the higher the temperature, the shorter the wavelength? Let's check on the Sun: its surface is 5772 K. Therefore, $\lambda = 2.898 \cdot 10^{-3} / 5772 \approx 5.0 \cdot 10^{-7}$ m, which is 500 nanometers. This is green light, the middle of the visible spectrum!

The sage: And this is not a coincidence: the human eye, as we mentioned in the book on optics, is tuned to the light that our star provides the most. And now, bodies that do not emit light. Let's take any object in the room, 293 K:

Young man: $\lambda = 2.898 \cdot 10^{-3} / 293 \approx 9.9 \cdot 10^{-6}$ m, almost 10 micrometers. This is infrared radiation, it is not visible to the eye!

Sage: Exactly: all warm objects constantly emit in the infrared range — both your palm and the table and the walls. A thermal imager, which "sees" heat, simply registers this invisible radiation.

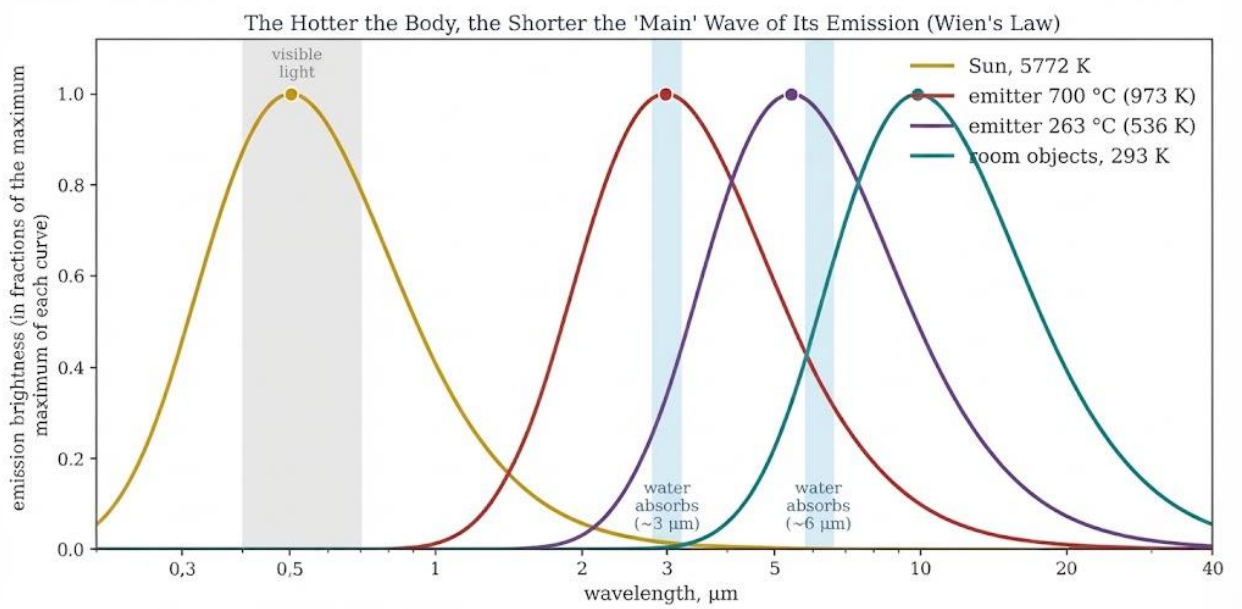


Fig. 2.6. Spectra of radiation from heated bodies (each curve is referenced to its maximum). The hotter the body, the further left its peak. The blue bands indicate areas where water strongly absorbs infrared radiation (the position is approximate).

HISTORICAL FACT

Water strongly absorbs infrared radiation in several regions, particularly near 3 and 6 micrometers. Therefore, engineers designing infrared heaters for drying try to adjust the temperature of the emitter so that the peak of its radiation falls within the water absorption band. According to Wien's law, this is simply a conversion: an emitter at a temperature of about 700°C (973 K) gives a peak at $\lambda = 2898 / 973 \approx 3.0$ μm, which is in the range of about 3 μm; an emitter at 263°C (536 K) has a peak at 5.4 μm; at 129°C (402 K) — at 7.2 μm. The calculation itself is standard physics, verifiable with a calculator. However, how much such tuning actually accelerates

drying needs to be verified experimentally for each design: we will discuss this in the chapters on heat and transfer.

The sage: Temperature is also measured in places where a thermometer cannot survive: in a smelting furnace, in a torch, on the surface of a star. The device that does this by **means of** radiation is called a pyrometer. In the large solar furnaces of the world, the temperature at the focus reaches several thousand degrees, and it can only be measured in this way: no substance can withstand contact with such heat to become a thermometer.

7. What temperature does not indicate

Sage: The final warning. Let's compare two things: a spark from a Bengal fire and a bath of water. The spark has a temperature of about a thousand degrees, the bath — forty. Which is more dangerous if it falls on your hand?

The young man: The spark is certainly hotter. But it's small, it only pricks me a little. The bath is hot — but it's big...

Sage: You have identified an important distinction. Temperature indicates how intensely the **molecules are moving, but** it does not indicate how many there **are and** how much total energy is in the body. A spark is tiny, containing little energy. A whole bath of warm water contains energy millions of times greater. The amount of heat transferred to a body depends on temperature, mass, and the substance. This is a completely different concept, and we will address it in Chapter 5.

LET'S REMEMBER

Temperature is the "degree of heat," answering the question "how intensely do the particles move?". The amount **of heat answers the** question "how much energy has been transferred?". Confusing them is one of the most common mistakes in physics, and from this confusion arose the famous error about "caloric."

The young man takes out his phone from his pocket and opens the thermal camera app that the Sage gave him. On the screen are multicolored spots: a hand, a mug, a cooled cup.

Young man: Now I see how everything around radiates. But I still don't understand: what is heat, if not temperature? And how does it "transfer"?

Sage: Ask about this a little later. First, we need to understand how the simplest material behaves: gas. It obeys the beautiful laws established by Boyle, Gay-Lussac, and Clapeyron. This will be the subject of the next chapter.

WHAT WE HAVE LEARNED

- The sensation of "warm-cold" is unreliable (Locke's experiment); a device is needed — a thermometer.
- Thermometers work due to thermal equilibrium and the zeroth law of thermodynamics.
- Scales: Fahrenheit (1714), Celsius (1742, originally "upside down"), Kelvin (1848). $T(K) = t(°C) + 273.15$.
- The temperature is proportional to the average kinetic energy of the chaotic motion of molecules: $\langle mv^2/2 \rangle = (3/2)kT$. At 20 °C, nitrogen molecules travel at a speed of about 510 m/s.

- Light molecules move faster than heavy ones at the same temperature; sound travels at about two-thirds the speed of molecules.
- The pressure of a gas at constant volume increases linearly with temperature; the continuation of the line leads to absolute zero (−273.15 °C). It cannot be reached, and even there, zero quantum motion remains.
- Hot bodies emit light; Wien's law $\lambda_{max} \cdot T \approx 2.898 \cdot 10^{-3}$ m·K relates temperature and color. This is the basis for pyrometers and the calculation of infrared emitters.
- Temperature is not the amount of heat: a spark is hotter than a bath, but it contains less energy.

QUESTIONS FOR REFLECTION

- At what temperature do the Celsius and Fahrenheit scales show the same number? (Hint: solve the equation $t = 1.8 \cdot t + 32$.)
- How many times will the average speed of molecules change if the air temperature is raised from 300 K to 1200 K? And what about the average energy?
- Calculate according to Wien's law at what wavelength the peak radiation of a steel workpiece emits at 1000°C (T = 1273 K). In what range is this radiation: visible, infrared? Why does the metal still glow with a visible orange color at this temperature?
- Why can one comfortably sit in a bathhouse at 90°C for minutes, while in water at the same temperature, one cannot last even a few seconds? (Hint: remember the spark and the bathtub, as well as the doorknob.)

CHAPTER 3

Gas made of tiny spheres

Pressure, Boyle's laws, Gay-Lussac's laws, Clapeyron and the Mendeléev equation

The sage holds a plastic syringe without a needle in his hands. A finger tightly covers the opening. He extends the syringe to the Youth.

Sage: Press the piston.

Young man: It gives in... but the further you go, the harder it gets. If you press twice as hard, the piston moves just a little more. It's as if there's a spring inside!

Sage: That's right: the air in the syringe behaves like a spring. But who is "springing" in it? There is nothing but air in there. Let's figure out what gas pressure is and why it is arranged this way.

TRY IT YOURSELF

Close the syringe opening with your finger (syringe without a needle!) and press the piston halfway. The pressure inside has increased approximately twice: therefore, the finger is "pushed out." The experiment is safe, and the law is one of the first quantitative laws of physics.

1. "Nature fears a vacuum"

Sage: To understand gas pressure, let's start with the largest "gas" that surrounds us: air. In the times of Aristotle, it was believed that air was weightless, and when you suck water through a straw, the liquid rises

because "nature fears a vacuum" and does not allow empty space to arise. This idea is called *horror vacui, and* it lasted for almost two thousand years.

Young man: So what's wrong with it? After all, the water really does rise behind the piston.

The sage: The observation is correct, but the explanation is not. The Florentine masters of the 17th century built pumps for the ducal gardens, and suddenly it became clear: water does not rise above approximately 10 meters, no matter how hard they tried. If "nature feared emptiness," it would fear it at any height! Galileo, who was asked to explain this, acknowledged that the force of emptiness has a limit, but he could not explain the reason. His student Evangelista Torricelli did that.

> **DELUSION**
> "The nature fears emptiness." In reality, the liquid rises behind the piston not because something "fears," but because it is pressed by the atmosphere. Air is invisible, **but it also** has weight: a column of air from the surface of the earth to the upper boundary of the atmosphere exerts pressure on each square meter with a force approximately equal to the weight of ten tons. The pressure of the air is limited, which is why the water does not rise above a certain limit.

The sage: In 1643, Torricelli took a glass tube about a meter long, sealed at one end, filled it with mercury, and inverted it open end down into a basin of mercury. The mercury did not completely spill out: part remained in the tube, a column about 76 centimeters high. Above the mercury, an empty space formed — the "Torricelli vacuum." This device is **a barometer.**

Young man: So, the atmosphere is pressing down on the mercury in the cup and holding the column?

Sage: Exactly. Let's check with numbers. The pressure of a liquid column at the bottom is equal to $p = \rho g h$ (density times the acceleration due to gravity times the height; we derived this back in mechanics). For mercury, the density is 13,595 kg/m³:

$$p = 13595 \cdot 9.81 \cdot 0.76 \approx 1.01 \cdot 10^5 \ \ Pa$$

Sage: For water (density 1000 kg/m³), the same calculation gives a height $h = p / (\rho g) = 101325 / (1000 \cdot 9.81) \approx 10.3$ meters. Here is the explanation for the ceiling of pumps: 10 **meters is not a mysterious limit, but the height of a water column that weighs the same as the column of the entire atmosphere!**

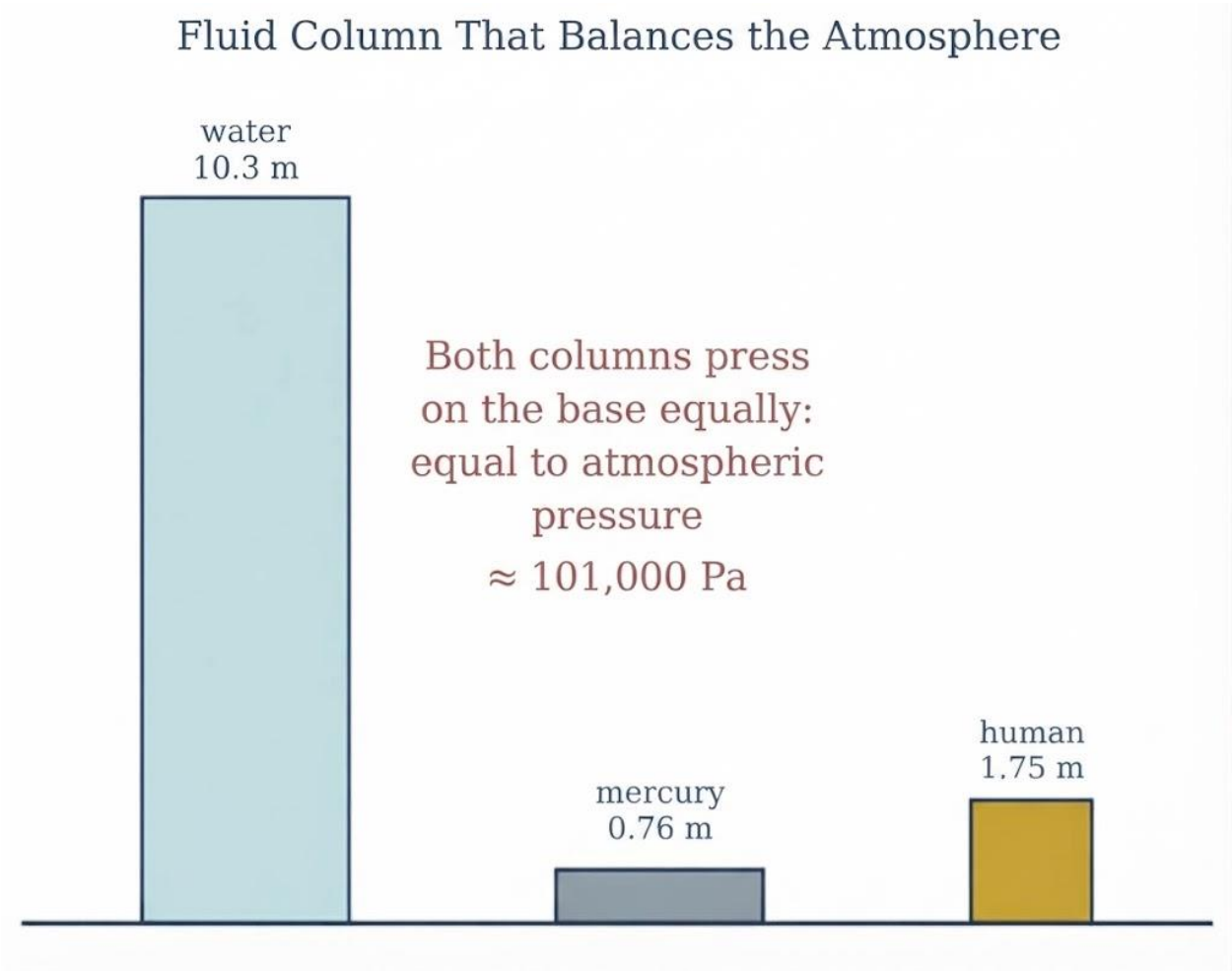


Fig. 3.1. The same column of atmosphere is balanced by a mercury column with a height of 0.76 m or a water column with a height of 10.3 m. Mercury is 13.6 times heavier than water, which is why the column is 13.6 times shorter.

2. Puy-de-Dôme: pressure decreases with altitude

The sage: If the atmosphere presses down with its weight, then at the top of the mountain there is less air above, and therefore the pressure is lower. This was predicted by the French scientist Blaise Pascal, who in 1648 asked his brother-in-law Florin Perrier to verify it. Perrier took two barometers. He left one at the bottom, in the city of Clermont, and took the other up to the Puy-de-Dôme mountain (the height difference is about a kilometer). The mercury column at the top turned out to be lower than at the bottom, by about three inches — that is, about eight centimeters.

Young man: Let's check! If the pressure decreases with height according to some law, then at one kilometer it should drop by...

Sage: For a simple model of an isothermal atmosphere, the law is as follows: $p(h) = p_0 \cdot e^{-h/H}$, where $H = RT / (M g)$ is called the height of the homogeneous atmosphere. At $T = 288$ K, with the molar mass of air $M = 0.029$ kg/mol, we get $H \approx 8.4$ km. Then, for each kilometer, the pressure drops to $e^{-1/8.4} \approx 0.888$ times, and the mercury column from 760 mm drops to 675 mm — a decrease of about 85 mm.

Young man: Eighty-five millimeters! Almost exactly what Perrier measured — about eight centimeters. Three centuries ago, two people with ordinary barometers confirmed the calculation we obtained without any instruments.

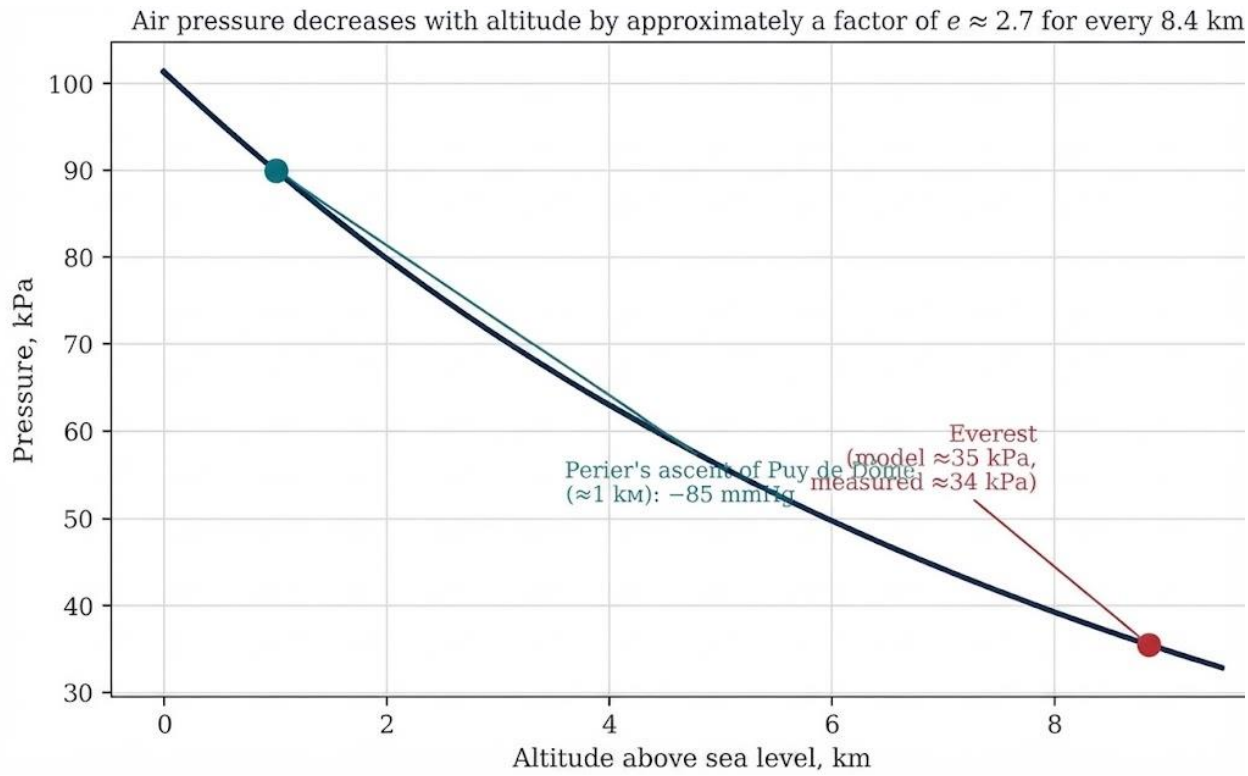


Fig. 3.2. Air pressure decreases with altitude. A simple model gives about 35 kPa for Everest; the measured value is about 34 kPa, because it is actually colder up there.

> **HISTORICAL FACT**
>
> By the way, at the summit of Everest, the pressure is only about a third of normal, which is why it is so hard to breathe there: there are three times fewer oxygen molecules in each breath. This is a good example of how pressure and the number of molecules in a unit volume are related: we will discuss this relationship now.

3. Magdeburg hemispheres

The most famous demonstration of air pressure was organized by the mayor of Magdeburg, Otto von Guericke — the inventor of the air pump. In 1654 (in Regensburg) and then in 1657 (in Magdeburg), he assembled two copper hemispheres, evacuated the air from them, and two teams of horses could not pull them apart. It is usually said that there were sixteen horses, eight on each side.

Young man: There is nothing between them — inside is emptiness. What held them together?

Sage: Atmospheric pressure. Let's estimate. Let the diameter of the sphere be 40 cm, which means the radius is 0.2 m. The area of the "circle" that the atmosphere presses on is $S = \pi r^2 = \pi \cdot 0.2^2 \approx 0.126$ m². Force:

$$F = p \cdot S \approx 1.01 \cdot 10^5 \cdot 0.126 \approx 1.3 \cdot 10^4 \text{ N (approximately 1.3 tons of force)}$$

Sage: The perfect vacuum is unattainable, but a force of over a ton is already a manageable load for a team of horses. Note: the larger the sphere, the more it is "gripped."

Young man: But then why aren't we being crushed? After all, there are ten tons of air sitting on me.

Sage: Because air pressure acts from all sides, **including from inside** our body, where there is also air and liquids under the same pressure. The forces are balanced. An object can only be crushed when the pressure on one side is significantly greater than on the other, as in the case of the Magdeburg hemispheres.

4. Why gas exerts pressure: the impacts of invisible particles

Young man: But where does the pressure itself come from? The weight of the air at height is clear. But in a closed syringe, where no air is pressing "from above"?

> **DELUSION**
> "The gas exerts pressure on the walls of the vessel due to its weight." No. The weight of the molecules is negligible: the pressure of the gas hardly depends on whether the vessel is lying on its side or standing **upright. Pressure is created by the** collisions of molecules with the walls. For the same reason, the gas exerts pressure equally in all directions: molecules move in all directions.

Sage: Let's derive the pressure ourselves. This is one of the most beautiful arguments in physics. Take a cubic vessel with an edge L, containing N molecules of gas. Consider one molecule with mass m, which is moving along the x-axis with a velocity v_x and collides with the right wall.

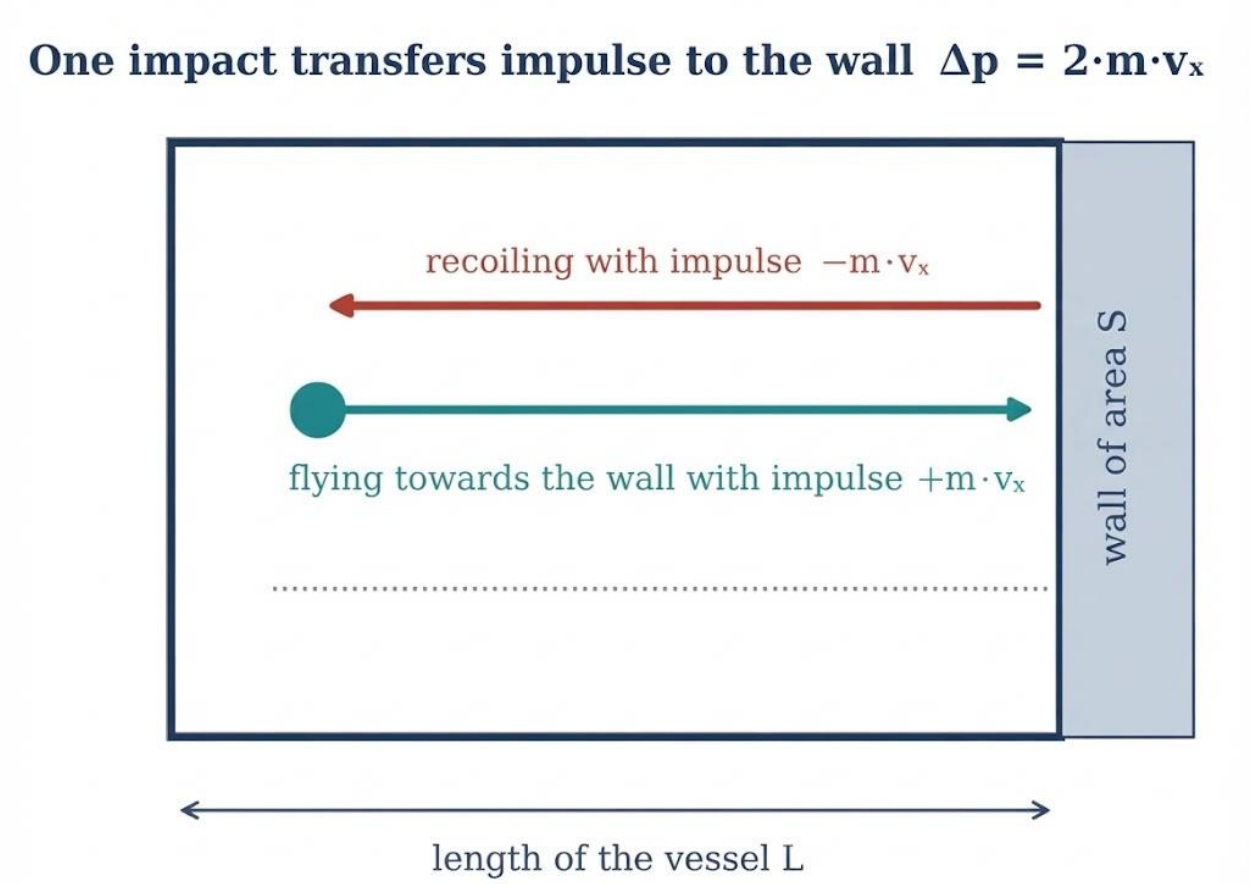


Fig. 3.3. One molecule, elastically colliding with the wall, transfers an impulse of $2mv_x$ to it.

The sage: The molecule flew towards the wall with a momentum of +m v_x and bounced off with a momentum of −m v_x. Therefore, its momentum changed by 2m v_x, and the wall received the same amount of momentum. How often does the molecule hit this wall? It travels to the opposite wall, returns — covering a distance of 2L at a speed of v_x. Thus, the number of impacts per unit time is: $v_x / (2L)$. The average force (momentum per unit time) from one molecule:

$$F_1 = (2m\,v_x) \cdot (v_x / 2L) = m\,v_x^2 / L$$

Sage: Now we consider all N molecules, each with its own v_x. We need to average: $\langle v_x^2 \rangle$. Since the directions are equal, $\langle v_x^2 \rangle = \langle v^2 \rangle / 3$. The force on the wall $F = N \cdot m \langle v^2 \rangle / (3L)$, and the pressure is the force per unit area (the area of the wall L^2):

$$p = F / L^2 = N m \langle v^2 \rangle / (3 L^3) = (1/3) \cdot n \cdot m \langle v^2 \rangle$$

The sage: Here n = N/V — the number of molecules per unit volume (concentration). This is the fundamental equation of the kinetic theory of gases. It was first derived in 1738 by the Swiss Daniel Bernoulli, but almost no one paid attention: the idea that air consists of bouncing balls seemed strange. In 1845, the Scot John Waterston submitted a paper with the same result to the Royal Society, and it was rejected as meaningless: the manuscript was found in the archive only in 1891 by Lord Rayleigh. It was only in the 1850s that Clausius and Maxwell turned this theory into an accepted one.

Young man: Let's check with numbers! In the air at 20 °C and normal pressure, how many molecules are in a cubic meter? And what is the pressure according to the formula?

Sage: First, let's find the concentration. For this, we need another relationship; we will see where it comes from. But let's take a ready-made one (in section 6 we will derive it ourselves): in one cubic meter of air at 20 °C and normal pressure, there are about $2.5 \cdot 10^{25}$ molecules. For a nitrogen molecule, $m = 4.65 \cdot 10^{-26}$ kg, $\langle v^2 \rangle = (511 \text{ m/s})^2 \approx 2.61 \cdot 10^5$ m^2/s^2. Substitute.

Young man: $p = (1/3) \cdot 2.5 \cdot 10^{25} \cdot 4.65 \cdot 10^{-26} \cdot 2.61 \cdot 10^5$. First, $2.5 \cdot 10^{25} \cdot 4.65 \cdot 10^{-26} = 1.16$; then I multiply by $2.61 \cdot 10^5$ — it results in $3.03 \cdot 10^5$; I divide by 3 — $1.01 \cdot 10^5$ Pa. This is exactly atmospheric pressure!

The sage: The theory in which air consists of small balls provided a number that coincides with the reading of the barometer. This is a very significant confirmation.

5. Equation of state: we combine

Sage: Now let's transform our equation. Recall from the previous chapter: the average kinetic energy of translational motion $\langle m v^2/2 \rangle = (3/2) k T$. Substituting: $m\langle v^2 \rangle = 3 k T$. Then:

$$p = n k T \text{ or } p V = N k T$$

Sage: Here it is **— the equation of state of an** ideal gas: pressure, volume, and temperature are not related arbitrarily. If we multiply and divide by Avogadro's number ($N = \nu N_A$, where ν is the number of moles, and $N_A k = R$ is called the universal gas constant), we obtain the form familiar to chemists:

$$p V = \nu R T, \quad R = N_A \cdot k \approx 8.314 \text{ J/(mol·K)}$$

The sage: This relationship for one mole of gas was discovered in 1834 by the Frenchman Benoît Clapeyron, combining the laws known up to that time. In 1874, Dmitri Mendeleev recorded it for any amount of gas with the universal constant R: therefore, in Russian textbooks, **it is called the Mendeleev-Clapeyron equation.**

> **HISTORICAL FACT**
>
> Long before Clapeyron, three laws were discovered separately. Robert Boyle in 1662 (and independently the Frenchman Edme Mariotte in 1676) established that at constant **temperature, the product pV is constant. Jacques Charles around** 1787, and then Joseph Louis Gay-Lussac in 1802, showed that at constant pressure, **the volume increases proportionally to the temperature; and at constant** volume, the pressure increases

proportionally to the temperature (this relationship was noted by Amontons as well). All three laws are special cases of one equation pV = νRT.

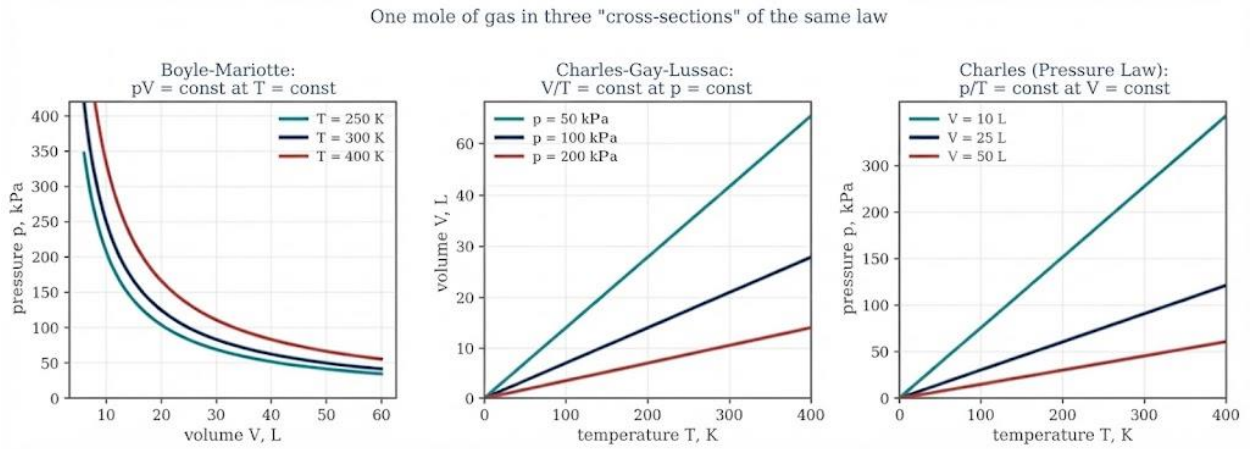


Fig. 3.4. One mole of an ideal gas: isotherms (Boyle), isobars, and isochors. All lines originate from zero — from absolute zero temperature, as we determined in the previous chapter.

Young man: Now I understand why the syringe resists twice as much: the volume has decreased by half, which means the pressure has increased by twice — Boyle's law. And the temperature remained the same because I pressed slowly.

Sage: And here is an important caveat. If you press sharply, the air inside will heat up, and the pressure will increase by more than double. This is how a bicycle pump works: the casing warms up at the end of the pumping. The condition "T = const" is mandatory.

6. We calculate using the Van der Waals equation.

Sage: Let's practice. First: what is the volume of one mole of gas at standard conditions?

Young man: $V = \nu RT / p = 1 \cdot 8.314 \cdot 273.15 / 101325 \approx 0.0224$ m³, that is, 22.4 liters.

Sage: Well: this is the famous molar volume at 0 °C. But at room temperature, 20 °C?

Young man: Then T = 293.15 K, and $V = 8.314 \cdot 293.15 / 101325 \approx 0.02406$ m³ — 24.06 liters.

Sage: Second: how many molecules are in one liter of air? We use p = n k T.

The young man: $n = p / (kT) = 101\,325 / (1.38 \cdot 10^{-23} \cdot 293.15) \approx 2.5 \cdot 10^{25}$ per cubic meter. Thus, in a liter (10^{-3} m³) — $2.5 \cdot 10^{22}$. Twenty-five thousand billion billion! And the distance between molecules is the cubic root of the volume per molecule: $(1 / 2.5 \cdot 10^{25})^{1/3} \approx 3.4 \cdot 10^{-9}$ m.

Sage: That is, about 3.4 nanometers — approximately ten diameters of the molecule itself (the diameter of nitrogen molecules is about 0.37 nm). And what is the volume fraction occupied by the molecules themselves? Each molecule has a volume of $(\pi/6)d^3 \approx 2.6 \cdot 10^{-29}$ m³, and there are $2.5 \cdot 10^{25}$ of them per cubic meter, so together they occupy about $7 \cdot 10^{-4}$ cubic meters, which is **0.07% of the** volume. Air is 99.9% empty space. Now it is clear why gas is so easy to compress and why Boyle's law works so well: the molecules hardly interfere with each other.

TRY IT YOURSELF

> Estimate the mass of air in your room. Let the room be 5 × 4 × 2.5 m (50 m³), the temperature is 20 °C, and the pressure is normal. The number of moles ν = pV / RT = 101325 · 50 / (8.314 · 293.15) ≈ 2080 moles. The molar mass of air is 0.029 kg/mol, so the mass of air ≈ 2080 · **0.029** ≈ 60 kg — this is about the weight of an adult person. Now think about what will happen to this mass if the room is heated to 40 °C with the window open (the pressure remains the same).

Young man: If heated to 40 °C… the density will drop, some air will escape through the window. In the calculation: ν ∝ 1/T, so 293.15 / 313.15 ≈ 93.6% of the air will remain, and 6.4% ≈ 4 kg will escape outside.

Sage: This way, without measuring anything, you calculated how much air leaves the room when it is heated. This is the power of physics.

7. Mixture of gases and Dalton's law

Sage: Air is a mixture. It contains about 78% nitrogen, 21% oxygen, and almost 1% argon. What about the pressure in the mixture? Dalton formulated a simple rule in 1801: the pressure of a **gas mixture is equal to the sum of the pressures that each gas would exert if it occupied the entire volume** alone. These pressures are **called partial pressures.**

Sage: If the air pressure is 101,325 Pa, then nitrogen accounts for approximately 79,000 Pa, and oxygen for 21,000 Pa. We breathe precisely the partial pressure of oxygen, not the total: at the summit of Everest, the total pressure is three times lower, and the amount of oxygen available for inhalation is three times less. Dalton's law also applies to water vapor, and this is already a direct path to the topic of drying.

1 gram of water becomes 1.7 liters.

Sage: Now let's pose a question that we will need for everything else. We poured 1 gram of water (that is, one cubic centimeter) into a vessel and heated it to 100 °C. The water boils and turns into steam. Steam is also a gas, and the equation of state applies to it. What volume will it occupy at normal pressure?

Young man: The molar mass of water is 18 g/mol, so 1 g of water is 1/18 mol, which is 0.0556 mol. V = νRT / p = 0.0556 · 8.314 · 373.15 / 101325 ≈ $1.7 \cdot 10^{-3}$ m³ — 1.7 liters!

Sage: From 1 milliliter of water, 1.7 liters of steam are produced: the volume **increases by** 1700 times. This explains both the power of steam engines and the danger of a closed vessel in which water has boiled. Moreover, this determines the design of drying installations.

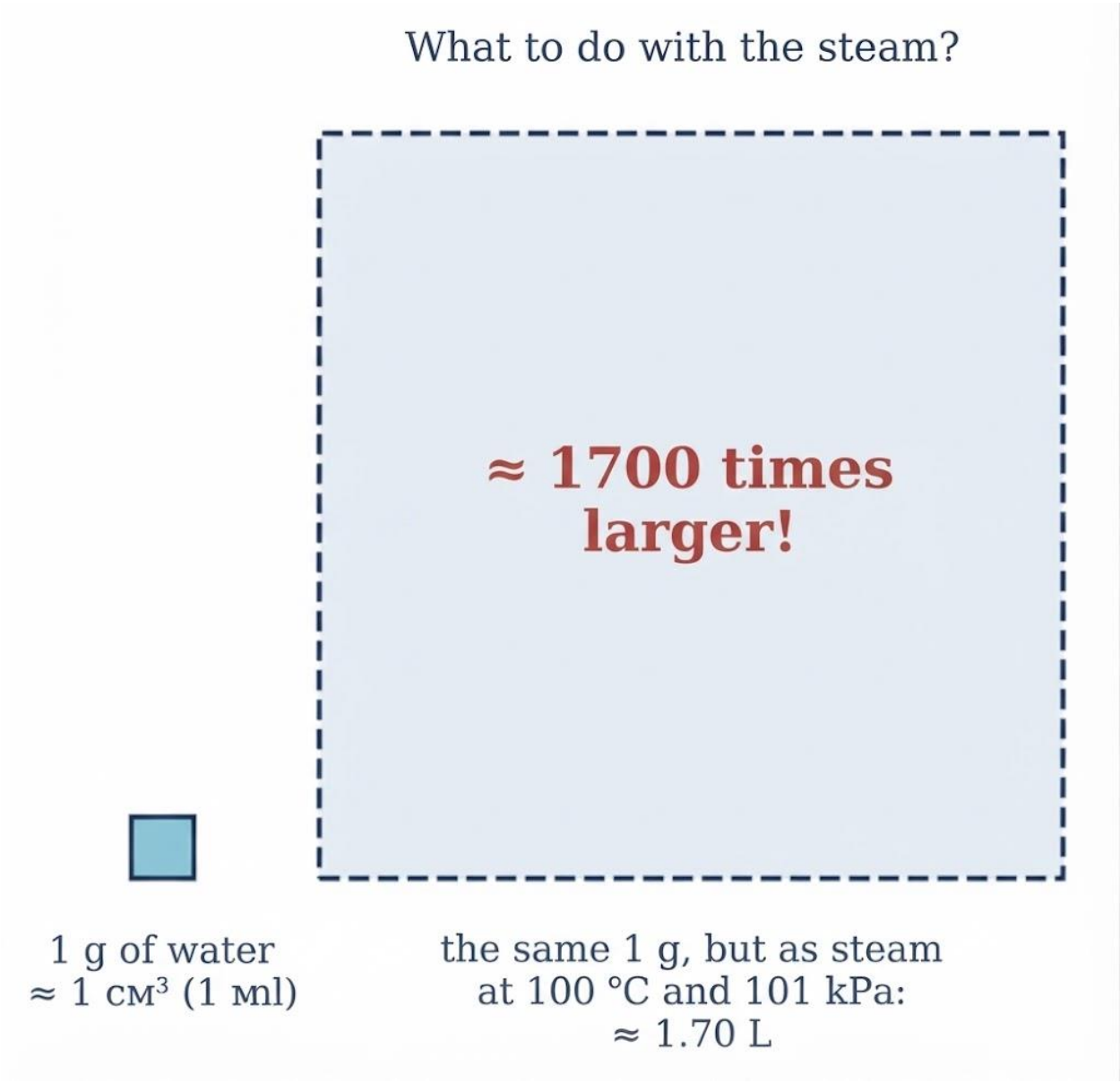


Fig. 3.5. One gram of water and the same gram of water vapor at 100 °C and normal pressure (the edges of the cubes are shown to scale). The volume of the vapor is approximately 1.7 L.

Sage: Think. In the dryer, water evaporates from the product. Every milliliter of water, turning into vapor, requires space — otherwise, the vapor pressure in the chamber increases, and further evaporation slows down. Therefore, vapor must be removed in dryers. In the drying **installations** developed by the author of this book, there are special units for this — an ejector and a labyrinth. And at temperatures below 100 °C, the volume of vapor is even greater: the saturation vapor pressure there is much lower than atmospheric pressure. For example, at 20 °C it is about 2.3 kPa, and one gram of water in the form of such vapor would occupy about 58 liters according to the Clapeyron-Mendeleev equation, which means almost 58,000 volumes of the original droplet.

> **LET'S REMEMBER**
>
> The calculation for 100 °C and normal pressure is exact. For other conditions, the volume is calculated using the same equation $pV = \nu RT$. As for how quickly water evaporates and what the rate depends on, we will explore this in the chapter on kinetic theory and in the chapter on phase transitions.

The young man closes the syringe again and presses the plunger.

Young man: Now I understand: this "air-springing" syringe is billions and billions of tiny balls, and each impact is almost imperceptible, yet together they support ten tons of atmosphere. But then why don't all these balls stick together and fall to the ground?

Sage: That's a good question. But to answer it, we need to know how exactly these molecules move: at what speeds, how often they collide, how far they travel. This will be covered in the next chapter.

WHAT WE HAVE LEARNED

- Air has weight and exerts pressure: normal atmospheric pressure ≈ 101,325 Pa; mercury column — 760 mm, water column — 10.3 m (“horror vacui” — misconception; Torricelli, 1643).
- Pressure decreases with height: $p = p_0 e^{-h/H}$, $H \approx 8.4$ km. Perier's experiment at Puy-de-Dôme (1648) confirmed Pascal's prediction.
- The pressure of a gas is created by the collisions of molecules, not by weight: $p = (1/3)\, n\, m \langle v^2 \rangle$ (Bernoulli, 1738; Clausius, Maxwell).
- From this and from $\langle mv^2/2 \rangle = (3/2)kT$ it follows that $p = nkT$, that is, $pV = \nu RT$, $R = N_A k \approx 8.314$ J/(mol·K) (Clausius, 1834; Mendeleev, 1874).
- The laws of Boyle - Mariotte, Charles - Gay-Lussac are special cases of this equation.
- In the air at 20 °C and normal pressure, there are approximately $2.5 \cdot 10^{25}$ molecules per m^3, the distance between them is approximately 3.4 nm; the molecules themselves occupy only about 0.07% of the volume.
- Dalton's law: the pressure of a mixture is equal to the sum of the partial pressures.
- 1 g of water when converted to steam (100 °C, 101 kPa) occupies ≈ 1.7 L, that is, 1700 times more: the steam needs to be removed.

QUESTIONS FOR REFLECTION

- The air from a bicycle tire with a volume of 2 L at a gauge pressure of 3 atm (absolute pressure of 4 atm) is released into a regular room. What volume will this air occupy at normal pressure if the temperature remains unchanged?
- Why does the law $pV = \nu RT$ work poorly for a highly compressed gas or for a gas near the liquefaction point? What did we miss in the derivation? (Hint: the size of molecules and the attraction between them; we will return to this in the chapter on real gases.)
- Estimate the force with which the atmosphere presses down on the page of this book (approximately 20 × 30 cm). Why doesn't the page get crushed?
- A pressure cooker operates at a pressure of about 2 atmospheres. How do you think this speeds up cooking? (We will discuss this in detail in the chapter on phase transitions.)

CHAPTER 4

Molecules in motion

Maxwell distribution, mean free path, evaporation, and diffusion

The sage opens a bottle of perfume and places it at the far end of the table. The young man, sitting three meters away, does not feel anything yet.

Young man: It doesn't smell like anything. But we just found out that air molecules travel at a speed of about 500 meters per second! They should cover three meters in a fraction of a second. So why doesn't the smell reach me instantly?

Sage: Congratulations: you asked the same question as the Dutch meteorologist Christiaan Bais-Balot in 1858. He wrote it to Rudolf Clausius, the creator of the kinetic theory of gases, and this objection prompted Clausius to refine the theory. Let's see how he responded.

1. Mean free path

The sage: The answer of Clausius: the molecule flies **fast, but** not far. The air is densely packed, and the molecule almost immediately collides with a neighbor, bounces in a random direction, and flies further — until the next collision. The average distance between two collisions is called the **mean free path λ.** Let's estimate it. A molecule of diameter d "sweeps out" a tube with a cross-sectional area πd^2; a collision will occur if the center of another molecule is within this tube. A strict calculation (taking into account that both molecules are moving) gives:

$$\lambda = 1 / (\sqrt{2} \cdot \pi \cdot d^2 \cdot n)$$

Sage: Let's substitute the numbers for nitrogen at 20 °C and normal pressure: $d \approx 0.37$ nm, concentration $n \approx 2.5 \cdot 10^{25}$ m^{-3} (we found it in the previous chapter).

Young man: $\pi d^2 = 3.14 \cdot (3.7 \cdot 10^{-10})^2 \approx 4.3 \cdot 10^{-19}$ m^2. I multiply by n: $4.3 \cdot 10^{-19} \cdot 2.5 \cdot 10^{25} \approx 1.1 \cdot 10^7$ m^{-1}. I multiply by $\sqrt{2} \approx 1.41$ — it results in $1.5 \cdot 10^7$. And $\lambda = 1 / 1.5 \cdot 10^7 \approx 6.6 \cdot 10^{-8}$ m — 66 nanometers!

Sage: Now the number of collisions. The average speed of nitrogen molecules (which is slightly less than the root mean square speed) is about 471 m/s. In one second, a molecule travels 471 m, while the distance between collisions is 66 nm. Therefore, the number of collisions per second is:

$$471 / (6.6 \cdot 10^{-8}) \approx 7 \cdot 10^9 \text{ per second}$$

Young man: Seven billion collisions per second! The molecule is not flying, but is darting around wildly.

Sage: Exactly. The mean free path is about 180 diameters of the molecule. This is considerable (we mentioned that molecules are far apart), but in comparison to a room, it is negligible. Here is what the path of a single molecule looks like:

45 collisions: path traveled ≈ 38 λ, but
from start the molecule moved only ≈ 9 λ

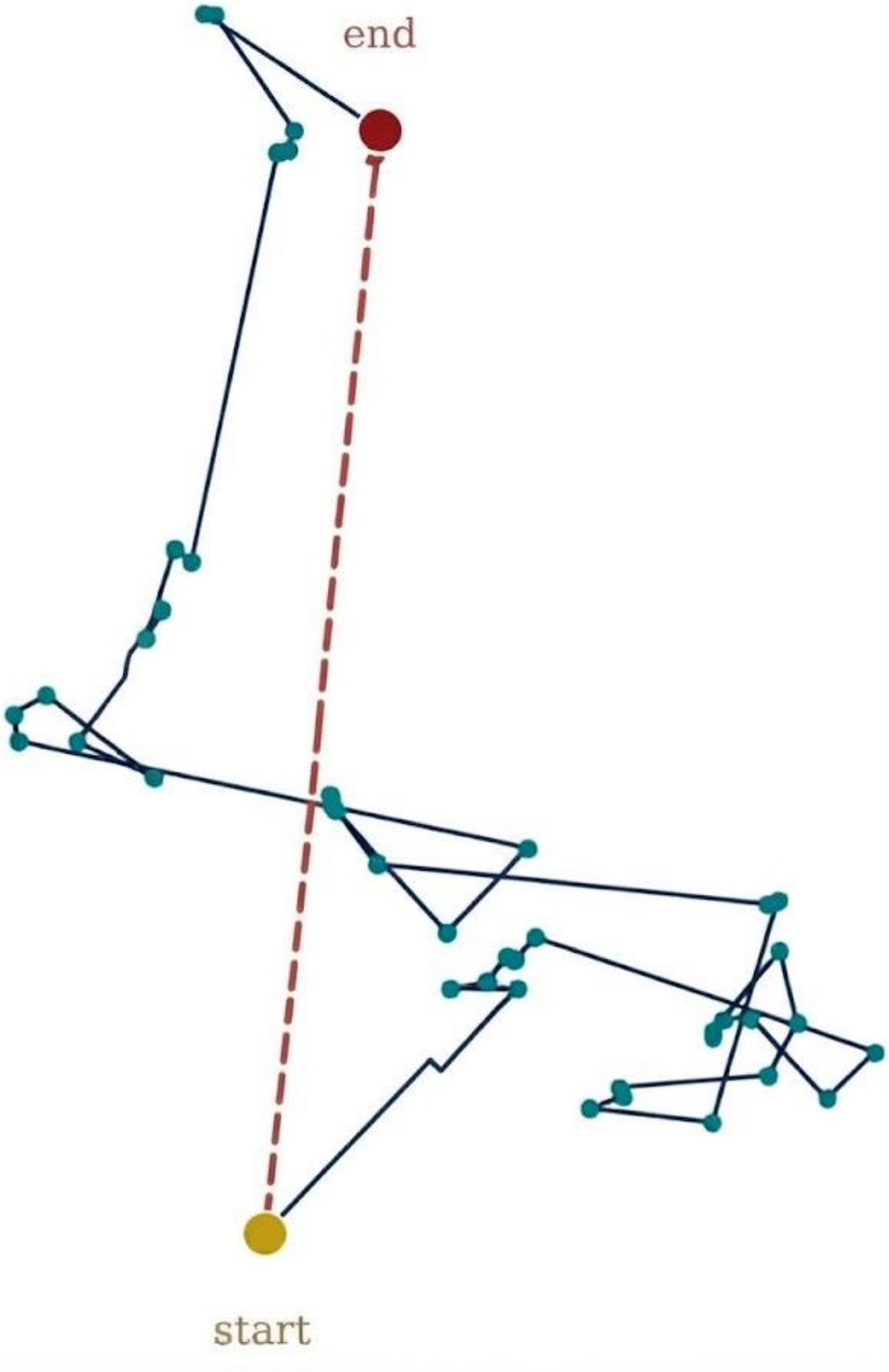


Fig. 4.1. The path of a gas molecule: it travels straight between collisions, and after each collision, it changes direction randomly. After 45 collisions, the molecule has traveled approximately 38 mean free paths, but has only moved away from the starting point by approximately 9.

The sage: It resembles a drunkard walking on a flat surface: each step is straight, but the direction is chosen at random. Recall the Brownian particle from the first chapter: it has the same type of motion. The distance from the start increases not proportionally to time, but as the square root of time. For gas, the estimate is as follows: the diffusion coefficient $D \approx (1/3) \cdot \bar{v} \cdot \lambda \approx (1/3) \cdot 471 \cdot 6.6 \cdot 10^{-8} \approx 1.0 \cdot 10^{-5}$ m²/s, and the time it takes for a molecule to travel a distance L is $t \approx L^2 / (2D)$.

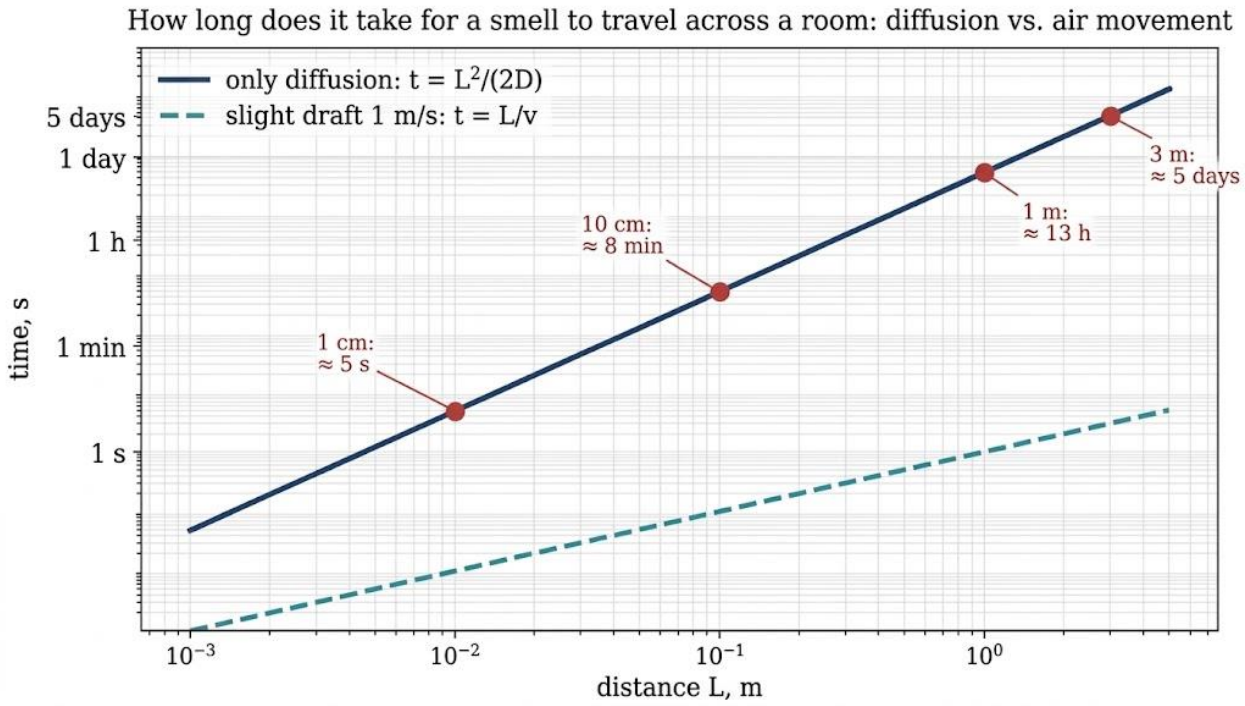


Fig. 4.2. The time required for molecules to "spread out" over a distance L solely due to diffusion. The dashed line represents transport by an air flow with a speed of 1 m/s.

The sage: Let's draw a conclusion. To the neighboring chair, which is one meter away, diffusion takes 13 hours! To the far end of the room, 3 meters away, it takes almost 5 days.

The young man: But the smell reaches faster. In a minute or two.

Sage: Yes, because in a real room, the air **does not remain** still: there are always convection currents from a warm body, from a warm lamp, from an open door. They carry smells hundreds and thousands of times faster than diffusion. This is **an important lesson: over large distances in a gas, wind operates, while diffusion works only on small scales, millimeters and centimeters.**

> **HISTORICAL FACT**
>
> Clausius published the idea of the mean free path in 1858. Then, James Clerk Maxwell estimated it in 1860 based on the viscosity of air — it turned out to be about 1/447000 inch, or approximately $6 \cdot 10^{-8}$ m (almost like ours). In 1865, the Austrian Josef Loschmidt, knowing the mean free path and the density of liquid air, first estimated the sizes of molecules and the number of molecules in a cubic centimeter. Thus, for the first time, a glimpse was gained into a world that no one could see at that time.

2. Not all molecules are the same.

Young man: You keep saying "average speed." Do all molecules fly at the same speed?

> **DELUSION**
>
> "At a given temperature, all gas molecules move at the same speed." No, and this is fundamental. Due to constant collisions, energy is redistributed from time to time: one molecule gains energy in a collision, while another loses it. At any moment, there are both very slow molecules and very fast ones in the gas, as well as a very small number of extremely fast ones. Temperature characterizes the average **but does** not imply that all are the same.

The sage: The question of how exactly the speeds are distributed was solved by Maxwell in 1860 when he was 29 years old. The result is **called the Maxwell distribution. Here** is its form:

$$f(v) = 4\pi \cdot (m / 2\pi kT)3/2 \cdot v^2 \cdot e{-}m^{v^2 / 2kT}$$

Sage: Do not be afraid: there are only two "details" in the formula. The multiplier v^2 indicates that there are few molecules with very low speed because the "velocity space" near zero is narrow. **The** exponent $e{-}mv^2/2kT$ is precisely the indicator that there are terribly few fast molecules: the greater the kinetic energy $mv^2/2$ compared to kT, the lower the chances. In both cases, the same exponent is visible, where the exponent contains energy **divided by kT. We** will encounter it many more times.

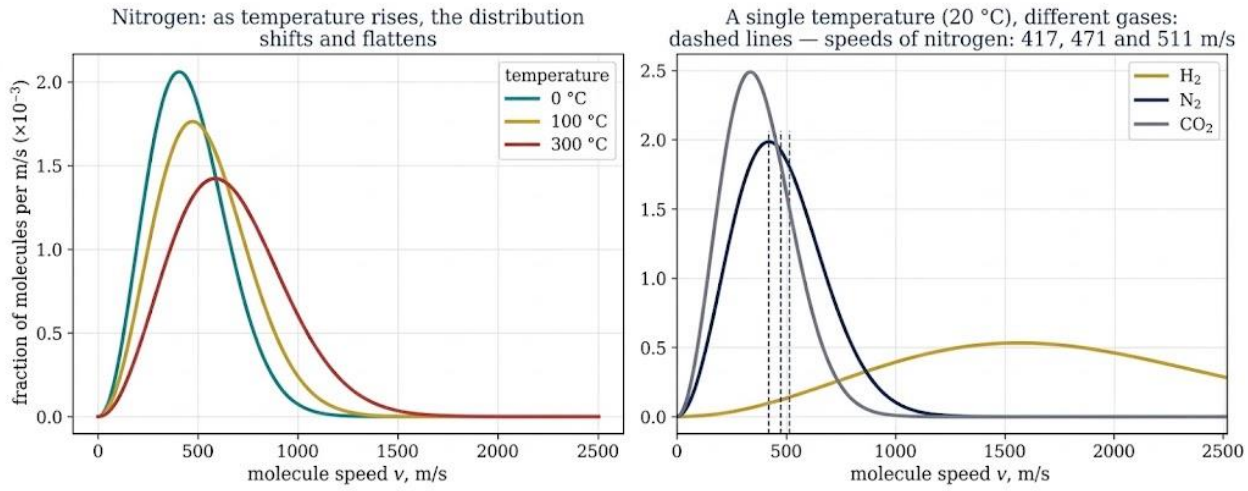


Fig. 4.3. On the left — the distribution of nitrogen molecules by speeds at three temperatures. On the right — different gases at one temperature: light hydrogen "spreads out" far to the right.

Sage: The curve has three "characteristic" speeds. The most probable (the peak of the curve), the average, and the root mean square (the one derived from energy, as we calculated earlier):

$$v_{вер} = \sqrt{(2kT/m)}, \quad \bar{v} = \sqrt{(8kT/\pi m)}, \quad v_{кв} = \sqrt{(3kT/m)}$$

Sage: For nitrogen at 20 °C, they are equal to 417, 471, and 511 m/s — they are in the ratio of 1 : 1.13 : 1.22. Now, how many nitrogen molecules at 20 °C are faster than 1000 m/s?

Young man: I don't remember the formula for the share...

The calculation shows: faster than 500 m/s — 41% of molecules; faster than 1000 m/s — only 0.93%; faster than 1500 m/s — about one in a hundred thousand ($1 \cdot 10^{-5}$). Notice how steeply the "tail" of the curve drops. For hydrogen at the same temperature, faster than 3000 m/s — already about 6%.

LET'S REMEMBER

In any gas, there is a small number of very fast molecules. It is these molecules that determine much: evaporation, chemical reactions, the leakage of light gases from the atmosphere. Later, we will see that this is one of the main characters of all thermodynamics — the **tail of the distribution.**

3. Stern's experiment: to see the speed of a molecule

Young man: Well, Maxwell derived it beautifully. But has anyone checked? After all, the speeds of the molecules are so enormous!

The sage: They checked, after sixty years. In 1920, the German physicist Otto Stern devised a clever experiment. He heated a silver wire in a vacuum, and the silver atoms flew out of it in a narrow beam through a slit. The beam hit the wall of a rotating drum. While an atom travels the distance R to the wall in the time R/v, the drum manages to rotate, and the silver deposit will be shifted. Fast atoms are shifted little, slow ones — significantly. From the distribution of the deposit, one can determine the distribution of speeds.

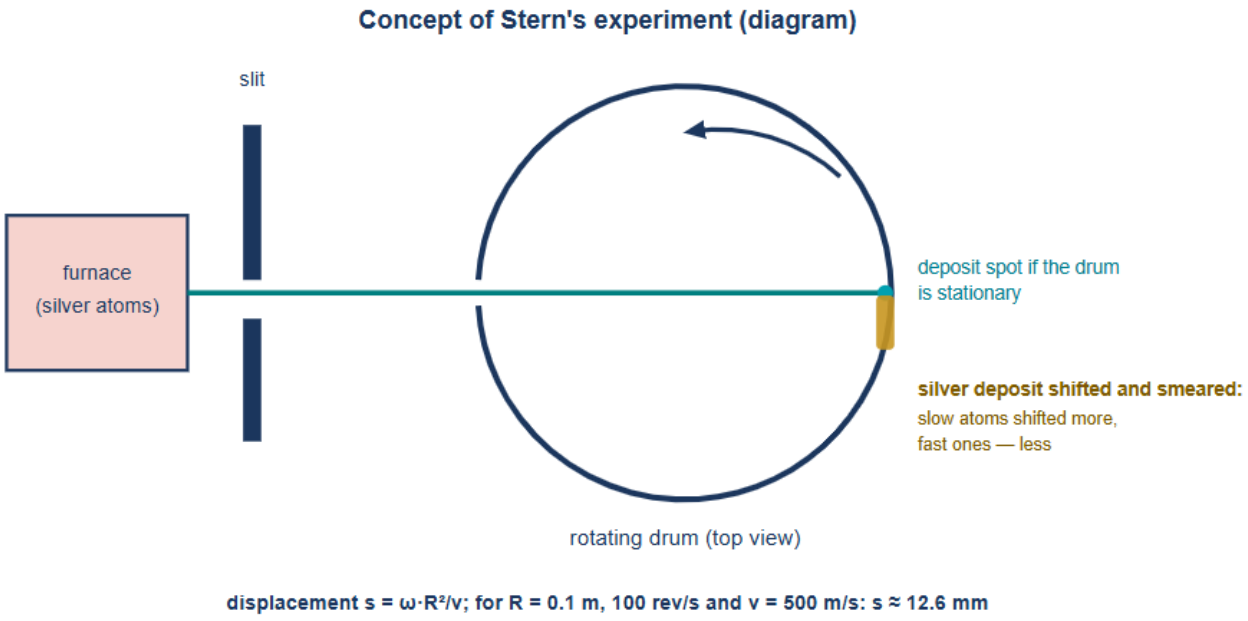


Fig. 4.4. The idea of the Stern experiment. The deflection $s = \omega R^2/v$ depends on the speed of the atom v.

Young man: Let's calculate. Let the radius of the drum be 0.1 m, it makes 100 revolutions per second, and the speed of the atom is 500 m/s. The flight time R/v = 0.1/500 = $2 \cdot 10^{-4}$ s. During this time, the drum rotates by an angle $\omega t = 628 \cdot 2 \cdot 10^{-4} \approx 0.126$ radians, and the wall shifts by $R \cdot 0.126 \approx 12.6$ mm.

The sage: Twelve and a half millimeters is perfectly measurable. The initial results of Stern were consistent with the theory, and the precise measurements of the 1950s (Miller and Kush in 1955) showed that the Maxwell distribution is fulfilled with great accuracy. Later, in 1943, Stern received the Nobel Prize for the molecular beam method.

4. Tail and evaporation

Sage: Now let's return to the tail of the distribution: it explains what you have seen a thousand times — the drying of a puddle. Pour water into a saucer and leave it on the windowsill. The water does not boil, the temperature is room temperature, yet the water still disappears. Why?

Young man: Because water molecules evaporate from the surface…

Sage: And why exactly them? Water in a dish is a liquid: molecules are attracted to each other. For a molecule to break free from the surface into the air, it must **overcome the attraction of its neighbors** — for this, a certain amount of energy is required. This is the very energy that we call the heat of vaporization. For water, it is about 42 kJ per mole. Let's find it for one molecule:

$$E = 42\,000\ J/mol\ /\ 6.02\cdot10^{23}\ mol^{-1} \approx 7.0\cdot10^{-20}\ J \approx 0.43\ eV$$

Young man: The average energy of a molecule at 20 °C is about 0.038 eV, as we calculated in the second chapter. This means the required energy is 11 times greater than the average!

Sage: When compared to kT (where kT at 20 °C is 0.0253 eV), the required energy is about 17 kT. There are few such molecules, but not zero: the fraction of molecules with energy above E at large E is determined by the **exponential** $e^{-E/kT}$:

$$e^{-17} \approx 3\cdot10^{-8}\ \text{at } 20°C; \quad e^{-13.5} \approx 1.3\cdot10^{-6}\ \text{at } 100°C$$

The sage: Only three out of one hundred million molecules have enough energy at 20°C. But there are so many molecules (remember 10^{21} in a drop!) that there are enough "runaways." And at 100°C, there are already forty times more.

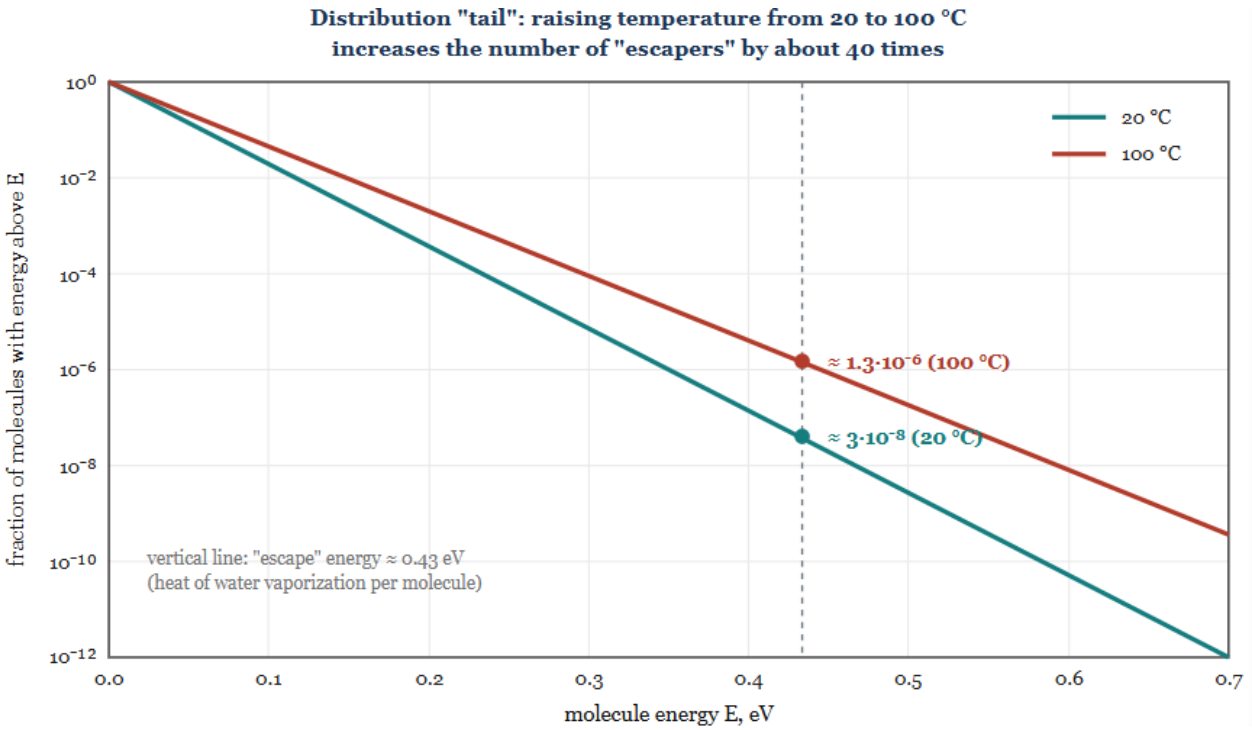


Fig. 4.5. The "tail" of the energy distribution: the fraction of molecules possessing energy sufficient for escape. The vertical scale is logarithmic.

Young man: Forty times! But can this be verified independently?

Sage: It is possible, and the result is beautiful. The saturation pressure of water vapor is proportional to the number of "runaways." It is known that at 20°C it is 2.34 kPa, and at 100°C it is 101.3 kPa, which is 43

times greater. Our rough calculation gave 40. This coincidence is not accuracy, but a good confirmation **of the idea: the fast tail of the distribution governs evaporation.** (The calculation is simplified: we used the average value of the heat of vaporization and did not take into account that not any molecules escape, but those directed outward. The exact theory gives a similar form.)

> **HISTORICAL FACT**
>
> The same exponent, but now for gravity, governs the distribution of particles by height. If we substitute potential energy mgh instead of kinetic energy, we get e−mgh/kT — and this is the barometric formula from the previous chapter: air pressure decreases with height according to the law $p = p_0 \cdot e^{-Mgh/RT}$. Jean Perrin used it in 1909: he counted how many resin balls were at different heights in a glass cuvette, and from this distribution, he found Avogadro's number — yet another independent method that yielded the same number as the Brownian motion of particles.

> **LET'S REMEMBER**
>
> Remember the main **point: everything that requires overcoming the energy barrier E occurs with a probability of** about e−E/kT. This is the law of evaporation, chemical reactions (Arrhenius, 1889), diffusion in solids, and the movement of electrons in semiconductors. As the temperature increases, such processes accelerate not linearly, but exponentially.

5. Evaporation cools.

Young man: Wait. If the fastest, most energetic molecules leave the water, then those that remain are, on average, slower. This means the remaining water should cool down!

Wise man: Brilliant — exactly so. Evaporation takes **away energy and** cools the liquid. That is why wet laundry in the wind is colder than the air, and a clay pot, soaking water through its pores, remains cool in the heat: in hot climate countries, including Central Asia, such pots have been used for centuries. Let's evaluate the effect. The heat of evaporation of water at 20 °C is about 2.45 MJ per kilogram, that is, 2.45 kJ per gram. How many grams need to be evaporated to cool one liter of water by 12 °C? The specific heat capacity of water is 4.18 kJ/(kg·K).

A young man: Cooling one liter by 12 K requires $4.18 \cdot 12 \approx 50$ kJ. Dividing by 2.45 kJ/g gives 20 grams. Two tablespoons of water are enough to cool one liter by 12 degrees!

Sage: This is the ideal limit, not taking into account the heat influx from the environment: in a real pitcher, the temperature drops less, but noticeably. Note the proportion: the energy carried away by one gram of evaporated water would be enough to heat approximately 586 grams of water by one degree. Evaporation is a very energy-intensive process.

Sage: Now a question for you, dryers. To dry a product, you need to turn water into steam. How much energy is required for this at a minimum?

Young man: 2.26 MJ per kilogram of water at 100 °C — the heat of vaporization of water is slightly less than at 20 °C. If converted to kilowatt-hours: $2.26 \cdot 10^6 / 3.6 \cdot 10^6 \approx 0.63$ kWh per kilogram of water.

> **LET'S REMEMBER**
>
> **The honesty scale for drying.** To evaporate one kilogram of water, it takes at least ≈ 0.63 kW·h of heat. Therefore, any dryer can be conveniently evaluated by the number of "kilowatt-hours per kilogram of

evaporated water": real installations consume more, as they also heat the product and the chamber, and lose heat. Less than 0.63 kW·h per kilogram can only be achieved if part of the heat is obtained for free (sun, heat recovery from exhaust steam). The claimed energy savings should always be compared with this scale: it immediately shows whether the number is plausible. In the chapter on the first law of thermodynamics, we will use it to analyze the energy balance of the dryer.

Young man: And one more thing. If in the dryer the steam escapes from the surface of the product only by diffusion, it is very slow, as we found out. This means that the air needs to be moved!

Sage: Correct. At the surface of a wet body, there is always a thin layer of stationary air saturated with vapor. Water molecules can only escape through this layer by diffusion, and it hinders drying. An airflow, jet, or ejector "blows away" this layer and brings the humidity at the surface closer to the humidity in the chamber. This is why the removal of vapor is not a trivial matter, but a central task in the design of the dryer, as we discussed in the previous chapter.

6. Why does the Moon have no atmosphere?

The Maxwell tail explains the fate of entire atmospheres. To leave Earth forever, a molecule at a high altitude needs a speed of about 11.2 km/s — the second cosmic velocity. In the upper layers of the atmosphere, where molecules hardly collide, the temperature reaches up to 1000 K. For hydrogen atoms at this temperature, the most probable speed is about 4.1 km/s. The escape velocity is three times greater, but the tail reaches there too: the fraction of atoms faster than 11.2 km/s is about 0.16%. Over billions of years, Earth's hydrogen has escaped into space.

Young man: And oxygen and nitrogen?

The sage: For oxygen under the same conditions, the probable speed is about 1.0 km/s, while the escape velocity is eleven such units. The exponent e−121 is approximately 10^{-5} ^{3}: throughout the history of the Universe, no oxygen molecule would have reached such a speed. That is why we breathe oxygen, and there is almost no hydrogen in the air. On the Moon, however, the escape velocity is only 2.4 km/s, and it has not retained anything: the gases have escaped. Jupiter, on the other hand, has an escape velocity of about 60 km/s, and it retains even hydrogen and helium.

The young man gets up and opens the window. A light draft catches the scent of perfume and carries it across the room.

Young man: Now I understand why a draft works, but diffusion does not. But I suddenly thought: in these discussions, I constantly say "gas," "molecules." But what is "heat"? When I heat water on the stove, what is being transferred from the fire to the water? After all, temperature is just a measure of motion.

The sage: This is the most important question of all. It gave birth to an entire science that we study throughout the book. But first, we need to address a historical misconception — one of the most persistent in physics: the idea of "caloric." This is what the next chapter is about.

WHAT WE HAVE LEARNED

- Molecules in a gas move quickly (hundreds of m/s), but collide approximately $7 \cdot 10^9$ times per second; the mean free path in air $\lambda \approx 66$ nm (Clausius, 1858).
- Therefore, diffusion is slow: 1 m takes about 13 hours; the smell in the room is carried by air currents.

- The speeds of molecules are distributed according to the Maxwell law (1860): there are slow, fast, and extremely fast ones; for nitrogen at 20 °C, the speed exceeds 1000 m/s for 0.9% of the molecules. Stern's experiment (1920) verified this.
- The probability of overcoming the barrier with energy $^{E} \sim e^{-E/kT}$: the "tail" of the distribution governs evaporation, reactions, and gas leakage from the atmosphere.
- Evaporation requires energy (water: ≈ 0.43 eV per molecule, 2.26 MJ/kg) and cools the liquid.
- The lower limit of drying energy: ≈ 0.63 kWh per kilogram of evaporated water.
- Light gases escape from planets: Earth has lost hydrogen, the Moon — everything; Jupiter retains everything.

QUESTIONS FOR REFLECTION

- How many times will the mean free path in air change if the pressure is reduced by 1000 times at the same temperature? What will happen to the pressure when λ becomes comparable to the size of the vessel? (Hint: this is how a technical vacuum works.)
- Why does a wet cloth cool down more in hot dry weather than in stuffy and humid conditions? What happens to the "tail" of the distribution when the air is already saturated with vapor?
- Calculate the minimum energy required to dry 200 kg of fresh apricots, from which 140 kg of water needs to be removed. How many kilowatt-hours is that? How much would it cost at a price of 0.1 dollars per kWh?
- Helium atoms are light and fast, but the Earth retains them much longer than hydrogen. How does this explain the Maxwell distribution? And what is happening with helium reserves on Earth?

CHAPTER 5

What is heat?

The misconception about "caloric," the experiments of Rumford and Joule, heat capacity, and latent heat.

The young man rubs his hands together vigorously, then presses them against his cheeks.

Young man: Warm! I rubbed my hands — and they got warm. But where did the heat come from? Nothing came from the fire.

Sage: A good question, and we are in good company: the best minds of two centuries have wrestled with it. First, tell me how you would explain: when you put a pot on the stove, what flows from the fire to the water?

Young man: Warmth. It seems like... it flows. From hot to cold.

Sage: This is exactly what the scientists of the 18th century thought. Let's analyze this idea — it was beautiful, useful, and incorrect.

1. "Thermal fluid": weightless liquid

The sage: In the 18th century, the Scotsman Joseph Black, followed by the French chemist Antoine Lavoisier, proposed a simple picture. Heat is a special **substance, a** weightless "fluid," which was called caloric. A **heated body** contains a lot of caloric, while a cold one contains little. When a hot body touches

a cold one, caloric flows from one to the other, like water from a full vessel to an empty one. It cannot be created or destroyed; it only transfers. Lavoisier even included "caloric" in 1789 in the table of simple substances next to oxygen and hydrogen.

Young man: Sounds logical! It "flows" — as we say: thermal flow.

The sage: And not only "flows." The theory was able to calculate. Black in the 1760s learned to **distinguish between the quantity of heat** and temperature: if equal masses of water at 80 and 20 degrees are mixed, then water at 50 will result. The caloric "shared equally." The theory explained the melting of ice (caloric "hides" in water, not raising the temperature — Black called it latent **heat). The French** engineer Sadi Carnot in 1824 derived the first laws of the operation of heat engines, still speaking of the fall of caloric "from a high level to a low one," like water on a mill wheel. And Carnot's results turned out to be correct!

> **HISTORICAL FACT**
>
> This is a **good lesson: an incorrect theory can yield correct** calculations as long as it is applied in the area where it works. The heat transfer model described heat exchange perfectly. But where heat arose from nothing, it collapsed.

2. Rumford and cannons

The sage: The one who began to dismantle it was a man of very unusual fate. The American Benjamin Thompson served as a military advisor to the Elector of Bavaria, received the title of Count Rumford, and later founded the Royal Institute in London. In 1797–1798, he oversaw the drilling of cannon barrels in the arsenal in Munich and noticed a strange thing: the barrel and the shavings became very hot. The drill became dull, yet the heat did not cease.

The sage: Rumford conducted an experiment. He placed a blunt drill in a barrel, and the barrel was immersed in a box of water. The drill was rotated by horses. The water, about 8.5 kilograms, boiled in two and a half hours — by itself, without fire.

Young man: If caloric is a substance contained in metal, then the metal should have "exhausted" it. Yet it kept getting hotter and hotter?

Sage: Exactly. The supporters of caloric argued that shavings have less heat capacity, and therefore caloric is expelled from them. Rumford checked: the heat capacity of shavings and solid metal is the same. Then he made a decisive conclusion: as long as the horses turn the drill, heat is produced continuously. A source that can be drawn from endlessly cannot be a finite supply of matter. Heat is associated **with motion.**

Young man: Let's estimate how much power that was. To heat 8.5 kg of water from 15 °C to almost boiling, about $Q = 8.5 \cdot 4186 \cdot 85 \approx 3.0$ MJ is needed. Over 2.5 hours, that is 9000 s, this is $3.0 \cdot 10^6 / 9000 \approx 330$ watts.

The sage: This is how much one horse can provide (horsepower — 735 W; and there was also energy spent on heating the metal). That is why this experiment is convincing: the mechanical work of the horses was converted into heat, and in an amount that can be calculated.

> **DELUSION**
>
> "Caloric is a weightless fluid that transfers from body to body and is neither created nor destroyed." Rumford's experiments (1798) showed that heat is generated by friction without a visible limit. Humphry Davy reported

in 1799 that ice melts by friction in a vacuum, but historians doubt the accuracy of these experiments. Nevertheless, caloric died slowly: it remained the standard explanation for another half-century.

3. Mayer and Joule: how many joules are in a calorie?

Sage: To kill a theory, it is not enough to say "heat is motion." It is **necessary to measure:** how much mechanical work is converted into how much heat. If conservation holds — the number will be constant. The first such number was obtained by the German physician Robert Mayer in 1842. He reasoned as follows. The gas was heated at constant pressure, and it expanded, lifting the piston, that is, doing work p ΔV. But for heating at constant volume, there is no work. The difference between the two heats is precisely the heat that went into work.

Sage: Let's repeat the calculation for air. The specific heat capacity of air at constant pressure $c_p = 0.240$ cal/(g·K), the ratio $c_p / c_V = 1.4$. The work of expansion of one gram of air when heated by 1 K is R/M = 8.314 / 28.96 = 0.287 J.

Young man: $c_V = 0.240 / 1.4 \approx 0.171$ cal/(g·K). The difference $c_p - c_V = 0.240 - 0.171 = 0.0686$ cal/(g·K). Thus, 0.287 J is equal to 0.0686 calories, and 1 cal = 0.287 / 0.0686 ≈ 4.18 J!

The sage: Indeed, 1 calorie is approximately 4.19 joules. Mayer, in 1842, based on the inaccurate data of the time, obtained about 3.6 J/cal — within 15% of the correct value. By the way, he came up with the idea on a ship while serving as a doctor in the tropics and noticed that the blood of sailors there was lighter than in Europe. He concluded that in a hot climate, the body "burns" less for its own heating. The idea evolved into the law of conservation of energy.

The sage: But the most accurate measurement was made by the English brewer and amateur physicist James Joule. In the 1840s, he submerged a paddle wheel in water and rotated it with falling weights. As they fall, the weights do work mgh, the paddles stir the water, and the water heats up.

Joule's experiment (diagram): falling weights spin the paddles, mechanical work is converted into heating water

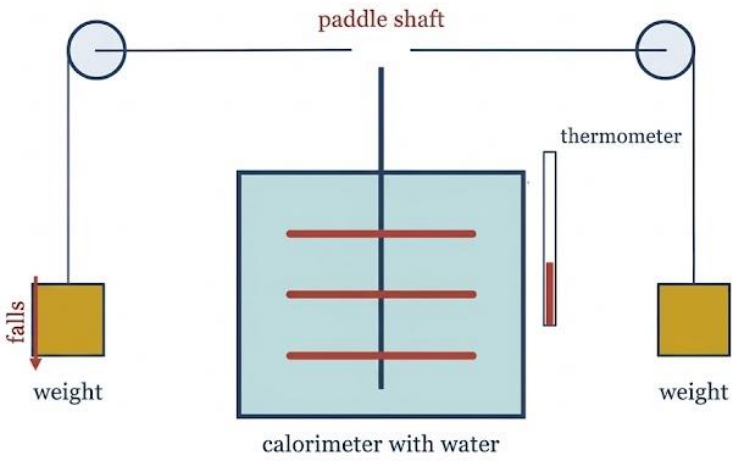


Fig. 5.1. Joule's experiment scheme: the mechanical work of falling weights is entirely converted into heat.

Young man: But the heating is negligible! Let's calculate. A 10 kg weight falls 1 meter — the work is mgh = 10 · 9.81 · 1 ≈ 98 J. And to heat 100 grams of water by 1 K, it takes 0.1 · 4186 ≈ 419 J…

The sage: That is, about four falls of the weight. And Joule measured the heating in hundredths of a degree with a thermometer he made himself, accurate to 1/200 of a degree Fahrenheit. For the 1840s — fantastic. In 1850, he obtained 4.15 J/cal, while now it is accepted that 1 cal = 4.1868 J.

> **HISTORICAL FACT**
>
> It is said that in 1847, during a honeymoon trip in the Alps, Joule couldn't resist and began measuring the temperature of the water at the base and at the top of the waterfall. Falling from 100 meters, the water should warm up by $\Delta T = g\,h / c = 9.81 \cdot 100 / 4186 \approx 0.23$ K. There he coincidentally met the young William Thomson (the future Lord Kelvin), who became interested in his work. It was Thomson who conveyed Joule's ideas to the scientific community. In the same year of 1847, Hermann von Helmholtz formulated the law of conservation of energy in general terms.

The sage: Caloric was not buried immediately. Carnot himself, in posthumous notes published only in 1878, doubted caloric and assessed the mechanical equivalent of heat, but died young in 1832, not having had time to publish.

4. So, what is heat?

Sage: Here is the modern answer. Heat **is the energy that is transferred from one body to another due to a temperature difference. It is not** a substance and not a reserve within the body, but **a means of energy transfer,** like work. The energy reserve in a body **is the internal energy** U: the sum of the energies of the chaotic motion and interaction of all its molecules.

> **DELUSION**
>
> "In a hot body, a lot of heat is stored." No. The body **stores internal energy.** Heat only appears when energy transfers from one body to *another. One* cannot say "there are 5 kJ of heat in the kettle," just as one cannot say "there are 5 rubles of salary in the wallet": money in the wallet is savings, while salary is what has come to you. The word "heat" is a legacy of caloric that we carefully carry in our language, but we understand it differently now.

Young man: Then the answer to my first question: when I rub my palms together, I do work, and the work turns into the internal energy of the skin — it heats up.

Sage: And nothing is created: the energy of your muscles has transitioned into the chaotic motion of skin molecules. Work and heat are two forms **of energy** transfer, internal energy is its reserve. We will formulate this thought as a law — **the first law of thermodynamics —** in the next chapter.

5. Heat capacity: how much heat is needed to heat

Sage: Now we will learn to calculate. To heat a body with mass m by ΔT degrees, it is necessary to transfer heat to it:

$$Q = c \cdot m \cdot \Delta T$$

The quantity c is called **specific heat capacity: how** many joules are needed to heat one kilogram of a substance by one kelvin. For water, it is extraordinarily high.

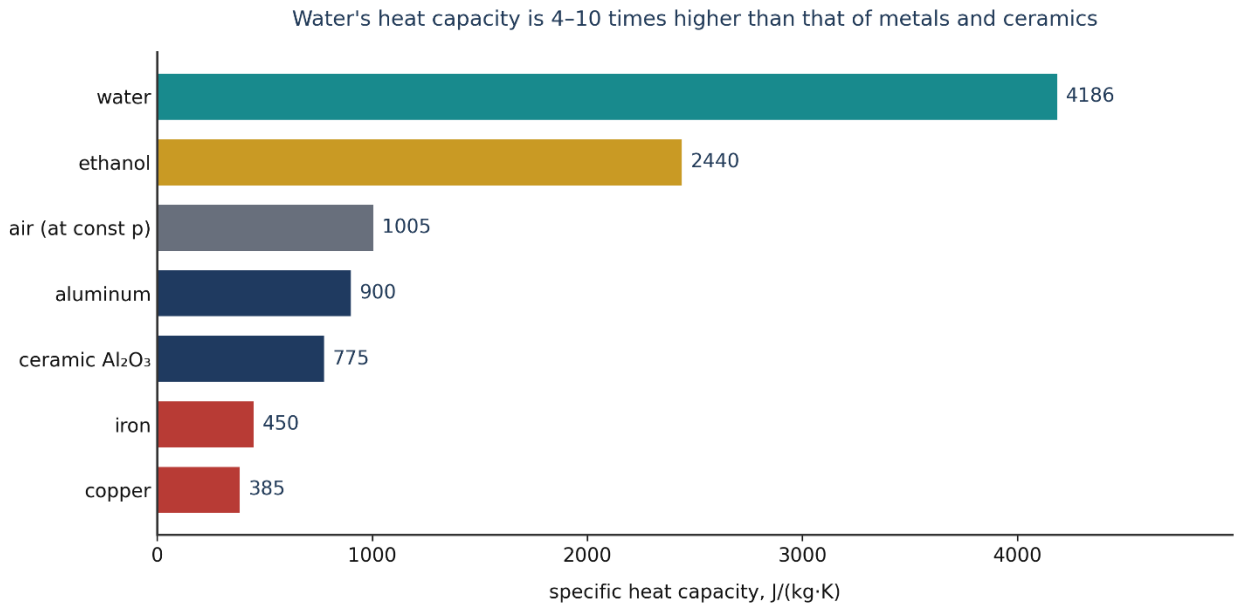


Fig. 5.2. Specific heat capacities of some substances. Water has the highest heat capacity of all common substances.

Sage: Because of this, water heats up slowly and cools down slowly, which is why the sea moderates the coastal climate, and in the heating system, it is advantageous to pump hot water. Let's solve the problem: 200 grams of water at 80 °C is mixed with 300 grams at 20 °C. What temperature will be established?

Young man: Hot water gives off, cold water receives, and according to the law of conservation of energy, the heat given off is equal to the heat received: $c \cdot 0.2 \cdot (80 - t) = c \cdot 0.3 \cdot (t - 20)$. I simplify c: $16 - 0.2\,t = 0.3\,t - 6$, thus, $22 = 0.5\,t$, and $t = 44$ °C.

Sage: This is calorimetry: **the mixture** "exchanged" heat, and no one lost anything. But why does water have such a large heat capacity, while metals have a small one? The answer is hidden in the molecules.

Sage: Let's remember that temperature is related to the average kinetic energy. For a molecule of a monatomic gas (helium, argon), the energy is only translational motion in three directions: (3/2) kT. For one mole, (3/2) R T, which means the molar heat capacity at constant volume $C_V = (3/2)\,R \approx 12.5$ J/(mol·K). This is precisely measured for helium and argon. For diatomic gases, like nitrogen, there is also rotational motion of the molecule, and $C_V = (5/2)\,R \approx 20.8$ — this also matches experimental data at room temperature.

The sage: In a solid body, atoms sit at the lattice nodes and oscillate in three directions. Oscillation is kinetic and potential energy, with (1/2) kT for each direction, which totals $6 \cdot (1/2)\,kT = 3kT$ per atom. For a mole: C = **3R ≈ 24.9 J/(mol·K) —** the same for all simple solids! This was discovered experimentally by the Frenchmen Pierre Dulong and Alexis Petit in 1819.

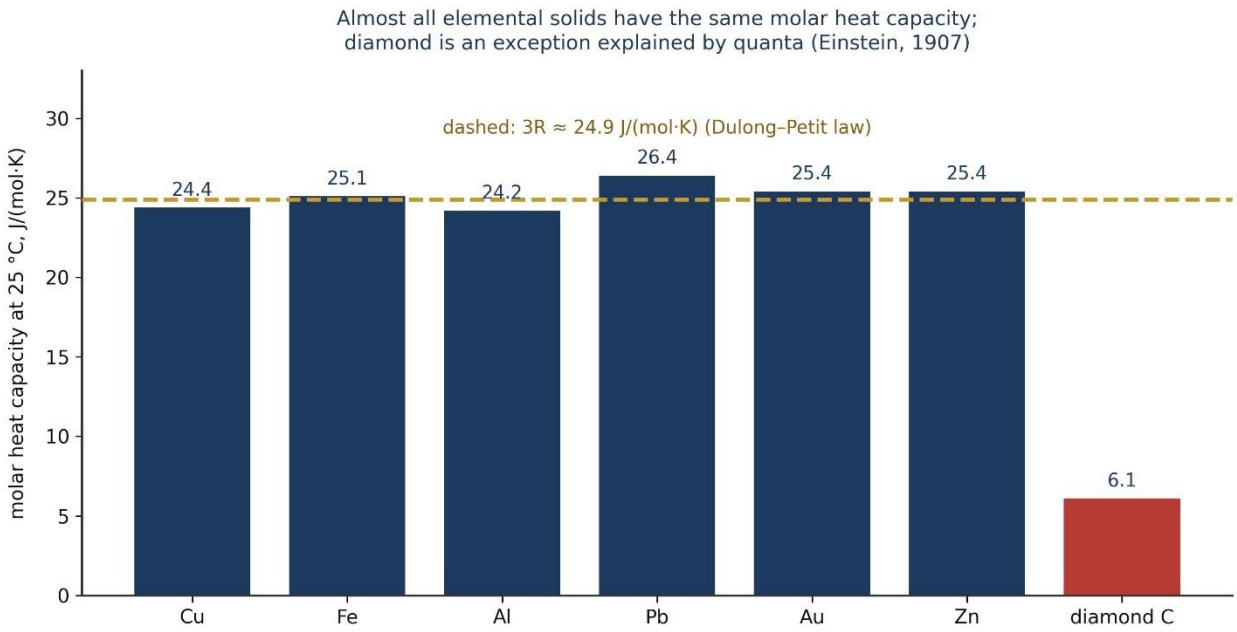


Fig. 5.3. Dulong-Petit law: the molar heat capacity of simple solids is close to 3R. Diamond is a striking exception.

Young man: But why does the diamond "not obey"? 6 instead of 25!

The sage: Because the vibrations of atoms in diamond are rigid and fast, and the energy of vibrations, as Einstein guessed in 1907, **is quantized** — it is released only in portions. At room temperature, the average energy kT is not enough to excite many vibrations of diamond: they are "frozen." This was one of the first confirmations of quantum theory outside of optics. The portions of vibrational energy were later called phonons — **a** term coined by the Soviet physicist Igor Tamm in 1930, by analogy with the photon.

6. Latent heat: when heat "disappears"

The sage: Black noticed another peculiarity. If ice is heated, the temperature rises to 0 °C, and then it stops for a **while, even though** heat continues to be supplied. All the energy goes into breaking the bonds between the molecules in the ice crystal. This heat is called the heat **of fusion; for** water, it is 334 kJ/kg. When all the water has melted, the temperature rises again. Then there is a pause at boiling: **the heat of vaporization** is 2260 kJ/kg.

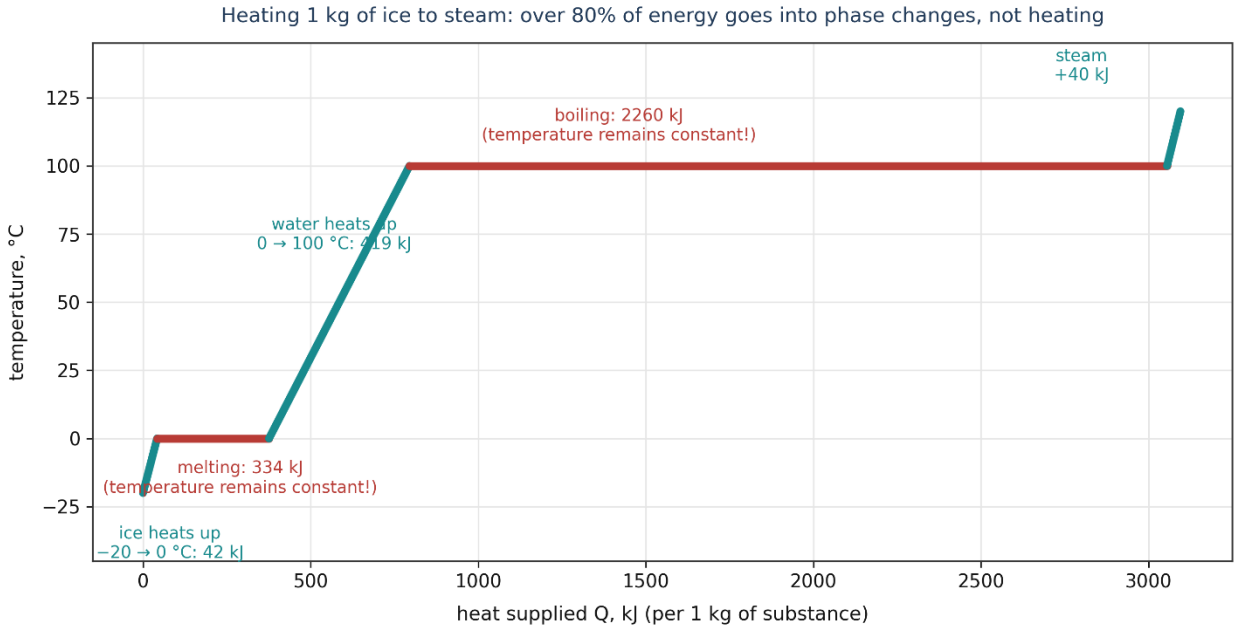


Fig. 5.4. Heating 1 kg of ice from -20°C to vapor at 120°C. The horizontal sections represent the heat of fusion and vaporization.

Young man: Look at how long the "boiling" is! To turn water at 100°C into steam, it takes 5.4 times more energy than to heat it from 0 to 100°C: 2260 / 419 ≈ 5.4.

Sage: Exactly. The amount of energy required to melt ice is the same as that needed to heat water by 80 degrees: 334 / 4.19 ≈ 80. Everyone takes advantage of this: ice in a drink cool more effectively than the same mass of cold water because it melts rather than just heats up. We have already seen in the previous chapter that at 20°C the evaporation energy is slightly higher — 2450 kJ/kg, while at 100°C it is 2260.

7. How heat enters the body

The sage: There are three ways, and you encounter them every **day. Conduction: heat** is transferred through a substance from molecule to molecule, like the handle of a spoon in hot tea. **Convection:** warm substance itself carries energy, like a stream of air from a hairdryer. Radiation: **energy** comes as light, with no substance between bodies, like heat from the Sun. Detailed information about them can be found in the chapter on transfer phenomena. But let’s mention one point now.

Young man: If radiation is light, then how does light turn into heat, that is, into the movement of molecules?

Sage: A good question. Light is an electromagnetic wave, and infrared radiation, as we discussed in the book on optics, is not visible to the eye. When it strikes a substance, the **photon is absorbed:** its energy is transferred to the energy of atomic vibrations, that is, to phonons. The internal energy increases, and the body heats up. In a solid body, heat is indeed the vibrations of the lattice; it spreads from the point of heating — thermal conductivity.

> **HISTORICAL FACT**
>
> In 1800, the astronomer William Herschel passed sunlight through a prism and began measuring the temperature in different parts of the spectrum. The strongest heating was found beyond **the red boundary,** where the eye sees nothing. This is how infrared radiation was discovered. This occurred in the very year when caloric was still considered a "substance": it turns out that heat rays are quite ordinary light waves, just invisible.

Sage: For your industry, the dryers, this is key. When an infrared emitter illuminates a product or ceramic surface, photons are converted into vibrations of atoms and molecules, and the substance heats up "from within" the radiation. Where exactly the energy goes — into water, into ceramics, into the walls — depends on **who absorbs the** radiation: water absorbs strongly in the bands around 3 and 6 μm (we saw this in the chapter on temperature), ceramics — in their bands. And what can happen to the accumulated vibrations in special functional ceramics — this is the author's hypothesis, which we will discuss in the chapter on entropy and probability, separating the verified from the presumed.

8. Where does the energy go during drying?

Sage: Now, armed with the formula Q = c m ΔT and latent heat, let us return to the "honesty scale" from the previous chapter. We will take 100 kilograms of wet product, from which we need to remove 50 kilograms of water, and raise the temperature from 20 to 60°C. The average specific heat capacity of the wet product is assumed to be 3.5 kJ/(kg·K).

Young man: Heating: $Q_1 = 3.5 \cdot 100 \cdot 40 = 14{,}000$ kJ = 14 MJ. Evaporation: at 60°C the heat of vaporization is about 2.36 MJ/kg, so $Q_2 = 50 \cdot 2.36 = 118$ MJ. In total, 132 MJ, which is 36.6 kW·h.

Sage: And what portion goes to evaporation?

Young man: 118 / 132 ≈ 0.89. Almost nine tenths!

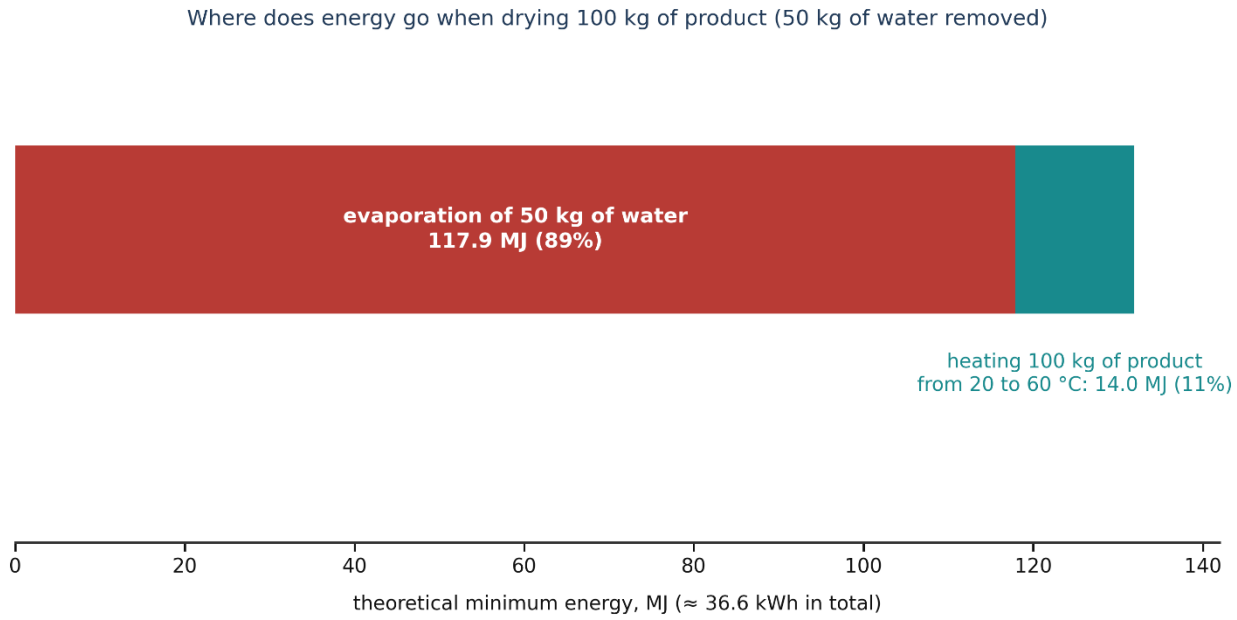


Fig. 5.5. Illustrative calculation of the theoretical minimum energy for drying 100 kg of product with the removal of 50 kg of water. The figures for the heat capacity of the product (3.5 kJ/kg·K) are taken as a typical value for moist food materials.

> **LET'S REMEMBER**
>
> For typical drying, **almost all the energy goes into evaporation** rather than heating the product. Therefore, firstly, the "ceiling" of savings is determined by the heat of vaporization: it is not possible to achieve less than 0.63 kWh per kilogram of water; secondly, the main reserves lie in losses (walls, escaping hot air) and in heat recovery. Any claim such as "our dryer saves 10 times" should be verified by comparing the stated consumption with the theoretical minimum, rather than with someone else's installation.

The young man rubs his hands together again and smiles.

Young man: Now I know that warm hands are not "thermal energy from friction," but my muscular work transformed into the chaotic motion of skin molecules. But if energy is always conserved, then where do heat engines come from? Why can't we build a machine that simply takes heat from the ocean and turns wheels?

Sage: You just asked a question that will require a whole two chapters. But first — the law that connects work, heat, and internal energy. This is the first law of thermodynamics. About it — in the next chapter.

WHAT WE HAVE LEARNED

- The caloric theory (Black, Lavoisier): heat is a weightless fluid. It explained heat exchange but did not explain friction.
- Rumford (1798): when drilling cannons, heat is generated without limit — it is associated with motion. The estimate of his experiment is about 330 W.
- Mayer (1842) and Joule (1840s): mechanical equivalent of heat 1 cal = 4.1868 J. According to our calculation for air, we get 4.18.
- Heat is the energy transferred due to a temperature difference; the stored energy in a body is internal energy.
- $Q = c\,m\,\Delta T$. For water, $c = 4186$ J/(kg·K) — the highest among common substances.
- Molar heat capacity: monatomic gas (3/2)R, diatomic (5/2)R, solid 3R (Dulong — Petit). Diamond exhibits the quantum nature of vibrations (Einstein, 1907).
- Latent heat: melting of ice 334 kJ/kg, vaporization of water 2260 kJ/kg: during the transition, the temperature does not change.
- Infrared radiation, when absorbed, is converted into lattice vibrations — phonons. Herschel discovered IR in 1800.
- During drying, approximately 89% of the theoretical minimum energy is used for evaporation (for example: 100 kg of product, 50 kg of water).

QUESTIONS FOR REFLECTION

- In a pot, there are 2 liters of water at 20 °C. How much energy is needed to bring the water to a boil? How many seconds will it take on a stove with a power of 2 kW, neglecting losses? And how much energy is needed for all the water to evaporate?
- Why does water in a bottle heat up more slowly than sand on the beach on a hot day? What is the connection between this question and the table of heat capacities?
- In a calorimeter with 500 g of water at 20°C, 200 g of iron at 100°C was thrown in. What temperature was established (the specific heat capacity of iron is 450 J/(kg·K))?
- Devise an experiment that could refute the caloric theory. Then find in the text of the chapter experiments that did so.

Chapter 6. The First Law of Thermodynamics

"Energy is not created from nothing and does not disappear into nowhere."

The kettle on the windowsill had long cooled down. The young man sat over a notebook where three words were listed in a column: "heat," "work," "energy." He drew arrows between them — and erased them, and drew them again.

1. Three main characters

Young man: I got everything mixed up, Sage. In the previous chapter, we parted ways with caloric: heat is not a fluid, Rumford drilled cannon barrels, Joule turned blades in water, and work turned into heating. So, is heat just energy? Then why three words instead of one?

Sage: Because these are three different roles. Energy in the body is a reserve, **like** money in an account. Heat and work are the **ways in** which money moves from one account to another. Imagine a piggy bank. It can be filled with cash or through a transfer. After being filled, there is simply "more money" in the piggy bank: it does not remember whether it was cash or a transfer.

Young man: So the gas in the vessel has a reserve — energy…

The sage: …which physicists call internal **energy and denote with** the letter U. This is the kinetic energy of the random motion of molecules plus the energy of their interactions. Heat Q and work A are the two "roads" along which energy enters or exits the gas. Heat is the transfer of energy when bodies at different temperatures come into contact, molecule by molecule. Work is the transfer of energy during movement, for example, of a piston.

> **[OS] Delusion / error — there is no "heat stored" in the body**
>
> The phrase "a lot of heat is stored in the heated gas" is a quiet fragment of caloric from the previous **chapter. Internal energy is** stored. Heat is the name of the process of transfer, not a thing that can be "possessed." Just as one cannot say "there is a translation in my wallet."
>
> Therefore, the state of a gas has internal energy U, but there is neither a "reserve of heat" nor a "reserve of work."

Young man: Alright. And how are they connected?

The sage: In one line — this **is the first law of thermodynamics:**

$\Delta U = Q - A$

Sage: The internal energy of the gas changed by as much as the heat it received, minus the work it performed on the surrounding bodies. Here A is the work done by the gas itself: **if the gas** expands and pushes the piston, $A > 0$ and it spends its energy. If the piston compresses the gas, $A < 0$. Some textbooks write $\Delta U = Q + A'$, where A' is the work done on the gas. **This** is the same thing, just counted from the other side. We will stick to the notation with A as the work of the gas.

Young man: Sounds like the law of conservation of energy. What's new here after mechanics?

The sage: The new aspect is in the word "heat" on the left side. In mechanics, the sum of kinetic and potential energies was conserved as long as there was no friction. And when there is friction, mechanical energy "disappeared." The first law states: it did not disappear; it transformed into internal energy — the bodies heated up. This is exactly what Rumford and Joule demonstrated. The law of conservation of energy became **complete:** it also accounts for heat.

2. How did this come about: a doctor, a brewer, and another doctor

Young man: Who understood this first? Joule?

The sage: The law "hung in the air" in the 1840s, and several people approached it independently at the same time. The first to publish the idea in 1842 was **the German physician Julius** Robert Mayer. According to his own account, the idea came to him while sailing to Java in 1840, when he was bleeding sick sailors: venous blood in the tropics turned out to be lighter than he was used to seeing at home. Mayer reasoned that in the heat, the body needs to burn less food to maintain its temperature — therefore, less oxygen is consumed from the blood. From this, he moved to the general idea: heat and motion transform into each other in strict proportion.

The sage: The English brewer **James Joule approached** it from the other side — through experience. For many years, he turned blades in water, measured heating, and obtained about 4.15 J per calorie. In 1847, the German physicist **and physician Hermann** von Helmholtz presented the law of conservation of energy as a universal principle of nature. Clausius recorded it in 1850 in the form that we use today.

Young man: Meyer, did the doctor from the tropics obtain a numerical value?

Sage: He received it — and in a very clever way. He noticed that air can be heated in two ways, and this affects how much heat is needed. Now we will repeat his reasoning. But first, we need to understand what the internal energy of a gas is equal to.

3. Internal energy of gas: Joule's experiment

Sage: Do you remember the kinetic theory from the fourth chapter? The average kinetic energy of a single molecule of a monatomic gas is (3/2)kT. If the molecules hardly interact with each other (ideal gas), then the gas has no other energy. For one mole:

$U = (3/2)\,R\,T$

Young man: Let's calculate. One mole of helium at 300 K: $1.5 \cdot 8.314 \cdot 300 \approx 3741$ J. A total of 3.7 kilojoules?

Sage: Is it a lot or a little?

Young man: If you divide by $g \approx 9.81$, then $3741 / 9.81 \approx 381$. This would be enough to lift a one-kilogram weight to a height of 381 meters. A little! For a whole mole — $6 \cdot 10^{23}$ molecules — it's very little.

Sage: Good estimate. Now the question that depends on a lot: does U of the gas depend on **its** volume? If the molecules do not interact with each other, then nothing should depend on the distances between them. But that's what we think. What does the experiment show?

The sage: In 1845, Joule connected two copper vessels: one contained air at a pressure of about 22 atmospheres, while the other had the air evacuated. He placed both vessels in a water bath and opened the valve. The air rushed into the empty vessel. The gas did no work in this process — it had nothing to push against. Heat did not escape — everything was in the bath. What did the bath thermometer show?

Young man: Nothing? The temperature hasn't changed?

The sage: Within the accuracy of the experiment — it has not changed. Therefore, $Q = 0$, $A = 0$, and $\Delta U = 0$, while the volume has doubled.

> **[UV] The established fact is that the internal energy of an ideal gas depends only on temperature.**
>
> For rarefied gases, U does not depend on volume (experiments by Gay-Lussac in 1807 and Joule in 1845). For an ideal gas, this is a strict statement; for real gases, it is a good approximation.
>
> Real gases do have a slight effect: when expanding through a porous barrier, they cool down slightly (the Joule-Thomson effect, 1852). This effect is the basis for gas liquefaction, which will be discussed in Chapter 11.

4. The work of gas and the path it has chosen

Young man: How do you calculate work A?

S·Δx = ΔV. Таким образом, работа A может быть записана как A = p·ΔV.

A = p · ΔV

Sage: If the pressure changes during expansion, you need to sum small pieces. On the "pressure-volume" diagram, this is the area **under the curve.**

Young man: So, can the area be different if the beginning and the end are the same?

Sage: This is exactly what you will see now. Let's take a mole of gas at 300 K, a volume of 24.6 L, and a pressure of 1 atm. We will expand it to twice the volume, up to 49.2 L, and return the temperature to the initial state, which means the pressure will ultimately be 0.5 atm. States "1" and "2" are defined. However, the path from one to the other can be chosen in different ways.

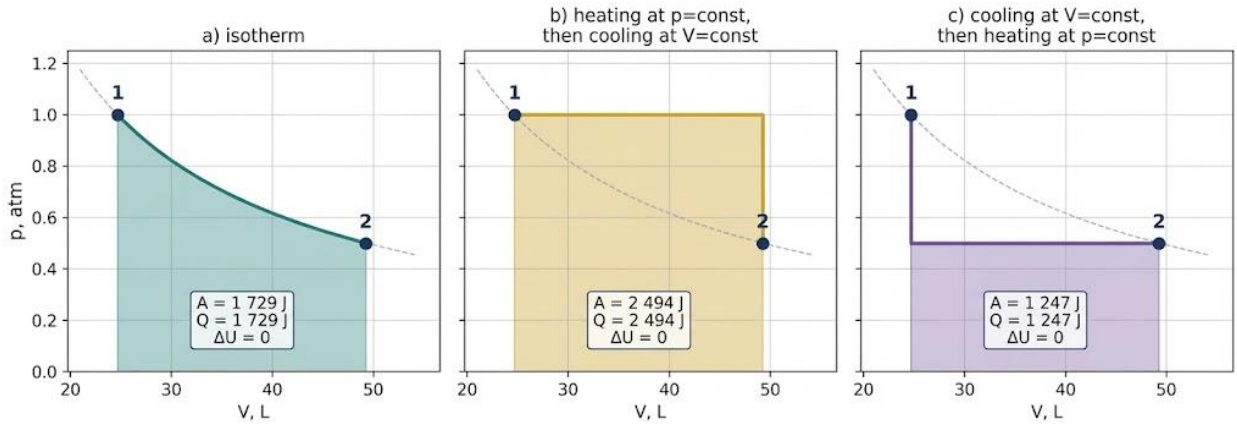


Fig. 1. Three paths from one state to another. Work and heat are different, but their difference Q − A is always equal to zero: the temperature at the end is the same, ΔU = 0.

Young man: The areas are indeed different: 1729, 2494, and 1247 joules! But ΔU is zero everywhere. This means that work and heat depend on the path, but their difference does not.

Sage: This is one of the most important thoughts of the **chapter. U is** a state function: it is defined by where the gas is now, not by how it got there. Q and **A are** process functions. Therefore, we write ΔU (change), but we do not write Δ for Q and A: "change of heat" is nonsense, just like "change of translation."

Young man: I will check on the second path: first heating at a constant pressure of 1 atm from 24.6 to 49.2 L. Work = pΔV = 101325 · 0.0246 ≈ 2494 J. It matches with RT = 8.314 · 300 = 2494. It's good that it matches: heating from 300 to 600 K at the same p is indeed an isobar.

Sage: Notice that you performed the check yourself, not trusting the drawing. That's how it should be.

5. Four typical processes

Sage: Now we will apply the first law to four simple processes. You will see: each of them "turns off" one of the terms.

Process	Condition	Work A	ΔU	Heat Q
Isochoric	V = const	0	(3/2)R·ΔT	equals ΔU
Isobaric	p = const	p·ΔV = R·ΔT	(3/2)R·ΔT	ΔU + A = (5/2)R·ΔT
Isothermal	T = const	$RT \cdot \ln(V_2 / V_1)$	0	equals A
Adiabatic	Q = 0	= −ΔU	(3/2)R·ΔT	0

Young man: Is an adiabatic process one without heat exchange?

Sage: Yes. Either the vessel is perfectly thermally insulated, or the process occurs so quickly that heat does not have time to escape. We will discuss this in detail soon. For now, let's see what these processes look like on a "bookkeeping" diagram: for one mole of a monatomic gas, heating by 100 K, expansion by a factor of two.

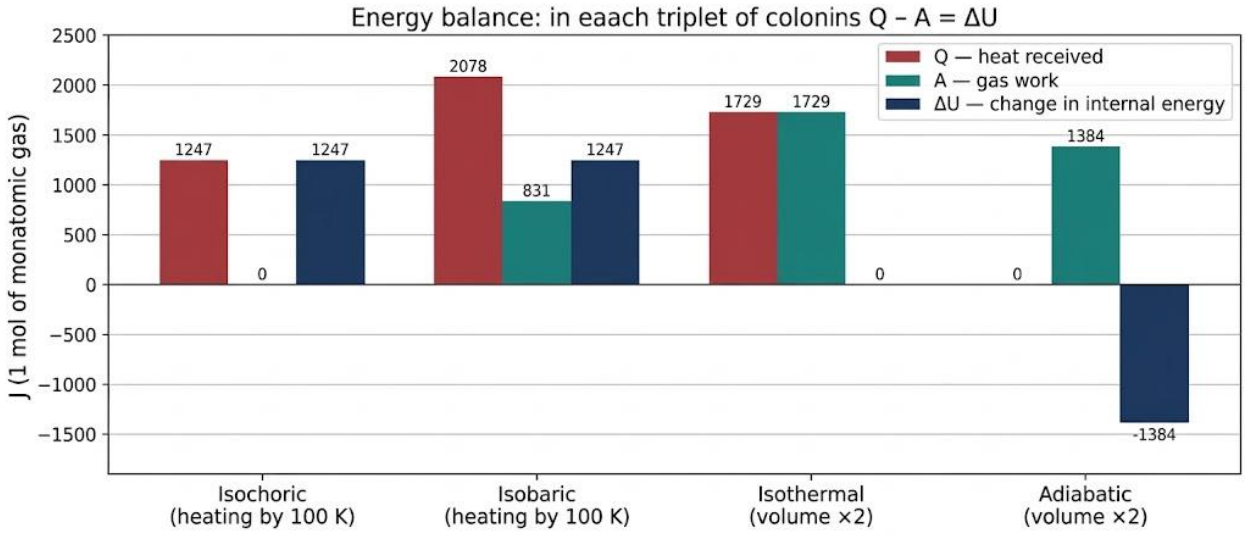


Fig. 2. Energy balance in four processes for one mole of a monatomic gas. In each group of columns, Q − A = ΔU.

Young man: Look, what's interesting: at the same heating of 100 K, the isobaric process requires more heat — 2078 J compared to 1247. The extra 831 J is exactly the work of expansion R·ΔT = 8.314·100.

Sage: That's right. And now pay attention: you have just seen **why** a gas requires more heat at constant pressure than at constant volume. Part of the energy goes not into heating, but into expanding the surrounding air. This leads us to the concept of heat capacity.

6. Heat capacity and Mayer's formula

Sage: Heat capacity is the amount of heat required to raise the temperature of a substance by one kelvin. For a gas, the method of heating is important. If at constant volume, we obtain CV, and at constant pressure — Cp. For one mole of a monatomic gas:

$_{CV}$ = (3/2)R, Cp = CV + R = (5/2)R

Young man: The second formula is what I just saw in the table: the excess R corresponds to the work of expansion. But why R? The work done during heating by 1 K: p·ΔV, and from pV = RT at p = const we get p·ΔV = R·ΔT. In total, R per kelvin per mole. Right?

Sage: That's correct, and this is the **general** formula for any ideal gas, not just monatomic. It is called Mayer's formula: Cp − CV = R. Mayer derived his "mechanical equivalent of heat" from it.

Young man: How is that? There are no calories in the formula.

Sage: In modern units — no, because both Cp and CV are measured in joules. But in 1842, heat was measured in calories, and work — in kilogram-meters. Mayer took the measured heat capacities of air in calories and compared their difference with the known work of expansion. Let's try to repeat this with modern data. For air, cp ≈ 0.240 cal/(g·K), the ratio of heat capacities γ = 1.40.

Young man: Then cV = cp / γ = 0.240 / 1.40 ≈ 0.171, and $_{\text{the difference}}$ cp − cV ≈ 0.0686 cal/(g·K). The work of expansion of one gram of air when heated by 1 K: R/M = 8.314 / 28.97 ≈ 0.287 J. Dividing: 0.287 / 0.0686 ≈ 4.18 J per calorie!

The sage: Modern significance. And now the most interesting part: Mayer in 1842 obtained 365 kgf·m per kilocalorie, which is about 3.6 J/cal — almost 15% less.

> **[OD] Justified addition — where Mayer's error comes from (reconstruction)**
>
> If we substitute into the same calculation the data that is usually provided for its era (cp of air about 0.267 cal/(g·K), γ about 1.42), we get cp − cV ≈ 0.079 and 0.287 / 0.079 ≈ 3.6 J/cal — close to Mayer's number. This means that the issue was not with the idea, but with the inaccurate measurements of the heat capacity of air at that time.
>
> This is our reconstruction based on commonly cited data, not a derivation from his manuscripts: we have not verified the exact numbers used by Meyer.

Young man: A beautiful story: the idea is correct, but the number is drifting due to poor instruments. And is γ the ratio of Cp to $_{CV?}$

Sage: Yes, γ = Cp / CV. For a monatomic gas, γ = (5/2)/(3/2) = 5/3 ≈ 1.67. For diatomic gases, such as nitrogen and oxygen, it is 7/5 = 1.40. Where does 7/5 come from?

Young man: From the previous chapters, I remember about degrees of freedom: for a diatomic molecule, in addition to three translational, there are also two rotational. Therefore, f = 5, and each degree of freedom contributes (1/2)R. Then CV = (5/2)R, Cp = (7/2)R, γ = 7/5.

Sage: That's right. The general rule is: γ = 1 + 2/f. Let's compare with experience.

Gas	f	γ according to the formula	γ measured (room temp.)
Helium, argon	3	1,67	1,66 – 1,67
Nitrogen, oxygen, air	5	1,40	≈ 1,40
Carbon dioxide	5 (according to the formula)	1,40	≈ 1,30

Gas	f	γ according to the formula	γ measured (room temp.)
Water vapor	6	1,33	≈ 1,33
Methane	6	1,33	≈ 1,31

Young man: Almost everywhere is good. But carbon dioxide is off: 1.30 instead of 1.40.

Sage: Do not rush to attribute this to inaccuracy. The CO_2 molecule has low-frequency vibrations of atoms that are already partially "activated" at room temperature. **Vibrations are** also degrees of freedom, and they take away some energy. The theory does not lie, but "the molecule is a rigid dumbbell" is too crude a picture for it.

Young man: But why then did the vibrations not activate for nitrogen?

Sage: A very good question, and classical physics stumbled on it. Look.

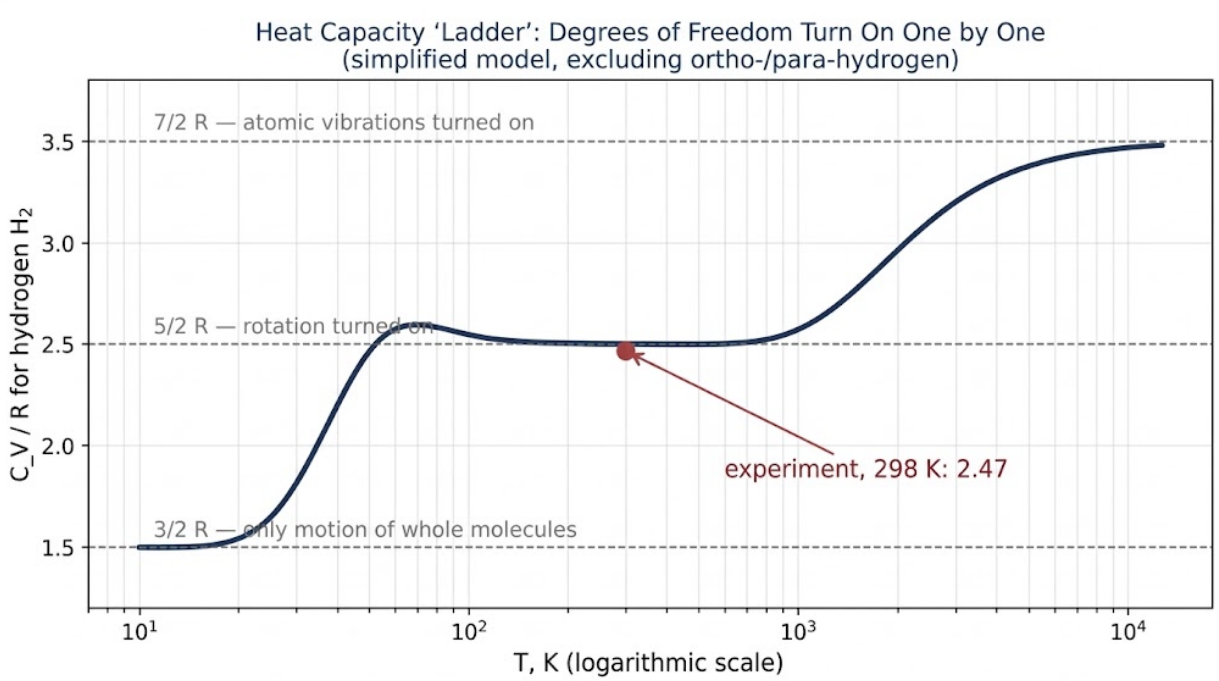


Fig. 3. Heat capacity of hydrogen at different temperatures: degrees of freedom are "activated" one after another (simplified calculation model; point — measurement at 298 K).

The sage: Classical physics states that for a diatomic molecule, vibrations should also contribute, and CV should be 7/2 R at any temperature. However, experiments show a "staircase": at low temperatures, hydrogen behaves like a monatomic gas (3/2 R), then rotation is activated (5/2 R), and only at thousands of kelvins do vibrations come into play.

Young man: As if the degrees of freedom sleep and wake up when the temperature reaches the threshold?

> **[UV] The established fact — the mystery of heat capacity and Lord Kelvin's "two clouds"**
>
> Maxwell also noted in the 1870s that the heat capacities of diatomic gases do not agree with the equal distribution of energy among degrees of freedom. In 1900, Lord Kelvin, in his famous lecture, referred to this as one of the two "clouds" on the clear horizon of physics (the first being the Michelson-Morley experiment).
>
> The solution came from quantum theory: the energy of rotation and vibrations takes on not any values, but "quanta." As long as the thermal energy kT is much smaller than the size of the quantum, the degree of freedom is "frozen." Einstein applied this idea to the heat capacity of solids in 1907. We will return to this story in the last chapter.

Young man: But what about the Dulong and Petit rule? We talked about it: for solids $C \approx 3R \approx 25$ J/(mol·K).

The sage: The rule was derived back in 1819 and applies to metals at room temperature. But here is diamond: its molar heat capacity at room temperature is only about 6 J/(mol·K), four times less. The same "frozen" degrees of freedom exist only in the crystal. This mystery, by the way, was also solved by quantum theory.

7. Adiabate: compression that ignites

Sage: Now the process with $Q = 0$. From the first law: $A = -\Delta U$. If the gas expands adiabatically, it does **work at the expense of its own** internal energy, and therefore it cools down. If compressed, it heats up. It can be shown (this is a short substitution exercise) that for an ideal gas during an adiabatic process, the product remains constant:

$$p \cdot V^{\gamma} = \text{const} \qquad \text{or} \qquad T \cdot V^{\gamma-1} = \text{const}$$

Young man: I'll try with numbers. Let's compress the air by 4 times. $T_2 = T_1 \cdot 40.4$. This means $40.4 \approx 1.74$. Therefore, $T_2 \approx 300 \cdot 1.74 \approx 522$ K, which is about 249°C! Just from the compression, without a heater.

Sage: And this is not a trick, but a practice. Quickly compress the air in an empty tube with a piston with cotton at the end — the cotton will ignite. Such a "pneumatic fire starter" has been used in Southeast Asia for a long time, and it became known in Europe in the early 19th century.

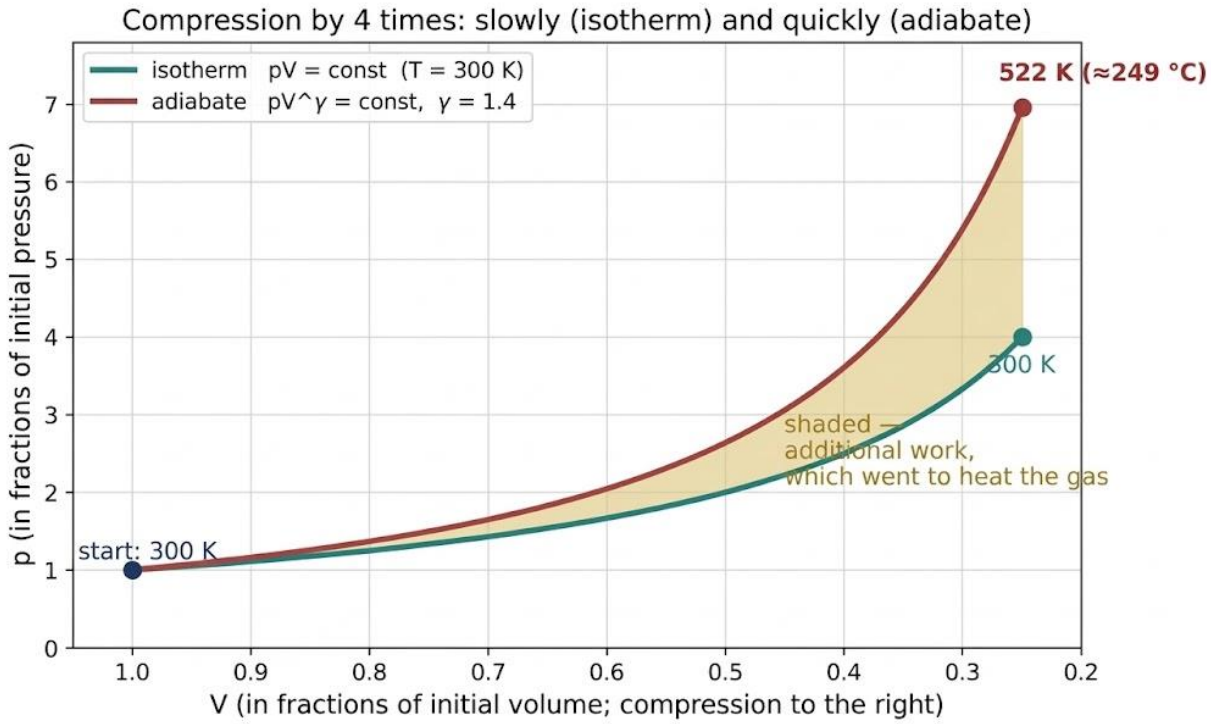


Fig. 4. Compression by four times: along the isotherm (slowly, the gas releases heat) and along the adiabate (quickly, heat has nowhere to go). The adiabate is steeper, and the gas heats up.

Young man: Is the shaded area the difference in work?

Sage: Yes, more work must be expended for adiabatic compression, and all this "extra" work remains in the gas in the form of increased temperature. Now about the engine.

Young man: Diesel? There are no spark plugs there.

Sage: Exactly: the air in the cylinder is compressed by about 15–20 times. Calculate for 16.

Young man: 160.4: ln 16 = 2.77, multiplying by 0.4, we get 1.109, the exponent ≈ 3.03. Then 300 K · 3.03 ≈ 909 K, which is about 636 °C. And diesel ignites at a temperature of about 200–250 °C. That's more than enough!

> **[NI] Not independently confirmed — the legend of Diesel and the spark plug**
>
> It is often said that Rudolf Diesel invented his engine (patent from 1892, working prototype — 1897) inspired by the pneumatic igniter. The calculation above is reliable, but the story of inspiration is passed down from book to book, and we have not verified it with the original source — we leave it as a legend.

Sage: And now a story about a famous mistake — about the speed of sound. What do you think, during which process does air oscillate in a sound wave: slowly, with heat exchange, or quickly, without it?

Young man: I don't know. The sound is very fast... Wait: if there is no heat exchange, then it's an adiabatic process!

[OS] Misconception / error — Newton's error and Laplace's correction

Newton suggested that air in a sound wave is compressed and rarefied at a **constant temperature (isothermally). The** formula $v = \sqrt{(p/\rho)}$ for air at 0°C gives about 280 m/s — noticeably less than the measured ≈ 330 m/s.

In 1816, Laplace pointed out: the oscillations occur too quickly for heat to transfer from the compressed layers to the rarefied ones — the process is adiabatic. **Then a** multiplier $\sqrt{\gamma}$ is added to the formula: $v = \sqrt{(\gamma p/\rho)}$. For air, $\gamma = 1.4$ and $\sqrt{1.4} \approx 1.183$.

Verification: $280 \cdot 1.183 \approx 331$ m/s — matches the experiment. Estimation of the heat exchange time: at half the wavelength ≈ 17 cm (sound at a frequency of 1 kHz) and a thermal conductivity coefficient of air of about $2 \cdot 10^{-5}$ m²/s, it takes about $(0.17)^2 / (2 \cdot 10^{-5}) \approx 1500$ s, that is, tens of minutes, while the oscillation period is one millisecond. Heat "does not have time" at all.

Young man: It's funny: Newton's mistake is not in mathematics, but in the thermal "performance" that the air plays out. By the way, does that mean the speed of sound depends on γ, that is, on the structure of the molecules?

The sage: Exactly, and this will be encountered again in the book on acoustics. And now — the weather.

Sage: An air mass, rising up a mountain, enters an area of lower pressure, expands adiabatically (the thermal **conductivity of** air is too low), and cools — by approximately 9.8 K for every kilometer of ascent. Verification: $g / c_p = 9.81 / 1005 \approx 9.8$ K/km.

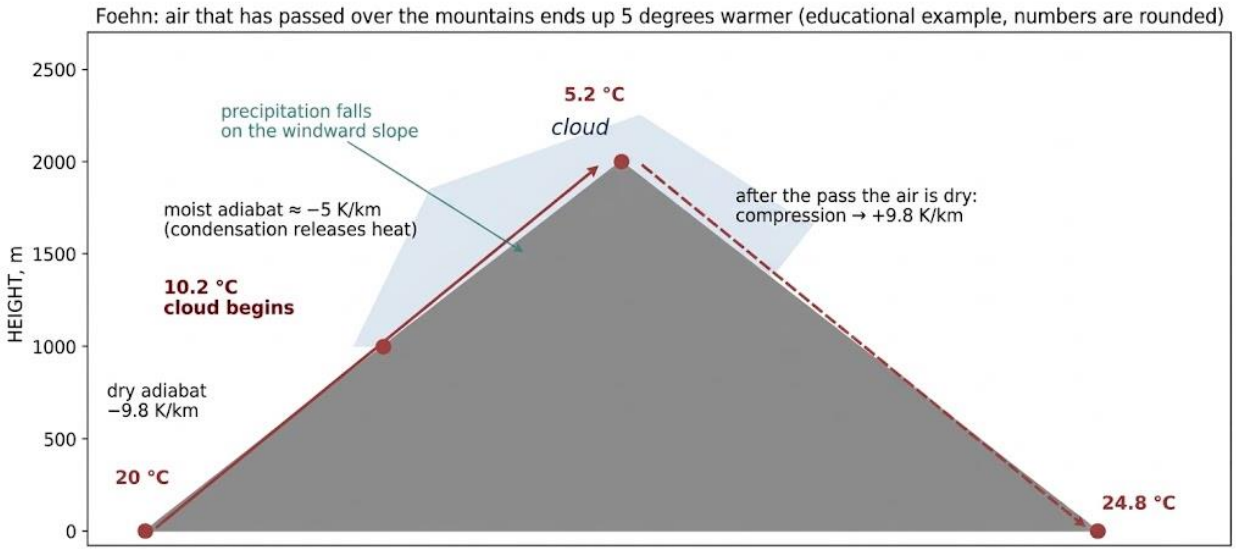


Fig. 5. Föhn: air rises, cools, forms a cloud, and loses moisture, then descends and heats up more than it cooled during ascent.

Young man: Wait, where did the difference come from? I went up and came down the same way — I should return to the original temperature!

Sage: A good question, to which the first principle answers. During ascent, when condensation begins, water vapor turns into droplets and releases heat (about latent heat in chapter 10). **Therefore, moist air** cools more slowly — about 5 K/km. On descent, the air is already dry: some of the water has fallen as rain or snow, and it heats up according to the dry adiabatic lapse rate — 9.8 K/km. Accounting: $20 - 9.8 - 5 + 19.6 \approx 24.8$°C, which means it is almost 5 degrees warmer on the leeward side. This is how föhn and chinook winds arise.

Young man: So, this "gift" is just the heat released during the formation of droplets. Energy did not come from nowhere again!

8. The perpetual motion machine of the first kind and the question "where from?"

The sage: The first principle can also be read as a prohibition: it is impossible to build a machine that performs work without receiving energy from outside. Such an imaginary machine is called a perpetual motion **machine of** the first kind. In 1775, the Paris Academy of Sciences announced that it would no longer consider designs for perpetual motion machines — too much time was being spent on refutation.

Young man: Why "of the first kind"? Does that mean there is a second kind?

Sage: Yes, but about it — in the following chapters. It will violate not the conservation of energy, but something more subtle.

Young man: What if one day they find an experiment where energy is not conserved?

The sage: This has happened before — and it is very instructive. In the 1920s, during beta decay, measurements showed that part of the energy was lost without a trace. Some physicists, including Bohr, were willing to concede that the law of conservation of energy holds only "on average." Wolfgang Pauli proposed a different idea in 1930: part of the energy is carried away by an invisible light neutral particle.

> **[UV] The established fact — neutrino: the law saved a new particle**
>
> Pauli proposed the existence of neutrinos (1930) to preserve the law of conservation of energy. The particle was only detected in 1956 (the Cohen and Reines experiment). When energy "disappears," physicists first look for not a violation of the law, but a new form of energy or a new particle.
>
> From the perspective of modern physics, the conservation of energy is related to the homogeneity of time (Noether's theorem, chapter 12 of the mechanics book): the laws of nature do not change over time — therefore, energy is conserved.

9. The first principle as an accountant

Young man: Sage, I always want to ask about our dryers and ceramics. Does the first beginning say anything about them?

The sage: Says, and it is very useful. Let's start with water. To turn one kilogram of water into steam, about 2.3–2.5 MJ is required — it depends on the temperature. This is the latent heat of vaporization, and no technology can bypass it: the energy needed to "tear" water molecules from the liquid must come from somewhere.

Young man: Then I can estimate. Suppose the dryer consumes 1 kW. In an hour, that's 3.6 MJ. Dividing by 2.5 MJ/kg, it results in approximately 1.44 kg of water per hour — no more! If the passport states "up to 3 kg of moisture per hour" at a power of 1 kW, does that violate the first law?

The sage: Either it violates, or the power in the passport is different, or moisture evaporates not only due to electricity (for example, heat from the Sun or outside air is supplied). But as a verification of passport data, this is a very useful tool.

> **[OD] Justified addition — the first principle as a check of the claimed "savings"**
>
> The minimum energy for the evaporation of 1 kg of water is about 2.4–2.6 MJ (≈ 0.7 kWh). If energy savings are claimed to be k times compared to the previous technology, then the previous technology must have consumed at least k such minimums per kilogram of water. This does not prove or disprove the claim — but **it shows** what needs to be verified: what the conventional technology spent the remaining energy on (heating the heavy furnace, molds, air, heat loss through the pipe).
>
> In Volume 5, we encountered claims of energy savings by factors of several times and hundreds of times. The first principle does not dismiss them in advance. It requires showing the energy balance: how much was spent on water, how much on everything else.
>
> This is our addition to the source materials, not a statement by the authors themselves.

Young man: Well, what about hydrogen? I remember there were high efficiency numbers — 85–95%, determined by gas chromatography.

Sage: Let's reason honestly. Chromatography measures the amount of hydrogen produced **and** the purity of the gas. This is the yield by substance. The energy efficiency is the ratio of the energy that can be obtained from hydrogen (upon combustion about 286 kJ per mole, or ≈ 142 MJ per kilogram) to **the** total energy supplied to the reactor.

> **[OD] Justified addition — output by substance and energy efficiency — are different things.**
>
> The value "85–95%" from the previous volumes refers to the efficiency based on hydrogen output or energy? The definition in these volumes is unknown to us; it affects how to interpret the figure. If it is based on energy, then the denominator should include the total supplied power (in the lamp circuit — the electrical power of the lamp, not just its luminous part), and in the numerator — the energy of the produced hydrogen (based on the higher heating value).
>
> This is a question of definitions, not an accusation: in scientific articles on photocatalysis, efficiency is typically defined explicitly and for specific conditions.

> **[GA] The author's hypothesis — the heat of the lamp: not lost, but participating**
>
> In the coaxial reactor, the author's halogen lamp provides both light and heat simultaneously, and the heat, by design, is not dissipated but participates in the reaction—as a second channel for phonon accumulation. For the initial start, this means a simple requirement: both channels must be accounted for in the balance.
>
> The contribution of each channel individually has not yet been measured. This is a blank line in the energy balance, and it is this that needs to be filled with an experiment.

Young man: And ITE? There are impulses there, and we said that they "fire" at a frequency of 434 Hz.

The sage: Here the first principle gives a beautiful estimate. The impulse can be very powerful — power and energy are not the same. Think about it: a pen-"squeaker," in which the spring is slowly compressed and then instantly straightens out.

Young man: So, the power at the moment of impulse can be huge, but the energy per second is still no more than what has been received. Let's calculate: suppose the emitter consumes 1 kW, and there are 434 impulses per second. Then, for one impulse, it is no more than 1000 / 434 ≈ 2.3 J.

> **[GA] The author's hypothesis is that momentum does not create energy.**
>
> The ITE hypothesis in any formulation — energy accumulation (photon → phonon → accumulation → secondary photon) and the "shot" of the accumulated energy — must pass this accounting test: the average output power cannot exceed the average input power, and the energy of a single pulse cannot be more than approximately 2.3 J for each kilowatt of supplied power at 434 pulses per second.
>
> This is an upper estimate, independent of the mechanism. If the measurement yields a higher value, it means there is an error in the measurement or another source has not been accounted for. If it is lower, the hypothesis with the first beginning does not conflict. The status of the ITE hypothesis remains [G].

Sage: Note that the first principle neither confirms nor refutes the ITÉ. It sets a framework within which any hypothesis has the right to exist. And that is normal: a good framework is also a contribution to science.

10. What we have learned and what remains

Sage: Let's summarize.

- **The internal energy** U is a reserve, a state function. Heat Q and work A are methods of energy transfer, process functions.
- **The first law:** $\Delta U = Q - A$. Energy is neither created nor destroyed; it transforms from one form to another.
- For an ideal gas, U depends only on temperature (Joule's experiment). For a monatomic gas, $U = (3/2)RT$.
- **Heat capacity: Cp** − CV = R (Mayer's formula); $\gamma = Cp / CV = 1 + 2/f$. The degrees of freedom are "frozen" — the first hint at quanta.
- **Adiabatic:** Q = 0, pV^Y = const. Compression heats (diesel, flint), expansion cools (clouds, fohn). Sound is an adiabatic process (Laplace correction).
- The first principle is the "accountant" for any technology, but it says nothing about the direction **in which the** processes are moving.

Young man: That's what worries me. If energy is simply conserved, then nothing prevents hot tea from heating itself up, while the cold air around it cools down. The energy would balance out. But no one has ever seen that!

Sage: You have grasped the main point. The first principle does not explain why hot things cool down rather than the other way around. To answer this, we will need heat engines, Carnot, and a new concept. The next chapter is about steam engines and why even an ideal engine cannot convert all heat into work.

Problems for independent solution

- 1 mole of helium is heated from 300 to 400 K at constant volume. How much heat is needed?
- The same, but at constant pressure. How much heat is there and how much of it will go to work?
- The air was compressed 10 times (adiabatically), the initial temperature is 300 K, $\gamma = 1.4$. What temperature did it heat up to?
- A kettle with a power of 2 kW heats one liter of water from 20 to 100 °C. How much time will it take to heat if there are no losses? (The specific heat capacity of water is 4.18 kJ/(kg·K).)

- A dryer with a power of 2 kW must remove 3 kg of water. What is the minimum time it will take to do this if it requires 2.5 MJ to evaporate one kilogram?

> **[UV] Established fact — answers to the problems**
> 1) Q = (3/2)·8,314·100 ≈ 1247 J.
> 2) Q = (5/2)·8,314·100 ≈ 2078 J, of which 831 J is for work.
> 3) T_2 = 300 · 100.4 ≈ 300 · 2.51 ≈ 754 K, that is about 481°C.
> 4) Q = 4180 · 1 · 80 ≈ 334 400 J; t = 334 400 / 2000 ≈ 167 s ≈ 2.8 min.
> 5) Q = 3 · 2.5 MJ = 7.5 MJ; t = $7.5 \cdot 10^6$ / 2000 = 3750 s ≈ 1.04 h.

Chapter 7. Heat Engines

"Fire drives the machines — but not all of its heat goes to work."

> *The young man brought a pot of boiling water from the kitchen. The lid was slightly bouncing, releasing steam. "Look, Sage," he said, "the steam lifts the lid. But what if we lift a weight instead of the lid?"*

1. The machine that drank fire

Sage: You have repeated a question that people have been asking for centuries. Even in the 1st century, Hero of Alexandria built the aeolipile — a hollow sphere that rotated from the jets of steam. But it was a toy: it did no useful work. The first real need arose in the mines.

Young man: Mines? Why?

The sage: They were flooded with groundwater, and there was no way to pump it out except by the muscle power of people and horses. In 1698, the Englishman Thomas Savery patented a steam pump, and in 1712, Thomas Newcomen built an "atmospheric" engine: steam was admitted into the cylinder, then it was cooled with cold water, the steam condensed, and the external atmospheric pressure pushed the piston down. The piston pulled the beam, and that — the pump.

Young man: So, it was actually air that was pressing, not steam?

Sage: Exactly. The steam only created a vacuum. The machine was bulky and insatiable: to cool down and heat up again, the entire cylinder was alternately cooled and heated. It was "fed" mountains of coal — its efficiency was around one percent or less.

The sage: In the 1760s, Scottish mechanic James Watt, while repairing a model of Newcomen's engine, came up with the idea of moving the **condensation to a** separate vessel so that the cylinder would remain hot at all times. This step and many subsequent improvements brought the steam engine to factories, ships, and locomotives.

[UV] Established fact — Newcomen and Watt

Newcomen — 1712, the first practical steam engine; Watt — separate condenser (idea of 1765, patent of 1769). Watt measured power in "horsepower": one mechanical hp is about 745.7 W (metric hp is 735.5 W).

The efficiency of Newcomen machines was a fraction of a percent or around one percent, while Watt machines had several percent. These are orders of magnitude, not exact figures: they vary in different sources.

[NI] Not independently confirmed — the kettle and young Watt

The popular story about the boy Watt, who supposedly saw the power of steam while looking at a kettle lid, is a beautiful legend. We have no confirmed evidence of this, and in this book, we use it only as an illustration, not as a fact of biography.

Young man: So, Watt improved the machine, but he did not **understand** why it still wasted a large part of the heat?

Sage: That's right. Engineers improved machines by feel. But the question "how much can they be improved at all?" was answered by a person who has never built machines himself.

2. The twenty-eight-year-old Frenchman and the water wheel

The sage: In 1824, the French engineer **Sadi Carnot** published a thin book titled "Reflections on the Motive Power of Fire." He was 28 years old. He pondered why the English, with their steam engines, were ahead of France in industry and decided to understand the very principle.

The sage: His main idea is that work is not provided by the **fire itself, but by the** temperature difference. He compared this to a water wheel: water falls from a height to a height, and the work depends on the height of the fall. Heat, Carnot said, "falls" from a hot body to a cold one, and it is precisely during this fall that work can be obtained.

Young man: It sounds beautiful. But didn't he have a caloric theory?

The sage: There was — and this is exactly how he reasoned: heat "flows," like water, and along the way its quantity does not change. This is incorrect. We know from chapter 6 that part **of the heat** is converted into work, and less reaches the refrigerator than was taken from the heater.

[OS] Delusion / error — incorrect theory — correct conclusion

Carnot based his theory on caloric (heat is not consumed, but only transferred), and he was mistaken in this. Nevertheless, his main result — the limit of efficiency, which depends only on temperatures — turned out to be correct. Later, Clausius and Kelvin revised the proof, preserving the essence and abandoning the concept of caloric.

In the posthumous notes of Carnot, published only in 1878, he himself already doubted the caloric and provided an estimate of the mechanical equivalent of heat, close to what Mayer and Joule obtained. Sadi died of cholera in 1832 at the age of 36. His work was hardly noticed during his lifetime; it was returned to science by Clapeyron, who presented it in 1834 using "pressure-volume" graphs.

3. Ideal cycle

Young man: And what did Carnot come up with?

Sage: He asked: what machine will be the best — without friction, without heat loss through the walls? Its cycle should consist only of two types of processes: isotherm (gas in contact with a heater or cooler) and adiabatic (gas is insulated). After all, any contact of bodies **at** different temperatures is a useless loss of potential, like water falling past a wheel.

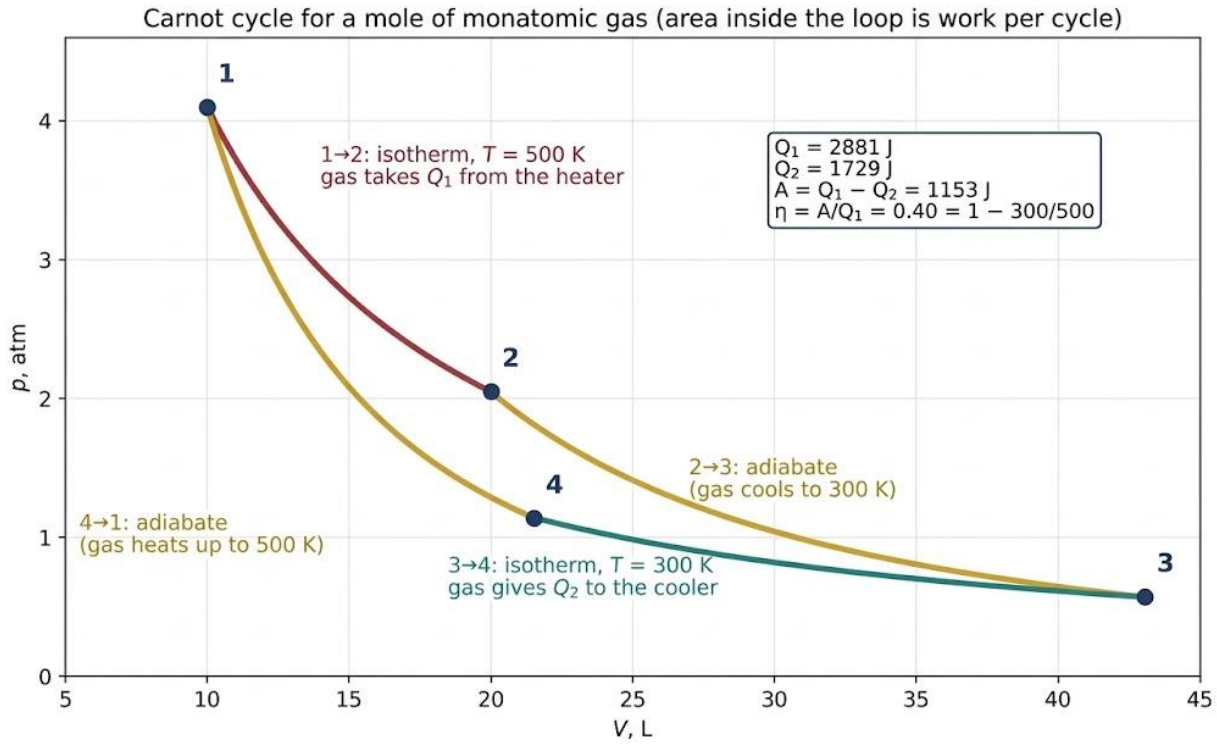


Fig. 1. Carnot cycle for one mole of a monatomic gas: two isotherms and two adiabats. The area enclosed by the loop represents the work done over the cycle.

Sage: Let's go through the cycle. In the section 1→2, the gas expands at the heater temperature T_1 and absorbs heat Q_1 from it. In 2→3, it expands without heat exchange (adiabatically) and cools down to T_2. In 3→4, the gas is compressed at the refrigerator temperature T_2, and it releases heat Q_2. In 4→1, it is compressed adiabatically, and it heats back up to T_1.

Young man: And the work? If we consider the first law: for a complete cycle, the gas returns to its initial state, which means $\Delta U = 0$, and $A = Q_1 - Q_2$. The area of the loop is this difference.

The sage: Correct. And the efficiency is what we obtained (work) from what we paid (heat from the heater):

$$\eta = A / Q_1 = (Q_1 - Q_2) / Q_1 = 1 - Q_2 / Q_1$$

Sage: Now let's express Q_1 and Q_2 in terms of temperatures. On the isotherm, $\Delta U = 0$, therefore $Q = A = RT \cdot \ln(V_{end}/V_{start})$.

Young man: Then $Q_1 = R \cdot T_1 \cdot \ln(V_2 / V_1)$, and $Q_2 = R \cdot T_2 \cdot \ln(V_3 / V_4)$. The ratio $Q_2/Q_1 = (T_2/T_1) \cdot [\ln(V_3/V_4) / \ln(V_2/V_1)]$. I need to show that the logarithms are equal.

Sage: Use [adiabats:] $T \cdot V^{\gamma-1} = \text{const}$ (chapter [6].) For section 2→3: $T_1 \cdot V_2^{\gamma-1} = T_2 \cdot V_3^{\gamma-1}$. For 4→1: $T_2 \cdot V_4^{(\gamma-1} = T_1 \cdot V_1^{\gamma-1}$.

Young man: I divide one by another: $(V_2/V_1)^{\gamma-1} = (V_3/V_4)^{\gamma-1}$. Thus, $V_2/V_1 = V_3/V_4$ and the logarithms coincide! Then $Q_2/Q_1 = T_2/T_1$.

The sage: So:

$$\eta_{Carnot} = 1 - T_2/T_1$$

Young man: I will check in Fig. 1: $T_1 = 500$ K, $T_2 = 300$ K, $\eta = 1 - 0.6 = 0.4$. According to the figure: $Q_1 = 2881$ J, $Q_2 = 1729$ J, A = 1153 J, 1153 / 2881 = 0.40. It matches!

Sage: Note two consequences. First: the temperatures here are absolute, in kelvins **(do you** remember the scale from chapter 2?). Second: the efficiency is equal to 100% only if $T_2 = 0$ K, that is, never.

> **[UV] The established fact is that for an ideal gas, the result has been verified by direct calculation.**
> The formula $\eta = 1 - T_2/T_1$ was derived above for an ideal gas. In the next chapter, we will see that it **is** valid for any working substance: steam, air, light — this follows from the second law.

4. Why it is better not to

Young man: But I can come up with another machine using a different gas and a different cycle. Why can't it be better?

Sage: This is **the Carnot** theorem: no heat engine operating between a heater T_1 and a refrigerator T_2 can have an efficiency greater than that of an ideal reversible machine. And all reversible machines have the same efficiency—regardless of what is poured into them.

The sage: The idea of the proof is as follows. Let there be a "supermachine" with an efficiency higher than that of a Carnot **machine. We connect it** to a Carnot machine running in reverse (like a refrigerator). The supermachine takes heat from the heater and does work, which drives the Carnot refrigerator that returns heat to the heater. The net result remains: heat transfers from a cold body to a **hot one by** itself, without any expenditure.

Young man: But the first beginning does not interfere with that! Energy is conserved.

Sage: It is precisely here that we again encounter the question on which we concluded chapter 6. The first law does not prohibit this. But the experience of all human life indicates: a cold body does not heat a hot body by itself. Therefore, the Carnot theorem is not a consequence of the first law, but a consequence of a **new law of nature. It** will be formulated by Clausius and Kelvin; it will be the subject of chapter 8.

> **[OD] Justified addition — what exactly does the "supermachine" prove**
> The reasoning above is a scheme, not a complete deduction: it relies on the postulate that spontaneous heat transfer from cold to hot is impossible. We do not prove it, but accept it as an empirical fact, and will justify it in Chapter 8. This caveat is important: a "proof" without a postulate would be circular.

5. The ceiling and reality

Young man: And what does this give in practice? Let's substitute. Newcomen engine: steam at 100°C, cold water at about 30°C. $T_1 = 373$ K, $T_2 = 303$ K. $\eta = 1 - 303/373 = 0.188$. Less than 19%!

Sage: Note that even an ideal machine with such a temperature difference would not yield more than 19%. The real Newcomen machine produced about one percent: the rest was lost to heating and cooling of the cylinder walls, friction, and steam leaks. Watt, by removing the condenser, increased the efficiency several times. It still could not be higher until the steam temperature was raised.

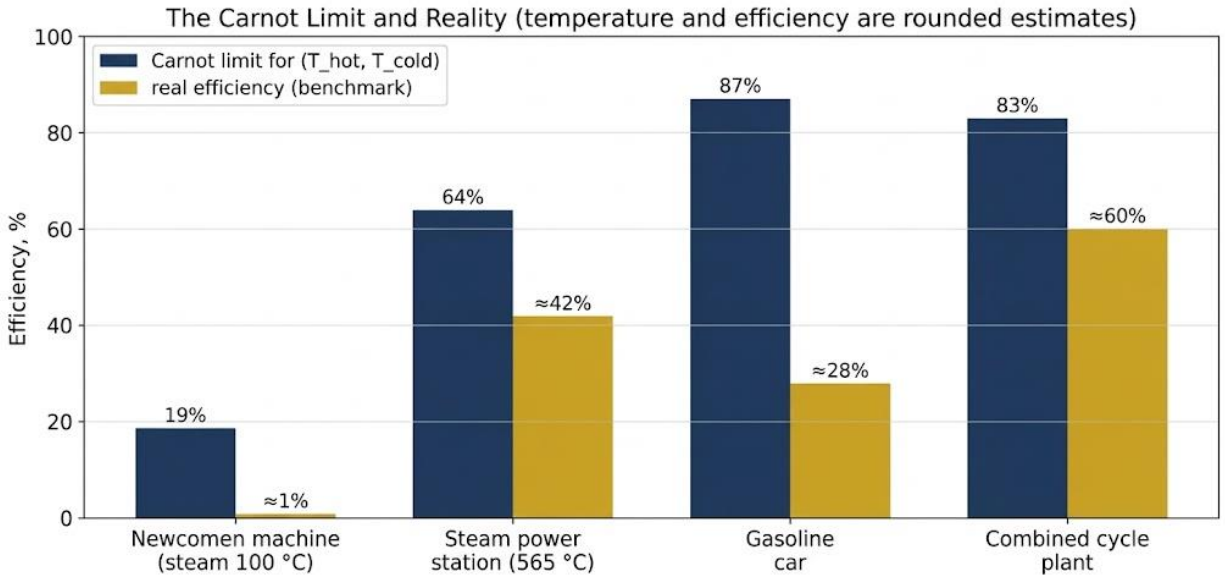


Fig. 2. Carnot limit and actual efficiency for four machines (temperatures and efficiency values are rounded estimates, not specifications).

The sage: The main point is clear from the diagram: the higher the temperature of **the** heater, the higher the limit. Therefore, modern steam power plants operate at 550–600°C, while gas turbines operate at gas inlet temperatures of about 1500°C. The actual efficiencies are much lower than the Carnot efficiencies, but they strive to reach them: approximately 40–45% for steam power plants and up to 60% for modern combined cycle plants.

Young man: And what about the gasoline engine? The combustion temperature is two thousand degrees, but the efficiency is only about one third!

Sage: That's a good question. A temperature of 2300 K is the peak temperature of the flare, lasting for a moment; the gas immediately expands and cools, the walls need to be cooled, and the exhaust carries away the hot gases. Therefore, it is more accurate to compare not with the Carnot map at peak temperature, but with the ideal cycle of the engine itself — the Otto cycle.

Sage: For him, the efficiency defines **the degree of** compression $r = V_1 / V_2$. From the adiabatic relations (chapter 6), it follows:

$$\eta_{Otto} = 1 - 1 / r^{\gamma-1}$$

Young man: For r = 10 and γ = 1.4: $10^{0.4} \approx 2.51$; $1/2.51 \approx 0.398$; $\eta \approx 1 - 0.398 \approx 0.60$. Sixty percent! But real gasoline engines have about 30.

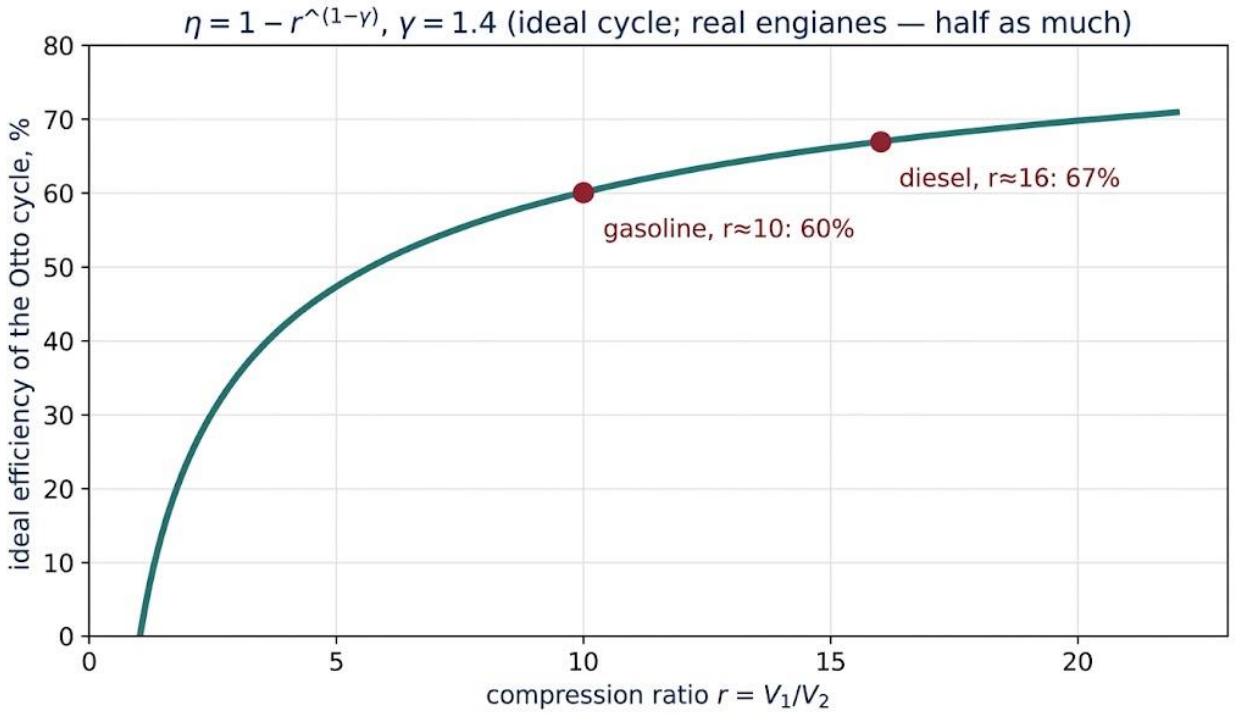


Fig. 3. Ideal efficiency of the Otto cycle as a function of the compression ratio.

Sage: The ideal cycle is a ceiling for this scheme, not a promise. In a real engine, there is friction, heat loss to the walls, incomplete combustion, and air throttling. About half remains. But the formula explains why engineers increase the compression ratio: an increase in r raises the ceiling. And why a diesel engine with $r \approx 16$ is more economical than a gasoline engine.

Young man: Then why not compress even more in the gasoline one?

Sage: Because the gasoline-air mixture self-ignites from compression too early — this is "detonation" (knocking in the engine). In a diesel engine, pure air is compressed, and fuel is injected at the end, so there is no such problem.

> **[OSH] Misconception / mistake — "lost heat" is not the fault of bad engineers.**
>
> A common thought: "if only the designers tried harder, the engine would convert all the fuel's energy into work." The limitation found by Carnot is not a technical shortcoming, but a law of nature: even an **ideal** machine at given temperatures must give a portion of $1 - \eta$ to the refrigerator.
>
> Engineers can approach the limit but cannot exceed it. They can raise the limit by increasing the temperature of the heater or lowering the temperature of the refrigerator.

6. The car in reverse

Young man: You said that the Carnot cycle can be run in reverse. What will happen?

Sage: Refrigerator! We expend work A (electricity) and transfer heat Q_2 from a cold place to a **warm one. More heat is** released in the warm place than taken in: $Q_1 = Q_2 + A$. The first law is upheld.

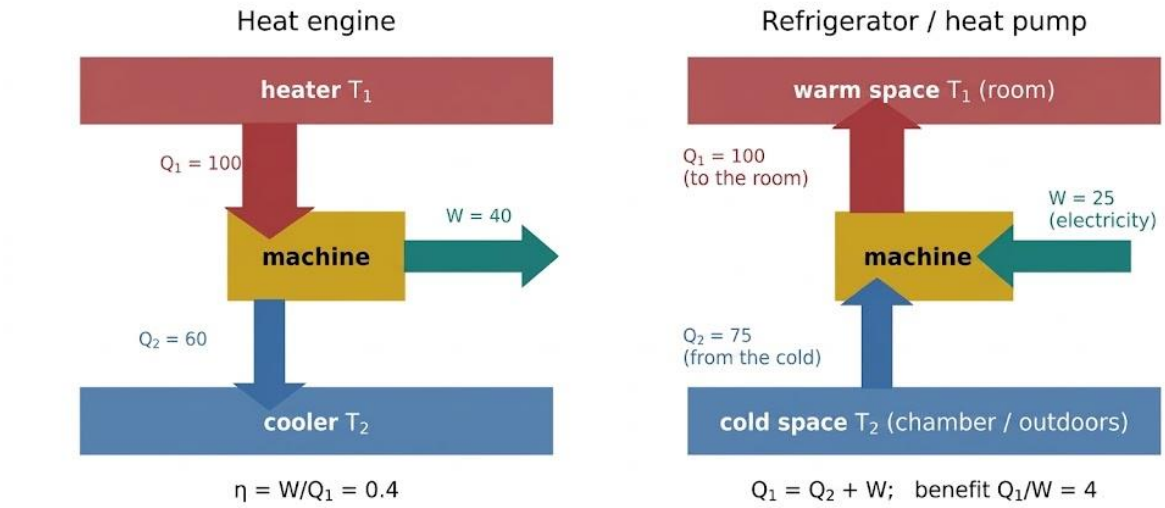


Fig. 4. On the left — a heat engine: heat flows from hot to cold and part of it is converted into work. On the right — a refrigerator or heat pump: work causes heat to flow "uphill."

Sage: And now let's place the "refrigerator" with its window facing the street: the cold place is the street, the warm one is the **room. This is** a heat pump: it heats the house by transferring heat from the street into it. An electric heater converts 1 J of electricity into 1 J of heat. And the pump?

Young man: From the scheme: spent 25, received in the room 100. That is, four times more than the electric heater. So where does the extra heat come from?

Sage: From the street. There is also thermal energy in the cold, because "cold" does not mean "zero." For an ideal pump, the gain (conversion coefficient) is equal to:

$$K = T_1 / (T_1 - T_2)$$

Young man: For a room at 20°C (293 K) and outside at −10°C (263 K): K = 293/30 ≈ 9.8. A real pump, say, is twice as bad — which means about 4–5 J of heat for every joule of electricity. And at −20°C: K = 293/40 ≈ 7.3. The warmer it is outside, the better.

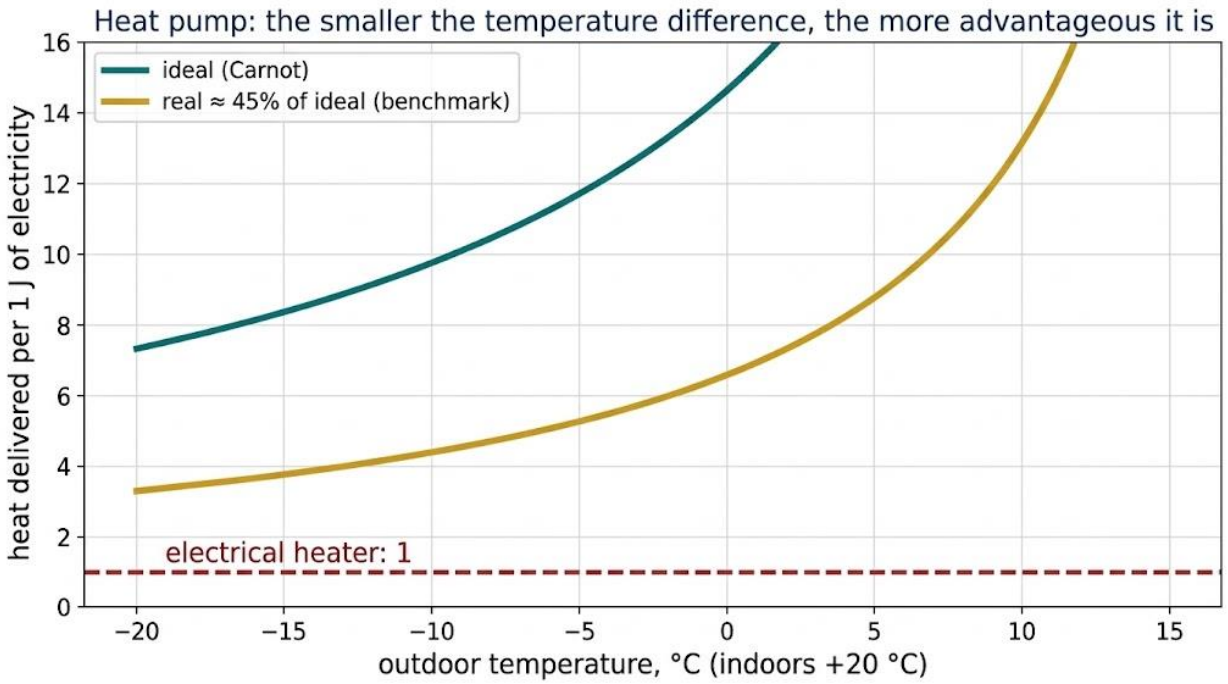


Fig. 5. Heat pump gain as a function of outdoor temperature. The "real" graph is a conditional benchmark (about 45% of the ideal), actual values depend on the model.

> **[OD] Justified addition is not a perpetual motion machine.**
>
> The heat pump delivers more heat than it consumes in electricity, but it does not violate the first law: the missing energy is taken from the environment. It would violate the law if it operated without electrical costs: then heat would flow from cold to hot on its own.

7. What does this mean for our work

Young man: Sage, I now look at our dryers differently. They emit humid hot air. Is it possible to convert its heat into electricity?

Sage: Let's calculate. Let the exhaust be 100°C (373 K), and the surroundings be 25°C (298 K).

Young man: $\eta_{max} = 1 - 298/373 \approx 0.20$. No more than 20%, and **this is for** an ideal machine. A real one will probably give 5–10%, and it will be expensive.

Sage: Therefore, low-temperature heat is more advantageous to use **as heat:** to preheat incoming air (recuperation), to heat water, and if desired, to use a heat pump. Rule: low-temperature heat poorly converts to work but serves excellently as heat.

> **[OD] Justified addition — a general principle of using low-potential heat**
>
> The Carnot ceiling shows: the value of heat depends not only on its quantity but also on temperature. 1 MJ at 1000 K is "worth" more than 1 MJ at 350 K because a lot of work can be obtained from the former, while little can be obtained from the latter. We are using these words intuitively for now; their strict meaning will be clarified in Chapter 8.
>
> This is a general conclusion from Carnot's theorem, not a statement from sources about any specific installation.

Young man: And solar furnaces? We talked about how CSP collects light into a very hot spot.

Sage: Yes, the concentration of light is needed precisely to raise T_1 : the hotter the receiver, the higher the Carnot limit. But there is a nuance here: a receiver heated to a high temperature emits losses itself, so any solar installation has an optimal temperature. This is already a discussion about radiation — chapter 12.

Young man: And hydrogen? We talked about the fact that the energy efficiency of the photocatalytic reactor should be calculated based on all the supplied energy.

The sage: It is worth noting an interesting thing here. Carnot limits **thermal** machines — those that convert heat into work. A fuel cell is a chemical source of electricity: it does not heat the gas but directly converts chemical energy, so the Carnot limit does not apply to it. Another limit, which we will discuss later, is for the reaction of hydrogen with oxygen at room temperature, about 83% of the heat of combustion.

> **[GA] The author's hypothesis — heat that is not emitted**
>
> In the coaxial reactor, the lamp provides both light and heat, and the heat is intended to participate in the reaction instead of simply dissipating. In thermal energy, there is a related **idea —** cogeneration: a power plant that delivers "waste" heat to urban networks utilizes up to 80–90% of the fuel energy, although the electrical efficiency remains around 40%.
>
> The lesson for our reactor: an honest calculation should show what fraction **of the** lamp's heat is actually used by the reaction, rather than dissipated. The separation of contributions from the two channels has not yet been measured, the status remains [G].

8. What we have learned and what remains

- **The heat engine takes** Q_1 from the heater, gives Q_2 to the refrigerator, and performs work $A = Q_1 - Q_2$. The efficiency $\eta = A/Q_1$.
- **The Carnot** cycle (two isotherms, two adiabats) is ideal: $\eta = 1 - T_2/T_1$, temperatures in kelvins.
- **Carnot's theorem: there** is no better reversible machine, and its efficiency does not depend on the working substance. This is not a consequence of the first law.
- **Efficiency can be** increased by raising T_1 or lowering T_2 . For the Otto cycle — by increasing the compression ratio: $\eta = 1 - r^{1-\gamma}$.
- **The reverse machine** — refrigerator and heat pump: $K = T_1/(T_1 - T_2)$.
- Low-temperature heat is poorly converted into work; it is better to use it as heat.

Young man: I still feel uneasy about one thing. Why can't we take the heat from the ocean and convert it into work — we don't need a refrigerator? The first law allows it: there's a lot of energy in the ocean.

Sage: This is indeed the question for the next chapter — you have almost formulated the second law of thermodynamics. It was expressed in different ways, but equivalently, by Clausius and Kelvin. And from it will arise one of the strangest concepts in physics — entropy.

Problems for independent solution

- Find the efficiency of an ideal Carnot engine between 127°C and 27°C.
- The machine received 1000 J from the heater and did 300 J of work. How much was given to the refrigerator? What is the minimum temperature of the heater at which this is possible if the refrigerator is at 300 K?
- Find the ideal efficiency of the Otto cycle with a compression ratio of 8 ($\gamma = 1.4$).

- An ideal heat pump heats a room (20°C) when the outdoor temperature is −10°C. How many joules of heat does the room receive for each joule of electricity? How much better is this compared to an electric heater?
- The inventor claims that his machine operates between 100°C and 20°C with an efficiency of 50%. Is this possible?

> **[UV] Established fact — answers to the problems**
>
> 1) $T_1 = 400$ K, $T_2 = 300$ K: $\eta = 1 - 300/400 = 0.25$, that is, 25%.
>
> 2) $Q_2 = 1000 - 300 = 700$ J; $\eta = 0.3$. The Carnot limit cannot be less than the achieved efficiency: $1 - 300/T_1 \geq 0.3$, from which $T_1 \geq 300/0.7 \approx 429$ K (≈ 156°C).
>
> 3) $80.4 \approx 2.30$; $\eta = 1 - 1/2.30 \approx 0.565$, that is about 56%.
>
> 4) K = 293/30 ≈ 9.8 J per 1 J of electricity; the heater gives 1 J, which means it is approximately 10 times better (ideal case; for real pumps — several times).
>
> 5) Carnot limit: 1 − 293/373 ≈ 0.21, that is, 21%. An efficiency of 50% is impossible.

Chapter 8. The Second Law and Entropy

"Energy is conserved, but events have a direction."

> *The young man poured hot tea into a glass and placed an ice cube next to it. A minute later, he said: "Look, Sage: the ice has melted, the tea has cooled down. I have never seen the opposite happen — for the tea to boil by itself, and the ice to return from the water."*

1. The question that the first beginning left open

Sage: It's good that you started with this. Let's recall how the last two chapters ended. The first beginning only requires that the energy balances. Therefore, it allows for both the cooling of tea and its spontaneous heating due to the cold room: energy is conserved in both cases. And nature chooses only one option.

Young man: So, we need another law — not about the amount of energy, but about the direction of processes?

Sage: Exactly. It is called **the second law of thermodynamics.** Historians of science like to note that it has two formulations, developed independently, which later turned out to be equivalent.

The sage: In 1850, the German physicist **Rudolf Clausius expressed** what you saw over tea: heat does not spontaneously transfer from a colder body to a hotter one. In 1851, the British scientist William **Thomson (future** Lord Kelvin) stated what we discussed in Chapter 7: it is impossible to completely convert heat taken from one source into work in a cyclic process without giving part of it back to the refrigerator.

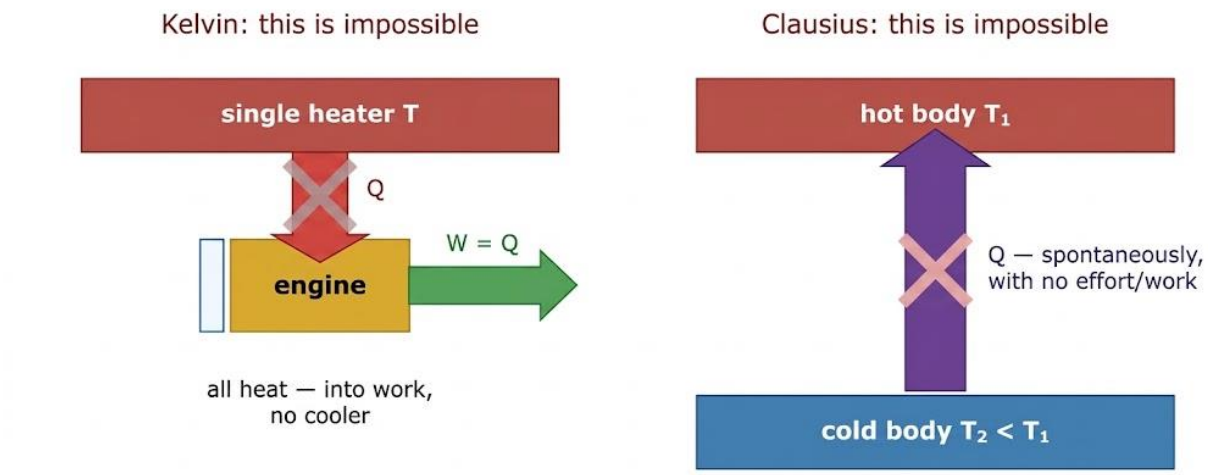


Fig. 1. Two forbidden paintings. On the left is Kelvin's formulation, and on the right is Clausius's formulation.

Young man: These formulations resemble different prohibitions. How do we know that they are the same?

Sage: Show that the violation of one leads to the violation of the other. Suppose a "Kelvin machine" is possible: it takes heat Q from the heater T_1 and completely converts it into work. We use this work to spin a regular refrigerator, which takes heat from the refrigerator T_2 and gives back to the heater more than Q. The result?

Young man: The heater lost Q, but received more Q — which means the remainder came from the cold body! The work inside the system was used up and disappeared, and the net result — heat transferred from the cold to the hot on its own. This is a violation of Clausius.

The sage: That's correct. And the converse — that a violation of Clausius implies a violation of Kelvin — is proven similarly. Therefore, the formulations are equivalent. Wilhelm Ostwald referred to the imaginary Kelvin machine as a perpetual engine of **the second kind — do you** remember I hinted at it in chapter 6?

> **[UF] The established fact — perpetual motion machines of two kinds**
>
> First kind: produces work from nothing, violating the first law. Second kind: converts heat from a single source (for example, the ocean) into work, not violating the first law but violating the second. The ocean contains a colossal amount of energy, but without a temperature gradient, useful work cannot be obtained from it.

Sage: Notice the important point: the second law is also a generalization of experience, not a theorem proven from the first. All attempts to build such a machine have failed. But its status is different from that of a "merely" observed fact: the entire thermodynamics is based on it. This is a law that is supported by the entire body of experiments.

2. What does "irreversibly" mean?

Young man: Let me make a list of things in nature that only go one way. Tea cools down. A ball dropped on the floor stops bouncing. A drop of ink spreads in water and never comes back together. Gas from an open cylinder fills the room. Right?

Sage: A good list. Such processes are called irreversible. **Note that** mechanics (Newton's laws) does not distinguish between the past and the future: a film of an elastic collision between two balls looks plausible

in either direction. However, a film of a ball bouncing higher and higher by itself is instantly recognized by us as reverse playback.

Sage: The ideal opposite case is a **reversible process:** so slow and "careful" (without friction, without temperature gradients) that at any moment it can be reversed, returning both the system and the surrounding bodies to their initial state. The ideal Carnot cycle and the isotherm from Chapter 6 are reversible. Real processes are only approximately reversible.

Young man: Let's recall Joule's experiment with the expansion of air into a vacuum: energy is conserved, temperature does not change. But no one has seen gas spontaneously gather back into one vessel!

Sage: This is indeed an example. The first principle does not prohibit self-compression. To explain this, a new quantity is needed — one that **increases** in irreversible processes and remains **unchanged in reversible** ones. It was discovered by Clausius.

3. Entropy: a new state function

Sage: Let's return to the Carnot cycle. From Chapter 7, we know that for a reversible machine $Q_2 / Q_1 = T_2 / T_1$. Let's rewrite it:

$$\mathbf{Q_1 / T_1 = Q_2 / T_2}$$

Sage: Heat taken at a high temperature is divided by that temperature; heat given off at a low temperature is divided by its own. These two numbers are equal. If the given heat is considered negative, then for the cycle the sum $\Sigma Q/T = 0$.

Young man: It seems like there is some quantity that returned to its original value after the cycle: as much as was "taken" — that much was "given back."

S = $\int$**(δQ/T)**

$$\mathbf{\Delta S = Q_{обр} / T}$$

The sage: In 1865, he named it **entropy** — from the Greek word "transformation." He chose the name to be similar to the word "energy" because, according to him, these quantities are closely related in meaning in physics. The unit of measurement is J/K.

Young man: I will check on our cycle from chapter 7. Q_1 = 2881 J at 500 K — this is 5.76 J/K. Q_2 = 1729 J at 300 K — also 5.76 J/K. It matches!

Sage: And now the most beautiful part. If we plot entropy on the X-axis and temperature on the Y-axis, the Carnot cycle will become a rectangle. The adiabats will be vertical lines (S = const), and the isotherms will be horizontal.

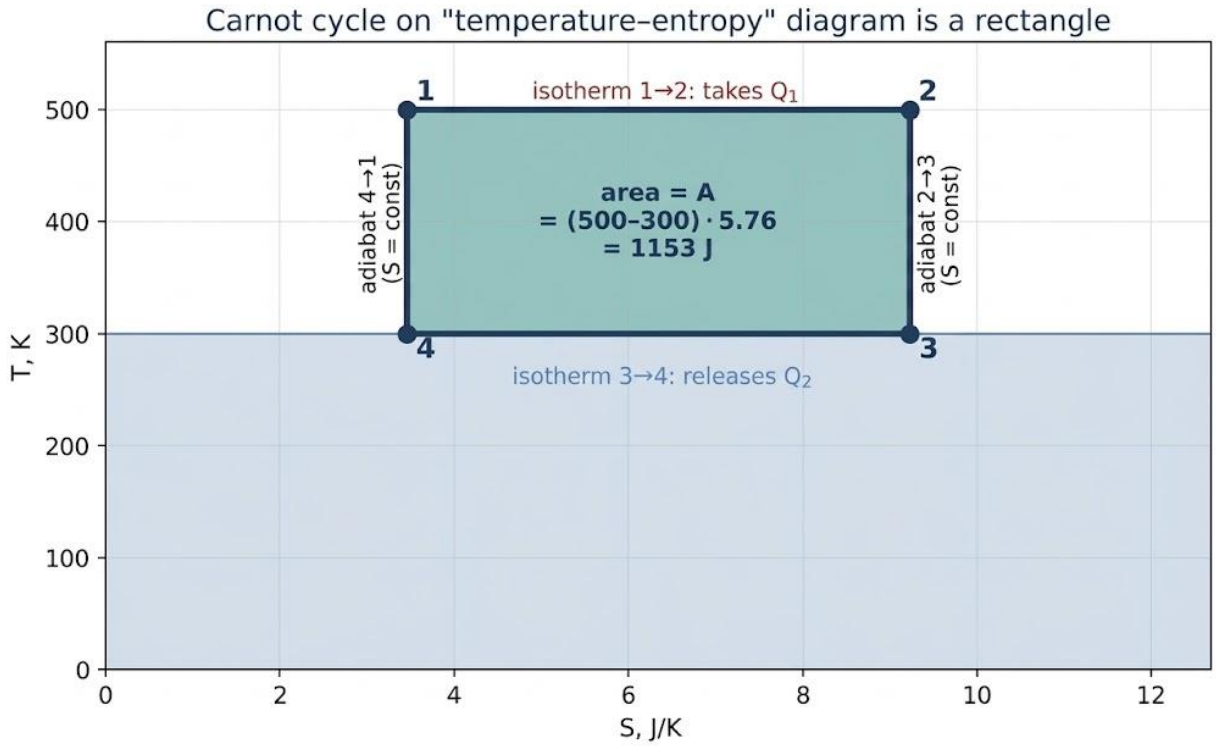


Fig. 2. Carnot cycle on the T–S diagram. The area of the rectangle equals the work done over the cycle. The same 1153 J as on the p–V diagram in Chapter 7.

Young man: The area is equal to (500 − 300) · 5.76 ≈ 1153 J. The same answer in another way.

Sage: One more remark: the adiabats in the Carnot cycle are called **isoentropic because the** entropy remains constant on them. When we said in Chapter 6 that an adiabatic is Q = 0, we were actually referring to the reversible adiabatic: Q = 0, and S does not change.

4. The law of increasing entropy

Sage: Now we use the formula for the irreversible case. Here is our tea. Let's assume a hot body at a temperature of 400 K has transferred heat Q = 1200 J to a cold body at 300 K. What happened to the entropy of each?

Young man: The hot one lost Q, so $\Delta S_1 = -1200/400 = -3.0$ J/K. The cold one received: $\Delta S_2 = +1200/300 = +4.0$ J/K. The sum: +1.0 J/K. The entropy of the entire system increased!

Sage: And now imagine the opposite: heat flows from cold to hot. All signs change — the sum equals −1.0 J/K.

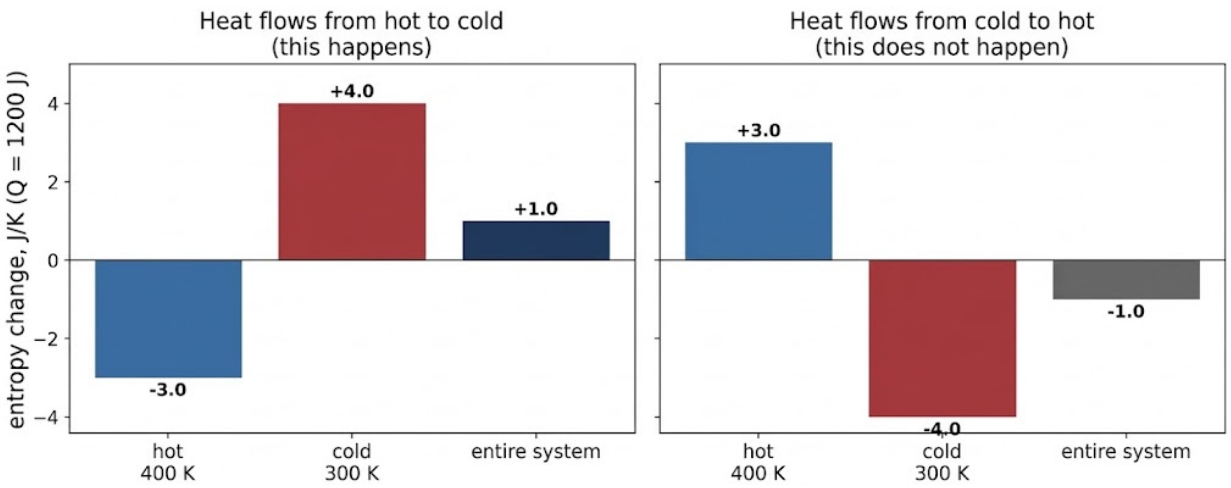


Fig. 3. Entropy accounting during heat exchange. On the left — the real process (the entropy of the system increases). On the right — the forbidden process (the entropy would decrease).

The sage: Thus, the second law is formulated in the language of **entropy: in an isolated system, entropy does not decrease:**

$$\Delta S \geq 0$$

The sage: The sign "=" is for reversible processes, the sign ">" is for irreversible ones. Clausius summarized this in 1865 with two phrases: the energy of the world is constant, and the entropy of the world tends to a maximum.

Young man: Now let's write this for mixing water to feel the numbers. One kilogram of water at 80°C was mixed with one kilogram at 20°C. The final temperature is 50°C (323 K). The entropy of the hot water: $c \cdot m \cdot \ln(T_к/T_н) = 4180 \cdot \ln(323{,}15/353{,}15) \approx 4180 \cdot (-0{,}0888) \approx -371$ J/K. The cold water: $4180 \cdot \ln(323{,}15/293{,}15) \approx 4180 \cdot 0{,}0974 \approx +407$ J/K. The sum ≈ +36 J/K.

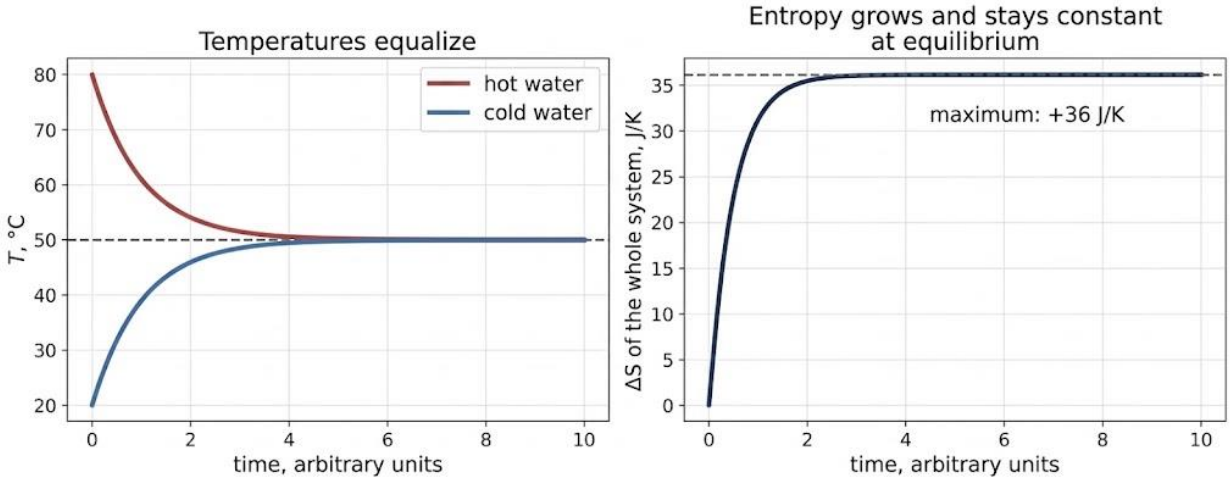


Fig. 4. Alignment of temperatures of two portions of water (1 kg each, 80°C and 20°C) and the increase of total entropy. The process stops when the entropy reaches its maximum.

Sage: Note that in calculating entropy, we replaced the actual irreversible process with an imagined reversible one: we **heated the** cold water infinitely slowly, using a series of bodies with slightly higher temperatures. This is permissible: S is a state function, and it does not matter by what path we arrived.

Young man: And Joule's experiment with expansion into a vacuum? There Q = 0, but the entropy increases?

Sage: It grows. Invent a reversible path from the same initial state to the same final state: isothermal expansion. For one mole, when the volume is doubled, $\Delta S = Q/T = R \cdot \ln 2$.

Young man: $8.314 \cdot 0.693 \approx 5.76$ J/K. The same number as in the Carnot cycle! Only there it was heat transfer, and here there was no heat at all.

> **[OD] Justified addition — entropy — not heat**
>
> Entropy increases even without the transfer of heat (expansion into a vacuum, mixing of gases, friction). Therefore, the statement "entropy is heat divided by temperature" is only true for a reversible path, while entropy itself is defined for any state.
>
> Another useful example: melting ice. 334 kJ is supplied per kilogram at 273.15 K; $\Delta S = 334000/273.15 \approx 1223$ J/K per kilogram. Liquid water is "more disordered" than ice, and its entropy is higher.

5. Arrow of time

The sage: Now we can answer the question we started with. When we "see" the direction of time, we are actually seeing the increase of entropy. The English astrophysicist Arthur Eddington called this the "arrow of time" in 1927.

Young man: But wait. I know from mechanics: Newton's laws are time-reversible. If each molecule moves according to the laws of mechanics, where does irreversibility come from? How can irreversible laws arise from reversible ones?

Sage: You have asked the very question that the Austrian physicist Josef Loschmidt posed in 1876, a friend and opponent of Boltzmann: if the velocities of all molecules are reversed, the system should retrace its path. And in 1896, Zermelo, relying on Poincaré's theorem of return, added: any closed system will sooner or later return almost to its initial state. How should we deal with this?

Young man: And how?

The sage: The answer will be given by Ludwig Boltzmann. He will show that the second law is not an absolute prohibition, but **a law of overwhelming probability.** This is the topic of chapter 9, and it will change the very understanding of entropy. For now, remember: Clausius's formulation is an exact law for macroscopic systems, **where the number** of particles is on the order of 10^{23}.

> **[G] Hypothesis — the heat death of the Universe**
>
> Already Kelvin in 1852 and Clausius in the 1860s noted: if the entropy of the "world" tends to a maximum, then the Universe must come to a state of complete temperature equilibrium, where no processes are possible. This was called "heat death."
>
> This is a hypothesis, and a very shaky one: the conclusion is based on the extrapolation of the law for isolated systems to the entire Universe, but it has extensions and gravity, which change the notion of "maximum entropy." Modern cosmology does not simply answer this question.

6. Where do order and life come from?

Young man: Now I have a question that seems paradoxical to many. Crystals grow, life emerges, order and complexity arise from chaos. However, entropy is supposed to increase. How is this possible?

> **[OS] Delusion / error — "order arises — therefore, the second law is violated"**
>
> The law $\Delta S \geq 0$ **applies to** an isolated system. A living cell, a crystal, a **refrigerator** — these are open or partially open systems: their internal entropy can decrease if more entropy is expelled to the outside. A refrigerator cools products (S inside decreases), but expels more heat into the room (S of the room increases more significantly).
>
> Therefore, the second law does not prohibit either life or crystals. It requires that the account be settled: "order here" is paid for by "disorder there."

The sage: The largest example is our Earth. Energy on Earth does not accumulate: as much as comes from the Sun, that much goes into space. But it arrives as "hot" photons (the surface of the Sun ≈ 5800 K), and it leaves as "cold" ones (the Earth radiates like a body with a temperature of about 255 K).

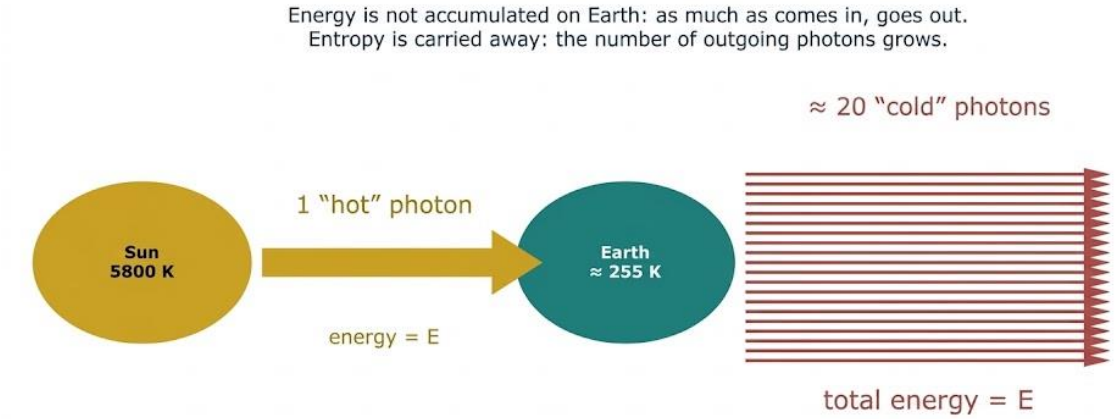


Fig. 5. The Earth is not an energy reservoir, but a flowing system: for every incoming "hot" photon, there are about twenty outgoing "cold" ones.

Young man: So instead of one photon, twenty are used. The number of particles, and thus the disorder, increases, and this "pays" for the order of life?

The sage: The number 20 is approximate: it is the ratio of temperatures, 5800/255 ≈ 23. The estimate is rough, but it shows the essence: the Sun provides the Earth with "low-entropy" energy, life uses it and releases high-entropy energy into space. Erwin Schrödinger expressed this in 1944 with the formula "life feeds on negative entropy."

> **[OD] Justified addition — clarification**
>
> Schrödinger's phrase "negative entropy" is witty, but strictly speaking, it is not accurate; modern authors prefer to speak of the flow of low-entropy energy (or free energy). The estimate "≈ 20 photons" is an order of magnitude, not an exact number.

7. The second aspect of our work: the quality of heat

Young man: Sage, let's return to what is important to us. In chapter 6, we calculated: to evaporate one kilogram of water, at least 2.5 MJ is needed, so a 1 kW dryer cannot evaporate more than 1.44 kg of water

per hour. And then you mentioned the heat pump. But does it mean that the heat pump can bypass this limitation?

The sage: You have identified a mistake in my statement — more precisely, its incompleteness, and it is worth correcting it honestly.

> **[OD] Justified addition — clarification to chapter 6**
>
> In Chapter 6, we estimated that 1 kW of electric heating produces no more than approximately 1.4 kg of evaporated water per hour. This estimate is valid if all thermal energy is derived from electricity and the heat of the steam is expelled. However, if the **heat of** condensation of the steam is returned (recuperation) or heat is added from the surrounding air (heat pump, Chapter 7), then less than 2.5 MJ of electricity is needed per kilogram of water. According to literature, the specific moisture extraction of heat pump dryers is approximately in the range of 1–4 kg per kilowatt-hour, meaning that for the best installations, it exceeds the "ceiling" of 1.44 kg/(kW·h) from Chapter 6, and this does not violate either the first or the second law (the figures are approximate; we did not verify them against specific models).
>
> Thus, the first principle sets the lower boundary of the total energy for evaporation, while the second sets the lower **boundary** of the paid energy, which depends on how the heat flows are organized. A heater with a filament temperature of thousands of degrees for drying at 60°C is a "fall from a height" (increase in entropy) that can be utilized rather than wasted.

Young man: So, "saving several times" can be honest: if heat was lost before, but now it is being returned.

Sage: Yes. The question is only how many times the heat returns and what portion goes "to waste." This is precisely why an energy and entropy balance is needed.

Young man: What does the second law say about the hydrogen reactor? Last time you mentioned the number 83%.

Sage: This promise is from chapter 7, and here it is. To decompose water into hydrogen and oxygen at 25°C, an energy of ΔH ≈ 286 kJ per mole of H_2 is required. However, not all of this energy needs to be supplied in the form of electricity or light ("work"). A portion, equal to TΔS ≈ 49 kJ/mol, can come in **the** form of heat from the surroundings — the reaction increases the entropy of the system (gas is produced from liquid), and this increase is "paid for" with heat.

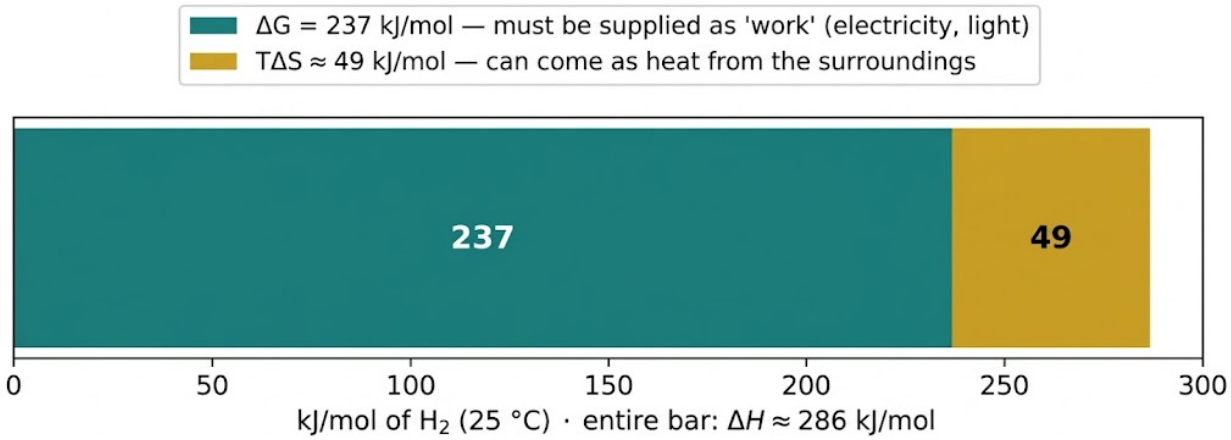


Fig. 6. Energy balance of water decomposition: "work" ΔG and "thermal" part TΔS.

Young man: Then the minimum "working" energy — ΔG ≈ 237 kJ/mol, and the ratio ΔG/ΔH = 237/286 ≈ 0.83. There you have it, 83%!

Sage: That's right. This is also the limit for a fuel cell operating on the reverse reaction. And now you see why the heat in the reactor does not have to be a "loss": it can legitimately pay for the share of TΔS.

> **[GA] The author's hypothesis — the heat of the lamp: how much can it pay for**
>
> In the author's reactor, the lamp provides light and heat, and the heat is intended to participate in the reaction. The second law clarifies how to check this: heat can legitimately cover only part of TΔS (at 25°C — about 17% of ΔH, and this fraction increases with temperature). The rest must come in as "work" — for example, the energy of absorbed photons.
>
> Verification: the declared energy of hydrogen should not exceed ΔH, and the contributions of light and heat must be separated in the balance. The separation of contributions from the two channels has not yet been measured — this is the very line that needs to be filled with experience. Status [G].

> **[GA] The author's hypothesis — a question to ITÉ: where does entropy go?**
>
> The ITE hypothesis describes the accumulation of energy in ceramics (phonons) and its "shot" by a secondary photon — in the form of a pulse. The second law poses a specific question: the transformation of disordered heat into an ordered pulse requires "payment" of the entropy difference, as with any heat engine. In what form and where is the "excess" entropy expelled — thermal radiation, heating of the casing?
>
> We do not claim that the hypothesis contradicts this: it is only through calculation and measurement that it will have to answer this question. Status [G].

8. What we have learned and what remains

- **The second law:** according to Clausius, heat does not flow spontaneously from cold to hot; according to Kelvin, it is impossible to completely convert the heat from a single source into work. The formulations are equivalent.
- **Irreversibility: real** processes occur in one direction; reversible ones are an idealization.
- **Entropy S:** a state function, $dS = \delta Q_{rev}/T$. For an isolated system $\Delta S \geq 0$.
- Entropy increases not only during heat exchange: expansion into a vacuum, mixing, friction. On the T–S diagram, the Carnot cycle is a rectangle.
- Order and life do not violate the law: open systems "throw out" entropy (Earth — into space).
- The second aspect of energy is **its quality:** heat at a high temperature is more valuable than at a low temperature.

Young man: But I feel that something is missing. I know how to calculate entropy, I know that it **increases. But what** is it? Why should the "disorder" of a gas give exactly $R \cdot \ln 2$? Where does this logarithm come from?

Sage: Now you are asking a question not about the formula, but about the meaning — and this is the path of Boltzmann. In the next chapter, we will descend from flasks and heat engines to molecules, calculate the number of ways to achieve the same state, — and see that entropy is the logarithm of the number of such ways. Along the way, we will encounter Maxwell's demon, who tried to deceive the second law.

Problems for independent solution

- The body transferred 600 J of heat from a body at 500 K to a body at a temperature of 300 K. Find the change in entropy of the system.
- Find the change in entropy of 2 moles of an ideal gas during isothermal expansion by a factor of 4.
- One kilogram of ice at 0°C melted in a warm room (20°C). The heat of fusion is 334 kJ/kg. Find the change in entropy of the ice, the room, and the entire system (consider the room to be enormous, its temperature does not change).
- The machine operates between 500 K and 300 K and receives 1000 J from the heater. At what efficiency does the total entropy not increase?
- The inventor claims that his device takes 1000 J from a cold body at 280 K and gives 1000 J to a hot body at 330 K without any work expenditure. Is this possible?

> **[UV] Established fact — answers to the problems**
>
> 1) $\Delta S = -600/500 + 600/300 = -1.2 + 2.0 = +0.8$ J/K.
>
> 2) $\Delta S = \nu R \cdot \ln 4 = 2 \cdot 8.314 \cdot 1.386 \approx 23.1$ J/K.
>
> 3) Ice: $+334\,000/273.15 \approx +1223$ J/K. Room: $-334\,000/293.15 \approx -1139$ J/K. Sum $\approx +84$ J/K. Greater than zero — the process is spontaneous.
>
> 4) The total entropy does not increase in a reversible machine: $\eta = 1 - 300/500 = 40\%$, work 400 J, to the refrigerator 600 J. Verification: $-1000/500 + 600/300 = -2 + 2 = 0$.
>
> 5) Entropy: $-1000/280 + 1000/330 \approx -3.57 + 3.03 = -0.54$ J/K < 0. Impossible — this is a violation of Clausius's statement.

Chapter 9. Entropy and Probability

"The law that is based not on prohibition, but on arithmetic."

> *The young man spilled a handful of coins on the table. Almost half landed heads up. "Strange," he said. "I didn't do anything to make it come out even."*

1. One perspective on entropy — from the inside

Sage: In chapter 8, you asked: what is entropy really and where does the logarithm in the formula for gas expansion come from. The answer was devised **by Ludwig Boltzmann.** Let's start with your coins.

Sage: Let's toss 100 coins. The state "exactly 50 heads" can be achieved in a vast number of ways, while the state "all heads" can be achieved in only one way. If each specific arrangement is equally likely, then "equally" occurs not because something tends toward equality, but because there are many orders **of magnitude more such arrangements.**

Young man: So the gas in the vessel is like coins? Each molecule is "heads" (in the left half) or "tails" (in the right)?

Sage: Exactly. The state that we describe with a few numbers (pressure, temperature, "equal" in both halves) is called the macrostate. **The specific** arrangement and velocities of each molecule are the microstate. **One macrostate** is realized by an enormous number of microstates. Let's denote this number as W.

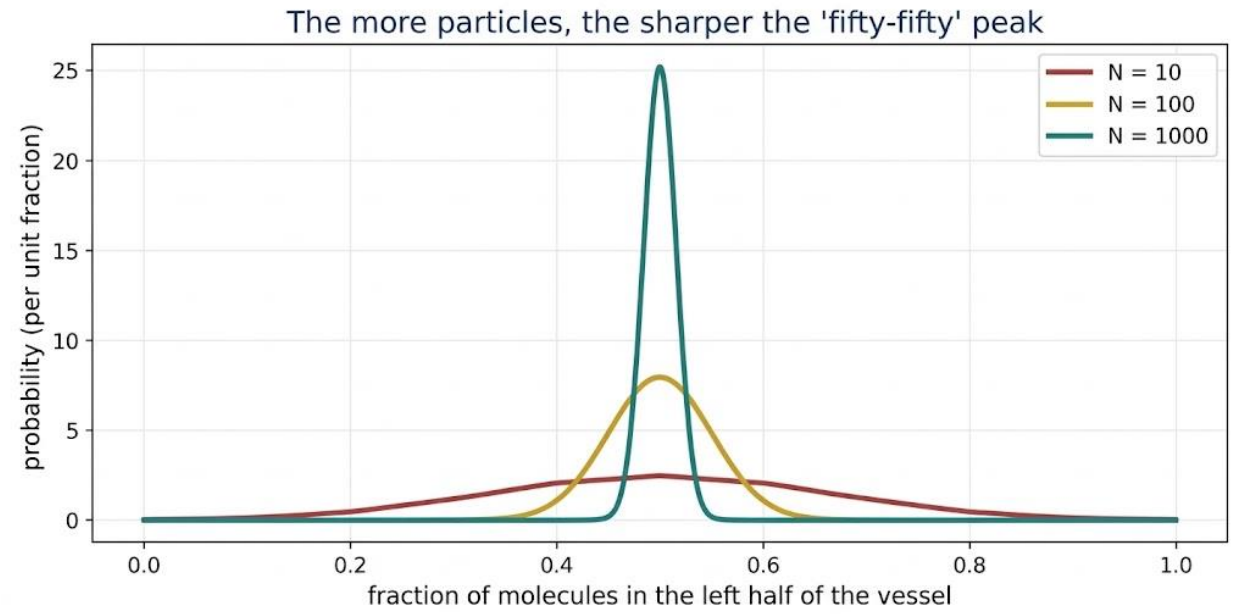


Fig. 1. The probability of finding a fraction of molecules in the left half of the vessel. As the number of particles increases, the "even" peak becomes sharper.

Young man: I will calculate for a hundred particles. All one hundred in the left half: the probability is $1/2^{100} \approx 8 \cdot 10^{-31}$. And suppose we peek into the vessel every nanosecond. Then we will have to wait $2^{100} \approx 1.3 \cdot 10^{30}$ nanoseconds $\approx 1.3 \cdot 10^{21}$ seconds, about $4 \cdot 10^{13}$ years — three thousand times longer than the age of the Universe!

Sage: And this is **only** a hundred molecules. In a cubic centimeter of air, there are about $3 \cdot 10^{19}$ of them. This is what is meant by the "law of overwhelming probability." It does not prohibit the gas from gathering in a corner — it makes this unimaginably rare. However, for ten particles, such "gatherings" are quite observable: the second law — the law of large numbers.

> **[UV] The established fact is the response to the objections of Loschmidt and Zermelo.**
>
> Loschmidt (1876): the laws of mechanics are reversible — therefore, a reversed system will return back. Zermelo (1896), relying on Poincaré's theorem: a closed system will sooner or later almost return to its initial state. Boltzmann's response: yes, but the probability of such events for 10^{23} particles is so small that the waiting time is incomparable to any scale. The second law — **a** statistical law: it holds not as an absolute prohibition, but with a probability indistinguishable from one.

2. The formula on the monument

S = k * ln(W)

Young man: Logarithm! $\ln(W_1 \ W_2) = \ln W_1 + \ln W_2$.

Sage: Therefore:

$S = k \cdot \ln W$

The sage: Here k = 1.380649·10^-23 J/K is **the Boltzmann constant. Its** value was introduced by Planck, and since 2019 it has been accepted as exact by the definition of SI units. The formula is inscribed on the monument to Boltzmann at the Central Cemetery of Vienna. Boltzmann died in 1906, a few years before

Perrin's experiments, which finally convinced skeptics of the existence of atoms — we discussed this in chapter 1.

Young man: I will check on the gas expansion. If the volume has doubled, each molecule has twice as many places. For N molecules, the number of ways has increased by 2N times. Then $\Delta S = k \cdot \ln(2N) = N \cdot k \cdot \ln 2$. For a mole, $N \cdot k = R$, we get $R \cdot \ln 2 \approx 5.76$ J/K. The same number from chapter 8!

Sage: This is what is called the "statistical meaning of entropy": it is a measure of the number of microstates that realize a macrostate. And the increase in entropy is the transition to the most probable macrostate.

> **[OS] Misconception / error — "entropy is disorder"**
>
> The popular phrase is true as an image but falls short as a definition. "Disorder" is a subjective thing: what looks like order depends on what we consider distinguishable. The precise definition is the number of microstates compatible with what we know about the system.
>
> The opposite can also occur: under certain conditions, an ordered system (for example, solid spheres in dense packing) has higher entropy because each sphere has more freedom to move. Therefore, it is more reliable to think in terms of "number of ways" rather than "disorder."

3. Boltzmann distribution: how nature divides energy

Sage: Now the second discovery. The system at temperature T finds states with low energy "advantageous," but thermal motion sometimes throws it into high energy states. The probability of a state with energy E is proportional to:

$$p(E) \propto \exp(-E / kT)$$

Sage: This **is the Boltzmann factor:** the higher the energy compared to kT, the exponentially rarer such a state is.

Young man: Where can it be "seen" with the eyes?

Sage: In the air above. Gas molecules at height h need additional energy mgh. Therefore, the air density decreases as $\exp(-mgh/kT)$. The characteristic height $H = kT/mg = RT/(Mg)$.

Young man: $H = 8.314 \cdot 273 / (0.029 \cdot 9.81) \approx 7980$ m — about 8 kilometers. Then at the summit of Everest (8.85 km), the pressure should be $\exp(-8.85/8) \approx 0.33$ atmospheres. It measures about 0.3!

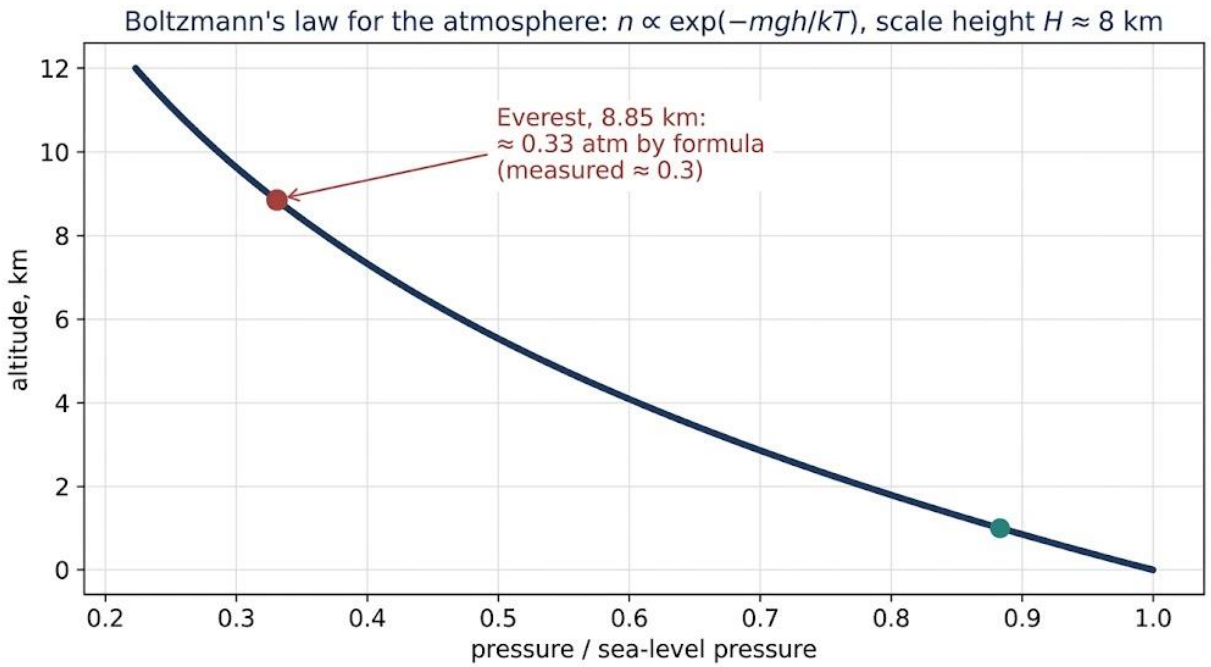


Fig. 2. Air pressure at different heights according to the Boltzmann formula (temperature is considered constant — approximation).

The sage: The approximation is rough (the temperature at height is lower, so the actual pressure is slightly less), but the order is correct. In 1908–1909, Jean Perrin applied this formula to tiny resin spheres weighed in water: their distribution at height under the microscope yielded Avogadro's number of about 6–7·10²³, consistent with other methods. This was one of the decisive arguments in favor of atoms. Perrin received the Nobel Prize in 1926 for his work.

4. Barriers and speed: Arrhenius law

The same Boltzmann factor explains why reactions and evaporation occur quickly or slowly. For a molecule to react or detach from the liquid, it must overcome the energy barrier E_a. **The** fraction of molecules with such energy is $\exp(-E_a/kT)$. Hence, the Arrhenius law (1889):

$$\text{speed} \propto \exp(-E_a / RT)$$

Young man: For a barrier of 50 kJ/mol and heating from 25 to 35°C: $\exp[(50\,000/8{,}314)\cdot(1/298 - 1/308)] \approx \exp(0.655) \approx 1.9$. The rule "the rate is twice as fast for every 10 degrees"! And for 100 kJ/mol — 3.7 times.

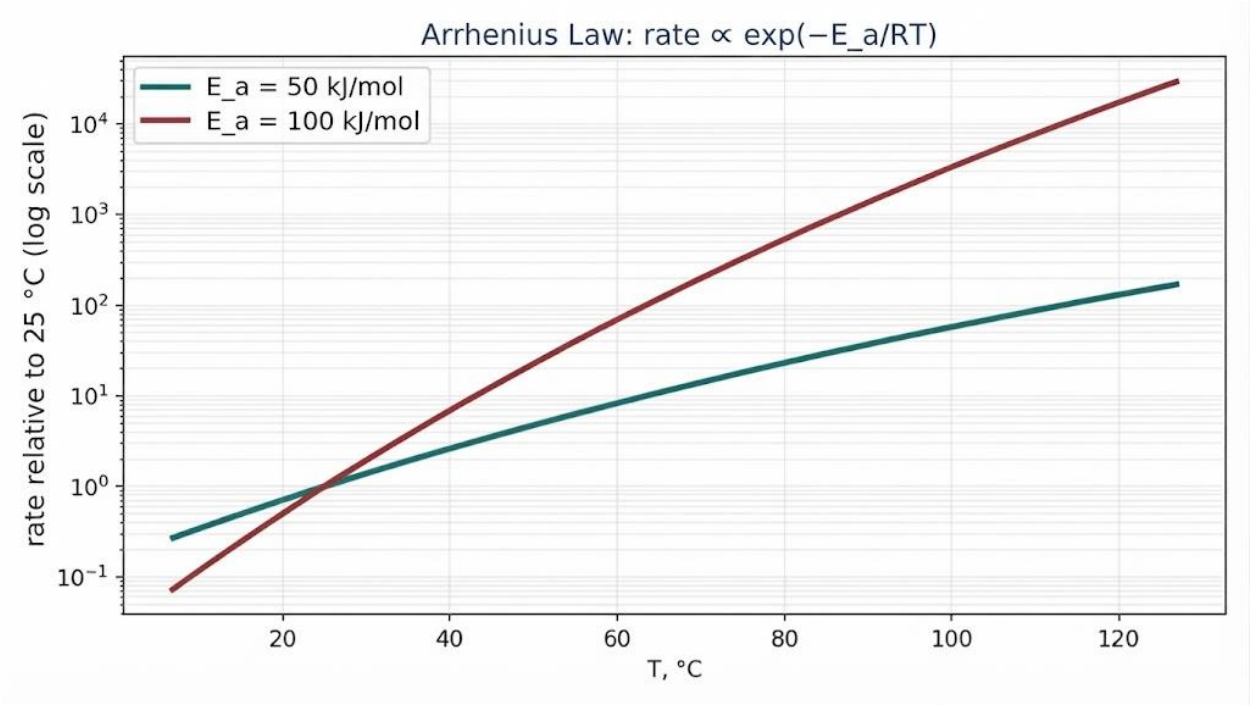


Fig. 3. Dependence of the rate on temperature according to the Arrhenius law for two barrier values.

The sage: Now the main distinction to keep from this entire chapter is **between thermodynamics and kinetics.**

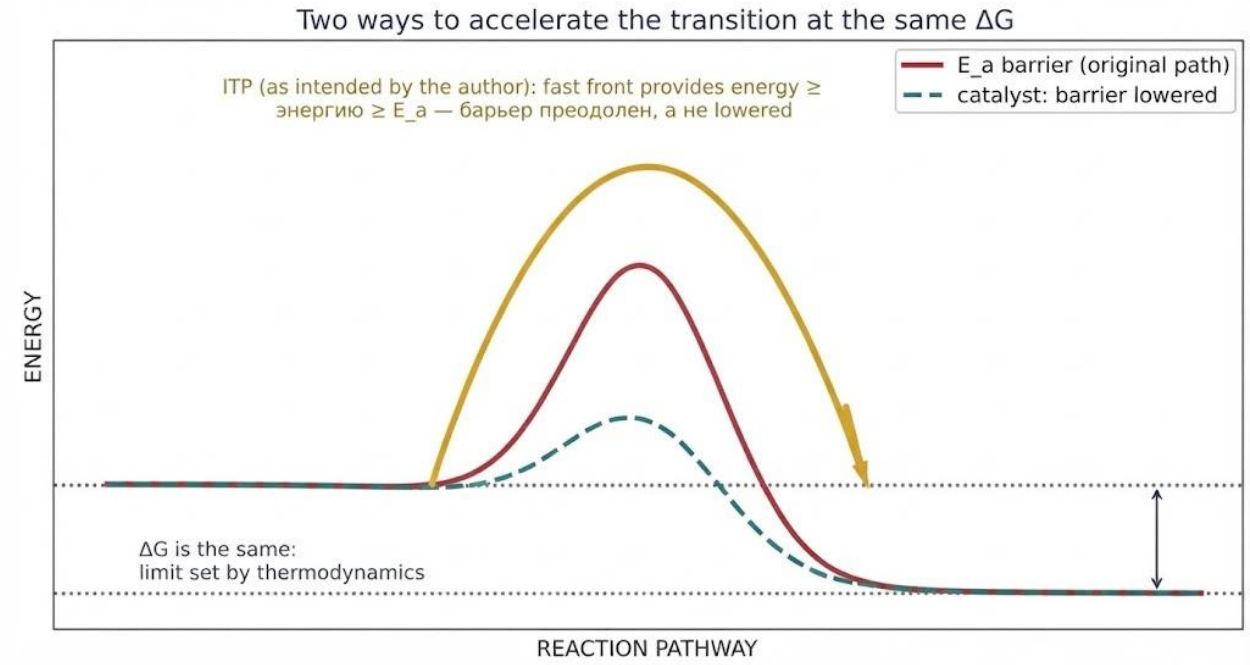


Fig. 4. The initial and final states are specified: the difference in levels (ΔG) determines the thermodynamics, while the height of the barrier (E_a) determines the kinetics. The catalyst lowers the barrier; the ITE, as intended by the author, keeps the barrier the same and overcomes it with the energy of the fast front.

> **[UV] The established fact — thermodynamics — "where to", kinetics — "how fast"**
> Thermodynamics indicates which state is more favorable and how much energy is minimally required (ΔG, ΔH, the second law). It says nothing about time. Kinetics tells us how fast the system moves there and depends on barriers. A catalyst lowers the barrier but does **not** change the position of equilibrium and does not negate thermodynamic limits.

5. Maxwell's Demon

The sage: Now a story that physicists have been unable to settle for a century and a half. In 1867, Maxwell, in a letter to Tait, conceived of a being later called a "demon" by Thomson. Suppose a vessel is divided by a partition with a tiny door, behind which stands the demon. It allows fast molecules to pass to the right and slow ones to the left. Without any work being done, one half heats up while the other cools down!

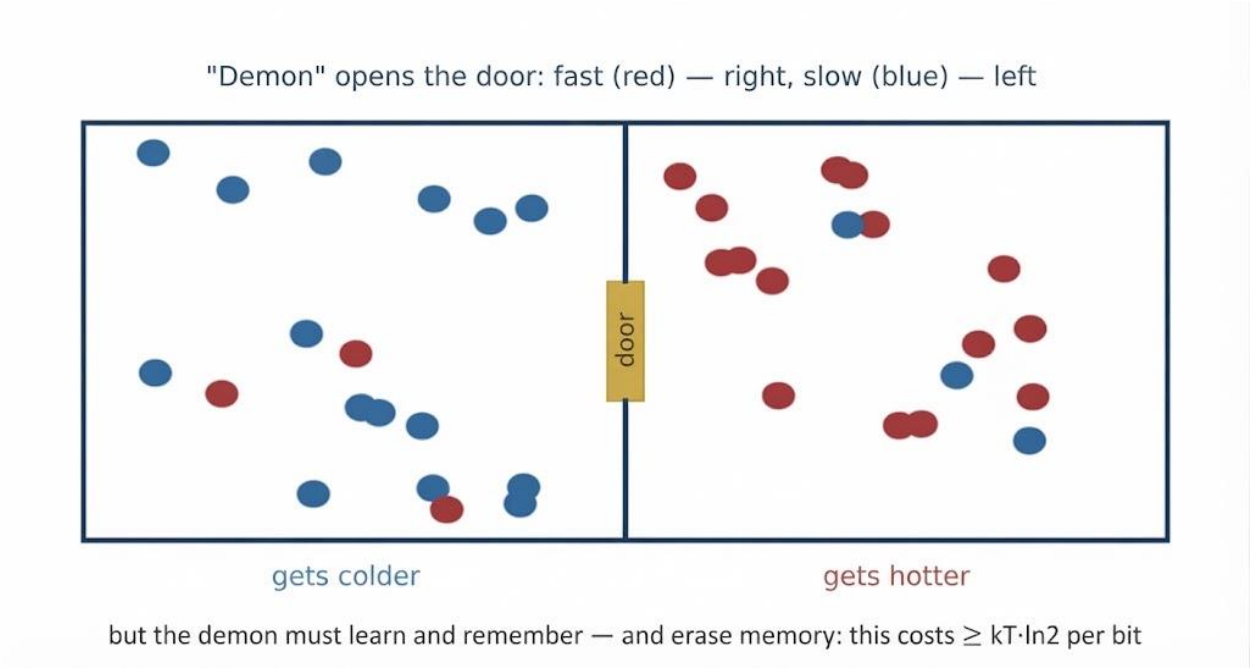


Fig. 5. Maxwell's Demon: if it could separate molecules by speed without expenditure, the second law would be violated.

Young man: This is a perpetual motion machine of the second kind: the temperature difference can be used for work! Where's the catch?

The sage: The catch lies in the demon itself. In order to choose, it must measure the **speed of** each molecule and remember the result. In 1929, Leo Szilard demonstrated that measurement must come at a cost. In 1961, Rolf Landauer, and later Charles Bennett (1982), clarified: one has to pay **for** erasing memory. Erasing one bit of information means releasing at least that much energy into the environment:

$$Q \geq k \cdot T \cdot \ln 2$$

Young man: At 300 K, this ^{is} 1.38·10^-23 · 300 · 0.693 $^{\approx}$ 2.9·10^-21 J per bit. Tiny!

The sage: Tiny — and therefore the verification was difficult. But in 2012, the group of Beru and colleagues confirmed in an experiment with a writable colloidal particle that this limit is real. The lesson: information is a physical quantity, and entropy is related not only to heat but also to knowledge about the system.

> **[UV] The established fact is that the demon does not violate the second law.**
> A working demon is a device with memory. The entropy "saved" by sorting molecules is returned with interest when erasing memory. The second law remains valid if the demon is considered as part of the system.

6. Drying and ITC: what can light do, and what is just arithmetic

Young man: Sage, I have a question for which I have traveled through these chapters. We talked about the limit: energy per kilogram of water is no less than 2.5 MJ, and no technology bypasses the second law. But in our books, drying with ITE is called more efficient than the usual one. What is the basis for this advantage?

The sage: This is the most important part of the book, and it needs to be formulated carefully. Thermodynamics sets the lower limit on energy expenditure. But between the limit and reality lies a wide band, and **kinetics operates** within it: how quickly moisture reaches the surface, how quickly it evaporates, how the vapor escapes, how much energy is simultaneously spent on heating the air, the walls, and the material itself. ITE operates precisely in this band: its advantage is not in breaking through the limit, but in **approaching it more closely.**

$$\eta_{drying} = E_{min} / E_{fact}, \qquad E_{min} \approx 2.5 \text{ MJ per kg of evaporated water (and more for bound water)}$$

Sage: By this measure, the method that wins is the one for which $E_{факт}$ is closer to $E_{мин}$. Below are eight factors that the author considers objective advantages of drying with ITÉ. For each, we will show what is known independently and what remains a hypothesis and how to verify it. This clearly illustrates where the advantage is already based on physics and where it is still awaiting measurements.

№	The claimed advantage (according to the author)	What is known independently	Status and how to check it
1	Target spectrum: energy falls on the absorption bands of water with low absorption by organic matter; the color is determined by the pulse front, not the temperature of the emitter.	Water strongly absorbs around 3 and 6 μm and in the near IR (≈1.45 and 1.94 μm). But organic matter also absorbs: N–H and O–H around 3 μm, amide I around 6.1 μm, carbohydrates 8–10 μm. As long as there is a lot of water (up to 80–90% by mass), it takes the majority of the radiation.	[G] for "minimal" absorption by the organic material. Measure the reflection and transmission spectra of the product at different stages of drying.
2	Volumetric energy input due to impulsiveness	The penetration depth ≈ 1/α is determined by the spectrum, not the pulse frequency (Fig. 6). Intermittent supply has long been known as a way to equalize humidity: during the pause, moisture moves from the inside to the surface.	[G]: depth mechanism. Thermocouples and layer cuts; compare continuous and pulsed modes at the same average power. Proven [1]
3	Photon transfer instead of conductive-convective.	Direct radiative transfer without a heat carrier is a established distinction. However, the absorbed IR photon in water is converted into molecular vibrations, that is, into heat.	Direct transmission — [UV]. The "principally non-thermal" effect — [G]: experiments are needed at the same surface temperature. Conducted and proven [1]
4	The used steam venting system	If the vapor is not removed, the vapor pressure at the surface increases and the evaporation	The solution is an engineering fact. Comparison with alternatives under equal conditions is to

№	The claimed advantage (according to the author)	What is known independently	Status and how to check it
		rate decreases. This is a general rule for any drying process.	measure. The automation system operates, reducing power [1].
5	Delicate objects at moderate temperatures due to high evaporation rates.	Evaporation cools the surface while moisture comes from below. When the surface dries, the temperature rises rapidly — this is the critical moment.	[OD]: limitation - the rate of moisture supply from the inside. Surface temperature control (IR camera). Conducted: Product temperature at 30°C is about 18-19°C [1]
6	Adaptability of the spectrum to the product	In ordinary thermal radiators, color is determined by temperature (Wien's law: $\lambda \cdot T = 2898\ \mu m \cdot K$). According to the author, in ITEs, color is determined solely by the steepness of the pulse front (the front can be considered as part of a sine wave) and does not depend on temperature; this is one of the advantages of the method.	[G]: measure the shape of the front and the spectrum of radiation at different temperatures of the radiator (more details in chapter 12)
7	Direct energy delivery is faster than convection; light is not "blown away" by ventilation.	The radiation does not depend on the air flow; in convective drying, a significant portion of heat is lost with the heated air. However, the rate is also limited by the diffusion of moisture within the product, especially towards the end of drying. The correct gas dynamics scheme	[G] for "significantly faster" at all stages. Drying curves (mass over time) for different methods. Proven [1]
8	Deeper dehydration without destruction	Residual moisture is determined by the equilibrium with air (sorption isotherm); bound water is held more strongly than free water. Heating the product and reducing the local air humidity allow for deeper penetration.	[G]: quality indicators (color, structure, composition) and residual moisture under the same conditions are needed. Conducted and proven [1]

The sage: In **the book "The Key** to Health...", the authors present conducted research and evidence for all points of the table, except for the first one. The first one is calculated.

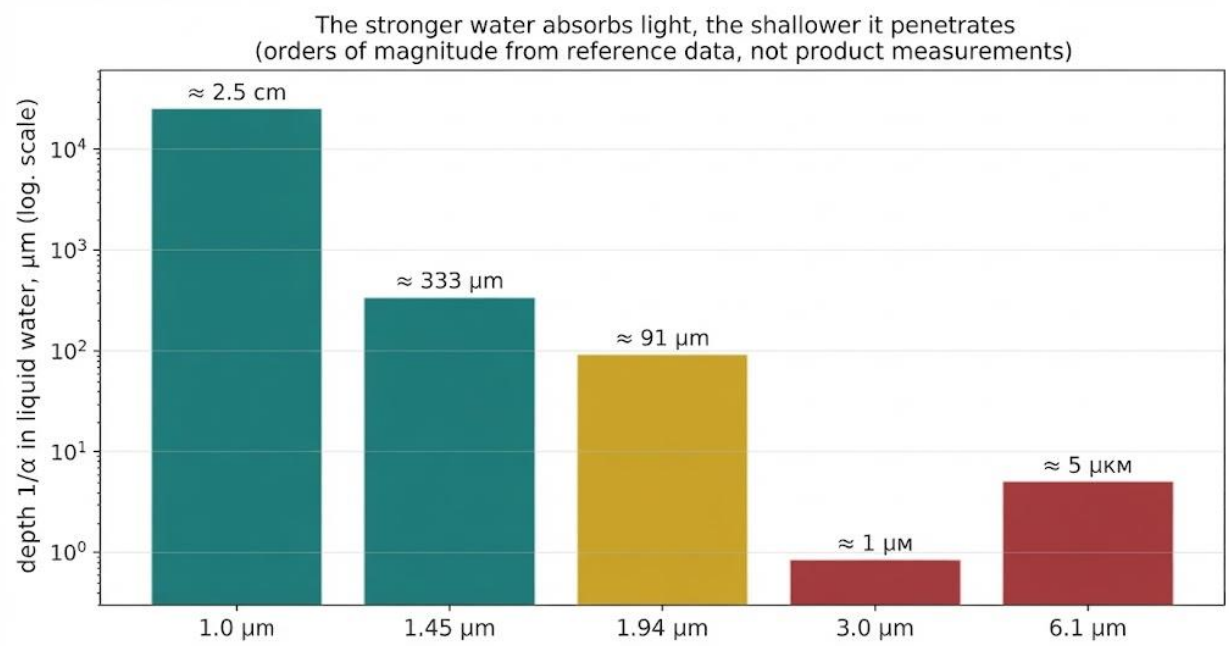


Fig. 6. The penetration depth of radiation in liquid water at different wavelengths (orders of magnitude). Strong selectivity (small depths) and voluminosity (large depths) pull in different directions.

Young man: Wait. If the spectrum is chosen so that water absorbs maximally, then the light will be absorbed in the first micrometers. Does that mean that points 1 and 2 contradict each other?

Sage: It's good that you noticed this: it's the most subtle point of the entire table. In ordinary (linear) optics — yes, they pull in different directions. Therefore, the ITE has three possible paths for alignment, and it's important to determine which one actually works.

- **The spectrum contains weakly absorbed bands** (near IR, 1–1.5 μm): they penetrate to millimeters and centimeters (Fig. 6), while strong bands quickly dry the surface.
- **Discontinuity. During the** pause between impulses, moisture equalizes within the product, and the surface does not dry out. This is an established effect of intermittent drying, and the impulse mode of ITÉ may be its fine, rapid version.
- **A sharp impulse front (author's explanation). The front** rises faster than water can rearrange hydrogen bonds (about 1 ps), so the energy goes to the water and does not have time to dissipate as heat throughout the volume. This is [G], and it needs to be checked separately.
- **In the book "The** Key to Health," the authors suggest selecting the spectrum so that water is absorbed by 5-10%. This allows for uniform drying throughout the volume, even with relatively thick layers of material. The automation system reduces power as drying occurs, preventing the temperature from rising above 35-40°C.

[1](Rakhimov R.Kh., Ermakov V.P. "The Key to Health or Functional Ceramics - What Is It?", Publishing House "Fargona", 2005. - 328 p.)

[GA] The author's hypothesis — deep penetration without damage: the author's explanation ("Praise for the C student")

The author explains how light penetrates deeply without damaging organic matter and almost without heating the product in three steps. (1) Windows of weak **water absorption: only about 5–105** of the energy is absorbed there, so the light does not "burn" at the surface. The existence of such windows is an established fact (near IR around 1 μm, Fig. 6; at a depth of $1/\alpha \approx 2.5$ cm, about 105 is absorbed in a layer of 2–3 mm). (2) Sharp **front of the flash: the** energy is transferred specifically to the water, not to vitamins and enzymes, and does not have time to spread as heat throughout the volume. The time for the restructuring of hydrogen bonds in water is on the order of picoseconds, as known from experiments, but the assertion of selective energy transfer to water **is [G]. (3) Energy** is lost with vapor: the object does not heat up but cools down — this is consistent with evaporative cooling from Chapter 10 at high evaporation rates.

All three steps together form a complete picture, but they can be verified separately: the absorption fraction in the "window" (the transmission spectrum of the product layer), the influence of the steepness of the front at the same average energy, the surface temperature during the process (IR camera, thermocouples).

[OD] Justified addition — why pulsation could accelerate evaporation even at the same average power

The evaporation rate is proportional to the saturated vapor pressure, which increases with temperature according to the law of the form $\exp(-L/RT)$ — essentially, the same Boltzmann factor with a "barrier" $L \approx 42$ kJ/mol. This dependence is **convex:** if the temperature of the thin surface layer oscillates around the average value, then the time-averaged saturated vapor pressure is greater than at a constant temperature. For an average value of 80°C and oscillations of ±20 K, the increase would be about 295, and for ±10 K — about 75.

But the condition is strict: at a frequency of 434 Hz, thermal oscillations penetrate into the water only to a depth of about 10 μm ($\sqrt{(D/\pi f)}$ with $D \approx 1.4 \cdot 10^{-7}$ m²/s), and their amplitude depends on the energy flux density and on how much heat has managed to escape. Therefore, this is a calculated possibility, not a result: the range of temperature oscillations at the surface needs to be measured.

[GA] The author's hypothesis — ITE as a tool of kinetics, not thermodynamics

The working formulation for this book. The energy limit is set by thermodynamics (≈ 2.5 MJ per kilogram of free water and more for bound water), and ITE does not change it, just as no other method does. The advantage of the method is **that it** overcomes "barriers" (evaporation, desorption, steam removal) — it quickly delivers sufficient energy for the transition with a fast front, rather than reducing them like a catalyst — thereby reducing the portion of energy that is not spent on water: on heating the air, walls, equipment, and on prolonged operation.

The consequence for claims of savings: the gain relative to conventional technology is limited by the ratio $E_{ordinary} / E_{min}$. If conventional convective drying consumes about 3–6 MJ per kilogram of water (a benchmark from the literature, depending on the product and installation), then a reasonable gain for the water itself ranges from 1.2 to 2.5 times; significantly larger gains (by factors and by tens of times) are possible where conventional technology expends energy on something else — for example, on prolonged heating of a heavy furnace or molds for a small amount of moisture. In each case, an honest answer is provided by the energy balance.

In summary: the advantage of drying with ITE is in a more effective approach to the theoretical limit compared to alternatives, which demonstrate lower efficiency under real conditions. The status of individual factors is shown in the table above.

Young man: So, the ITE advantage is not "against" the second law, but "within" it — the limits and equilibrium are the same, only the path to them changes. The catalyst lowers the barrier, while the ITE, according to the author's intention, overcomes it by delivering energy with a fast front.

Sage: Yes. And this is a strong position: it cannot be refuted by any thermodynamic calculation, and it can be verified by simple measurements — of mass, temperature, power, and time, which has been done by the authors of the book [1].

7. What we have learned and what remains

- **Macro- and microstates. One** macrostate is realized by W microstates; the most probable is the one for which W is maximal.
- **The Boltzmann formula** $S = k \cdot \ln W$ explains $R \cdot \ln 2$ when the volume is doubled. The second law is the statistical law of large numbers.
- **The Boltzmann factor** $\exp(-E/kT)$ gives the barometric formula and the Arrhenius law. The rate of processes increases exponentially with temperature, and the barrier is the most important factor.
- **Maxwell's demon does** not violate the second law: erasing information costs at least $kT \cdot \ln 2$ per bit.
- **Thermodynamics is about "where" and "how much minimum," kinetics is about "how fast" and "how much above the minimum."** For drying, the advantage of the method lies in approaching the limit.

Young man: Now I understand what equilibrium thermodynamics is. But all these gases have been ideal so far, and around us are liquids and crystals. Why does water hold together as a drop, rather than dispersing like a gas?

Sage: This is already the next chapter: the interaction of molecules. Surface tension, evaporation, boiling, and the strange critical point where liquid and vapor become indistinguishable.

Problems for independent solution

- How many ways are there to distribute 4 distinguishable molecules into two halves of a vessel so that 2 are on the left? And how many ways are there for all 4 to be on the left? What is the probability of each?
- Find ΔS for one mole of gas when the volume is tripled using the formula $S = k \ln W$. Compare it with $R \cdot \ln 3$.
- The height of a homogeneous atmosphere H = 8 km. At what height does the air density decrease by half?
- How many times will the reaction rate increase with a barrier of 60 kJ/mol when heated from 20 to 30°C?
- How much energy is minimally required to dissipate into the environment when erasing 1 GB ($8 \cdot 10^9$ bits) of information at 300 K?

[UV] Established fact — answers to the problems

1) Ways "2 to the left" — 6 out of 16 (probability 6/16 = 0.375); "all 4 to the left" — 1 out of 16 (0.0625).

2) $\Delta S = N \cdot k \cdot \ln 3 = R \cdot \ln 3 \approx 8.314 \cdot 1.0986 \approx 9.13$ J/K.

3) $\exp(-h/H) = 1/2$: $h = H \cdot \ln 2 \approx 8 \cdot 0.693 \approx 5.5$ km.

4) exp[(60 000/8.314)·(1/293.15 − 1/303.15)] = exp(7217 · 1.125·10^-4) ≈ exp(0.812) ≈ 2.25 times.

5) $Q \geq 8 \cdot 10^{9}$ · 2,9·10-21 J ≈ 2,3·10-11 J — negligible compared to the actual consumption of computers; the Landauer limit is still far from technical practice.

Chapter 10. Liquids, Vapors, Crystals

"When molecules stop being invisible to each other"

> *The young man carefully lowered a paperclip onto the water. It did not sink — it lay on the surface, slightly bending it, like a heavy person bends a trampoline. "But steel is heavier than water!" he said.*

1. Where do liquids come from?

Sage: Until now, our gases have been ideal: the molecules did not notice each other. But as soon as the gas is cooled, it turns into a liquid. This means that the molecules **are attracted to** each other. On the other hand, liquids are almost incompressible. This means that at small distances they repel **each other.** How can this be illustrated?

The sage: The graph of the interaction energy of two molecules as a function of distance is drawn. It is most often described by the Lennard-Jones formula.

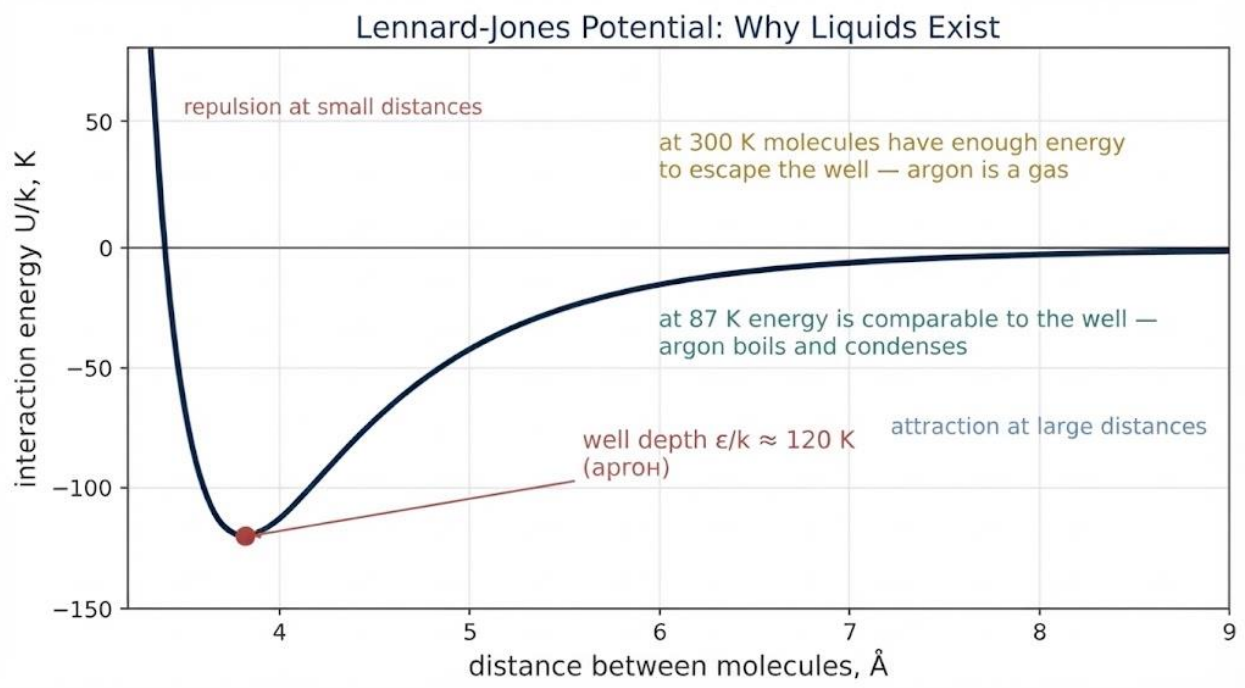


Fig. 1. Interaction energy of two argon atoms. The depth of the "well" is about 120 K in units of k; the distance at the minimum is about 3.8 Å.

Young man: A well! So, if a molecule's energy is less than the depth of the well, it is "stuck" next to its neighbor, and if it is greater — it escapes to freedom?

Sage: Exactly. And we can measure the energy of molecules with temperature: the typical energy of thermal motion is on the order of kT. Compare: the depth of the argon well is about 120 K. At room temperature (300 K), there is more than enough energy to escape: argon is a gas. But at 87 K, the energy is comparable to the depth of the well, and argon boils.

The boiling temperature should then be proportional to the depth of the pit. For argon, 87/120 ≈ 0.7.

Sage: This is a general rule for simple substances, and it is confirmed: the stronger the attraction between molecules, the higher the boiling point. Water, with its hydrogen bonds, boils at 100°C, even though the molecule is light. Now recall chapter 8: the system chooses a compromise between energy (wants "wells") and entropy (wants freedom). At low temperatures, energy prevails — resulting in a liquid or crystal, while at high temperatures, entropy prevails — resulting in a gas.

> **[UV] Established fact — numbers for argon**
>
> Lennard-Jones potential parameters for argon: ε/k ≈ 120 K, σ ≈ 3.4 Å. The normal boiling point of argon is 87.3 K. The parameters are adjusted based on experimental data; this is a model, not an exact formula for any pair of molecules.

2. Surface tension

Young man: And the paperclip?

Sage: Look at the water molecule in depth: it is pulled by neighbors from all sides, the forces are balanced. The molecule on the surface has no neighbors on one side — part of the "favorable bonds" is lost. The surface is more expensive: creating each square meter of surface requires energy σ. This quantity is called surface **tension; it is measured in** N/m or in J/m² — they are the same.

Young man: And what is σ for water?

Sage: At 20°C about 0.0728 N/m. The liquid tends to minimize the surface, so droplets in weightlessness are spherical (a sphere is the shape with the least surface area for a given volume), and the paperclip lies on the film, which bends but does not break.

Sage: Due to curvature, the surface exerts pressure inward. A bubble or droplet of radius r creates additional pressure (Laplace's formula):

$\Delta p = 2\sigma / r$

Young man: I will check on a droplet with a radius of 1 mm: 2 · 0.072 / 0.001 = 144 Pa — very little. And for a pore with a radius of 1 μm: 144,000 Pa — one and a half atmospheres!

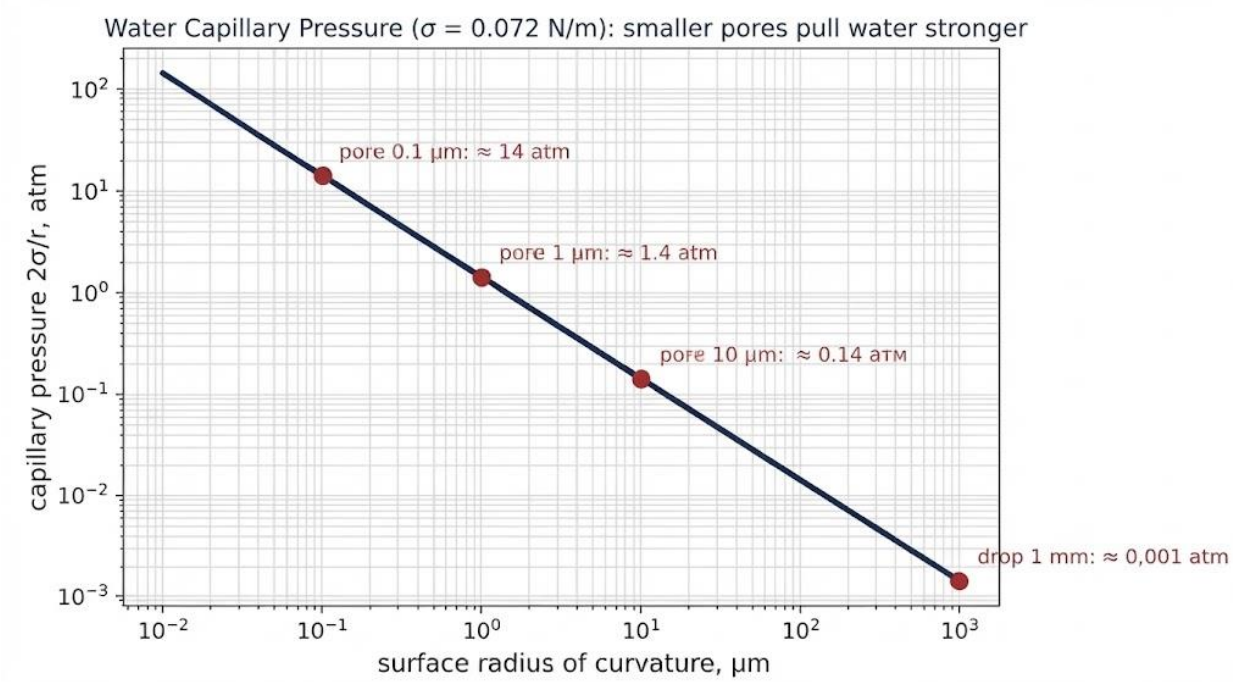


Fig. 2. Capillary pressure of water as a function of the radius of curvature of the surface (σ = 0.072 N/m).

Sage: Now capillaries. If a thin tube is immersed in water, the water that wets the glass will rise. The height of the rise (Jurin's law, 1718):

$h = 2\sigma \cdot \cos\theta / (\rho \cdot g \cdot r)$

Young man: For a tube with a radius of 0.5 mm and complete wetting: $2 \cdot 0.072 / (1000 \cdot 9.81 \cdot 0.0005) \approx 0.029$ m — about 3 cm. And for a radius of 1 μm: 14.7 m.

The sage: And here is the trap: trees grow taller than 100 meters, and their vessels have a radius of micrometers. This cannot be explained by capillarity alone. Today, the theory of cohesion and tension is accepted (Dixon and Joly, 1895): water in the vessels of trees is under **negative pressure — it is** pulled by evaporation from the leaves, and the cohesion of water molecules with each other prevents the column from breaking.

> **[G] Hypothesis — water under tension**
>
> The theory of "cohesion and tension" is accepted by most researchers, but the state of stretched water is metastable and experimentally challenging: debates continue about the details (column breaks, the role of bubbles). For our book, the consequence is sufficient: water is capable of withstanding tension because the molecules are attracted to each other.

> **[UV] Established fact — capillary forces and drying of porous bodies**
>
> In a porous material (wood, clay, ceramics before firing), water is retained and moves in the pores due to capillary pressure. As long as water fills the pores at the surface, capillary forces pull it from the inside to the outside, and the surface remains wet. When the water boundary moves inward, only diffusion allows vapor to move through the dry layer. This change in mechanisms determines the shape of the drying curve (see section 7).

3. Evaporation and saturated vapor

Young man: Why does the puddle dry up even though the water is far from boiling?

The sage: Water molecules in a liquid move at different speeds (do you remember the Maxwell distribution?). The fastest ones, which have enough energy to escape from the "well," fly away — this is evaporation. The remaining molecules are, on average, slower, and therefore the evaporating liquid cools down **(hence a** wet cloth is cooler than a dry one).

Sage: If the vessel is closed, then at some point the number of molecules escaping will equal the number of molecules **returning. This is dynamic** equilibrium, and the vapor in **it is** called saturated. The pressure of saturated vapor depends only on temperature and increases very rapidly: by the same Boltzmann factor that we discussed in Chapter 9.

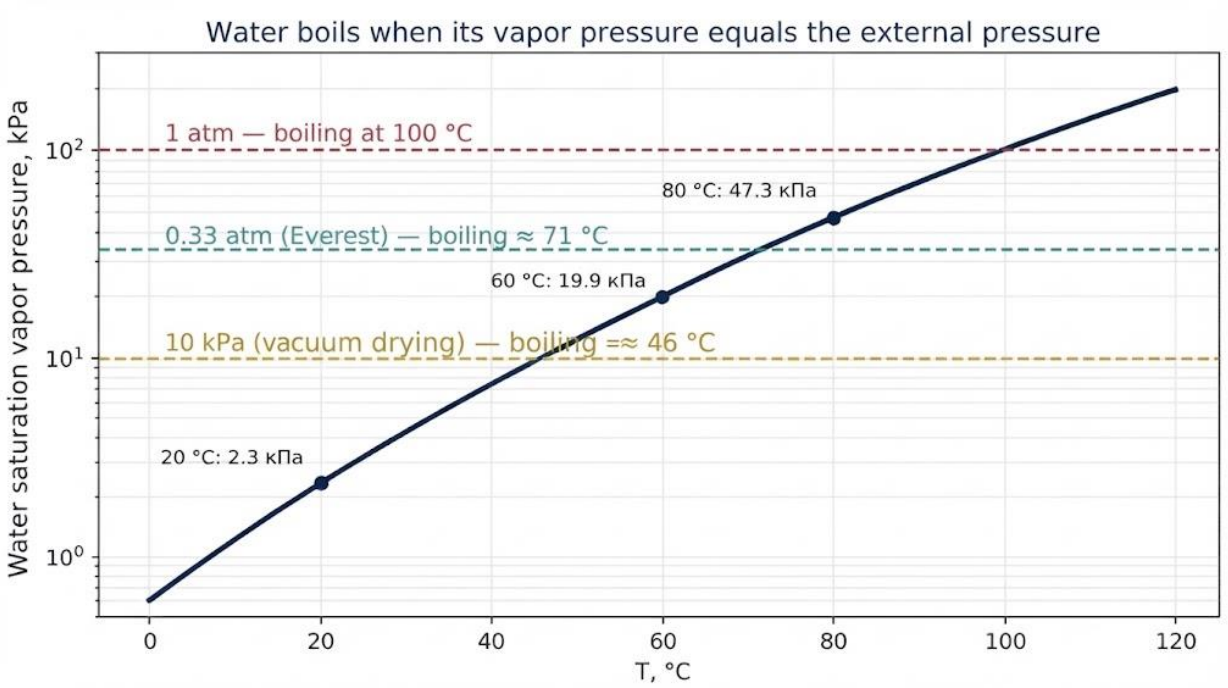


Fig. 3. Saturated vapor pressure of water (according to Antoine's equation). The horizontal lines represent the external pressure at which water boils at the specified temperature.

Young man: I look at the diagram: at 20°C — 2.3 kPa, at 60°C — 19.9 kPa, at 80°C — 47 kPa, and at 100°C exactly 101 kPa, that is, the atmosphere. The increase is very rapid!

Sage: Now the humidity of the air. Relative humidity is the ratio of the actual vapor pressure of water in the air to the saturation vapor pressure at a given temperature. At 20°C and 50% humidity, the vapor pressure is about 1.2 kPa. As long as it is less than the saturation pressure, water evaporates; when it equals — the process halts.

> **[OD] Justified addition — evaporation from the free surface**
>
> The evaporation rate is approximately proportional to the difference between the saturated vapor pressure at the surface (at the surface temperature) and the actual vapor pressure in the air away from it. To dry faster, one must raise the surface temperature, remove vapor from the surface, or do both. This is a general rule, not a feature of a specific setup; in Chapter 9, we have already encountered it as "vapor removal" or "gas dynamics."

4. Boiling

Young man: And boiling? How is it different from evaporation?

The sage: Evaporation occurs only from the surface and at any temperature. Boiling occurs in the volume: bubbles of vapor form inside the liquid. A bubble can grow only if the vapor pressure inside it is not less than the external pressure on the liquid. Thus, the liquid boils when the saturated vapor pressure equals the external pressure.

Young man: So, the boiling point depends on external pressure! At Everest, at 0.33 atm, according to Fig. 3, water boils at a temperature of about 71°C.

Sage: Measurements at the summit of Everest give about 70°C — it matches. Therefore, it is difficult to cook porridge in the mountains: the water "boils," but it is not hot enough. The reverse example is a pressure cooker: at a pressure of about 2 atm, the boiling temperature rises to about 120°C. And if you need to dry at low temperatures, you reduce the pressure: at 10 kPa, water boils at 46°C. This is the idea **of vacuum drying.**

> **[ERROR] Misconception / mistake — "boiling water always has 100°C"**
>
> 100°C is the boiling point at normal pressure of 101.325 kPa. At different pressures, the boiling temperature is different. Additionally, in perfectly pure water without nucleation sites, superheating **is possible:** water can exceed 100°C without boiling and then suddenly erupt into vigorous boiling (the so-called "explosive" mode). Therefore, heating pure water in a smooth glass in a microwave to boiling is dangerous: sudden vigorous boiling may occur.

5. Critical point

Young man: If you heat a closed test tube with liquid, the vapor pressure increases and boiling is still prohibited — the vapor just exerts more pressure. What will happen next?

Sage: A very beautiful thing. A liquid expands when heated and becomes less dense, while vapor compresses and becomes denser. Densities converge. Finally, at a certain temperature, they become equal, the **boundary** between the liquid and vapor disappears, and only one phase remains.

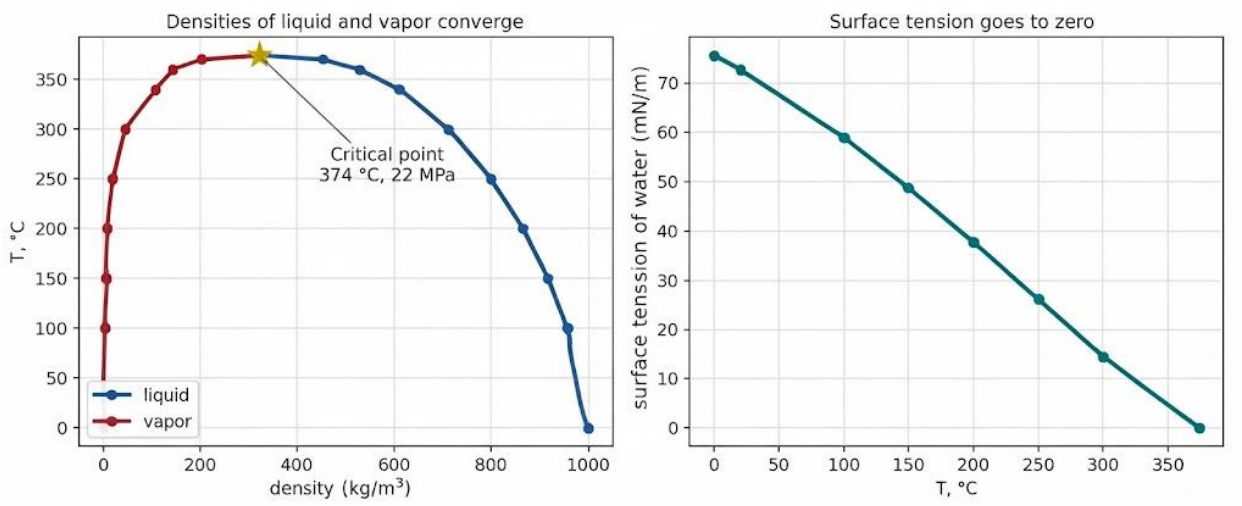


Fig. 4. On the left — the densities of liquid water and saturated vapor along the equilibrium line (data from steam tables) converge at the critical point. On the right — the surface tension of water approaches zero.

The sage: This point is called **the critical** point. For water, Tк = 374°C, pressure 22 MPa (about 218 atmospheres), density about 322 kg/m³. For carbon dioxide: 31°C and 7.4 MPa. The first observation of the disappearance of the boundary was made by Cagnard de la Tour in 1822, and the term "critical point" was introduced by Thomas Andrews in 1869 after experiments with CO_2 .

Young man: Then above the critical temperature, there is no liquid, no matter how much you compress it!

Sage: That's right. This is an astonishing result: **gas and liquid are not two fundamentally different things,** but two wings of one state, and one can transition between them smoothly, bypassing the critical point. A substance above the critical point is called **a supercritical fluid: it flows** like a gas and dissolves like a liquid. For example, it is used to extract caffeine from beans (supercritical CO_2).

> **[UV] The established fact is what happens at the critical point.**
> The phase boundary disappears, the surface tension approaches zero (Fig. 4), and the latent heat of vaporization vanishes. The heat capacity and compressibility increase sharply. When looking into the test tube near the critical point, the liquid "opalesces" — it becomes cloudy due to giant density fluctuations, which we discussed in Chapter 9: there are many molecules, but the system has "no preferences."

6. Crystals and glasses

Sage: Upon further cooling, the liquid solidifies. In the crystal, **atoms** are arranged in a strict order — the crystal lattice — and oscillate around their positions. The order is long-range: knowing the position of a few atoms allows one to predict the position of any other. Hence, the characteristic properties: a definite melting **temperature, faceting,** and anisotropy (properties depend on direction).

Young man: But order is low entropy. How does it win over disorder?

Sage: Through the compromise of energy and entropy: at low temperatures, the lattice is more favorable in energy, and this is enough to "pay" for the loss of entropy. At high temperatures, entropy prevails, and the lattice melts. Details are in Chapter 11.

Sage: There **are also** amorphous bodies: glass has no long-range order, it is a "frozen liquid." Its viscosity increases so much upon cooling that the molecules stop rearranging.

> **[OS] Misconception / error — "old window glass flows down"**
> There is a belief that the glass in medieval windows is thicker at the bottom because glass flows slowly. This is not true: the thickness depended on the method of production (sheets were blown and rolled, and when installed, the thick edge was placed down). Calculations show that at room temperature, glass flows so slowly that over the age of the Universe, the deformation would be imperceptible.

Young man: Sage, you mentioned ceramics. How are surface tension and crystals related to it?

Sage: Through sintering. **The** powder consists of numerous microscopic particles, which have a vast surface area. The surface energy of oxides is on the order of 1 J/m² (reference), which is an order of magnitude greater than that of water. If the particles are heated, the atoms begin to move, connecting neighboring grains: the total surface area decreases, and the energy drops. This is the same idea as in a drop of water, only the "drops" are solid.

[OD] Justified addition — sintering strength and speed

The driving force of sintering is the reduction of surface (and intergranular) energy: thermodynamics **states that** the powder "wants" to densify. The rate is **determined by** the diffusion barriers of atoms in the crystal, which depend on temperature according to the Arrhenius law (Chapter 9). For a powder made of 1 µm grains (alumina, density about 3.95 g/cm³), the specific surface area is about 1.5 m²/g, and the stored surface energy is about 1.5 kJ/kg, which is negligible compared to the heating energy. This suggests that the decisive factor is not the energy reserve, but the rate of overcoming the barriers.

7. Drying: two periods and where ITE operates in them

Sage: Now let's gather the chapter and return to drying. For wet porous material, the drying curve usually breaks down into two periods.

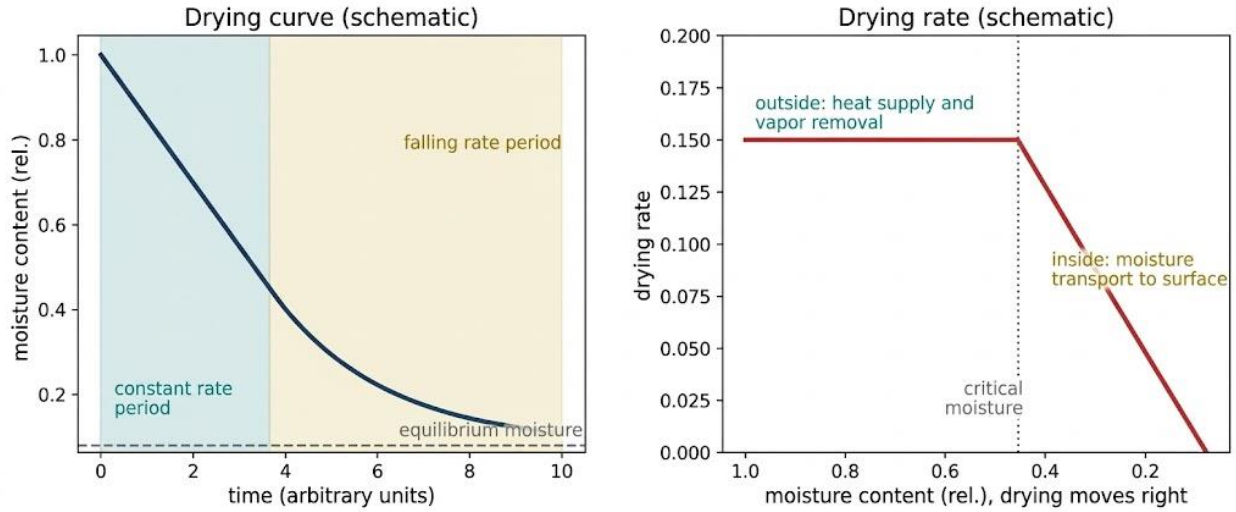


Fig. 5. Schematic curve and drying rate: constant rate period (limited by external heat supply and vapor removal) and falling rate period (limited by moisture transfer from the interior). Diagram, not measurement.

Sage: In the first period, the entire surface is wet, capillary forces draw up water, and the rate is limited by external conditions: how much heat has been supplied and how quickly the steam has been removed. Here, factors 3, 4, and 7 from the table in chapter 9 are clearly visible: direct radiation input and exhaust steam removal. This part is the most predictable.

Sage: In **the second** period, the boundary of water moves inward, and steam must be passed through the dry layer. Now internal transfer is determined — **which** means barriers: diffusion, desorption of bound water. In conventional drying, heat must be transferred to the boundary by conduction through the dry layer, which conducts heat poorly.

Young man: And does the radiation that penetrates deep heat the boundary directly?

Sage: If it really reaches the water boundary — yes, and that would be a real advantage in the second stage: heat is delivered where it is needed, bypassing the dry layer. But whether it "reaches" depends on the transparency of the dried layer in the chosen area of the spectrum — and this is measured, not assumed.

[GA] The author's hypothesis — where ITÉ can win: the second drying period

Hypothesis: the main advantage of ITE drying manifests not in the first period (where the method is limited by the general laws of evaporation), but in the second — due to the direct supply of energy to the moisture boundary and overcoming transfer barriers with a fast front (factors 2, 5, 8 from the table in chapter 9). This aligns with the view of ITE as a tool of kinetics.

Verification: drying curves (mass over time), temperature and humidity profiles through thickness, and measurement of the transmission spectrum of the dried layer. The first two items can be accomplished with simple instruments. Status — [G].

8. What we have learned and what remains

- **Liquids** exist because there is attraction between molecules; the boiling point is higher the deeper the "well."
- **Surface tension σ is** the energy per unit area. The pressure under a curved surface $\Delta p = 2\sigma/r$, the capillary rise $h = 2\sigma \cdot \cos\theta/(\rho g r)$.
- **Saturated vapor** is a dynamic equilibrium; its pressure increases exponentially with temperature. Boiling occurs when it equals the external pressure.
- **Critical point: the** distinction between liquid and gas disappears. Above it is the supercritical fluid.
- **A** crystal is long-range order, while glass is "frozen" disorder. Sintering of ceramics: the driving force is surface energy, and the rate is diffusion barriers.
- **Drying** consists of two periods: external (heat supply and vapor removal) and internal (moisture transfer to the boundary).

Young man: I have a question about phase transitions. How exactly does ice turn into water, and why does water not freeze instantly at 0°C? And you mentioned van der Waals gases.

Sage: Correct. The next chapter is about phase transitions: melting, phase diagrams, the Van der Waals equation, and how gases are liquefied.

Problems for independent solution

- Find the excess pressure inside a water droplet with a radius of 0.1 mm ($\sigma = 0.072$ N/m).
- To what height will the water rise in a glass tube with a radius of 0.1 mm under complete wetting?
- At what pressure does water boil at 80°C (Fig. 3)? What is the approximate boiling temperature at a pressure of 0.5 atm?
- How much energy is needed to evaporate 100 g of water from wet fabric at 25°C, if the latent heat of evaporation is 2.44 MJ/kg?
- Is it possible to turn CO_2 into a liquid by compression at 40°C? Why?

[UV] Established fact — answers to the problems

1) $\Delta p = 2 \cdot 0.072 / 10^{-4} = 1440$ Pa.

2) $h = 2 \cdot 0.072 / (1000 \cdot 9.81 \cdot 10^{-4}) \approx 0.147$ m ≈ 15 cm.

3) According to Fig. 3, at 80°C the saturation vapor pressure is approximately 47 kPa ≈ 0.47 atm. At 0.5 atm (50.7 kPa), water boils at a temperature slightly above 80°C, around 81–82°C — significantly lower than 100°C.

4) $Q = 0.1 \cdot 2.44$ MJ = 0.244 MJ = 244 kJ.

5) No. The critical temperature of CO_2 is about 31°C; at 40°C, no matter how much you compress it, the boundary between liquid and gas will not exist (a supercritical fluid will be obtained).

Chapter 11. Phase Transitions

"The substance changes its form when the gain in energy is compared to the gain in entropy."

> *A last piece of ice floated in the glass of water. The young man looked at the thermometer: zero degrees, and the readings did not change until the ice completely disappeared. "Strange," he said. "I've been heating the glass all the time, but the temperature remains the same."*

1. What is a phase transition?

Sage: You noticed the main point. As long as there is both ice and water in the glass, the supplied heat goes not to heating **but to** restructuring: it breaks the bonds in the crystal. This is the latent heat of fusion — 334 kJ per kilogram. Ice and water are two **phases** of the same substance: each is homogeneous and separated from the other by a boundary.

Sage: Why does a substance choose one phase or another? From Chapter 8, you remember that the system seeks to minimize not just energy, but free energy **G = H** − TS: energy H wants to be lower, entropy S wants to be higher, and temperature T decides whose voice is louder.

Young man: So, at low temperatures, energy — the lattice wins, and at high temperatures — entropy: liquid, vapor. And the transition occurs at the temperature where G of both phases is compared.

Sage: Exactly. At the transition point $\Delta G = 0$, that is, $\Delta H = T \cdot \Delta S$, from which the transition temperature $T = \Delta H / \Delta S$. Check on ice: $\Delta H = 6010$ J/mol.

Young man: $\Delta S = \Delta H / T = 6010 / 273.15 \approx 22.0$ J/(mol·K). And for the boiling of water: 40,660/373.15 ≈ 109 J/(mol·K). The entropy during evaporation increases much more: the vapor has much more freedom.

[UV] The established fact is the Trutnov rule.

For many "simple" liquids, the evaporation entropy at normal boiling temperature is close to 85–90 J/(mol·K) (Trouton's rule). For water, the value is higher (≈ 109) due to hydrogen bonds that organize the liquid. This is an empirical rule, not a law.

The sage: Transitions that involve latent heat and a jump in density are called **first-order transitions: melting,** evaporation, sublimation. There are also others—without latent heat, where only derivatives change: second-order transitions (their general theory was developed by **Landau in** 1937). Examples: the

loss of magnetism in iron at 770°C (the Curie point), the transition of helium to a superfluid state at 2.17 K—this will be discussed in the last chapter.

2. Melting and strange ice

Sage: For most substances, the solid phase is denser than the liquid phase, and under compression, the **melting** temperature increases. For water, on the contrary: ice is lighter than water, it floats. Therefore, compression should **lower** the melting temperature. Quantitatively, this is given by the Clapeyron-Clausius equation:

$$dp/dT = L / (T \cdot \Delta v)$$

Young man: For ice: L = 334,000 J/kg, T = 273.15 K, and the change in specific volume $\Delta v = 1/1000 - 1/917 \approx -9.1 \cdot 10^{-5}$ m³/kg. Then $dp/dT \approx 334{,}000 / (273.15 \cdot (-9.1 \cdot 10^{-5})) \approx -1.35 \cdot 10^{7}$ Pa/K, that is, about −135 atmospheres per degree. To melt ice at one degree lower, a pressure of 135 atmospheres is needed!

> **[OHS] Misconception / error — "the speed skater glides because the pressure of the blade melts the ice"**
> This is a popular explanation. The assessment shows: the pressure of the blade under a person's weight is about 7 MPa (≈ 70 atm), the melting temperature decreases by only about 0.5 K. At a frost of -10°C, this is far from sufficient. Modern explanations rely on a thin liquid-like film on the surface of the ice ("premelting") and on heating due to friction; the exact ratio of contributions is discussed and remains a subject of research — hence the formulation here is cautious.

3. State diagram

Sage: Now let's draw a "map" of the substance: on the axes — temperature and pressure, and on the map — regions where a certain phase is stable. The lines between the regions represent the conditions for the coexistence of two phases.

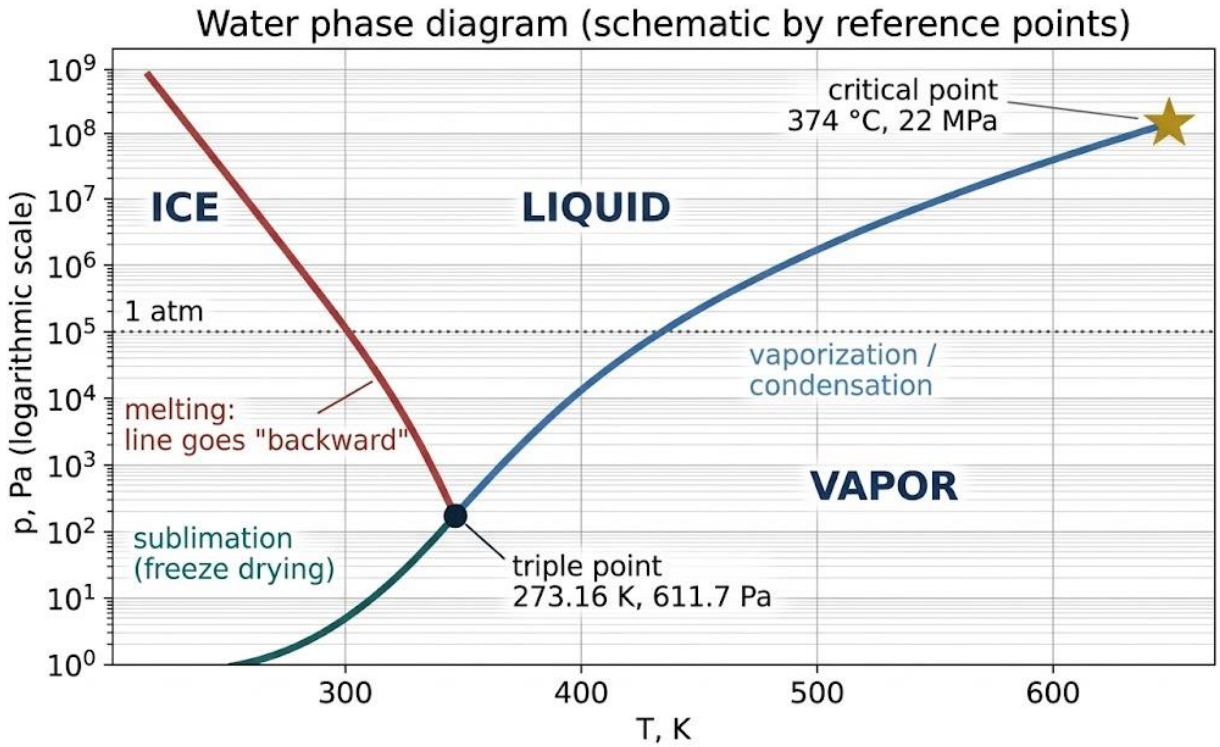


Fig. 1. Phase diagram of water (diagram based on reference points, pressure in logarithmic scale).

Young man: I see three lines converging at a single point. Is this a triple point?

Sage: Yes: at 273.16 K and 611.7 Pa, ice, water, and vapor coexist simultaneously. Since 1954, the triple point of water has been used as a reference for the Kelvin scale (until the SI revision in 2019). Below the pressure of the triple point, liquid water does not exist at all: ice transitions directly to vapor upon heating. This **is sublimation.**

Young man: So, if the pressure is below 611 Pa, then ice does not melt, but evaporates. Freeze-drying!

Sage: That's right. The frozen product is placed in a vacuum (pressure below 611 Pa), a little heat is applied, and the ice sublimates, preserving the structure of the product: there is no liquid water that would damage the tissue or dissolve the components. One has to pay the price: the latent heat of sublimation of ice (≈ 2.83 MJ/kg) is greater than that of water evaporation (≈ 2.5 MJ/kg), and in a vacuum, heat transfer is poor, so the process is slow.

> **[OD] Justified addition — sublimation drying according to the laws of this chapter**
>
> The first principle: the minimum energy per kilogram of ice is about 2.83 MJ (approximately 165 more than 2.44 MJ/kg for the evaporation of water at 25°C). This is not a "shortcoming of the apparatus," but rather the heat of transition. Additionally, energy is required for freezing, a vacuum pump, and a vapor condenser. **The method** excels in quality but not in energy costs per kilogram of water—comparisons with other drying methods should be conducted fairly, based on both indicators.

Sage: For comparison: the triple point of carbon dioxide is at a pressure of 5.1 atm (216.6 K), which is above atmospheric pressure. Therefore, "dry ice" does not melt at normal pressure, but sublimates at −78.5°C.

4. Real gases: Van der Waals

Sage: To describe both gas and liquid with a single formula, the Dutch physicist Johannes **Diderik van der Waals proposed** a simple modification of the Clapeyron-Mendeleev equation in 1873. He took into account two facts. The first: molecules have their own size, so the free volume is smaller: $V \to V - b$. The second: molecules attract each other, and the pressure on the wall is weaker: $p \to p + a/V^2$. For one mole:

$$(p + a/V^2) \cdot (V - b) = RT$$

Sage: The constants a and b are taken from experience. For carbon dioxide, $a \approx 0.364$ Pa·m^6/mol², $b \approx 4.27 \cdot 10^{-5}$ m³/mol. The critical point can be calculated from this equation: $T_к = 8a/(27Rb)$, $p_к = a/(27b^2)$.

Young man: I calculate: $T_к = 8 \cdot 0.364 / (27 \cdot 8.314 \cdot 4.27 \cdot 10^{-5}) = 2.912 / 0.009585 \approx 304$ K. And $p_к = 0.364 / (27 \cdot (4.27 \cdot 10^{-5})^2) \approx 7.4 \cdot 10^6$ Pa = 7.4 MPa. This is the critical point of CO_2 (304 K, 7.4 MPa)!

Sage: For a simple equation with two constants — an excellent result. But let's not idealize: the formula gives a value for the critical volume that diverges from experience, and the ratio $p_к \cdot V_к / RT_к$ is equal to 0.375, while for real substances it is about 0.27. Van der Waals is a qualitative model, not an exact law.

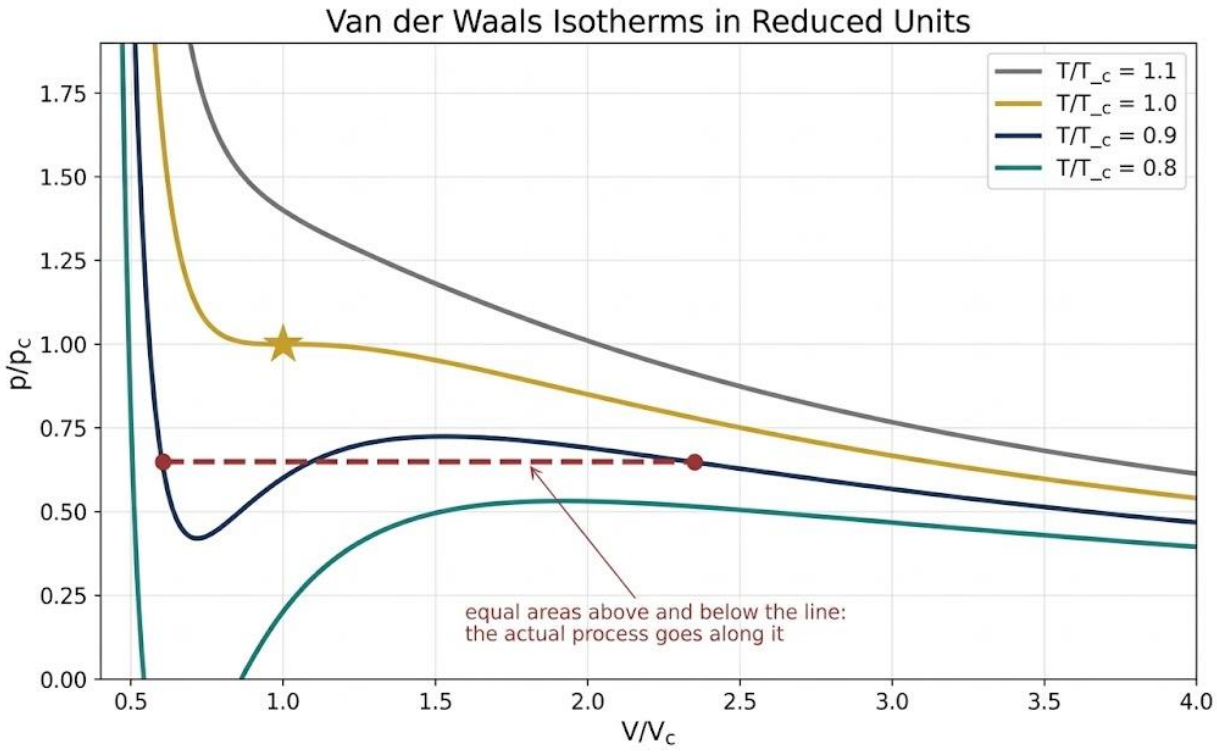


Fig. 2. Van der Waals isotherms in reduced variables (based on critical values). Below the critical temperature, a "loop" appears; the actual transition occurs along a line drawn according to the rule of equal areas (Maxwell, 1875).

Young man: What are the lower curves with a loop? Pressure increases during expansion, that doesn't happen!

Sage: It is true: the region where **dp/dV** > 0 is unstable, such a state does not occur. Instead, the substance undergoes "phase separation": part becomes liquid, part becomes vapor, and the pressure remains constant — this is a horizontal line (saturated vapor from Chapter 10). Its level was found by Maxwell using the rule of equal areas: the areas above and below the line are equal. The sections of the loop at the edges (between **the binodal** and spinodal) correspond to metastable states: superheated liquid and subcooled vapor.

Young man: So that's where the overheating of water from chapter 10 comes from! The state "liquid above the boiling point" exists as long as there are no vapor nuclei.

The sage: Yes. And if the curve passes through the inflection point — this is a critical isotherm: the loop disappears, and the distinction between liquid and gas vanishes. Van der Waals showed something more general: in the reduced variables ($p/p_к$, $V/V_к$, $T/T_к$) the equation is the same for all substances — the "law of corresponding states." For this, he was awarded the Nobel Prize in 1910.

> **[UV] The established fact is the answer to the promise from chapter 6.**
>
> In Chapter 6, we discussed that the internal energy of a real gas also depends on volume. For a Van der Waals gas, $U = C_V \cdot T - a/V$ (per mole): the attraction between molecules "stores" energy. Therefore, during free expansion without work and heat exchange, the **gas cools** down: $\Delta T = -(a/C_V) \cdot (1/V_1 - 1/V_2)$.
>
> Estimation for a mole of CO_2 ($C_V \approx 28.8$ J/(mol·K)) during expansion from 1 L to 2 L: $\Delta T \approx -0.364 \cdot (1000 - 500) / 28.8 \approx -6$ K. For regular air at atmospheric pressure, the effect is negligible: the ideal approximation is justified.

5. How "permanent" gases were liquefied

The sage: Now a story from a hundred years ago. In 1823, Michael Faraday liquefied chlorine and ammonia by pressure. But oxygen, nitrogen, and hydrogen stubbornly remained gases no matter how much they were compressed, and they were called "permanent."

Young man: Now I understand why. The critical temperature! Above it, you cannot obtain a liquid by compression.

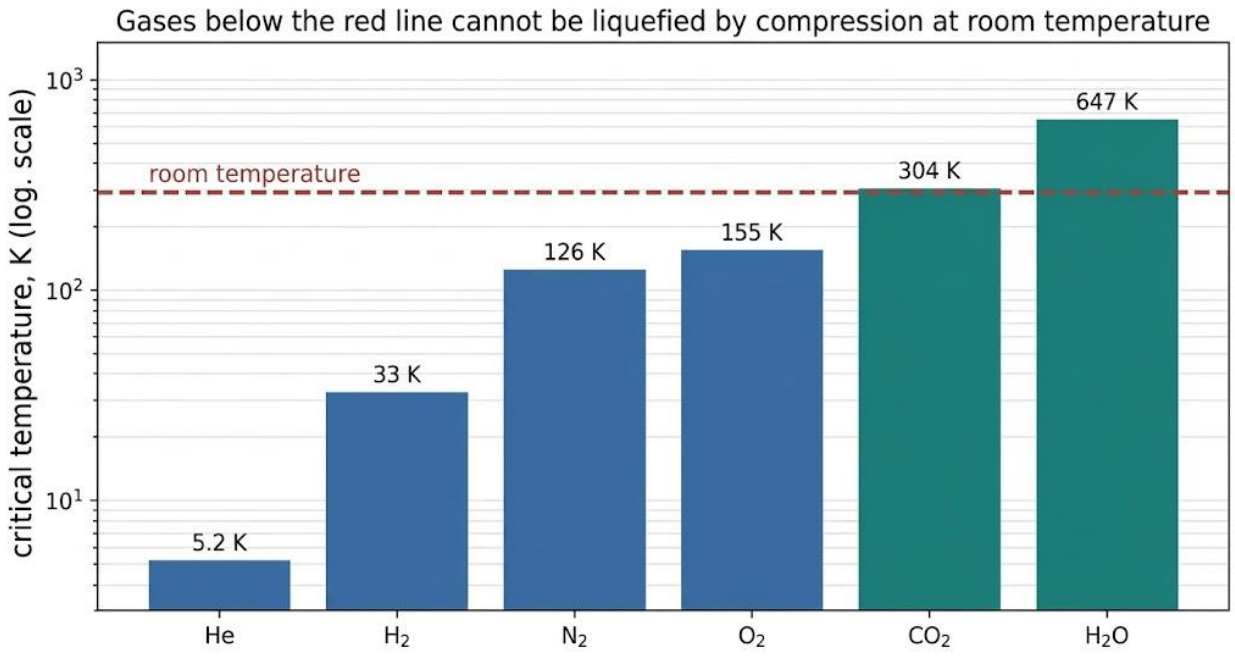


Fig. 3. Critical temperatures of some substances. Gases with a critical temperature below room temperature need to be cooled first.

The sage: This is exactly what Andrews understood in 1869. And in 1877, the Frenchman Cayette and the Swiss Pictet obtained the first drops (mist) of oxygen, and in 1883, Wroblewski and Olszewski in Krakow obtained liquid oxygen and nitrogen. Dewar in 1898 — liquid hydrogen (20 K), and Kamerlingh Onnes in Leiden in 1908 — liquid helium (4.2 K).

Young man: But how to cool the gas below room temperature? After all, the refrigerator itself does not cool below a certain limit.

The sage: Two techniques are used. The first **is expansion with work done: the** gas expands, pushes the piston or turbine, and cools down due to its own energy (adiabatic from chapter 6). Such a "detander" was used by Georges Claude in 1902. The second is **the Joule-Thomson effect: the gas** is throttled through a narrow opening, and due to the attraction of molecules (we just calculated this using CO_2 as an example), it cools down. Carl Linde added the idea of **counterflow in** 1895: the cooled gas at the outlet cools the incoming gas, and cold accumulates until a liquid is formed.

> **[UV] Established fact — inversion temperature**
>
> Throttling cools the gas only below a certain "inversion temperature": for nitrogen, it is about 620 K, for hydrogen about 200 K, and for helium about 40 K. Above this **temperature,** the gas heats up during throttling (molecular repulsion predominates). Therefore, hydrogen and helium must be pre-cooled by other means and only then throttled.

6. Phases in our case: ceramics and barriers

Young man: Sage, how is all this related to our ceramics?

Sage: In the most direct way. Many oxides have several crystalline modifications. Titanium dioxide TiO_2 exists as anatase and rutile. Under normal conditions, rutile is thermodynamically stable; however, anatase can persist for years: the transformation into rutile requires overcoming a barrier, and upon heating (depending on purity and grain size, approximately from 500 to 800°C), it proceeds irreversibly.

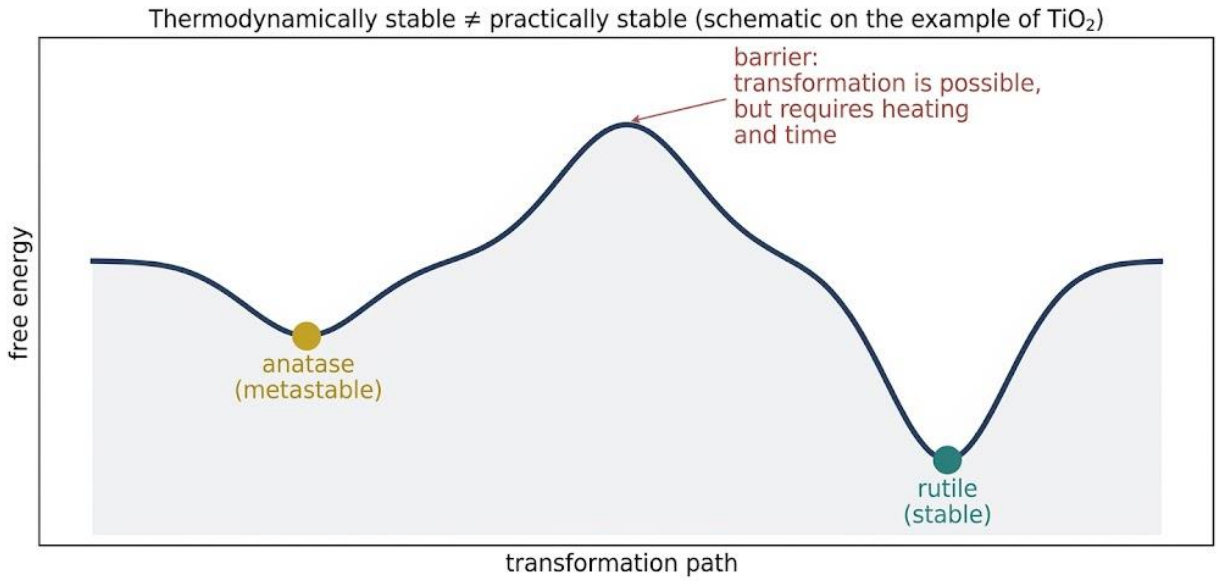


Fig. 4. Stable and metastable phases: as long as the barrier is not overcome, the system remains in the higher "well". Diagram, not calculation.

Sage: This is important for photocatalysis: anatase is often more active than rutile. Therefore, during synthesis, one has to follow a narrow path: hot enough to achieve good crystallinity, but not so hot as to convert anatase into rutile. The same principle applies to zirconium dioxide: at ≈ 1170°C, a transition occurs from the monoclinic to the tetragonal form with a volume change of 4–55, which causes products to crack until the phase is stabilized by the addition of yttrium oxide.

[UV] Established fact — energy sustainable ≠ practically sustainable

Thermodynamics determines which phase has the minimum free energy. Kinetics — barriers — determines whether the system will reach this minimum in a reasonable time. Metastable phases (anatase, diamond at room temperature, glass) exist precisely because of barriers. The same principle as in Chapter 9, only now for structural transformation.

[GA] The author's hypothesis - a testable question for ITÉ: does impulse feeding help overcome the transformation barrier?

If the ITE overcomes barriers by supplying energy with a fast front (the barrier does not decrease, as with a catalyst; working formulation from Chapter 9), then there is a directly verifiable effect in **ceramics:** with the same average supplied power, the pulsed mode could initiate a phase transition (for example, anatase → rutile) at a lower average temperature or in less time than continuous heating. The final stable phase is determined by thermodynamics and does not change.

The check is simple and inexpensive: X-ray phase analysis of samples processed at the same average power in two modes, with temperature control of the sample. If there is a shift, this is independent evidence that the front helps to overcome the barrier; if not, the hypothesis for this barrier is not confirmed. Status — [G].

7. What we have learned and what remains

- **A** phase is a homogeneous part of a system. The transition occurs where the free energies of the phases are equal: $T = \Delta H/\Delta S$.
- **First kind** — with latent heat and density jump (melting, evaporation, sublimation); second **kind** — without them (Curie point, superfluidity).
- **The phase diagram shows** the stability regions of phases; the triple point of water is at 273.16 K and 611.7 Pa. Below this pressure, ice sublimates.
- **Van der Waals:** $(p + a/V^2)(V - b) = RT$. It provides the critical point, metastable states, and the cooling effect during expansion. The model is qualitative.
- **Gas liquefaction:** it is necessary to cool below the critical temperature; expansion work, Joule-Thomson throttling, and Linde counterflow are helpful.
- **Metastable phases exist** due to barriers — a key principle of ceramic materials science.

Young man: I expected that thermodynamics would provide all the answers, but now I see: it tells us where the system is heading, but not how fast and by what path. Thus, there remains one more major topic: how do matter, energy, and momentum move through the material?

Sage: This is chapter 12: transport phenomena — diffusion, thermal conductivity, viscosity, and thermal radiation. There we will finally take "kinetics" seriously and discuss how light carries energy.

Problems for independent solution

- Find the molar entropy of melting of ice ΔS at 273.15 K, if $\Delta H = 6.01$ kJ/mol.
- What pressure is needed to lower the melting temperature of ice by 2 K? (Use $dp/dT \approx -135$ atm/K.)
- Using $a = 0.364$ Pa·m⁶ /mol² and $b = 4.27 \cdot 10^{-5}$ m³/mol, find the critical temperature using the formula $T_k = 8a/(27Rb)$.

- Is it possible to liquefy nitrogen by compression at 200 K? The critical temperature of nitrogen is 126 K.
- How much heat is needed to sublimate 2 kg of ice at low pressure (latent heat of sublimation 2.83 MJ/kg)? Compare this with the evaporation of 2 kg of water at 25°C (2.44 MJ/kg).

> **[UV] Established fact — answers to the problems**
>
> 1) $\Delta S = 6010 / 273.15 \approx 22.0$ J/(mol·K).
>
> 2) $\Delta p \approx 135 \cdot 2 = 270$ atm.
>
> 3) $T_k = 8 \cdot 0.364 / (27 \cdot 8.314 \cdot 4.27 \cdot 10^{-5}) \approx 2.912 / 0.009585 \approx 304$ K.
>
> 4) No: 200 K above the critical temperature (126 K), the liquid-gas boundary at this temperature does not exist. First, nitrogen must be cooled below 126 K.
>
> 5) Sublimation: $2 \cdot 2.83 = 5.66$ MJ. Evaporation: $2 \cdot 2.44 = 4.88$ MJ. Sublimation is approximately 165 more expensive.

Chapter 12. Transport Phenomena

"How molecules and light transport matter and energy from one place to another"

> *On the table stood a glass of water, and the Young Man carefully dropped a drop of ink into it. The drop hung like a cloud, then began to slowly spread. He timed it: after ten minutes, the colored area had grown by a couple of centimeters. "Why so slowly? — he asked. — Molecules are moving at hundreds of meters per second."*

1. Fast molecules, slow path

Sage: You have caught a paradox. Air molecules at room temperature travel at a speed of about 460 m/s. However, the scent of perfume opened in one corner of the room reaches the other corner not in milliseconds, but in minutes. Why?

Young man: From chapter 4: the mean free path! A molecule travels only about 70 nanometers and collides, changing direction.

Sage: That's right. In the air, a molecule experiences about 7 billion collisions per second. Each one changes its direction, so the path is a random walk. A **wandering particle moves away** from the starting point much more slowly than it would if it were flying in a straight line.

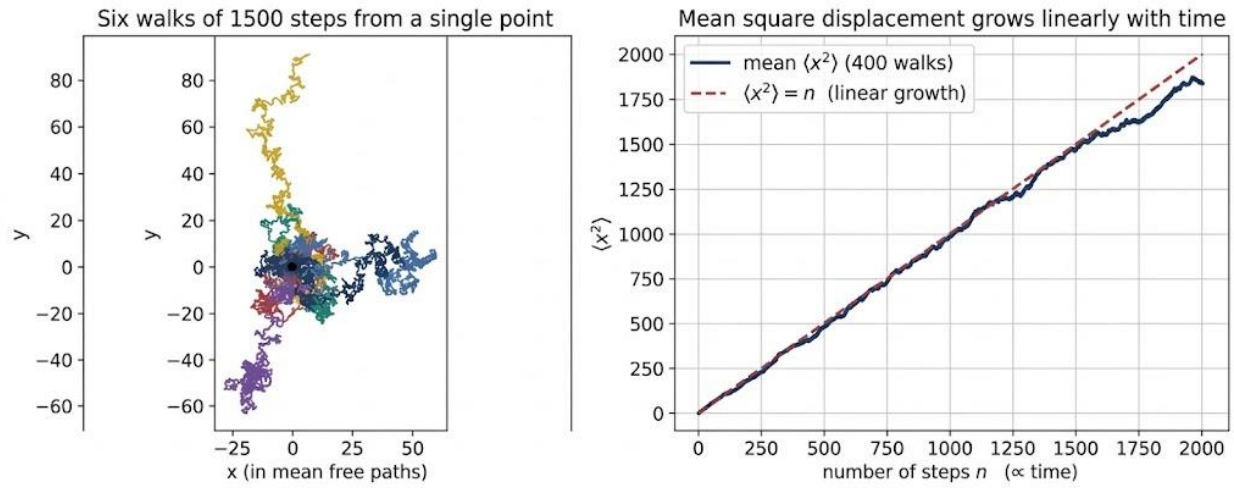


Fig. 1. On the left — random walks (six trajectories from a single point). On the right — the mean square displacement grows linearly with the number of steps, and thus with time.

The sage: From the diagram, the main point is clear: the distance that a wandering particle travels increases not with time, but **as the square root** of time: $\langle x^2 \rangle = 2Dt$. The quantity **D is called the** diffusion coefficient. This formula was derived by Einstein in 1905 for Brownian motion — the very same that we discussed in the first chapter.

Young man: So, in twice the time, the particle will not go twice as far, but only by the square root of two times?

Sage: Exactly. And conversely: to cover twice the distance, you need four **times** as much time.

> **[UV] Established fact — diffusion: what is known and how it is measured**
>
> Fick's law (1855): the flux of substance is proportional to the concentration gradient, $J = -D \cdot dc/dx$. Einstein (1905) related D to temperature and viscosity: for a spherical particle of radius r in a liquid $D = kT/(6\pi\eta r)$ (Stokes-Einstein relation). For a particle with a radius of 0.5 μm in water at 20 °C, $D \approx 4 \cdot 10^{-13}$ m²/s; Perrin tested such predictions experimentally, which became one of the decisive pieces of evidence for the existence of molecules.
>
> Typical values of D: gases (water vapor in air) about $2 \cdot 10^{-5}$ m²/s; small molecules in water about 10^{-9} m²/s; moisture within dense materials can be orders of magnitude lower (10^{-10} m^2/s and below; the value strongly depends on the material and humidity).

Young man: I will calculate the time for the plate. If $D = 10^{-9}$ m^2/s, then for a layer thickness of 1 mm: $t \approx L^2/D = 10^{-6} / 10^{-9} = 1000$ s, about 17 minutes. And for a layer of 1 cm — it's already 10^5 seconds, more than a day!

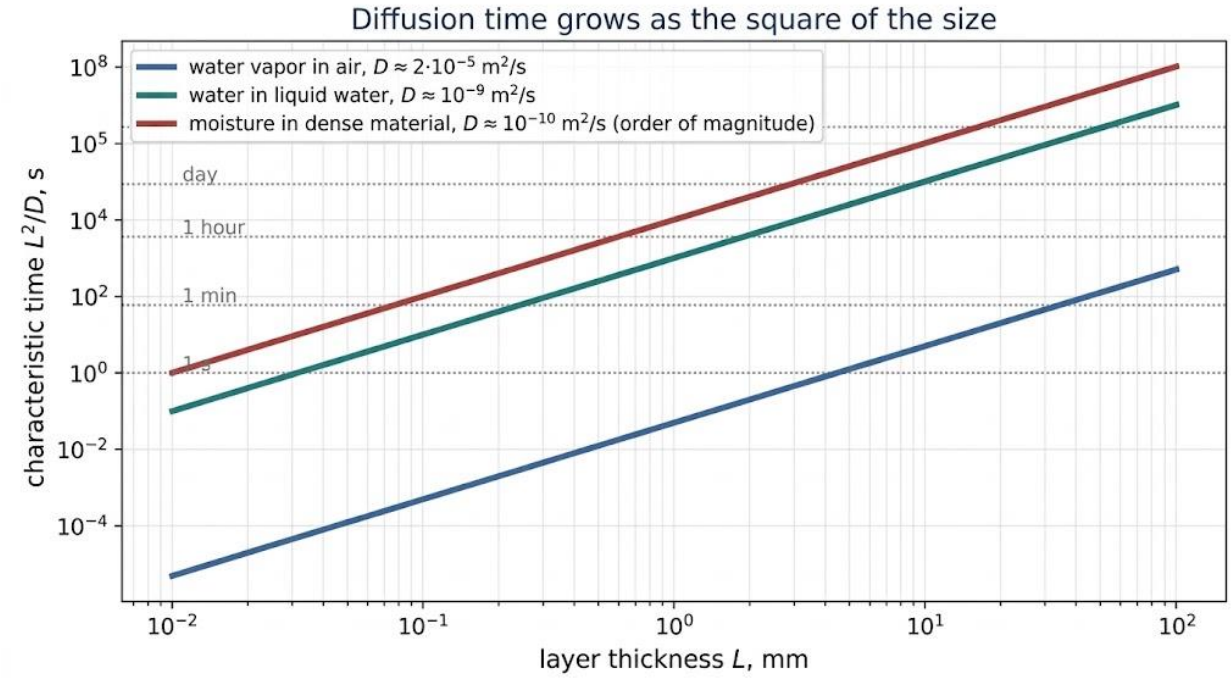


Fig. 2. Diffusion time through a layer of thickness L: $t \approx L^2/D$. Each reduction in thickness by a factor of 10 reduces the time by a factor of 100.

The sage: This is the answer to the question about drying thick materials. When drying is limited by moisture transfer within **(the** second period from Chapter 10), the time increases as the square of the thickness. Half as thick — four times faster. No dryer design, unless it changes D or the length of the path, will circumvent this law.

Sage: But D can be increased: with temperature, it grows according to the Arrhenius law — the same one from chapter 9: $D = D_0 \cdot \exp(-E_a/RT)$. For a barrier of 30 kJ/mol (a typical order for water in porous materials), heating from 40 to 60 °C approximately doubles D, and the time is halved.

Young man: So, heating **the interior** is the most effective way to accelerate internal transfer? Not just the surface?

The sage: It is to this place that we will return at the end of the chapter when we talk about radiation.

2. Thermal conductivity

Sage: Heat also travels by wandering: fast (hot) molecules migrate to the cold area, slow ones to the hot. Fourier's law (1822):

$$q = -\kappa \cdot dT/dx$$

Sage: Here q is the heat flux per unit area, κ is the thermal conductivity. Its values vary by tens of thousands of times.

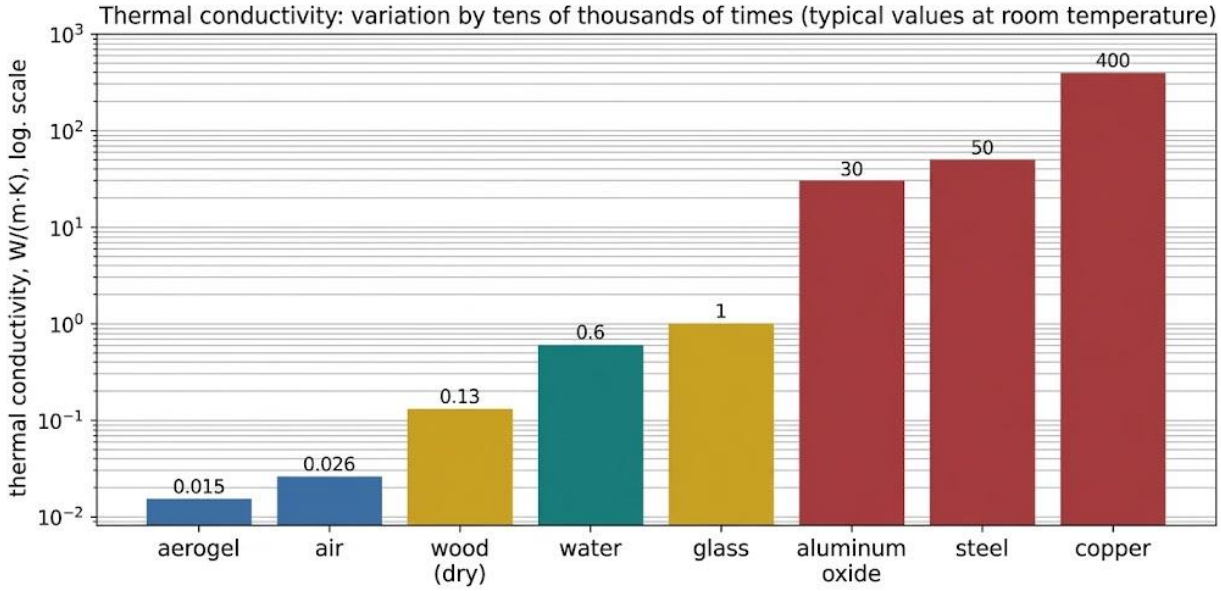


Fig. 3. Thermal conductivity of different materials at room temperature (typical values).

Young man: Air conducts heat 15,000 times worse than copper! That's why a down jacket keeps you warm — it contains air.

Sage: Yes. And that is why the dried layer of porous material is an excellent insulator: it consists mainly of air. Remember chapter 10: in **the** second drying period, heat must be transferred to the moisture boundary precisely through such a layer.

> **[UV] The established fact is the Maxwell's paradox.**
>
> The kinetic theory provides for gases: thermal conductivity and viscosity are proportional to the product of the concentration of molecules and the mean free path. However, the mean free path is inversely proportional to the concentration, and they cancel out: the **viscosity and thermal conductivity of a gas do not depend on** its density. Maxwell predicted this in 1860, and the result seemed absurd: it should be easier in a rarefied gas, not the same. He himself tested it experimentally in 1866 and confirmed it.
>
> The law ceases to work when the travel length becomes comparable to the dimensions of the vessel: in a deep vacuum, heat is almost not transferred. This is how a thermos is designed.

Sage: When heat is supplied in "pulses," a characteristic is needed for how deep the thermal wave penetrates. For oscillations with frequency f, the depth is approximately $\sqrt{(a/\pi f)}$, where $a = \kappa/(\rho c)$ — thermal diffusivity. For water, $a \approx 1.4 \cdot 10^{-7}$ m²/s, at f = 434 Hz, this results in about 10 μm.

Young man: Only ten micrometers! This means that the thermal "ripple" from impulses at such a frequency does not penetrate deeper than the thinnest surface layer.

Sage: Exactly. And during this time (period ≈ 2.3 ms), moisture will advance at $D = 10^{-9}$ m²/s by $\sqrt{(2Dt)} \approx 2$ μm. We will use these estimates later.

3. Viscosity

Sage: The third transfer — **of** momentum: the fast layer of fluid drags along the slow one. Newton's law: shear stress $\tau = \eta \cdot du/dy$, where η **is viscosity.**

The viscosity of water at 20°C is about 1 mPa·s, that of air is 0.018 mPa·s, glycerin is 1400 mPa·s, and resin (the experiment with "dripping resin," started in 1927 in Australia) is on the order of 10^8 Pa·s, which is 200 billion times greater than that of water. Drops in it fall once every few years.

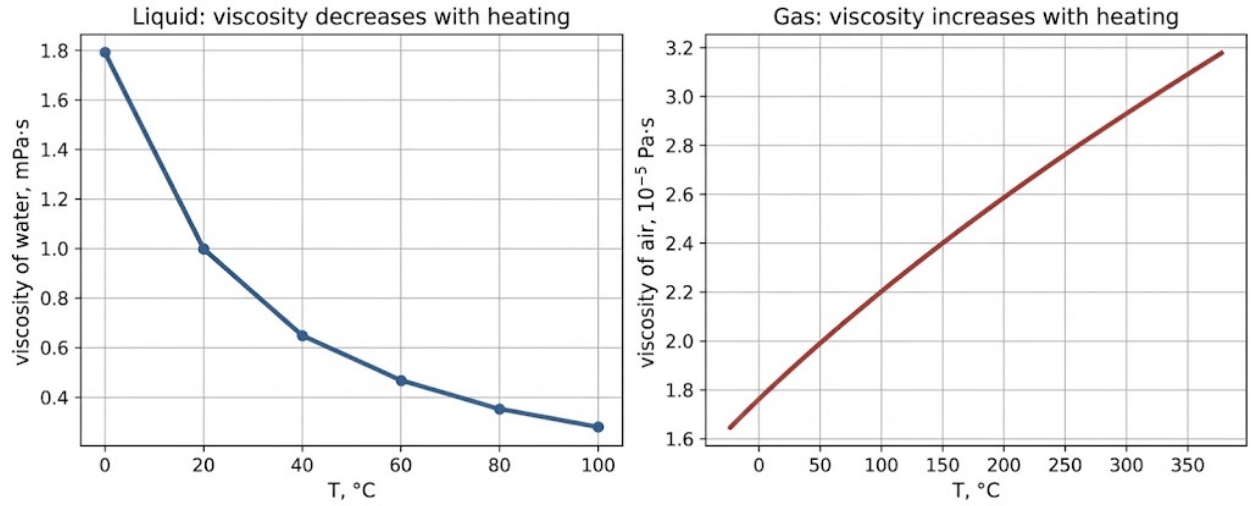


Fig. 4. The viscosity of the liquid decreases with heating, while the viscosity of the gas increases.

Young man: Why is it the opposite for gas?

The sage: In a gas, momentum transfer occurs through the molecules themselves: the hotter it is, the faster they wander between layers, and the greater the "friction." In a liquid, the molecules are strongly bound, and viscosity is determined by the barrier of "jumping" between neighbors, and when heated, the barrier is more easily overcome — again, the Boltzmann factor.

The sage: Viscosity is also needed in drying: it determines how air flows along the surface and what the thin stationary layer of air at the wall looks **like — the boundary** layer through which vapor must exit by diffusion.

> **[OD] Justified addition — boundary layer and "steam outlet"**
>
> If the thickness of the boundary layer δ, then the mass transfer coefficient is approximately D/δ. For steam in air ($D \approx 2.5 \cdot 10^{-5}$ m²/s) and δ = 1 mm, we obtain 0.025 m/s. The density of saturated steam at 60°C is about 0.13 kg/m³, so the limiting evaporation flux in dry air is $0.025 \cdot 0.13 \approx 3.3 \cdot 10^{-3}$ kg/(m²·s), that is, ≈ 12 kg/(m²·h).
>
> If δ is halved (for example, by injection or blowing), the flow will double. If the steam does not escape, the flow decreases. This is the quantitative justification for the factor of 4 from the table in chapter 9: the steam discharge system directly affects the speed. The numbers are an order of magnitude estimate.

4. Thermal radiation

Sage: There is also a fourth way of heat transfer: without a medium and without intermediary molecules. The Sun warms the Earth through the void. Any body at a temperature above absolute zero emits electromagnetic waves.

The sage: In 1859, Kirchhoff introduced the **concept of an absolute** black body — an ideal absorber and, therefore, an ideal emitter: a body that absorbs well also emits well. In 1879, Stefan found an empirical law, and in 1884, Boltzmann derived it from thermodynamics: the power of radiation per unit area is proportional to the **fourth power of** the temperature.

$$j = \varepsilon \cdot \sigma \cdot T^4 , \quad \sigma = 5.67 \cdot 10^{-8}\ W/(m^2 \cdot K^4)$$

Young man: For 1000 K: $5.67 \cdot 10^{-8} \cdot 10^{12} \approx 57{,}000$ W/m², that is, 57 kilowatts per square meter! And at 300 K — only 460 W/m².

Sage: The fourth power makes radiation a very cool function of temperature: doubling the temperature increases the flux by 16 times. And one **more thing:** for thermal radiation, the color shifts **with** temperature. Wien's displacement law (1893):

$$\lambda_{max} \cdot T = 2898\ \mu m \cdot K$$

Young man: For the Sun (5772 K): 2898/5772 ≈ 0.5 μm — green color! For the human body (310 K): ≈ 9.3 μm — infrared. And a radiator at 1000 K: 2.9 μm — right in the water absorption band.

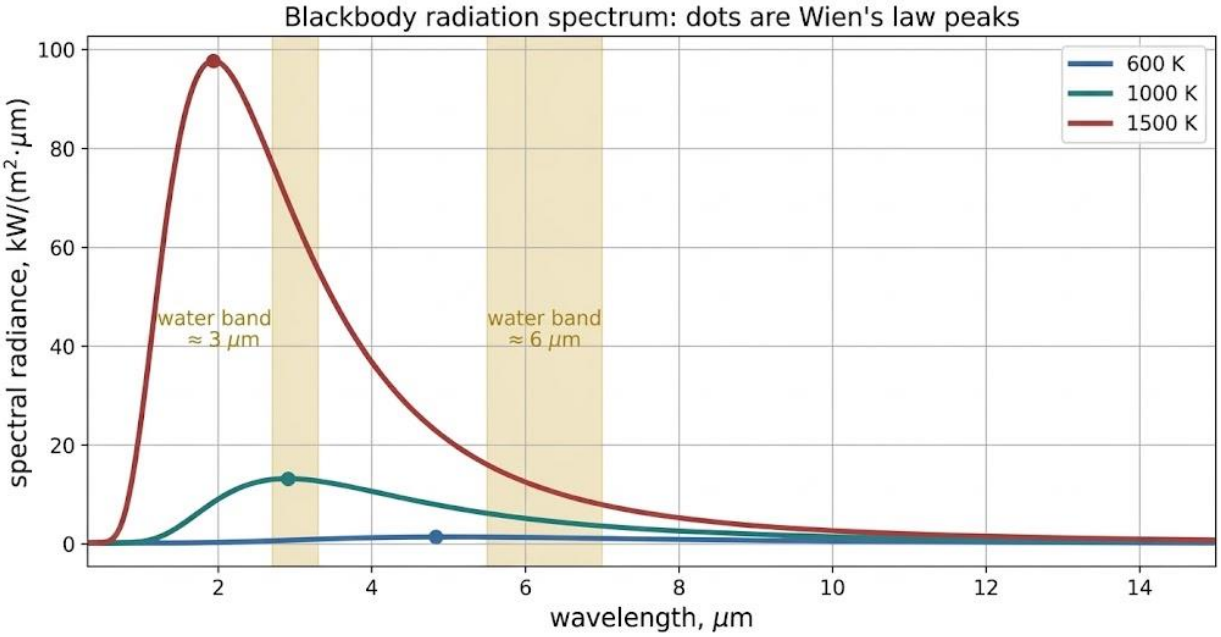


Fig. 5. Spectrum of radiation of a black body (ordinary thermal radiator) at three temperatures. Points — maxima according to Wien's law. Bands — regions of strong absorption of liquid water.

The sage: This is how the color of ordinary thermal emitters is determined: if a band around 3 μm is needed, an emitter at around 1000 K is used; if one wants to reach the near IR (1.45 and 1.94 μm), it is heated more, around 1500 K. But in the case of ITÉ, as intended by the author, everything is arranged differently — see the next block.

The author's hypothesis is that the color of the ITE impulse determines the front, not the temperature.

Wien's law describes the thermal **radiation** of a body in equilibrium: color is determined by temperature. According to the author, the spectrum of the ITD pulse is determined only by the steepness of its front: the rise front can be considered as part (a quarter of the period) of a sine wave, and the color does not depend on the temperature of the emitter. This is essentially a common property of non-thermal radiation: the frequency of a radio transmitter or laser is also not determined by the temperature of the casing. The advantage of the method: the desired "color" can be selected independently of temperature, without heating the emitter to 1000–1500 K for the sake of the desired band.

Quantitative relationship: if the rise time lasts t_ϕ, then the equivalent frequency $f \approx 1/(4 \cdot t_\phi)$, and the wavelength $\lambda \approx 4 \cdot c \cdot t_\phi$. For t_ϕ = 2.5 fs, λ = 3 μm; for 10 fs — 12 μm; for 1 ps — 1.2 mm. The time for the restructuring of hydrogen bonds in water (on the order of 1 ps) and the absorption band of water around 3 μm lie at different ends of this scale.

It is useful to indicate in the book: what duration of the front is accepted and how it is measured (fast photodiode, oscilloscope) and whether it concerns an IR carrier with a steep envelope or longer wavelengths; the emission spectrum — IR Fourier spectrometer at different temperatures of the emitter. Status — [G].

[UV] The established fact — the ultraviolet catastrophe and the birth of quanta

Classical physics (Rayleigh, Jeans) provided a formula for the radiation spectrum that diverged in the short wavelength region: the radiator would "emit" infinite energy. This was inconsistent with experimental results. In 1900, Max Planck found a formula that agreed with the experiments, and to derive it, he had to assume that energy is emitted in **quanta of** hν. The constant $h = 6.626 \cdot 10^{-34}$ J·s. Thus, quantum physics emerged, which we will discuss in the final chapter.

5. Radiation versus convection: a fair comparison

Sage: Now let's return to drying. It is often claimed that radiation transfers energy "faster." Let's check with numbers. Suppose the product has a surface temperature of 60 °C. A typical thermal radiator at 600 °C (873 K) with an emissivity of 0.9. What is the flux?

The young man: $j = 0.9 \cdot 5.67 \cdot 10^{-8} \cdot (873^4 - 333^4) \approx 0.9 \cdot 5.67 \cdot 10^{-8} \cdot 5.69 \cdot 10^{11} \approx 29{,}000$ W/m². This is 29 kilowatts per square meter. If all the energy goes to evaporation at 2.4 MJ/kg, we get $29{,}000 / 2.4 \cdot 10^6 \approx 0.012$ kg/s, that is about 43 kilograms of water per hour per square meter!

Sage: This is the upper limit: it assumes that the radiation is completely absorbed at the surface and all the energy goes to evaporation (and that the water has time to reach the surface). Now let's compare it with hot air: convection with a heat transfer coefficient $h \approx 25$ W/(m²·K) (a typical order for forced ventilation) at an air temperature of 150°C gives $25 \cdot 90 \approx 2{,}250$ W/m² — 13 times less.

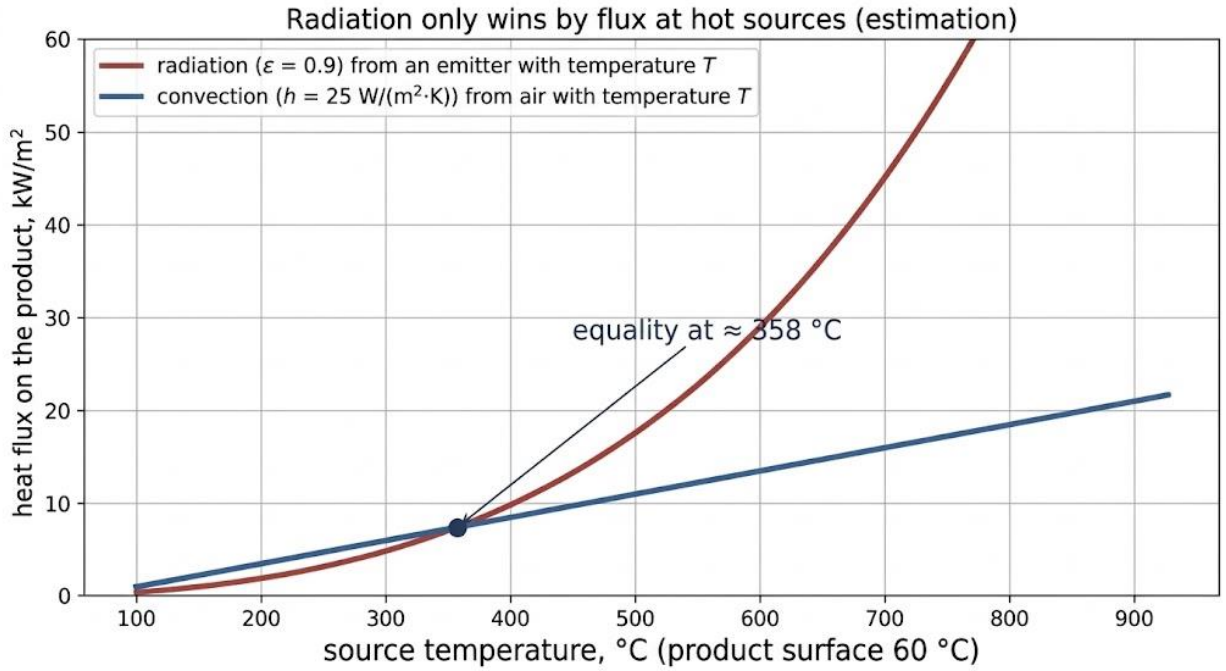


Fig. 6. Heat flow to the product (60°C) from a conventional heat emitter and from hot air as a function of the source temperature. Estimate: ε = 0.9, h = 25 W/(m²·K).

Young man: But if the source is colder — for example, air at 200°C — then convection gives more radiation!

Sage: It's good that you pointed that out: up to a temperature of about 360°C, the radiation flow from such a **source does not** exceed the convective one. The advantage of radiation is not in its nature, but in the fact that its source can be heated to hundreds of degrees without heating the entire air of the dryer. Hot air at such temperatures would destroy the product, while a radiator would not, if the spectrum is selected and the flow is regulated.

> **[OD] Justified addition — what exactly objectively benefits from radiation**
>
> (1) The flow increases as T^4 and does not require an intermediate carrier — radiation does not depend on the air flow (factor 7). (2) The spectrum can be adjusted: for thermal radiators at a certain temperature, for ITEs, according to the author's design, the impulse front is independent of temperature (factor 6). (3) Energy is delivered directly to the substance, and there is no need to heat and change the air; in convective drying, a significant portion of heat is lost with the hot exhaust. (4) With penetrating radiation, not only the surface is heated (but the depth is determined by the spectrum: see Fig. 6 of Chapter 9).
>
> Limitations that must be honestly remembered: the flow may exceed the material's ability to conduct moisture from the inside (then the surface dries out and overheats — critical mode); the hot emitter loses part of its energy heating the air and the casing; without steam removal, the velocity decreases (boundary layer).

6. What limits drying: summary

Sage: Now we can bring everything together. At each stage of drying, the speed is limited by its "narrow section"; depending on which one it is, the advantage of the method varies.

What limits	Law of transfer	What factors of ITE (Chapter 9) can help
Heat supply to the surface	Radiation $j = \varepsilon\sigma T^4$; convection $q = h \cdot \Delta T$	3 (direct transfer), 6 (spectrum), 7 (does not depend on air flow)
Vapor output from the surface	Diffusion through the boundary layer, flow $\propto D/\delta$	4 (steam vent system)
Moisture transfer within the product	Diffusion, time $\propto L^2/D$; D increases according to Arrhenius.	2 (volumetric underwater), 5 (moderate temperature), 8 (deep dehydration)
Final moisture content	Equilibrium with air and bound water	8

Young man: Then it is clear why it is important to heat **from the inside** in the second period: if D increases according to Arrhenius, heating the material by 20 degrees doubles the moisture transfer.

The sage: And this is the strongest physical argument in favor of penetrating radiation. But here is also the most delicate test. Do you remember the estimates: the thermal wave from impulses at a frequency of 434 Hz penetrates to ≈ 10 μm, while moisture penetrates in one period to ≈ 2 μm. This means that the impulses themselves cannot "mix" the moisture to a depth of millimeters: for a layer of 1 mm, the equilibration time is thousands of seconds.

> **[GA] The author's hypothesis — what does "volumetricity due to impulsiveness" mean in light of the assessments of this chapter?**
>
> The precise formulation of the hypothesis, consistent with the laws of transfer: (a) the depth of heating is determined by the radiation spectrum (Figure 6 of Chapter 9), not by the pulse frequency; (b) the frequency of 434 Hz is itself too high for moisture to redistribute between pulses; (c) if there is a beneficial effect of "intermittency," it should manifest on scales comparable to the moisture equilibration time (seconds to minutes for millimeter layers), that is, through the **envelope of** the on/off mode, rather than through the carrier frequency; (d) the direct mechanism discussed in the series (energy accumulation and overcoming the barrier with a fast front) is a separate hypothesis, it does not follow from these estimates and requires measurements. Status — [G].
>
> This does not weaken the method: some advantages (radiation, spectrum, independence from airflow, steam removal) follow from the laws of this chapter, while others are hypotheses with a clear testing scheme (profiles of T and humidity with thickness, modes with different envelopes at the same average power).

7. What we have learned and what remains

- **Wandering: the** distance grows as √t. Diffusion: $\langle x^2 \rangle = 2Dt$, time $\propto L^2/D$; Fick's law $J = -D \cdot dc/dx$.
- **Thermal conductivity (Fourier's** law) differs by thousands of times among different materials; a dry porous layer is an insulator. Maxwell's paradox: κ and η of a gas do not depend on density.
- **Viscosity: it** decreases with heating in liquids and increases in gases. The boundary layer determines the vapor output.
- **Radiation: j** = $\varepsilon\sigma T^4$ and $\lambda_{max} \cdot T = 2898$ μm·K. For thermal radiators, the spectrum is governed by temperature (for ITEs, according to the author, by the pulse front), and the flux increases as T^4 . The advantage over convection arises in hot sources (above ≈ 360°C in our estimate).

- **Drying:** at each stage, its own transfer limits; this determines which properties of the method actually work.

Young man: The last chapter remains. And you have often said: "quantum physics" and "ahead." What is there?

Sage: Limits and Continuity: the third law and absolute zero, low temperatures and superfluidity, the thermodynamics of black holes and information, and the connection with quanta. This will be a large chapter — because all threads converge in it.

Problems for independent solution

- What time will a molecule in water ($D = 10^{-9}$ m²/s) take on average to travel 1 mm? (Use $t = L^2/(2D)$.)
- Diffusion-limited drying of a plate with a thickness of 1 cm takes 6 hours. How much time will it take to dry a plate of the same material with a thickness of 2 cm and a thickness of 5 mm?
- Find the wavelength of the maximum radiation and the total power from 1 m² of a black body at 1000 K.
- Find the wavelength of the maximum radiation of the human body (310 K).
- Find the radiation flux ($\varepsilon = 1$) between a wall at 100°C and a body at 20°C.

> **[UV] Established fact — answers to the problems**
>
> 1) $t = (10^{-3})^2 / (2 \cdot 10^{-9}) = 500$ s ≈ 8 min.
>
> 2) Time ∝ L^2: for 2 cm — 4 times more, 24 hours; for 5 mm — 4 times less, 1.5 hours.
>
> 3) $\lambda_{max} = 2898/1000 \approx 2.9$ μm; $j = 5.67 \cdot 10^{-8} \cdot 10^{12} \approx 5.7 \cdot 10^{4}$ W/m² = 57 kW/m².
>
> 4) $\lambda_{max} = 2898/310 \approx 9.3$ μm (infrared radiation).
>
> 5) $j = 5.67 \cdot 10^{-8} \cdot (373.15^4 - 293.15^4) \approx 5.67 \cdot 10^{-8} \cdot 1.2 \cdot 10^{10} \approx 680$ W/m².

Chapter 13. Limits and Continuation

"Where the classical picture ends and the new one begins"

> *The young man laid out all twelve chapters on the table and stared at them in silence for a long time. "Wise one," he finally said, "we have passed so many prohibitions: one cannot create energy, one cannot take heat from the ocean for work... Is there any prohibition left?"*

1. The third law: absolute zero is unattainable

Sage: It remains. Remember, in chapter 2 we talked about absolute zero as the lower limit of temperatures? Now we can say more precisely about it. In 1906, the German physical chemist **Walther Nernst** discovered that as the temperature decreases, the change in entropy in any processes approaches zero. Max Planck formulated this more strongly in 1911: the entropy of a perfect crystal at $T \to 0$ is equal to zero.

$$S \to 0 \text{ as } T \to 0$$

Sage: Here is a direct consequence, which is **called the principle of** unattainability: no finite number of steps can cool a body to absolute zero.

Young man: But why? If it is possible to cool down to 1 K, why can't we cool down a little more, and a little more, and a little more?

Sage: Because of entropy. Any cooling method is structured as a cycle: first, we reduce the entropy of the body (for example, by compressing it in contact with a reservoir), then we isolate it and "change the parameter," the temperature drops at constant entropy. But according to the third law, the entropy curves at different parameter values converge at the point T = **0 (there** S is the same and equal to zero). The closer to zero, the smaller the gain at each step — and an infinite number of steps are needed.

Young man: And what about heat capacity? We know that $S = \int C\, dT/T$. If C did not tend to zero as $T \to 0$, entropy would diverge.

Sage: You hit the nail on the head: the **third** law requires that the heat capacity of all substances approaches zero. Classical physics cannot provide this: according to Dulong and Petit, C = 3R at any temperature. The solution was provided by quantum theory: in 1907, Einstein, and in 1912, Debye calculated the heat capacity of a crystal, considering the lattice vibrations as quantized.

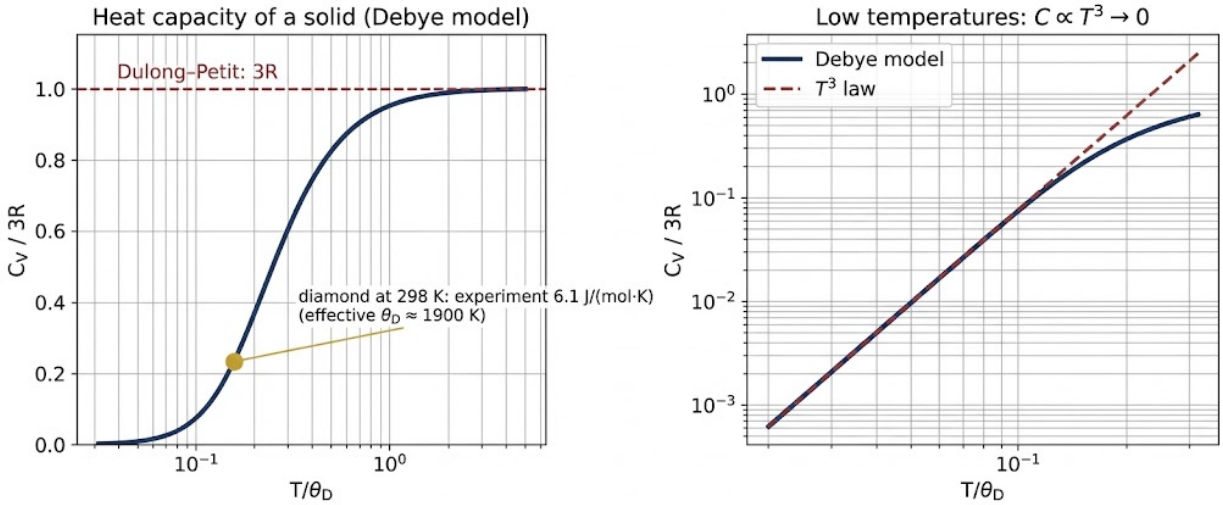


Fig. 1. Heat capacity of a solid according to the Debye model. At high temperatures — the value of Dulong and Petit 3R, at low temperatures — the T^3 law. The point — diamond at room temperature.

Young man: Here is the solution for diamond from chapter 6! At room temperature, its heat capacity is only about 6 J/(mol·K), four times lower than 3R ≈ 25: diamond has a very rigid lattice, and the vibrations at 300 K are still "frozen."

> **[UV] Established fact — the law T^3 and residual entropy**
>
> At low temperatures, the molar heat capacity of a dielectric crystal is proportional to T^3 (Debye's law) and approaches zero, as required by the third law. The ratio of heat capacities at different temperatures can be easily verified: when T is doubled from 10 to 20 K, the heat capacity increases by 8 times.
>
> There are exceptions to the "residual entropy" — when disorder remains in the crystal, "frozen" during cooling. A classic example is ordinary ice: the positions of protons in hydrogen bonds allow for many variations. Pauling calculated the number of configurations in 1935 as W = (3/2)N, from which $S_0 = k \cdot \ln W = R \cdot \ln(3/2) \approx 3.37$ J/(mol·K). Calorimetric measurements by Giocca and Stout (1936) yielded about 3.4. This is a beautiful confirmation of Boltzmann's formula from chapter 9.

2. How to cool almost to zero

The sage: The practice of achieving low temperatures is a story of ingenuity. Liquid helium (4.2 K) was obtained by Kamerlingh Onnes in 1908. By pumping out helium vapor, the temperature is lowered to 1 K. Further **down are dilution refrigerators** using a mixture of isotopes He-3 and He-4: down to several millikelvins (the idea of London in 1962, the first installations in the 1960s). **Even lower — adiabatic demagnetization (the** idea of Debye and Gioc, 1926–27): magnetic moments are ordered in a field, then the field is slowly removed, and the substance cools down.

The sage: Since the 1980s, laser **cooling of atoms** has been added (Chu, Cohen-Tannoudji, Phillips, Nobel Prize 1997): an atom slows down by absorbing an incoming photon. In 1995, the first Bose-Einstein condensate was obtained in this way, **and then through evaporative** cooling, at a temperature of about 170 nK (Cornell, Wieman, Ketterle, Nobel Prize 2001). In experiments with ultracold atomic gases, temperatures on the order of tens of picokelvins are achieved.

Young man: And what about superconductivity? I heard it also occurs at low temperatures.

The sage: In 1911, Kamerlingh Onnes cooled mercury in liquid helium and discovered that at 4.2 K its electrical resistance **disappears** — abruptly. This is superconductivity. The theoretical explanation (Cooper pairs of electrons) was provided by Bardeen, Cooper, and Schrieffer only in 1957. This is yet another example of a second-order phase transition and another manifestation of quantum physics on macroscopic scales.

3. A fluid that knows no friction

Sage: The strangest substance on Earth is helium. At normal pressure, it **does not** freeze even at absolute zero. Why?

Young man: From chapter 10: helium atoms are attracted very weakly — their potential well is only about 10 K. And the atom is very light.

Sage: Correct. From quantum mechanics, it follows that a particle confined in a region of size a cannot have zero energy — it has a zero-point **energy of the** order of $\hbar^2/(m \cdot a^2)$. For light atoms with a small well, this energy is so large that the lattice "blurs." To obtain solid helium, a pressure of at least 25 atmospheres (about 2.5 MPa) is required.

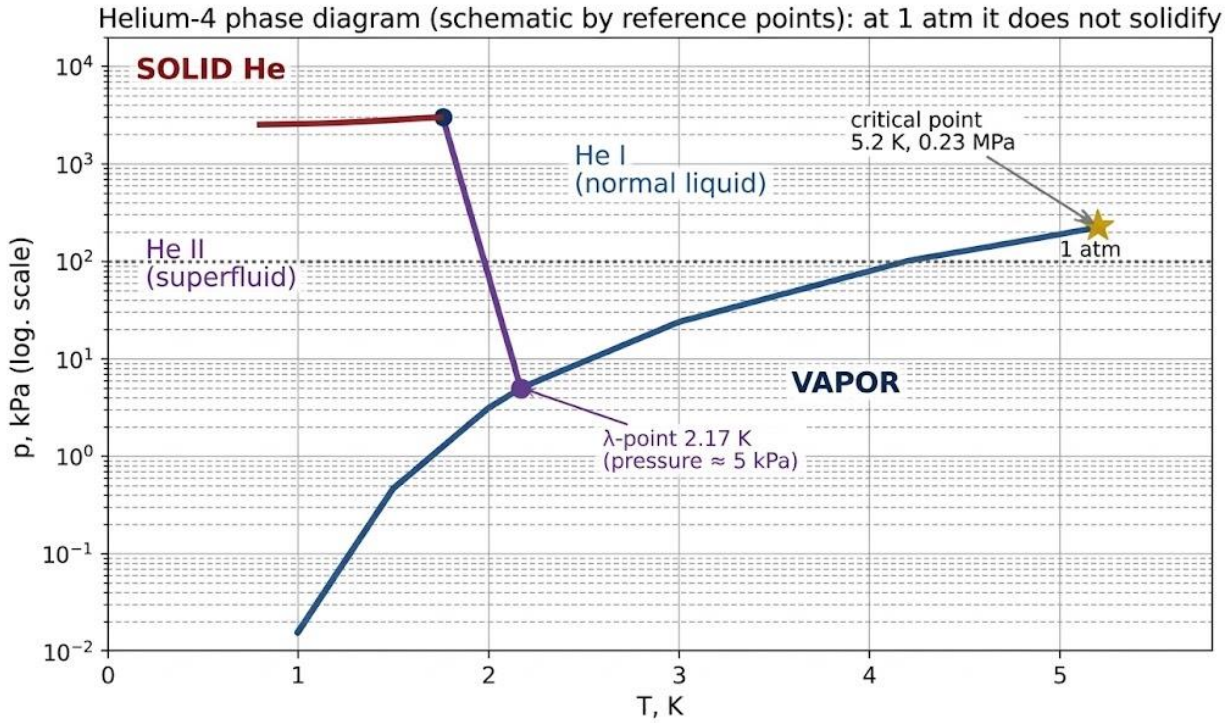


Fig. 2. Diagram of the state diagram of helium-4 based on reference points. The liquid remains liquid at a pressure of one atmosphere down to zero temperature; below 2.17 K it becomes superfluid.

The sage: In 1938, Pyotr Leonidovich Kapitsa (and independently Allen with Misner) discovered that below 2.17 K, liquid He-4 flows through the thinnest slits without viscosity. **It leaks** from the vessel, crawling along the walls in a thin film; in it, perpetual vortex currents are possible. **This is superfluidity.**

> **[UF] The established fact - what is known about superfluidity**
>
> In 1941, Lev Landau developed a theory: helium II behaves like a mixture of two "liquids" — normal (with viscosity and entropy) and superfluid (without viscosity and without entropy). Fritz London in 1938 linked the phenomenon to Bose-Einstein condensation: He-4 atoms are bosons, and at low temperatures, many of them occupy the same quantum state. Strictly speaking, in liquid helium at T → 0, there are only about 10^5 atoms in the condensate (due to interactions), so the picture of the "ideal Bose gas" is only an approximation.
>
> The isotope He-3 (its atoms are fermions) is superfluid only at millikelvin temperatures (around 2 mK, Osherov, Richardson, Lee, 1972): He-3 atoms (fermions) at such temperatures form pairs similar to Cooper pairs of electrons in a superconductor. Nobel Prizes: Landau (1962), Kapitsa (1978), for He-3 (1996).

4. Where does quantum physics come from?

Young man: You keep saying "quantum theory has solved." But where did it even come from? Not from nothing, right?

Sage: Exactly from thermodynamics! Let's gather the three "cracks" in classical physics that we have seen throughout the book.

- **Black body radiation (chapter** 12): the Rayleigh-Jeans formula diverges in the ultraviolet. Planck derived the correct formula in 1900 by assuming that energy is emitted in quanta of hν. Einstein went further in 1905: light **consists** of such quanta (photons).

- **Heat capacity** (chapters 6 and 13): the degrees of freedom are "frozen" because the energy of vibrations and rotations does not take on arbitrary values.
- **Gases at low temperatures: Bose** (1924) and Einstein (1924–25) developed statistics for particles with integer spin, while Fermi and Dirac (1926) developed statistics for particles with half-integer spin that obey the Pauli principle.

Sage: The criterion when quantum effects become noticeable is very simple: compare the thermal **de Broglie wavelength** $\lambda = h/\sqrt{(2\pi mkT)}$ with the distance between particles. If λ is much smaller than the distance — the gas is classical, if comparable — quantum effects.

System	T	Wavelength λ	The distance between particles	Mode
Nitrogen, 1 atm	300 K	0.019 nm	≈ 3.4 nm	classical
Liquid helium-4	4 K	0.44 nm	≈ 0.36 nm	quantum
Electrons in copper	300 K	≈ 4.3 nm	≈ 0.23 nm	strongly quantum (degenerate)

Young man: The electrons in copper at room temperature are quantum! Is that why metals have such a small electronic heat capacity?

Sage: Yes. Electrons obey the Pauli principle: two cannot occupy the same state, and the entire "crowd" fills the levels up to the Fermi energy. For copper, it is about 7 eV, which corresponds to a Fermi temperature of about 80,000 K. At 300 K, the thermal energy is only present in a small fraction of electrons near the upper level, and the contribution to the heat capacity is small — exactly what classical Drude theory could not explain.

[OD] Justified addition — how many quanta in the mode: Bose-Einstein distribution

For the oscillatory (or electromagnetic) mode of frequency ν, the average number of quanta at temperature T is given by n = 1/(exp(hν/kT) − 1). When hν ≪ kT, this simplifies to n ≈ kT/hν — the classical limit (equal distribution). When hν ≫ kT, the mode is "frozen" (n ≈ 0). Quantum effects become noticeable when hν is on the order of kT or greater.

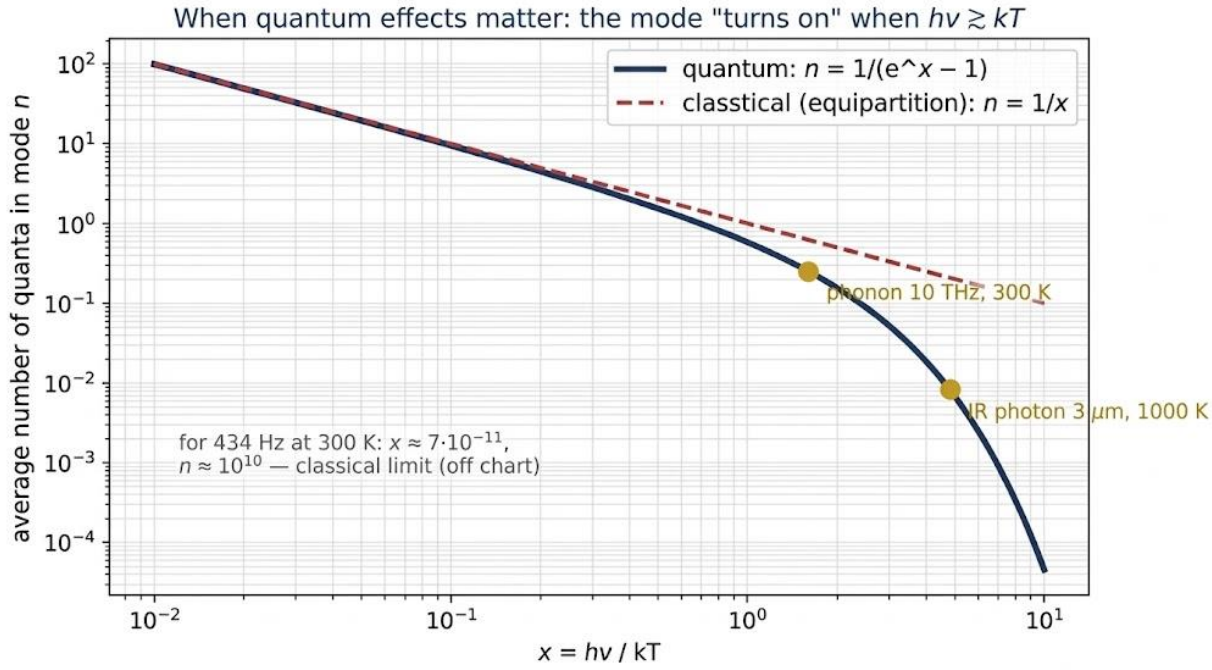


Fig. 3. The average number of quanta in the mode as a function of hν/kT. For ordinary vibrations of the crystal at room temperature, the number of quanta is on the order of one.

Quantum	Frequency ν	x = hν/kT (300 K)	The average number of quanta n
Pulse repetition frequency	434 Hz	≈ $7 \cdot 10^{-11}$	≈ $1.4 \cdot 10^{10}$ (classical)
Hypersonic phonon	1 GHz	≈ $1{,}6 \cdot 10^{-3}$	≈ 600
Optical phonon of the crystal	10 THz	≈ 1,6	≈ 0,25
IR quantum 3 μm	100 THz	≈ 16 (at 1000 K: 4.8)	≈ 10^{-7} (at 1000 K: ≈ 0.008)

[GA] The author's hypothesis - what does "accumulation of phonons" mean for ITE - a clarification question.

The pulse repetition frequency of 434 Hz is not the frequency of phonons: lattice vibrations lie in the terahertz range, and at room temperature, in the optical mode, it is on average about 0.25 quanta (and significantly more at high temperatures). Therefore, "accumulation" should be described as an increase in the number of quanta in **the necessary modes or as** a reserve of energy in lattice vibrations; the repetition frequency of 434 Hz should be considered as the rate at which this reserve is released.

The first principle limits such a scheme from above: the energy of the impulse is no more than the average power divided by the repetition frequency (≈ 2.3 J per kW at 434 Hz, chapter 6). What exactly is accumulated, in which modes, and up to what number of quanta — this is what the author should specify in the text and confirm with measurement. Status — [G].

5. Black holes: thermodynamics at the edge of physics

The sage: Now the most unusual place on our journey. In the early 1970s, Jacob Bekenstein and then Stephen Hawking showed that black holes have entropy and temperature.

Young man: Black holes? But they don't emit anything, they're "black"! How can they have a temperature?

The sage: Hawking in 1974 took into account quantum effects near the event horizon and concluded that a black hole **emits as** a body with a temperature inversely proportional to its mass:

$$T_H = \hbar c^3 / (8\pi G M k)$$

The sage: Entropy is proportional to the area **of the horizon** A:

$$S = k c^3 A / (4 G \hbar)$$

Young man: I will calculate for the Sun: $M = 2 \cdot 10^{30}$ kg. $T_H \approx 6 \cdot 10^{-8}$ K — 60 nanokelvins! And the relic radiation is 2.7 K. Thus, a black hole of such mass currently absorbs more from space than it emits.

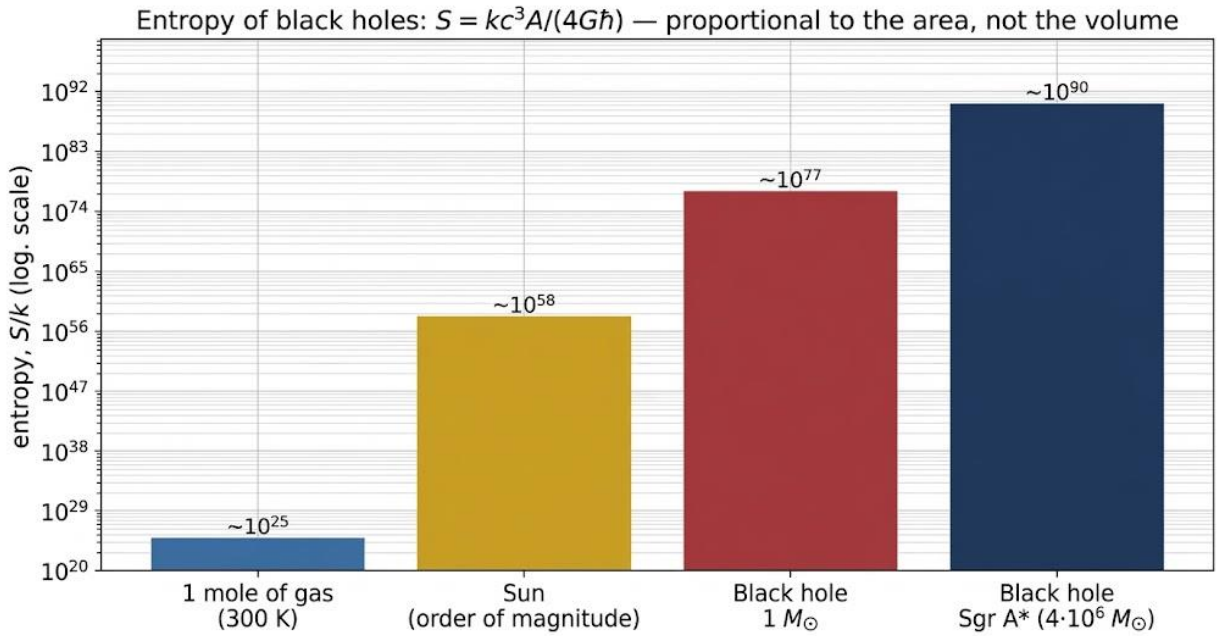


Fig. 4. The entropy of black holes (in units of k) compared to the entropy of a mole of gas and the Sun.

The sage: Indeed, and the entropy is impressive: for one solar mass $S/k \approx 10^{77}$, which is 10^{19} times greater than the entropy of the Sun itself. Black holes are among the most "entropic" objects known to physics. Another strange property: a black hole loses mass — and **heats up: it** has **negative heat capacity. The lifetime** of a solar black hole due to radiation is on the order of 10^{67} years.

> **[G] Hypothesis — the status of these results**
>
> The Bekenstein-Hawking formulas have been derived theoretically and constitute an accepted part of modern theoretical physics, but Hawking radiation for astrophysical black holes has not been **observed (it** is too weak), and there is no direct experimental verification of it. Laboratory analogs (for example, in acoustic "black holes") only test the mathematical aspect.
>
> This is an example of a domain where thermodynamics, quantum theory, and gravity converge, and where physicists do not yet know the definitive answers.

Young man: What is the "information paradox" that was mentioned in the program?

The sage: Hawking noted in 1976: the radiation of a black hole is thermal, that is, "random," and does not carry a trace of what fell into it. But quantum mechanics requires that information about the initial state does not disappear without a trace. What happens to the information when the black hole has evaporated?

Sage: The connection with chapter 9: entropy is a measure of missing information, and Landauer showed the cost of erasing information. A black hole compels us to pose this question in the most stringent form. From entropy, which is proportional to area rather than **volume, the holographic idea** was born (t'Hooft and Susskind, 1990s): information about everything within a region is recorded on its boundary.

> **[G] The hypothesis - the paradox is not closed**
>
> In recent years (since 2019), theoretical works have produced results consistent with information preservation (the so-called "Page curve," "islands"), but these are calculations in models, not experiments, and the discussion continues. For a school book, a fair formulation is: the question remains a subject of research.

6. Three beginnings and what they do not say

Sage: Let's gather the entire book into one table.

Beginning	What is stated	What is prohibited
First	Energy is conserved: $\Delta U = Q - A$	Perpetual motion machine of the first kind: work from nothing
Second	The entropy of an isolated system does not decrease: $\Delta S \geq 0$	Perpetual motion machine of the second kind; spontaneous transfer of heat from cold to hot.
Third	As $T \to 0$, entropy approaches zero, $C \to 0$.	Achieving absolute zero in a finite number of steps

Young man: Three principles. And what we discussed in the last chapters: barriers, speed, transfer — do they also relate to that?

Sage: No, and this is the most important conclusion. The three **principles state** what **is possible** and what the limit is. They **say nothing** about how quickly the system moves **towards equilibrium** and by what means. This is the realm of kinetics: barriers, diffusion, transfer, radiation. It is precisely here that technologies have room to invent ways to approach the limit set by thermodynamics more quickly and economically.

> **[OD] Justified addition — the outcome for our technologies**
>
> For drying, ceramic synthesis, and photocatalytic water decomposition, thermodynamics sets energy minima (heat of vaporization, ΔG, ΔH). A specific technology benefits when it approaches the minimum: it spends less on heating "excess" mass, air, equipment, and operates faster. The claimed advantages of the method need to be verified by two questions: does the claim violate the limits, and in what *way is the method closer to the limit than the alternatives? This is precisely how we* constructed the tables in Chapters 9 and 12.

7. Instead of an epilogue

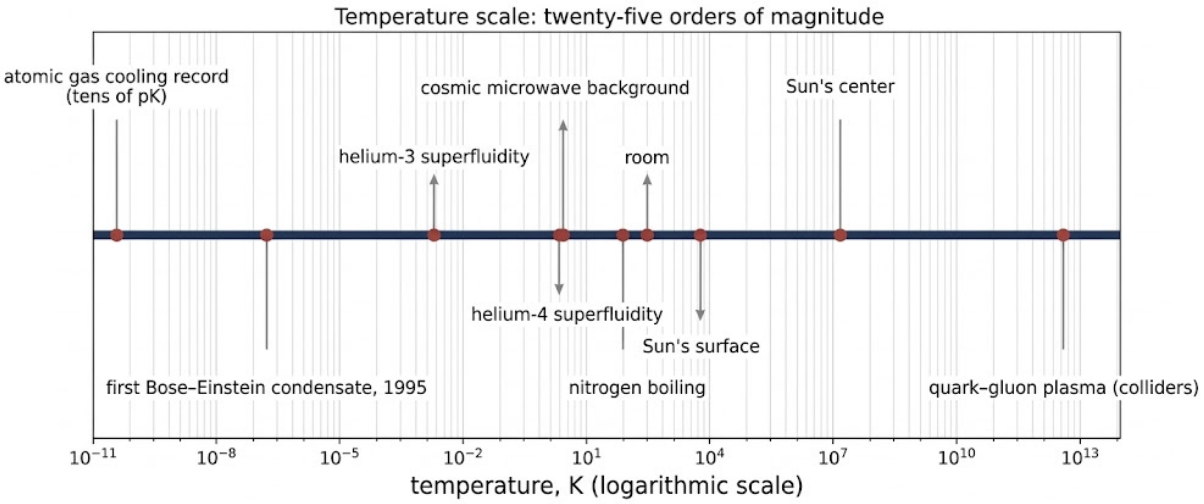


Fig. 5. Temperature scale: twenty-five orders of magnitude — from atomic gases cooled to nearly zero, to plasma in accelerators.

Young man: We started with the question "What is everything made of?" — with Democritus and atoms. And now it turns out that even simple heat and temperature lead to quanta, black holes, and information.

Sage: Notice how this happened. Each time, someone stopped in front of a fact that did not fit the picture: in front of a cannon barrel that heated up without fire; in front of heat capacity that "froze"; in front of a spectrum that diverged in the ultraviolet. They did not say, "probably a mistake in the experiment." They asked, "but why exactly this way?"

Young man: So, the best questions arise precisely where the formulas stop working?

Sage: Exactly. Don't be afraid of twos for extra questions. You have many "unclear places" ahead of you — and perhaps one of them is yours.

Problems for independent solution

- By what factor will the heat capacity of a dielectric crystal change when cooled from 20 K to 10 K (according to the T^3 law)?
- Find the residual entropy of ice using the Pauling formula $S_0 = R \cdot \ln(3/2)$.

- The Hawking temperature for a black hole with a mass of 1 M⊙ is approximately $6.2 \cdot 10^{-8}$ K. Find it for a black hole with 10 M⊙.
- The thermal wavelength of He-4 at 4 K is 0.44 nm. What is it at 1 K?
- The average number of quanta in the mode when hν/kT = 2.

> **[UV] Established fact — answers to the problems**
>
> 1) It will decrease by 8 times ($C \propto T^3$: $(20/10)^3 = 8$).
>
> $S_0 = 8.314 \cdot \ln 1.5 = 8.314 \cdot 0.4055 \approx 3.37$ J/(mol·K).
>
> 3) $T \propto 1/M$, therefore $T = 6.2 \cdot 10^{-8} / 10 = 6.2 \cdot 10^{-9}$ K = 6.2 nK.
>
> 4) $\lambda \propto 1/\sqrt{T}$: at 1 K, $\sqrt{4} = 2$ times greater, that is, ≈ 0.87 nm.
>
> 5) $n = 1/(e^2 - 1) = 1/6.39 \approx 0.16$.

Conclusion

What we have learned, what remains to be verified, and where to go next.

> *The rain had stopped, and it became bright in the yard. The young man flipped through the book from the very beginning — from Democritus to black holes — and closed it, holding the last page with his finger.*

1. What did the Young Man understand?

Young man: If I were to tell everything in one paragraph... Everything is made of atoms that move randomly. Temperature is a measure of their average energy. Energy is conserved — this is the first law. But processes have direction because there are infinitely more disordered states than ordered ones — this is the second law and Boltzmann's entropy. Absolute zero cannot be reached — the third law. And the speed at which a system reaches equilibrium depends on barriers, and here thermodynamics is silent.

Sage: You have said all the most important things. And notice what path you have traveled: in chapter 1 we argued about whether atoms exist, and in chapter 13 we were already discussing how atoms behave at a temperature of billionths of a degree above absolute zero.

2. The Path of Heat Science

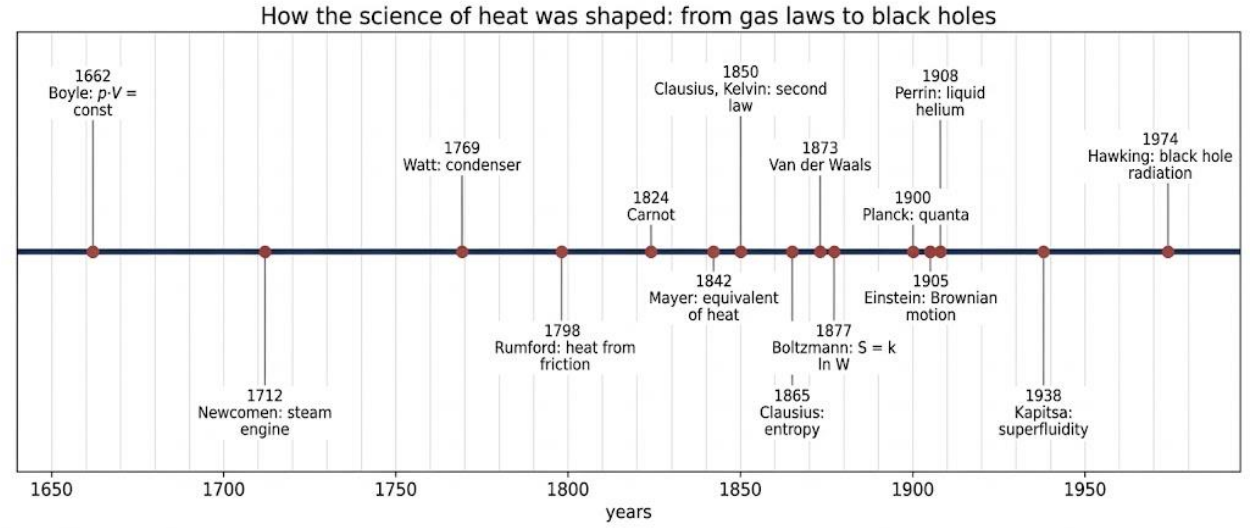


Fig. Z.1. Milestones: from the gas laws to black hole radiation.

3. Main ideas by chapters

Ch.	Topic	The main idea and formula
1	What everything is made of	The atomic hypothesis from Democritus to Perrin; Brownian motion as evidence
2	Temperature	Scales and absolute zero; $T = t + 273.15$
3	Gas from balloons	Pressure is the collisions of molecules; $pV = \nu RT$
4	Molecules in motion	Kinetic theory; $E_{kin} = (3/2)kT$; Maxwell distribution; mean free path
5	What is heat?	Heat is not a substance (caloric), but the transfer of energy; equivalent ≈ 4.18 J/cal
6	First principle	$\Delta U = Q - A$; $C_p - C_V = R$; adiabate $pV\gamma = \text{const}$
7	Heat engines	$\eta_{Carnot} = 1 - T_2/T_1$; Carnot cycle; heat pump $K = T_1/(T_1-T_2)$
8	The second law and entropy	$dS = \delta Q_{обр}/T$; $\Delta S \geq 0$ for an isolated system; irreversibility and the arrow of time
9	Entropy and probability	$S = k \cdot \ln W$; Boltzmann factor $\exp(-E/kT)$; Arrhenius law; Maxwell's demon
10	Liquids, vapor, crystals	$\Delta p = 2\sigma/r$; saturated vapor; critical point; crystals and glasses
11	Phase transitions	$T_{пер} = \Delta H/\Delta S$; phase diagrams; Van der Waals equation; liquefaction of gases
12	Transport phenomena	$\langle x^2 \rangle = 2Dt$; Fourier's law; viscosity; $j = \varepsilon\sigma T^4$; $\lambda_{max} \cdot T = 2898\ \mu m \cdot K$ (for thermal radiators)

Ch.	Topic	The main idea and formula
13	Limits and Continuation	Third law; superfluidity; quanta; entropy of black holes $S = kc^3A/(4G\hbar)$

4. What is established and what remains a hypothesis

The sage: An honest conclusion requires dividing the book into two parts. The first is conventional physics: laws, formulas, experiments. It has been validated by thousands of experiments, and we relied on sources and calculations in it. The second is the author's ideas about the ITE method, which we examined throughout the book against the backdrop of this physics.

> **[UV] The established fact is that in the book it pertains to established science.**
>
> The three laws of thermodynamics; the statistical meaning of entropy; the Arrhenius law and the Boltzmann factor; transfer regularities (diffusion, thermal conductivity, radiation); phase transitions and state diagrams; heat capacity and its "freezing" at low temperatures. Here, discrepancies are possible only in details and numerical values.

> **[GA] The author's hypothesis — what pertains to the author's hypotheses about ITE**
>
> In drying and other applications (according to the author's design): the color of the pulse determines the steepness of the front, not the temperature of the emitter; a fast front allows the energy to reach the water, not the **organic matter; the** ITE overcomes the energy barrier by supplying enough energy for the transition, rather than lowering it like a catalyst; the heat of the lamp participates in the reaction, rather than being wasted.
>
> These statements do not contradict the three principles: ITE, like any method, is limited by a theoretical boundary, and its advantage lies in a more efficient approach to the boundary than alternatives. However, they remain hypotheses until confirmed by independent measurements. Below is what exactly needs to be measured.

5. What to measure: a summary of checks from the chapters of the book

In different chapters, we formulated questions that the experiment will answer. Here they are compiled into a single list:

№	Question	Chapter	What to measure	What does it check?
1	How much energy per kilogram of water is consumed by drying with ITÉ and traditional methods?	6, 9	Power · time · mass of removed water under the same conditions	The approximation to the limit is ≈ 2.5 MJ/kg.
2	What does "efficiency" of a hydrogen reactor mean: output by substance or by energy?	6, 8	Volume of H_2 (chromatography), total supplied power	Energy efficiency
3	What are the contributions of light and heat from the lamp in the reactions?	6, 8	Experiments at different ratios of light and heat	Filling in the balance sheet
4	Transmission spectrum of the product at different drying stages; absorption share in the "windows"	9	Reflection and transmission spectra of the product layer	Explanation of deep penetration
5	Temperature and moisture profiles through thickness; continuous and pulsed mode at a constant average power	9, 12	Thermocouples by layers, slices with weighing	Volume, the role of the envelope
6	The range of temperature fluctuations on the surface under impulses	9	IR camera, fast sensors	Thermal contribution of pulsation
7	Shape and duration of the front; spectrum at different temperatures of the emitter	12	Fast photodiode, oscilloscope, IR Fourier spectrometer	Independence of "color" from temperature
8	Phase transition shift (anatase → rutile) during pulsed heating	11	X-ray phase analysis of samples processed in two modes	Overcoming the barrier with the front
9	What quanta accumulate and in which modes?	13	Spectrum and dynamics of radiation, energy per pulse	Correct formulation of the mechanism
10	Where does entropy go when heat is converted into ordered impulses?	8	Thermal balance of the radiator and the casing	Agreement with the second law

Sage: Notice: some of these measurements are simple — mass, time, power, temperature. It is these that will provide the first reliable answers. This is honest science: not to prove an idea with words, but to propose an experiment whose result may refute it.

6. What remains beyond

We left many topics outside the scope of the book: nonequilibrium thermodynamics and self-organization (Prigogine), Shannon's information theory, detailed theory of phase transitions and critical phenomena, non-ideal solutions and electrochemistry, thermodynamics of biological systems. Any of these topics could have become a separate book. Some of them we only touched upon hinting.

Continuation in related fields — in other books of the series: on mechanics, acoustics, optics, quantum mechanics, chemical kinetics and catalysis, electrodynamics, and the history of misconceptions in physics and chemistry (series plans).

7. Instead of the last word

Young man: Sage, what should I do next?

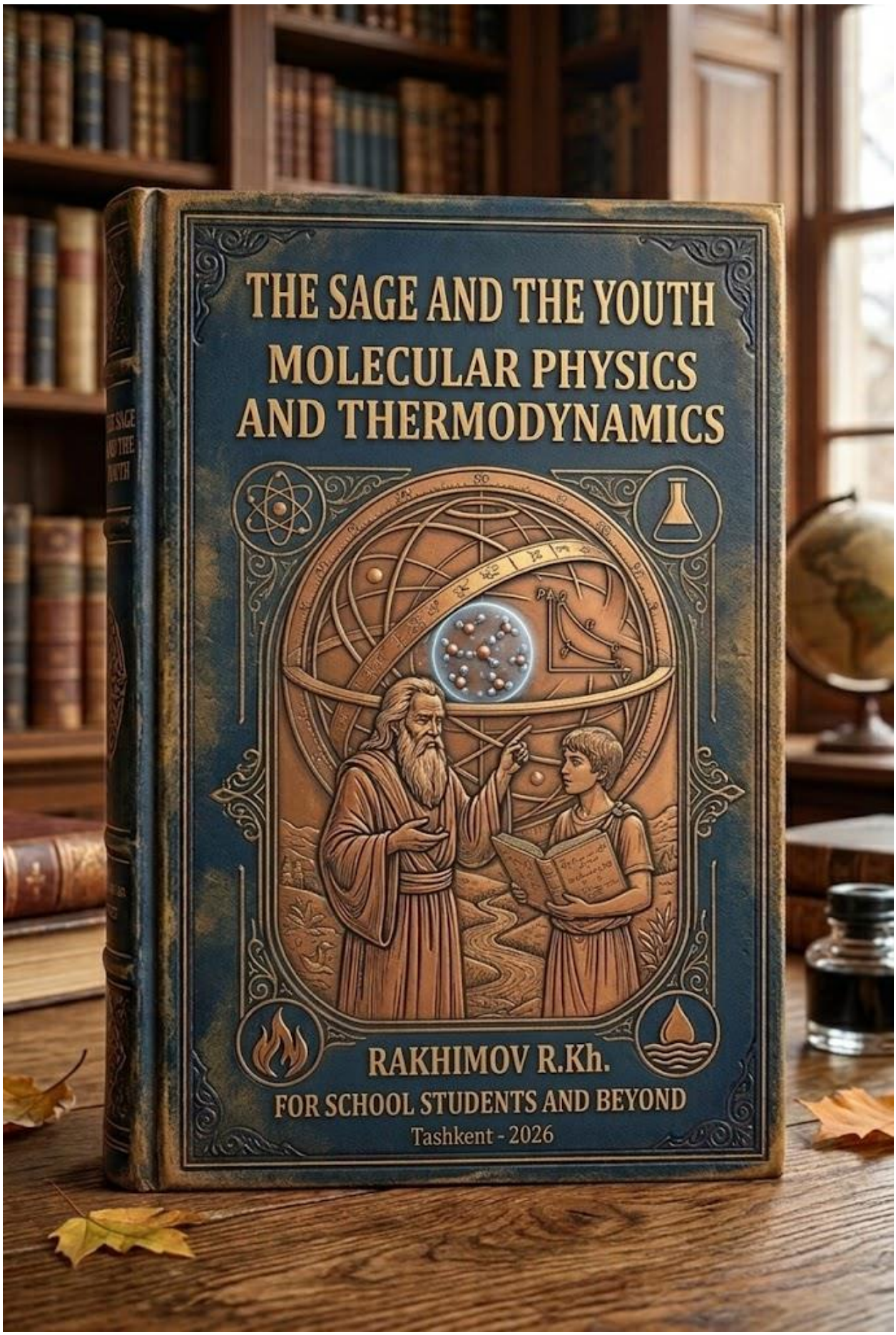
Sage: Keep asking "unnecessary" questions. But now you have a tool: before believing a beautiful statement, ask whether it violates the limits, and in what way it is closer to the limit than everything else. If an answer is found — that is new knowledge. If not — that is the next research.

Young man: Sounds like a lifelong assignment.

Sage: That's right. Don't be afraid: every great discovery once had its "failing student" who stopped where everyone else moved on.

Appendix: constants and formulas

Magnitude	Designation	Value
Boltzmann constant	k	$1.380649 \cdot 10^{-23}$ J/K
Avogadro's number	N_A	$6.02214076 \cdot 10^{23}$ mol^{-1}
Universal gas constant	$R = N_A \cdot k$	8.314 J/(mol·K)
Planck's constant	h	$6.62607015 \cdot 10^{-34}$ J·s
Speed of light	c	$2.99792458 \cdot 10^{8}$ m/s
Stefan-Boltzmann constant	σ	$5.670 \cdot 10^{-8}$ $W/(m^2 \cdot K^4)$
Wien's law constant	$\lambda_{max} \cdot T$	2898 μm·K
Calorie	1 cal	4,184 J
Acceleration due to gravity	g	9.81 m/s²
The heat of fusion of ice	L	334 kJ/kg
The heat of vaporization of water (100 °C / 25 °C)	L	2257 / 2442 kJ/kg
Specific heat capacity of water	c	4.18 kJ/(kg·K)


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